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विद्युत मंत्रालय / Ministry of Power
पूर्वी क्षेत्रीय विद्युत समिति / Eastern Regional Power Committee

सं/NO. ERPC/EE/OPERATION/2026/ 569

DATE:19.06.2026

सेवा में /To
संलग्न सूची के अनुसार /As per list enclosed.

विषय : 12.06.2026 (शुक्रवार) को ईआरपीसी सचिवालय, कोलकाता में भौतिक रूप से आयोजित 240वीं ओसीसी बैठक के कार्यवृत्त – के संबंध में

Sub: Minutes of 240th OCC Meeting held on 12.06.2026 (Friday) physically at ERPC Secretariat, Kolkata – reg

महोदय/महोदया,
Sir(s)/Madam,

कृपया अपनी जानकारी और आवश्यक कार्रवाई के लिए 12 जून 2026 (शुक्रवार) को आयोजित 240वीं ओसीसी बैठक के कार्यवृत्त संलग्न देखें। यह ईआरपीसी वेबसाइट (www.erpc.gov.in) पर भी उपलब्ध है।

Please find enclosed **Minutes of 240th OCC Meeting** held on 12.06.2026 (Friday) physically at ERPC Secretariat, Kolkata at 10:30 hrs. This is for your kind information and necessary action. The same is also available at ERPC website (www.erpc.gov.in).

टिप्पणियाँ, यदि कोई हों, कृपया यथाशीघ्र इस कार्यालय को अग्रेषित करें।
Observations, if any, may please be forwarded to this office at the earliest.

इसे सदस्य सचिव के अनुमोदन से जारी किया जाता है।
This issues with the approval of Member Secretary.

भवदीय /Yours faithfully


(B.S. Ray)

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Eastern Regional Power Committee

MINUTES OF 240th OCC MEETING

Venue: ERPC Secretariat, Kolkata

Date: 12.06.2026

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EASTERN REGIONAL POWER COMMITTEE

MINUTES OF 240th OCC MEETING HELD ON 12.06.2026 (FRIDAY) AT 10:30 HRS

Member Secretary, ERPC welcomed all OCC members and other participants who were present physically as well as those who had joined virtually to the **240th OCC Meeting**.

Thereafter he requested all the participants to give brief introduction.

□ He outlined the performance of ER grid during **May-2026** highlighting the following points:

• In **May-2026**, energy consumption of ER was **19578 MU** which is **2.9 % more** than **May-2025**.

• In **May-2026**, Peak demand met of ER was **34875 MW** which is **9 % more** than **May-2025**.

❖ Thermal **PLF** of **ER** during **May -2026** was **75 %** whereas National average is **71.5 %**.

❖ Generating stations whose PLF was **90%** or more during **May-2026**:

Utility	Generating station	PLF (%)
WBPDC	SANTALDIH TPS	90
IPP	HALDIA TPP (HEL)	97

❖ During **May-2026**, **77.3 %** of the time, the ER grid frequency was within IEGC Band (49.90Hz-50.05Hz).

Coal stock position (As on 07.06.2026) is as follows:

SL.	Name of States/Power Stns.	% of Actual Stock vis-à-vis Normative Stock
1.	Jharkhand (TVNL)	197%
2.	Odisha (IB TPS)	78%
3.	West Bengal (WBPDC)	67%
4.	D.P.L. TPS	30%
5.	DVC	80%
6.	NTPC	75%

He also stressed that during rainfall deficits, thermal generation is the main dependable resource, particularly in non-solar hours, and thus, must be available with maximum capacity.

ED, ERLDC delivered a comprehensive presentation (**Annex-A**) delineating historical trend analysis of monsoon demand w.r.t summer demand amid rainfall deficit and touched upon the following key facets:

□ Higher demand is generally observed in Monsoon season of years experiencing rainfall deficit

□ Around 25.4 GW of Thermal outages at All India level are presently scheduled directly in the vulnerability window (i.e. July to September 2026)

□ Around 9.7 GW of shortage at All India level is projected in the month of August 2026.

*List of participants attached at **Annex-B**.*

Member Secretary, ERPC requested S.E (Operation) to take up the agenda for discussion.

1. PART-A: CONFIRMATION OF MINUTES

1.1. Confirmation of Minutes of 239th OCC Meeting held physically at ERPC Secretariat on 13th May 2026

The Minutes of 239th Operation Coordination Sub-Committee meeting held on 13.05.2026 was circulated vide letter dated 18.05.2026.

Deliberations in the meeting

Since no comments have been received so far, the Minutes of 239th OCC meeting were confirmed.

2. PART-B: ITEMS FOR DISCUSSION

2.1 Update on follow up agenda: ERPC

a) Intrastate Transmission Network Assessment & Mitigation-Odisha

Reference:

Implementation of the Under Voltage Load Shedding (UVLS) scheme in the Odisha system has been under review since the 231st, 232nd, 233rd, and 234th OCC Meetings held on 22.09.2025, 24.10.2025, 22.11.2025, and 23.12.2025 respectively.

The matter was also discussed in the recently concluded 55th TCC/ERPC meeting held on 16.12.2025 and 17.12.2025 at Kalimpong, West Bengal. As per deliberation in 55th TCC Meeting, **300MW load** has been identified and it will be operationalized before **Summer'26**.

The continued delay in implementation is posing increasing risks not only to the Odisha system but also to the Eastern Region as a whole during the forthcoming Summer-2026 period.

- During a review meeting held on 30.04.2026, OPTCL and SLDC Odisha informed that the quotation from the existing ADMS vendor had been received and was under examination. Further, 31 feeders of TPCODL and 23 feeders of TPNODL have been identified under the scheme. The target date for completion was indicated as June-2026.

As per 239th OCC:

Odisha representative updated:

The vendor of ADMS has been entrusted for implementation of the UVLS in Odisha system and the same will be operational by end of June 2026.

239th OCC Decision

- OCC expressed serious concern over inordinate delay in implementation of UVLS in Odisha system, especially amid demand surge in ongoing Summer that renders intra-state network of Odisha vulnerable to low voltage condition.

- SLDC Odisha was advised to share weekly update on the status of implementing UVLS for identified loads around Bhubaneswar to ERLDC & ERPC
- OCC referred the matter to TCC for deliberation.

As per 56thTCC meeting:

Odisha updated that 400 MW load has been identified for UVLS scheme & the scheme will be implemented by June-2026.

During deliberation, ERLDC highlighted the transmission constraints in Odisha network and stated that the reconductoring of 400 kV OPGC-Lapanga-Meramundali section shall be completed as long-term solution to mitigate the transmission constraint issue in Odisha system.

CEA also pointed out that in transmission RA plan of Odisha, the reconductoring of intra-state lines will be completed by OPTCL in the timeframe of 2026-27. However there is no progress in this regard. CTUIL also submitted the delay in strengthening of intra-state networks by OPTCL will result in ISTS system vulnerable in future.

56th TCC Decision:

Forum advised OPTCL to implement the UVLS scheme by June-2026 without any further delay.

OPTCL would submit a timeline for carrying out reconductoring of their intra-state networks as finalized in the resource adequacy plans.

As per 56th ERPC meeting:

Odisha officials were not present in the preceding (ERPC) discussion.

ERLDC highlighted the urgency of early completion of this scheme for voltage stability of Bhubaneswar region.

56th ERPC Decision:

ERPC forum advised OPTCL to implement the UVLS scheme by June-2026 without any further delay. Detailed work plan for reconductoring of intrastate transmission line shall be shared with ERPC & ERLDC within two weeks.

Deliberations in the meeting

OPTCL informed that the work was under progress and would require approximately one additional month for completion. It was stated that the execution is being carried out through the on-going ADMS project.

OCC Decision

Forum strongly urged OPTCL to complete the UVLS implementation by **June 2026** by treating UVLS as a priority project for the capital region of Bhubaneswar.

b) Status of ER ULDC Phase-III SCADA/EMS Up-gradation Project (SCOD 18 months from date of LOA)

POWERGRID is executing the Project for up-gradation of Main and Backup SLDCs under implementation of ULDC Phase-III Project in Eastern Region. NOA is issued to M/s. GE Vernova T&D India Ltd. on **02.08.2024** for execution of this work.

Supply is completed for majority of control centers except WBSETCL control centers which are undergoing Functional & Integrated FAT with ERLDC System by Grid-India. Installation works are also completed at some of the sites and undergoing at the remaining sites based on site readiness by constituents.

19th TeST Deliberation:

ERPC enquired about the progress of the SCADA upgradation work in the Eastern region. M/S GE Vernova presented the updated progress status as given below:

ERLDC:

- ERLDC apprised the forum regarding the continuous delay in execution of the project and conflicts concerning allocation of responsibilities within the project implementation team. It was observed that ambiguity and misunderstanding regarding the scope of work at the time of project award have been the major reasons for the delay in project execution.
- ERLDC further emphasized that upgradation of the ERLDC system is required on priority for seamless integration with all entities and proper functioning of the overall system. Since the AMC of the existing system is not being renewed further, it is essential to execute the work in mission mode to ensure commissioning of the upgraded system before complete stoppage of the current system.
- POWERGRID and M/S GE Vernova expressed willingness to explore the possibility of establishing ICCP link connectivity with the existing ERLDC system till the ERLDC system upgradation is completed by GE.
- In this regard, ERLDC apprised that such an arrangement may lead a contractual deviation. However, feasibility of the same may be explored, ensuring that the existing system which is running without OEM support must remain operational without impacting real-time grid operation activities.
- It was informed that at the Backup Control Centre, NR-UPS is already commissioned, following which load shifting activities shall be carried out by 20th May 2026.. Subsequently, ER-UPS is targeted to be commissioned by 5th June 2026.
- The forum requested ERLDC to ensure site readiness at both Main Control Centre (MCC) and Backup Control Centre (BCC), before May 2026.
- M/S GE Vernova is directed to bring the system live by 31st Aug, 2026 without further delay.
- M/S GE Vernova requested for timely payment may be provided so that the project execution can be completed within the stipulated timeline.

WBSLDC:

- M/S GE Vernova appraised the forum that the MCC system is dispatched on 11th May 2026 and must be arrived by 20th May, 2026 at the site.
- Installation & commissioning, ICCP integration with ERLDC, SAT are targeted to be finished by Aug, 2026.
- WBSLDC stressed to integrate weather forecasting API with newly upgraded system which was committed to be finished by Aug' 2026.
- It is also suggested to incorporate skilled software team along with residential team under AMC to the site for each state.

Sikkim: M/S GE Vernova updated that MCC will be functional by July, 2026 & BCC by Aug 2026, as work is in advanced stage without major challenges at present.

DVC: M/S GE Vernova informed that the MCC at Kolkata was inaugurated on 30th March 2026 and the complete handover is likely to be done by August 2026.

Bihar:

Bihar raised query regarding migration of historian data to the new system for smooth & efficient operation.

GE, Vernova apprised that team is working on conversion tool which can translate old historian data from OSI to new SCADA system of GE. The work is in progress and shall be confirmed to all states and ERLDC within fortnight.

Odisha: raised queries about deployment of efficient skilled manpower at the site for expedition of the work. Further, availability of DG set at site shall also be ensured by end of May, 2026.

19th TeST Decision

- ✓ The forum advised POWERGRID, M/S GE Vernova, and ERLDC to jointly examine the feasibility of the proposed arrangement and take necessary action in a coordinated manner without affecting real-time grid operations.
- ✓ The forum further emphasized that no further extension shall be granted to M/S GE Vernova for completion of the project, as the execution is already significantly delayed beyond the stipulated timeline by around six months (half a year) and must be operational by 31st August, 2026.
- ✓ M/S GE Vernova was advised to deploy adequate skilled manpower at site for faster execution of the work.

As per 56th TCC meeting

TCC raised serious concern over the inordinate delay in commissioning of SCADA at ERLDC and at SLDCs.

56th TCC Decision:

- ✓ TCC Forum advised Powergrid to form a working level group consisting of representatives from ERPC, Powergrid, M/s GE, ERLDC, concerned SLDC & Chemtrol for speedy implementation of the project and smooth transfer of data from old system to new system.
- ✓ Forum suggested that the matter may be followed up in the forthcoming OCC meetings.

As per 56th ERPC meeting

Member GO&D highlighted the importance of ULDC-III for grid security and reliability. Further, MS, ERPC emphasized the requirement of new SCADA system for all SLDCs/ERLDC as the existing AMC is going to expire soon.

56th ERPC Decision:

- ✓ ERPC Forum advised Powergrid to form a working group and ensure completion of new SCADA system before the expiry of the existing AMC and to submit the progress report to Member Secretary ERPC on weekly basis.
- ✓ Further, Forum strongly recommended that M/s GE Vernova to ensure deployment of sufficient manpower at ERLDC and SLDCs for speedy completion of the SCADA system and smooth transition to new system.

Deliberations in the meeting

Concerned Vendor (GE Vernova) informed:

- ✓ *Functional patch is complete.*
- ✓ *100-hour performance test for ERLDC main and backup systems has started at GE Noida.*
- ✓ *Performance test is expected to be completed by **15th June 2026** followed by dispatch of equipment.*
- ✓ *Dispatch from factory expected by **16th June 2026**, packing and logistics will take 5-7 days; hardware expected to reach ERLDC by **end of June 2026**.*
- ✓ *Conduct of SAT for DVC is in final stage and shall be completed within a month.*
- ✓ *Progress of backup systems and RTU reporting at various state utilities was reviewed.*

State-wise status is as follows:

DVC: *Local RTUs reporting is completed, calculations & validation ongoing.*

Odisha: *All RTUs reporting has been completed.*

Bihar: *All RTUs reporting has been completed.*

West Bengal: *Hardware installed, UPS charged; some site issues being resolved jointly by WBSETCL and GE.*

- ✓ Backup SCADA systems for some states (OPTCL, BSPTCL) are installed and ready except for ICCP integration with ERLDC.
- ✓ Battery performance issues and replacement of deficient cells were also discussed.

Detailed status of ongoing and pending activities i.r.o. ER ULDC Phase-III project attached at **Annex B.2.1.b**

OCC Decision

- The Forum expressed serious concern over repeated slippages in timelines and highlighted the cyber-security and operational risks associated with continued operation of the old system.
- OCC affirmed that Performance test of ERLDC SCADA systems must be completed by 15th June, 2026 and the SCADA system should be dispatched by 16th June 2026 so that it may be installed at ERLDC by end-June 2026. Further, Powergrid was advised to ensure strict adherence to this revised target to make ER ULDC Phase-III project operational at the earliest.
- OCC also reiterated the decision of 56th ERPC for formation of working level coordination groups involving ERPC, ERLDC, Powergrid and respective state utilities for close monitoring of the milestone-wise targets (installations, ICCP, FAT/SAT etc.). Weekly progress reports on ER ULDC Phase-III project shall be submitted by Powergrid to ERPC.
- OCC also suggested for replacement and retesting of Battery cells failing acceptance criteria.

c) Upgradation of 220kV Network in Kolkata Region

Vide **217th OCC** dated 24.07.24 Due to persistent N-1 violation, Upgradation of these 220 KV feeders to be planned:

- ☐ 220kV Subhasgram (WB)-Lakshmikantpur D/C
- ☐ 220 kV Jeerat-Barasat D/C
- ☐ 220 kV Barasat-Kasba D/C

- In **229th OCC** Meeting, WBSETCL updated that the approval for PSDF grant has not yet been received.

They also apprised that only the following lines have been listed for upgradation.

- ✓ 220 kV Jeerat-Barasat D/C
- ✓ 220 kV Barasat-Kasba D/C
- ✓ 220 kV Subhasgram-Kasba
- ✓ 220KV Kolaghat-Foodpark
- ✓ 220KV Foodpark-Jangalpur

- OCC advised WBSETCL to expedite the upgradation work of above lines since it has already been delayed by more than 2 years and in case PSDF grant is not possible then other avenues of funding maybe explored since the loading of the above lines are going to be critical in the coming summer months.

As per 55th TCC meeting:

WBSETCL updated that tender has been done for upgradation of following lines:

- ✓ 220 kV Barasat-Kasba D/C
- ✓ 220 kV Subhasgram-Kasba D/C
- ✓ 220 kV Rajarhat-KLC S/C

- ♦ For remaining lines, the upgradation work will be taken up progressively as per the system requirement.
- ♦ They updated that the tentative timeline to complete the upgradation work of above three lines is **9(nine) months**.

As per 239th OCC:

WBSETCL updated:

- The tendering process was delayed due to restrictions during the State Assembly election period.
- Tendering process has already been issued for 220 kV Barasat–Kasba, 220 kV Subhashgram–Kasba, and 220 kV Rajarhat–KLC.
- Upgradation of 220 kV Kodalghat–Food Park and 220 kV Food Park–Jamalpur lines is still under internal approval.

239th OCC Decision

- ✓ WBSETCL was advised to submit the exact tender issuance dates along with revised timeline for upgradation i.r.o all the concerned 220 kV lines.
- ✓ Progress of the reconductoring works shall be henceforth reviewed in OCC meetings.

Deliberations in the meeting

WBSETCL updated the HTLS up-gradation status of feeders as follows:

SI No.	Feeder	Status
1	220 KV Barasat-Kasba D/C	LoA placed on 6/2/26
2	220 KV Subhasgram-Kasba D/C	LoA placed on 6/2/26
3	220 KV Jeerat-Barasat D/C	Ready for placement in next Board of Directors meeting for approval
4	220 KV Kolaghat-Foodpark	Ready for placement in next Board of Directors meeting for approval
5	220 KV Foodpark-Jangalpur	Ready for placement in next Board of Directors meeting for approval

OCC Decision

In view of criticality of these 220 kV lines for ensuring reliability in Kolkata region, WBSETCL was advised to expedite ongoing reconductoring work. Further, a detailed execution plan along with tentative timeline of completion to be submitted i.r.o reconductoring of lines for which LoA has been placed.

d) Spare Reactor procurement under Eastern Regional Pool as per CEA norms

In **52nd ERPC** meeting, estimated expenditure of **Rs. 55.67 Crores** (exclusive of GST but including transportation cost) was concurred towards procurement of spare reactors in ER pool by Powergrid ER-II as per CEA spare norms.

STATE	VOLTAGE	SIZE	STORAGE PLACE
WEST BENGAL	400 KV	125 MVAR	DURGAPUR SS
		80 MVAR	BINAGURI SS
SIKKIM	220 KV	31.5 MVAR	NEW MELLI SS

JHARKHAND	400 KV	125 MVAR	NEW RANCHI SS
ODHISSA	400 KV	63 MVAR	ROURKELLA SS

In **54th TCC** meeting, Powergrid intimated that the tender evaluation is on progress.

As per 55th TCC meeting:

- ✓ POWERGRID updated that tender has been awarded for the reactors except 31.5 MVAR reactor.
- ✓ Since the 31.5 MVAR reactor is seldom used at present, hence they have segregated it from the rest of the population and it will be done separately.

As per 239th OCC:

Powergrid appraised the details as follows:

STATE	VOLTAGE	SIZE	STORAGE PLACE	UPDATE
WEST BENGAL	400 KV	125 MVAR	DURGAPUR SS	Evaluation under progress and final award to be placed shortly.
		80 MVAR	BINAGURI SS	Procurement to be completed by July 2026
SIKKIM	220 KV	31.5 MVAR	NEW MELLI SS	In bidding stage.
JHARKHAND	400 KV	125 MVAR	NEW RANCHI SS	No update received
ODHISSA	400 KV	63 MVAR	ROURKELLA SS	No update received

239th OCC Decision

- ✓ Powergrid was advised to expedite procurement and update on progress to be shared henceforth in OCC meetings.
- ✓ Powergrid was also advised to share the current status in procurement of 125 MVAR & 63 MVAR reactors at New Ranchi & Rourkella substations respectively.

Deliberations in the meeting

Powergrid updated the status of procurement as follows:

STATE	VOLTAGE	SIZE	STORAGE PLACE	UPDATE
WEST BENGAL	400 KV	125 MVAR	DURGAPUR SS	LOA is awarded with delivery schedule of 2 years
		80 MVAR	BINAGURI SS	
SIKKIM	220 KV	31.5 MVAR	NEW MELLI SS	Purchase order has been placed for procurement. LOA to be placed by 30 th June 2026.
JHARKHAND	400 KV	125 MVAR	NEW RANCHI SS	
ODHISSA	400 KV	63 MVAR	ROURKELLA SS	

OCC Decision

OCC advised Powergrid to keep all spare reactors at designated substations of respective states as per submitted delivery schedule, i.e. within 2 years. At the same time procurement of 63 MVAR reactor at Rourkela S/s needs to be expedited Monthly update on spare reactor procurement to be shared with ERPC.

e) Restoration of 220KV FSTPP LALMATIA Line.

The line is out of service since long due to tower collapse. Presently 220 kV Farakka-Lalmatia line is charged (from loc no 241 to loc 84) at 132 kV voltage level for anti-theft purpose by tapping at loc. No. 100-101.

As per 239th OCC:

JUSNL updated:

- The delay in restoration of the line is mainly attributed to ROW issues, administrative constraints, and impacts during the COVID period.
- Line charging has been completed up to location no. 24. Foundation works is pending at one location while erection of three towers remains pending.
- Approximately 3.74 km of stringing work is left for execution.
- The line is expected to be ready by June 2026.

239th OCC Decision

- ✓ JUSNL was advised to ensure completion of the balance works and commissioning of the line within the targeted timeline (June 2026).
- ✓ This was referred to upcoming TCC meeting for information.

As per 56th TCC:

- ✓ JUSNL informed that the line will be completed by 10th June.
- ✓ ERLDC suggested that JUSNL may take all requisite FTC clearances beforehand so that the line may be charged at the earliest.

56th TCC Decision:

TCC noted the work progress by JUSNL and advised NTPC to check healthiness of the bay equipment of the line at Farakka end.

56th ERPC Decision:

ERPC noted.

Deliberations in the meeting

JUSNL updated:

- ✓ *Erection of one tower and stringing over a span of 450 meters is pending. Expected timeline of completion is 25th June 2026.*
- ✓ *Bay upgradation is pending at NTPC Farakka end.*

OCC Decision

- *OCC advised JUSNL to adhere to the schedule of line restoration i.e. by 25th June 2026.*
- *NTPC was advised to keep the bay ready at Farakka STPS end to facilitate charging of the 220kV FSTPP-LALMATIA Line.*
- *Implementation of Farakka islanding scheme is to be taken up after restoration of 220KV FSTPP LALMATIA Line.*

f) Update on islanding schemes in ER.

The list of existing Islanding schemes in ER is as under:

Eastern Region (ER)		
1	Kolkata (CESC) IS	City/Major Town/ Strategic Load
2	Howrah (Bandel) IS	City/Major Town/ Strategic Load
3	Bakreswar TPS IS	Industrial and Railway load
4	Haldia (Tata Power) IS	Industrial areas of Haldia and Port
5	Chandrapura (CTPS DVC) IS	Industrial areas

As per 238th OCC:

OCC suggested for formation of a sub group committee under chairmanship of S.E. (Operation), ERPC comprising members from ERPC, ERLDC, concern SLDC and Generating company for regular monitoring of islanding scheme formation and implementation in the ER States. **The committee shall coordinate with stakeholders and submit progress report in OCC.**

A special sub-group has been constituted for the identification and implementation of islanding schemes in the capital cities of ER states.

56th ERPC Decision:

The forum advised that the matter may be regularly followed up in OCC meetings.

1. Ranchi Islanding Scheme

- ☐ Nomination for ER sub-group of islanding scheme implementation was received from JUSNL.
- ☐ The kick-off meeting for the implementation of the Ranchi City Islanding Scheme was held on **23rd April 2026** under chairmanship of ERPC.
- ☐ To accurately define the electrical boundaries and load behaviour of the island, SLDC Jharkhand was requested to share:
 - **Island Boundary & Feeder Logic:** Clearly define the islanded area and provide a comprehensive list of feeders that must be disconnected to ensure successful island formation.
 - **Load Profiles:** Node-wise load data (Maximum, Minimum, and Average) for the identified area to ensure generation-load balancing.
 - **Defense Mechanism Data:** Current node-wise loads of Jharkhand that are already covered under **AUFLS** (Automatic Under Frequency Load Shedding) and **ADMS** (Automatic Demand Management System).
 - **Infrastructure Timeline:** Confirmation of the expected commissioning date for the **400kV Latehar New – Patratu New D/C** line, as this must be integrated into the islanding model.

As per 239th OCC:

SLDC Jharkhand apprised:

- Approximately 500 MW minimum islanded load zone has already been identified.
- Patratu generating station has been identified as the source generation station for the islanding scheme.

239th OCC Decision

- ✓ SLDC Jharkhand shall submit all pending details of line feeders and feeder segregation plan by 18.05.2026. Essential loads should not be included in UFLS operation.
- ✓ The proposal shall be prepared and to be shared with ERPC at the earliest.

As per 56th TCC meeting

JUSNL informed that they have already provided data regarding Ranchi islanding scheme.

2. Patna Islanding Scheme under PSDF

- In 54th TCC meeting held on 23.06.2025, BSPTCL had proposed to implement Patna Islanding Scheme through Internal Resource Fund.
- However, a meeting was held on 24th June 2025 under the chairmanship of the Hon'ble Minister of Power and Housing & Urban Affairs, wherein the matter of Islanding Patna city was discussed. In the meeting, it was decided that the State of Bihar would submit a proposal for funding the Islanding scheme by the Ministry of Power).
- In compliance to minutes of the meeting held on dt. 24.06.2025, Board of Directors, BSPTCL has approved for implementation of Patna Islanding Scheme through PSDF in 131st BOD meeting held on dt. 17.07.2025 vide its resolution no. 131-06.
- In line with the above, a proposal has been submitted for Implementation of Patna Islanding Scheme under PSDF to NLDC vide this office letter no. 549 dt. 18.07.2025 along with all the required documents in compliance to minutes of meeting held on dt. 24.06.2025.
- Further, Chief Engineer (Generation), CEA has requested the recommendation of ERPC for implementation of Patna Islanding Scheme through PSDF.

As per 239th OCC:

SLDC Bihar apprised:

- L1 bidder has already been finalized.
- Revised Board approval for 70% PSDF funding was obtained on 16 March.
- The proposal has been submitted to NLDC on 26 March for sanction of funds.

239th OCC Decision

OCC opined that the issue may be pursued with PSDF committee to expedite approval process. This is referred to TCC for information.

As per 56th TCC meeting

SLDC Bihar informed that the PSDF Appraisal committee approved the PSDF grant proposal of Patna islanding scheme and the approval from the PSDF monitoring committee is awaited.

3. IB Valley TPS Islanding Scheme (Odisha)

IB valley TPS Islanding scheme has also been put on hold for long time. The status regarding the same has been sought on urgent basis by Ministry of Power (Govt of India).

As per 55th TCC meeting

Director(Op), OPTCL stated that they need a clarification regarding load quantum required for islanding operation and as per their assessment load of 140-160 MW can be arranged at Budhipadar end for islanding operation.

OPGC representative replied that minimum load of 150 MW is required for islanding operation with one unit of IB TPS.

55th TCC decision

- After deliberation, it was finalized that the scheme will be implemented with minimum load quantum of 140 MW at Budhipadar along with one unit of IB TPS generation.
- TCC pointed out the inordinate delay in implementing the scheme and opined that OPTCL & OPGC shall take necessary steps to implement the scheme within six months.

As per 239th OCC:

OPTCL updated:

An MoU has been signed with OPGC. Approximately 140 MW load has been identified. Discussions regarding some minor loads are also underway.

ERLDC informed that base case for the revised study of IB Valley TPS islanding scheme with one-unit configuration has been sought from SLDC Odisha.

239th OCC Decision

OCC advised ERLDC & SLDC Odisha to share the revised study in upcoming OCC.

As per 56th TCC meeting

OPGC informed that those necessary lines have been identified for Ib valley islanding scheme but base data is required from SLDC Odisha for further action.

4.Bhubaneswar Islanding Scheme (Odisha)

Nomination for sub-group of islanding scheme implementation in ER is still awaited.

As per 239th OCC:

Odisha representative updated that:

- BPBL (Bhubaneswar Power Limited) units (65X2=130 MW) have been identified for islanding of

the capital city of Bhubaneswar. Bhubaneswar islanding is slightly challenging due to absence of nearby generating stations.

- PRDC is carrying out feasibility studies in connection with the proposed Bhubaneswar Islanding Scheme.

239th OCC Decision

OCC advised SLDC Odisha and OPTCL to send nominations for recently constituted islanding sub-group of ER positively by the next week.

5. Farakka islanding scheme

This is presently not in service due to long outage of 220 kv Farakka-Lalmatia line.

□ Continuous monitoring & Testing of Islanding schemes.

- Ensuring the uninterrupted operation of critical services during emergencies is of paramount important, Islanding Schemes are one of the measures which prevent total blackout and enable quicker restoration of grid at the time of grid disturbances.
- As per Central Electricity Authority (Grid Standards) Regulation, 2010, "(1) *The Regional Power Committees shall prepare Islanding Schemes for separation of systems with a view to save healthy systems from total collapse in case of grid disturbance. (2) The Entities shall ensure proper implementation of the Schemes referred to in sub-regulation (1).*"
- During the Conference of Power Ministers held in 2025, all regions agreed upon the following measures regarding islanding schemes and system protection mechanisms:

a) Periodic review and testing (simulation-based or field-based) of all schemes shall be undertaken.

b) AUFLS and df/dt protection system shall be tested to ensure reliability.

c) Real-time visibility and monitoring of system parameters for the islanding schemes shall be concerned RLDCS/SLDCS.

D) Embedded generation forming part of islanding schemes shall be operated as required, including during off-merit situations. Appropriate compensation mechanisms for such minimum generation levels shall be developed, with the SERCS/CERC determining the methodology to ensure financial viability.

In view of the above, the following actions required to be done on priority:

a) **A Comprehensive review** of all the Islanding Schemes and LGB to be monitored continuously with the participating generators and loads. Specifically, the critical loads such as Airport, Defence & Critical loads within the Islands are to be reviewed.

b) **Testing and Validation of Islanding Schemes:** Periodic testing of the implemented islanding schemes must be carried out to ensure their readiness and functional health.

Deliberations in the meeting

☐ **Ranchi Islanding Scheme(Jharkhand)**

ERLDC informed:

Required data from the concerned SLDCs has been received. A base case has been prepared for dynamic/transient study.

OCC Decision

ERLDC was advised to carry out dynamic study on the base case and share relevant inputs with SLDC Jharkhand.

After the finalization of logic, the concerned utility i.e. JUSNL to prepare DPR and implementation plan.

☐ **Patna Islanding Scheme**

SLDC Bihar apprised that LOA shall be placed once sanction of PSDF funds is received from NLDC.

OCC Decision

OCC noted.

☐ **IB Valley TPS Islanding Scheme (Odisha)**

OCC Decision

ERLDC was advised to carry out revised study on the modified base case recently submitted by OPGC. OPTCL was advised to submit identified loads and detailed implementation logic in the next OCC.

☐ **Bhubaneswar Islanding Scheme (Odisha)**

OCC Decision

OPTCL was advised to finalize the load and generation philosophy (essential loads, startup power, etc.) and submit detailed technical feasibility including unit capabilities, protection scheme, and control logic within one month.

☐ **Farakka islanding scheme**

Implementation of Farakka islanding scheme shall be taken up after restoration of 220KV FSTPP LALMATIA Line.

☐ **Continuous monitoring & Testing of Islanding schemes.**

OCC Decision

- *OCC opined for continuous testing, validation and monitoring of in-service islanding schemes in ER. (Latest Simulation test results of islanding schemes attached at **Annex B.2.1.f**). Real-time visibility and monitoring of system parameters for the islanding schemes should be made available to ERLDC/ concerned SLDCs.*
- *As per decision during Conference of Power Ministers in 2025, OCC also suggested for continuous service of embedded generation forming part of islanding schemes including during off-merit situations and compensation mechanisms may be devised by*

SERCs/CERC.

g) Review of AUFLS in Eastern Region: SCADA Integration & Data Updation

- ♦ Based on the recommendation and decisions in 14th NPC meeting held on 05.02.24, 214th OCC meeting and special meeting on 10.07.2024, a load relief quantum of 6916 MW was finalized for Eastern Region. UFR Feeders real time monitoring has been discussed in NPC as well as various fora of ERPC.
- ♦ Further, with new IEGC 2023 the same has been mandated as quoted below: IEGC 2023, Clause 13.d: "SLDC shall ensure that telemetered data of feeders (MW power flow in real time and circuit breaker status) on which UFR and df/dt relays are installed is available at its control centre. SLDC shall monitor the combined load in MW of these feeders at all times.
- ♦ SLDC shall share the above data with the respective RLDC in real time and submit a monthly exception report to the respective RPC. RLDC shall inform SLDCs as well as the concerned RPC on a quarterly basis, durations during the quarter when the combined load in MW of these feeders was below the level considered while designing the UFR scheme by the RPC. SLDC shall take corrective measures within a reasonable period and inform the respective RLDC and RPC, failing which suitable action may be initiated by the respective RPC."

As per 239th OCC meeting:

Both Bihar & Jharkhand updated to Complete AUFLS implementation as well as SCADA integration by end of May 2026.

As per 56th TCC meeting

Both Bihar & Jharkhand updated that they will Complete AUFLS implementation as well as SCADA integration by end of June 2026.

56th TCC Decision:

TCC urged early completion of AUFLS implementation as well as SCADA integration.

56th ERPC Decision:

ERPC advised concerned ER states for early completion of AUFLS as well as SCADA integration.

Update:

The load relief quantum to be implemented at each stage of AUFLS for FY 2026-27 is computed by NPC based on the regional peak demand met in FY 2025-26. Details enclosed in [Annex B.2.1.i](#).

Stage-wise Load relief= Stage-wise relief (%) * RPRegion * AP/RP

AP: All India Peak Demand Met in MW

RP: Sum of Regional Peak Demand Met

As per this methodology, total load relief quantum for ER comes out as 7195 MW against 6916 MW in preceding year.

This 7195 MW of load relief has to be distributed among all ER states as per methodology to be finalized in OCC forum.

Deliberations in the meeting

- ✓ SLDC Jharkhand updated that pending 35 MW of AUFLS quantum to be implemented by July 2026.
- ✓ SLDC Bihar updated that out of 1578 MW quantum, 1000 MW has been integrated with SCADA while rest 578 MW is still pending. (details at **Annex B.2.1.g**)

OCC Decision

- Bihar and Jharkhand utilities shall expedite RTU rectification and SCADA integration works. Completion target is latest by July 2026.
- OCC consented to the methodology of distributing the stage-wise AUFLS load quantum among individual states based on respective contribution in annual energy consumption and peak demand met. All ER states are advised to note the revised AUFLS allocation as communicated in **Annex B.2.1.g**, identify and configure additional AUFLS load quantum and the same needs to be integrated with SCADA accordingly.

2.2 Issues for follow-up: ERLDC

a) Replacement of CLR insulators of PMJTL lines

- Replacement of faulty Decan make CLR insulators in the 765 kV New Ranchi–Medinipur–New Jeerat corridor and associated 400 kV PMJTL lines has been under discussion in various OCC/TCC meetings owing to the strategic importance of these transmission corridors for reliable power supply to Kolkata and adjoining areas of South Bengal. While replacement activities in the 400 kV system have substantially reduced outage incidences, frequent shutdown requests continue to be received for the 765 kV corridors.
- During FY 2025-26, POWERGRID availed 63 shutdowns on the 765 kV New Ranchi–Medinipur–New Jeerat corridor for replacement of failed CLR insulators. Further, during the ongoing summer period, several additional emergency shutdown requests have already been processed. Such frequent outages, particularly during periods of high demand in West Bengal, pose operational challenges and increase the risk to reliability and security of power supply to Kolkata and surrounding areas. Shutdown of one circuit followed by any contingency on the parallel circuit may lead to a highly stressed system condition.

As per 239th OCC meeting:

ERLDC submitted:

- During the last financial year, 63 emergency shutdown requests were processed while from March onwards (this year), around 15 emergency shutdown requests had already been received.

- Frequent emergency shutdowns are creating operational difficulties during high-demand conditions in West Bengal. Moreover, coordination with NLDC is required for every 765 kV shutdown.
- Shutdown of any one circuit of the 765 kV New Ranchi – Medinipur – New Jeerat corridor and subsequent forced outage of the parallel ckt (during the ongoing summer period when West Bengal demand is consistently above 10 GW), poses a serious threat to the reliability/ security of power supply to Kolkata and adjoining areas of S. Bengal.
- As a **short term measure**, it was suggested that the entire 765 kV New Ranchi – Medinipur – New Jeerat corridor may be patrolled on a fast-track basis and list of defective insulators/strings requiring urgent replacement may be detailed and shared with ERLDC / ERPC to facilitate a more structured and coordinated shutdown planning process.

SLDC West Bengal reiterated the concerns raised above and commented that several shutdown requests are being sought on emergency basis under near-breakdown conditions citing damaged insulator strings may collapse the line at that point any time. It was further highlighted that simultaneous outage of both circuits could seriously impact reliability of supply to Kolkata and South Bengal areas.

POWERGRID clarified that most shutdowns are being sought based on patrolling and identification of damaged insulators before actual tripping/failure occurs. It was informed that immediate shutdowns are sometimes unavoidable due to safety concerns arising from damaged suspension/tension strings and the possibility of conductor snapping.

As a **long term measure**, POWERGRID informed:

- Around 5000nos. Of faulty (Decan make) CLR insulators in various 400 kV lines have already been replaced and subsequently, tripping/shutdown incidences of those 400 kV lines have reduced significantly compared to last year.
- Supply of new insulators for 765 kV lines expected by March 2026, has been delayed due to limited availability of suppliers in India and is now expected by June 2026. However, POWERGRID confirmed that replacement of faulty insulators in the identified 765 kV corridors shall be completed by October 2026.
- Replacement of faulty Decan make insulators in both 400 kV and 765 kV systems is being carried out using Decan make insulators from different manufacturing batches to avoid recurrence of similar failures.

239th OCC Decision

- Powergrid shall carry out detailed patrolling & online condition monitoring of lines and shall submit a detailed report to ERLDC/ERPC/SLDCs, encompassing work already completed, identified defective locations along with remaining locations vulnerable to the impact of Norwester storms/rain, thermovision scanning details, PID assessment of the entire length of line and planned replacement methodology. Tower footing resistance checking should be integrated with insulator replacement activities.

- POWERGRID was also advised to expedite the insulator replacement activities and adhere to the committed timelines for completion of balance works in 400 kV and 765 kV systems.
- Shutdown planning should be improved by Powergrid to reduce frequency of emergency shutdowns. Instead of seeking shutdowns for isolated locations on a day-to-day basis, POWERGRID may identify and compile all faulty insulator locations requiring emergency shutdowns over a weekly horizon and plan such shutdowns preferably during weekends or low-demand periods, so as to minimise system reliability risks.
- Multiple locations should be covered during any shutdown period by deploying multiple gangs to maximise replacement works and expedite completion of activities within the allowed shutdown opportunity. Railway crossings and public safety-sensitive locations should continue to receive priority.
- Weekly planned shutdowns should be preferred over repeated emergency shutdowns wherever feasible while Emergency shutdowns can be availed only for genuinely critical cases. To avail emergency shutdowns during the summer months, efforts shall be made to consolidate such requirements on a weekly basis and to avail the same preferably during weekends or other low-demand periods.

Update:

- ✦ No comprehensive assessment & condition report received to ERLDC.

Similar, simultaneous outage of 765 kV New Jeerat – Medinipur-1 (04:28Hrs-16:15hrs) & 400 kV Subhasgram-New Jeerat-2 (06:01hrs-15:34hrs) had happened due to insulator damage on 05.06.2026. At the same time, 765 kV bus 2 at New Jeerat (11:28 hrs to 02:15hrs of 6th June) & 765/400 kV ICT – 1 (11:28 hrs to 02:20hrs of 6th June) were also under outage due to damaged Jack bus jumper of ICT-1, which made the system around Kolkata more vulnerable.

Deliberations in the meeting

Powergrid informed:

- ✓ *Approximately 28,000 insulators (400 & 765 kV combined) are planned for replacement. For 400 kV system: about 10,000 insulators to be replaced and for 765 kV system: about 17,000 insulators to be replaced.*
- ✓ *Total 10273 Nos (400KV) and 17785 Nos (765KV) CLR insulators are under procurement. As of 30th June 2026, 700 (at 400 kV level) and 850 (at 765 kV level) insulators have been supplied at site and progressive supply will continue.*
- ✓ *Replacement work will be prioritized on critical lines (e.g. 765 kV D/C Ranchi- Medinipur) and shall be carried out progressively, aiming to complete both supply and replacement by October-November 2026.*

*Detailed assessment report of damaged CLR insulators and detailed plan for procurement as well as replacement of defective CLR insulators attached at **Annex B.2.2.a***

OCC Decision

OCC opined that a special meeting may be convened by ERPC involving stakeholders from Powergrid, WB & ERLDC to finalize a detailed plan for replacement of CLR insulators in a phased manner.

b) Review of UFR/PFR Performance during Low Frequency Event of 13.05.2026

- On 13th May 2026, the Indian grid experienced its largest generation loss event so far, following outage of evacuation corridors associated with the Khavda RE Complex (KPS-1, 2 & 3) in Gujarat. The disturbance resulted in sudden loss of more than 9 GW of generation, causing grid frequency to dip sharply from 49.95 Hz to 49.39 Hz. Nation-wide automatic under frequency relay installed for shedding load under stage-1 (UFR Stage $f \leq 49.4\text{Hz}$) operated resulted load disconnection of approx. 6.5 GW. Frequency subsequently settled to 49.59 Hz, due to the combined action of primary response, automatic under frequency relay (UFR) based load shedding and AGC (SRAS) response. Analysis of constituent wise response during the low-frequency event indicated noticeable load relief due to AUFLS/UFR operation and Primary Frequency Response (PFR) by generators across the country.
- ERLDC highlighted analysis of the event and performance of AUFLS/UFR, PFR and AGC schemes installed at different constituents in the Eastern Region in 56th TCC meeting, dated 24.05.26. However, the event highlighted the need for further strengthening of defence mechanisms, improvement in frequency response from generating stations and enhancement of response via AUFLS.
- The forum advised regular follow-up in OCC regarding the readiness of UFR and PFR schemes in ER and emphasized compliance of LVRT/HVRT requirements by RE generators.

Deliberations in the meeting

- ✓ ERLDC presented the UFR operation details of all constituents along with SCADA data availability and Demand Response by respective states during the event. (**Annex B.2.2.b**)
- ✓ During this low frequency event, around 440 MW quantum in feeders with installed UFR did not operate.
- ✓ JUSNL & BSPTCL have lowest feeder wise data availability in SCADA.

SLDC Bihar clarified that SCADA data reporting of UFR feeders has increased after this low frequency event

*The issue of primary frequency response (PFR) is detailed in **Agenda B.2.9**.*

OCC Decision

OCC advised all SLDCs to ensure any upcoming RE Generators to be LVRT/HVRT compliant in accordance with CEA technical standards and FTC may be issued accordingly.

c) SPS for ACCP-II/JSPL

- Following the 233rd OCC meeting, a special meeting was held on November 26, 2025, to discuss the implementation of System Protection Schemes (SPS) for the new ACCP-II plant, during which OPTCL was advised to assess existing protection adequacy at the JSPL site.

- Although the site visit was eventually conducted on January 11, 2026, the overall implementation remains significantly delayed. While JSPL has implemented one SPS for injection control, a second scheme designed to restrict drawal at the Meramandali end is currently pending the transmission of Talcher-Meramandali power flow data from OPTCL. Consequently, OPTCL has apprised that installing the necessary RTU at Meramandali for data visibility will require approximately two months.
- In 235th OCC meeting, forum directed both JSPL and OPTCL to expedite the finalization and implementation of the remaining scheme to ensure grid stability. 237th OCC meeting, it was advised to SLDC/OPTCL to expedite the implementation.

Deliberations in the meeting

OPTCL informed that the data transmission-related works including procurement and installation of RTUs have been completed. However, JSPL informed that issues related to data integration and communication are still persisting.

OCC Decision

OCC advised JSPL to implement the SPS at the earliest in coordination with OPTCL. Both utilities were advised to resolve the persistent issues bilaterally.

2.3 Shutdown of Talcher-Kolar bi-pole: ERLDC

- Simultaneous shutdown of the HVDC Talcher–Kolar Bipole was initially requested by POWERGRID, Odisha Projects from 04.03.2026 to 18.03.2026 (15 days) for diversion of transmission lines associated with the NH Bypass Road project at Nayagarh, Odisha. Since the shutdown pertains to the ER–SR inter-regional corridor, the proposal was deliberated in the 235th SR OCC Meeting held on 10.02.2026, wherein it was decided that the shutdown could be facilitated only after May 2026, considering the prevailing system conditions, election-related constraints and anticipated peak demand scenario in the SR.
- Subsequently, in the 239th SR OCC outage Meeting held on 09.06.2026, revised shutdown proposal for the period 01.07.26 to 15.07.2026 (15 days) has been proposed by POWERGRID. As gathered, the shutdown has been concurred from 17.07.2026 to 31.07.2026 along with replacement of Pole-I Bph converter transformer at Talcher. It was also discussed that the shutdown would coincide with the planned outage of NTPC Talcher Unit #6 scheduled from 10.06.2026 onwards.
- In this regard, it is pertinent to note that NTPC Talcher (6 × 500 MW) is a major generating station in ER, with approximately 2000 MW of its generation normally evacuated to the SR through the HVDC Talcher–Kolar link. In the absence of the HVDC link, the remaining generation (considering outage of Unit #6) would have to be evacuated through the associated AC transmission network in the ER, resulting in significant loading on the 400 kV Talcher–Meramundali D/C lines.
- System studies indicate that, following the outage of the HVDC Talcher–Kolar Bipole, substantial generation curtailment would be required to maintain N-1 security for the Talcher–Meramundali corridor. Required curtailment quantum could, at times, necessitate backing down NTPC Talcher units up to their technical minimum limits. Such a measure may not be desirable considering that Talcher is a low-cost generating station and the All-India demand during June–July 2026 is

expected to remain in the range of 240–250 GW. Further, to facilitate the shutdown, the ER–SR TTC/ATC may need to be curtailed by around 2000 MW, which could impose additional stress on other inter-regional corridors and adversely impact overall grid operation and transfer flexibility.

- Additionally, almost all beneficiaries in the ER have allocation from NTPC Talcher, the impact of such curtailment would be shared proportionately among them. Share allocation in % of Talcher is as follows:

TSTPP Stg I	BIHAR	ODISHA	WB	Jharkhand	DVC	Sikkim
2x500MW	41.404	33.065	10.085	6.831	0.2	2.438

In view of the above operational challenges and system security considerations, the shutdown proposal placed in the 240th ER OCC Outage Meeting for July 2026 may be deliberated further to arrive at an appropriate way forward.

Deliberations in the meeting

Powergrid presented the requirement of shutdown of Talcher–Kolar HVDC link (for a period of 15 days) for diversion works near Angul associated with National Highway construction. It was apprised that SRPC/SRLDC has already approved shutdown for 15 days between 16th to 31st July 2026, subject to ER concurrence.

*ERLDC requested for rescheduling the shutdown to a lower-demand period, presented the implications of the continuous shutdown of the HVDC Talcher–Kolar Bipole and its impact on the 400 kV network in and around the Bhubaneswar area (details at **Annex B.2.3**) and highlighted the following:*

- ✓ *Demand is expected to remain high during July, when the shutdown is planned, owing to the anticipated rainfall deficit, as indicated in the updated Long-Range Forecast for the Southwest Monsoon Seasonal Rainfall (June–September 2026) issued by IMD on 29 May 2026.*
- ✓ *Subsequent to the shutdown of the HVDC Talcher–Kolar Bipole, network around Bhubaneswar, catering to a load of around 2,500 MW, would remain vulnerable, particularly when Odisha is experiencing an all-time high demand exceeding 8,000 MW.*
- ✓ *Facilitating the shutdown would require generation curtailment at both stages of NTPC Talcher to ensure N-1 criteria, resulting in a proportionate reduction in the ISGS share allocation to ER states. This may not be prudent from the resource adequacy perspective, especially during a period of anticipated high demand.*

Representative of SLDC Odisha acknowledged reliability concern and gave consent for the shutdown.

Representative of SLDC Bihar expressed concerns regarding reduced generation availability during Bihar's peak demand period (July–August), as any generation shortfall during this period could adversely affect portfolio management.

Representative of SLDC West Bengal informed that they would discuss the matter with state DISCOM and revert.

NTPC Talcher clarified that operating the Talcher units at Technical Minimum for an extended period would not pose any operational issues.

OCC Decision

Forum decided to convene a separate outage coordination meeting by ERPC secretariat involving SRPC, SRLDC, ERLDC, Powergrid, NTPC Talcher & concerned beneficiaries of TSTPP at the earliest.

2.4 400 kV Bus-1 & Bus-2 Shut down at Malda SS for CRP Upgradation work under ADDCAP of Chukha Transmission System: POWERGRID

Reason for Shutdown - Shutdown of 400 kV Bus-I & Bus-II at Malda SS is required for commissioning of 400 kV Bus Bar Protection system along with various associated works requiring shutdown of 400 kV Bus-I and Bus-II.

Following Works are planned to be carried out during shutdown of 400 kV Bus-I & Bus-II :

- a) CT primary injection for Busbar stability checking.
 - b) Testing and commissioning of 400KV Busbar -I&II and Transfer Bus.
 - c) Removing old cables from 400KV Bus -I & II CVT JB.
 - d) Termination and dressing of new cables in 400 KV Bus-I&II CVT JB.
 - e) Sag adjustment for 400KV Bus Conductor for isolator alignment of Bus-2 Isolator (408 89B) of 400kV Farakka Ckt-I.
 - f) Isolator alignment and adjustment of fixed hanging contact of Bus-1 Isolator (404 89A) of 400KV Farakka Ckt-II.
 - g) Removal of old jack bus of 400 KV Malda-Farakka-1 along with removal of hardware fittings (Require both bus shut down).
 - h) Replacement of 01 span of bus conductor (400KV Bus-I).
 - i) Completion of wiring of mutual compensation for D/C 400KV Malda-Purnea Ckt-I&II and Malda-Farakka Ckt-I&II.
- ☐ After Completion of CRP upgradation work of all the 400 kV Bays with last being 400 kV Bus Coupler Bay which was taken into service on 21.03.2026, the shutdown of 400 kV Bus-I & Bus-II were earlier planned on **22.03.2026**. However, due to last minute cancellation of the 400 kV Bus-I & Bus-II shutdown, all the planned work associated with 400 kV Bus -I & Bus-II Shutdown had to be deferred.
- ☐ Since the 400 kV Bus -I & Bus-II shutdown was not approved, the 400 KV Bus Coupler at Malda has been kept open after due concurrence from ERLDC/SLDC, w.e.f, 24.03.26. Further, the Z-4 timing of all the 400 kV Line feeders and 400 kV ICTs have been kept 250ms considering the 400 kV Bus bar protection is out of service at Malda SS.
- ☐ The ongoing Split Bus operation of 400 kV Bus -I & Bus-II at Malda SS needs to be restored at the earliest possible considering system reliability.

- ☐ Accordingly, it is proposed that shutdown of 400 KV Bus-I & Bus-II at Malda SS may kindly be approved for 01 day during the month of July-2026 for CRP Upgradation work under ADDCAP of Chukha Transmission System.

Work Flow-March SD details- Last minute cancellation- Split Bus operation- Earliest Restoration.

Name of Element	S/D Start Date	S/D End Date	Remarks	
Busbar Protection of 220 kV System was kept out of service from <u>26.11.2025</u> during commencement of CRP upgradation of individual 220 KV Bays.				
220/132 KV ICT-II	26.11.25	11.12.25	CRP upgradation work.	
220 KV TBC				
132 KV Manikchak-1	12.12.25	13.12.25	BCU Software upgradation.	
220/132 kV ICT-I	13.12.25	19.12.25	CRP upgradation work.	
220/132 kV ICT-4	19.12.25	24.12.25		
132 kV Manikchak-2	26.12.25	26.12.25	BCU Software upgradation.	
132 kV WBSETCL-1	27.12.25	27.12.25		
132 kV WBSETCL-2	28.12.25	28.12.25		
220 kV Bus Coupler	27.12.25	30.12.25	CRP upgradation work.	
220 kV Gazole-2	02.01.26	02.01.26	For PLCC dismantling.	
220 kV Gazole-1	04.01.26	08.01.26	CRP upgradation work.	
220 kV Gazole-2	10.01.26	14.01.26		
400 KV Bus Bar Protection kept out of service after necessary Z-4 setting changes on 16.01.26.				
400 kV TBC	16.01.26	19.01.26	CRP upgradation work.	
400/220 kV ICT-III	21.01.25	25.01.26		
400/220 kV ICT-V	27.01.26	01.02.26		
Testing & Commissioning of 220 KV Bus Bar Protection carried out 30.01.2026 and 220 KV Bus Bar protection taken into service.				
400 kV Farakka-2	09.02.26	13.02.26	CRP upgradation work.	
400 kV Purnea-1	14.02.26	25.02.26		
400 kV Farakka-1	27.02.26	07.03.26		
400 kV Purnea-2	09.03.26	17.03.26		
400 kV Bus Coupler	18.03.26	21.03.26		

- ❑ After Completion of CRP upgradation work of all the 400 kV Bays with last being 400 kV Bus Coupler Bay which was taken into service on 21.03.2026 , shutdown for 400 kV Bus- I & Bus -II was applied on 22.03.2026 for testing & commissioning of 400 kV Bus Bar Protection. However, the shutdown was denied by ERLDC due to upcoming West Bengal Election.
- ❑ Considering, non-allowing the bus sd, 400 KV Bus Coupler at Malda kept open after due concurrence from ERLDC/SLDC, w.e.f, 24.03.26.
- ❑ Further, in subsequent discussions in 238th OCC SD meeting, both ERLDC/WB-SLDC, disagreed for above bus bar SD during high demand period uptill June-26 and will be reviewed during load relief in Monsoon period.
- ❑ Considering high power flow at Malda corridor, it is expected that after peaking of hydro generation, 01 Day SD will be given by Grid Operator for commissioning of 400 KV BUS BAR PROTECTION (S/D time period is flexible and can be done in night hours also).

Deliberations in the meeting

Powergrid apprised:

- ✓ *Complete replacement and upgradation of CRPs, including busbar protection, is in progress at Malda, for which multiple shutdowns have already been availed with RLDC/SLDC support.*
- ✓ *The earlier requested shutdown on 22 March for bus-bar protection commissioning could not be granted due to elections and related constraints.*
- ✓ *Presently 400 kV Malda bus is being operated in split bus mode. Such sustained operation may result in uneven loading pattern of ICTs in event of tripping of any line.*
- ✓ *Shutdown was requested with the onset of monsoon and consequent relief in demand. Maximum 7 to 8 hours of shutdown is required for control, protection & testing related works, with flexibility to avail the shutdown even during night hours.*

WB SLDC submitted:

- ✓ *As per the Malda ICT loading data in July of Last year, the proposed S/D may be availed in August 2026.*
- ✓ *Even in morning hours, minimum ICT loading is around 350 MW.*
- ✓ *It was apprehended that even shifting of the entire load to Dalkhola (PG) may not be sustainable.*

OCC Decision

OCC agreed that the proposed shutdown may be availed in August, 2026 during low demand period.

2.5 Establishment of 400kV substations at Katihar and Chopra under ISTS: ERPC

As per information received from CTU:

In order to strengthen the ISTS system in the Siliguri – Kishanganj – Dalkhola – Malda corridor and to meet the upcoming load demand of West Bengal, a new 400kV substation is being planned at Chopra, West Bengal. Along with the same, 400kV substation at Katihar shall also be required.

Accordingly, BSPTCL and WBSETCL may confirm/provide the following:

- a. **BSPTCL:** Need for establishment of 220kV level at this stage or only space provision is required to be kept. If 220kV level is to be established, details of 220kV downstream system may be provided.
- b. **WBSETCL:** Kishanganj – Dalkhola line shall be disconnected from Dalkhola and connected to Chopra. WBSETCL may indicate the details of 220kV downstream system from Chopra to meet its upcoming demand in the surrounding area including TCF.

After receipt of inputs, updated study files shall be shared for further deliberations in the joint study meeting.

Deliberations in the meeting

OCC Decision

BSPTCL and WBSETCL were advised to share the requisite data with CTU. Further, a joint study may be conducted involving concerned stakeholders.

2.6 Present status of the configuration update in Secure Meter Midnight Reactive high/low data parameter logic (from Export – Import to Import – Export): POWERGRID

- ☐ In the 238th OCC meeting, it was highlighted regarding certain logic mismatch in the newly supplied Secure Meters. In ER, the net midnight reactive high/low data logic is followed as Import - Export (I-E) and the same logic has been implemented in all the Meters of L&T and Genus make. However, in the Secure make Meters, the logic of Net reactive data calculation was initially mapped as Export – Import (E-I).
- ☐ After this issue was highlighted and discussed at various forums, POWERGRID had conducted multiple meetings with the Meter OEM and in a joint meeting, OEM demonstrated on how to change the logic of Net Reactive calculation. Further PGCIL conducted training sessions at RHQ, and the concerned station Engineers are being trained to carry out the Meter logic change activities at Substation Level. This activity started in April 2026.
- ☐ Total number of Secure meter count was 176, where the logic correction was required. As on 09th Jun 26, the count of Meters where the configuration has been changed successfully is 136. After completing the configuration changes, all the stakeholders are being informed about the change, over email. M/S TCS team checked and confirmed the applied changes in Meter by analyzing the midnight data. Further checking was done by ERLDC MO team.
- ☐ Following is the utility wise status of the Secure Meter logic change in ER.

Utility	Total Secure Meter	Logic Changes done	Logic Changes pending
Powergrid	157	134	23
West Bengal	4	0	4
Sikkim	4	0	4
Jharkhand	1	0	1
Odisha	1	0	1
IPP/Generator	9	2	7
Total	176	136	40

**** the detailed information with Meter SL no. & Feeder-wise is mentioned in Annexure- B.2.6.**

- ❑ POWERGRID initially conveyed the expected completion date as 15th Jun 26. The required activities have been nearly completed & expected to be completed very shortly.

It is requested to ERLDC to verify concerned Reactive data further, if any meter is/are, still to be attended.

Deliberations in the meeting

Powergrid ER-II informed:

- ✓ *Most of the pending meters pertains to NTPC S/S, and for completion of the logic change, help of concerned NTPC establishment is required. 07 no's Meters in Lakhisarai (PG) and Arah(PG) have completed logic change but are yet to be integrated in AMR.*
- ✓ *16 no's Meters in Dhanbad (NKTL), Dumka (JUSNL), Govindpur (JUSNL) and PVUNL (NTPC) have not confirmed whether logic change done from Substation.*
- ✓ *Activity at Jodda & NTPC Talcher will be completed in coordination with Powergrid Odisha.*

Updated status is as below:

Completed. Meter available in AMR	195
Completed (Meter not in AMR, Logic change confirmed by SS/Utility)	7
Work In Progress	5
Meter not in AMR(Logic change confirmation pending from SS/Utility)	16
Total	223

- ✓ *ERLDC was requested to check the meters and update for further modification, if any.*
- ✓ *Logic modification in all Secure meters will be completed latest by 20th June, 2026.*

OCC Decision

- *Powergrid was advised to carry out logic modification in meters as per submitted timeline (20th June 2026).*
- *With regard to logic correction in meters at Bhutan end, a separate meeting may be convened by ERLDC/NLDC involving concerned stakeholders.*

2.7 AMR system Data Sharing with SLDCs & DVC- present status: POWERGRID

- ❑ *In the Eastern Region, the AMR system is in operation for the last 12 years and the SEM data is being accessed by ERLDC team. It was a requirement from all SLDCs to have access to this SEM data to help reduce their DSM penalties and other operational improvements.*
- ❑ *In the 54th ERPC meeting, the AMR data sharing project was approved for 05 SLDCs (West Bengal, Sikkim, Odisha, Bihar & Jharkhand) and DVC. The scopes and requirements were finalized by the respective utilities. LOA was awarded to M/S TCS on 17.09.2025. The LOA*

scopes were procurement of two Servers and Application development for 6 utilities, with a timeline of 09 months implementation and 06 months warranty support for stabilization.

- ☐ The 1st phase pilot development was initiated for West Bengal utility. The required developments testing and UAT was conducted with the team concerned. Application was hosted in the Live environment of WB SLDC. Subsequently the development was started for the other utilities. Servers' delivery was completed in Nov-2025.
- ☐ The present status of the project is as below:

Utility	Application Development Completed	Separate NW port arranged for application access	Live Demo Completed	Live Demo Date	System arranged	Go-Live declared
WB	YES	YES	YES	12-Dec-2025	YES	YES
DVC	YES	YES	YES	19-Feb-2026	YES	YES
SIKKIM	YES	YES	YES	27-Feb-2026	YES	YES
JHARKHAND	YES	YES	YES	26-Feb-2026	WIP	TBD
BIHAR	YES	YES	YES	09-Mar-2026	WIP	TBD
ODISHA	YES	YES	YES	12-Mar-2026	WIP	TBD

- ☐ Considering the baseline requirements which were initially agreed upon, the application development part was completed. Live demonstration was given to the respective utilities physically at respective offices.
- ☐ Till date, system arrangements and the Go-Live have been declared for three utilities i.e. WB, SIKKIM & DVC. It is further requested to the other three utilities to expedite the system arrangement process and plan for the application Go-live.
- ☐ Considering the original LOA timeline, the development and Go-live was supposed to be completed on 17th Jun 2026. Due to unavailability of required infrastructure, Go-Live is under progress for three utilities. Hence it is proposed to extend the timeline till 31-Aug-26. Remaining three utilities are requested to complete the required arrangements and the go-live process within this period.
- ☐ As this is an external dependency which is not in control of the system integrator (M/S TCS), it is hereby requested to allow further time extension of the placed LOA till 30-08-2026, such that all remaining SLDC's can provide requisite system for online display and package could be completed.

Deliberations in the meeting

Powergrid updated:

- ✓ Live Demo is completed and system is Go-Live in SLDC Bihar on 10th June 2026.
- ✓ At present, system is Go-Live in all SLDCs except Jharkhand and Odisha.

SLDC Odisha submitted that the system will be available by 30th June 2026 while SLDC Jharkhand updated that system procurement is under progress.

DVC submitted that data is not being fetched from many L&T meters.

OCC Decision

- SLDC Jharkhand and SLDC Odisha were advised to arrange the required systems as per submitted timelines so that AMR data can be Go-Live by Powergrid.
- DVC was advised to submit fresh requisition to Powergrid for replacement of the L&T meters.

2.8 Security and Redundancy enhancements of the AMR system solution in ER: POWERGRID

- In the existing solution architecture of AMR system, the entire data communication is happening through the Fiber Optical Network. The initial level of Layer2 network has been upgraded to the higher Level of Layers (Layer3/Layer4) to enforce more security protocols. Zone based routers are installed and configured to isolate and protect utility level AMR system from the other zones/utilities. This Layer3/Layer4 network upgradation project went live in Jun-25.
- When the AMR Data Center refreshment was being carried out, one CISCO Firewall was procured and configured with the policies and protocols. This was done in Jun23 time. After that, the AMR Network upgradation project started where Network Routers are installed, and this has been in operation since Jun25.
- As members may appreciate, security measures and policies are very dynamic and those need to be upgraded in a regular interval. Also, we must think through the redundancy of the system to be prepared with any un-precedent scenarios. Considering the criticality, necessity and complexity of the AMR solution, it is recommended to adopt the following additional requirements in the AMR solution.

Item	Current Version	New Requirement	New Version	Advantages
AMR Firewall	CISCO-FPR 1140	Procurement of 01 additional Firewall.	CISCO-FPR 3105	• One Backup Firewall. • High Availability Pair. • Host Standby. • Additional Maleware protection.
AMR Router	CISCO 8200L	Procurement of 04 additional Routers.	CISCO 8200L	• Backup Redundancy of Routers, of AMR upgraded network. • Hot standby, any issue with the existing Routers happens, will be replaced immediately.
AMR Firewall – Threat protection licenses upgradation	FTD 7.0.1 (build 84) VDB:338	Procurement of new licenses	FTD 7.4+ or 7.6	• This is kind of patch upgrade. • The same is suggested in the recent VAPT assessment. • More stringency in the Network policies.

AMR Switch OS upgrade	HP Aruba 6200F	Procurement of new licenses	Aruba OS-CX 10.13.0001	• This is kind of patch upgrade. • The same is suggested in the recent VAPT assessment.
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- ☐ The above-mentioned items are required on an immediate basis. The new Firewall which will be procured will be the higher version of the existing one. This will give more redundancy and flexibility to ensure security and smooth AMR operations. There will be an additional cost implication for this, and the detailed cost breakup will be shared in subsequent meetings. (an approximate estimate is 80-95 lacs including hardware & services).

It is requested for technical concurrence at OCC level and subsequently details pertaining to financial implications shall be provided in the next CCM.

Deliberations in the meeting

*Powergrid presented the details on proposed AMR architecture, additional requirements for AMR Router setup and specifications of the proposed firewall in compliance to CERT-in guidelines.(attached at **Annex B.2.8**)*

The additional cost for proposed enhancement in Eastern regional AMR network is estimated around Rs.1 crore.

OCC Decision

- OCC forum agreed with the proposed enhanced security and redundancy architecture for AMR network by Powergrid.
- Powergrid was advised to place the proposal in next CCM for financial concurrence along with a detailed cost estimate and implementation plan.

2.9 Frequency Response Performance of Eastern Region for May 2026 events: ERLDC

- ☐ In accordance with Clauses 30.8 and 30.10(a) to 30.10(q) of the IEGC, 2023, generating stations in the Eastern Region are required to submit the prescribed information to ERLDC within two days of notification of a reportable frequency event.
- ☐ Further, as stipulated under Clause 30.10(n), all control areas in the Eastern Region are required to evaluate their frequency response characteristics and submit the assessment, along with high-resolution operational data, to ERLDC. The frequency response performance of generating stations and control areas during the reportable frequency events of May 2026 is summarized below:
- ✓ Event 1: Reporting of Generation loss event of 8613 MW in RE Generation Complex, NR at 14:09 hrs. on 13.05.2026.
 - ✓ Event 2: Reporting of Generation loss event of 1698 MW in JSW Mahanadi, WR at 16:16 hrs. on 13.05.2026.
 - ✓ Event 3: Reporting of Generation loss event of 2394 MW in 220kV Bhadla, NR at 11:06 hrs. on 15.05.2026.

□ Generator wise performance summary is given below:

Name of Generator	Event 1			Event 2			Event 3		
	Percentage response as per 5% droop considering saturation at 5% of MCR	FRP	Grading	Percentage response as per 5% droop considering saturation at 5% of MCR	FRP	Grading	Percentage response as per 5% droop considering saturation at 5% of MCR	FRP	Grading
FSTPP I&II	94.60 %	1.63	Excellent	88.12 %	7.32	Excellent	181.33 %	10.58	Excellent
FSTPP III	104.14 %	1.83	Excellent	22.34 %	1.77	Excellent	272.56 %	12.75	Excellent
KhSTPP I	-14.17 %	-0.27	Poor	40.14 %	3.64	Excellent	-35.14 %	-1.90	Poor
KhSTPP II	110.98 %	2.55	Excellent	70.26 %	5.57	Excellent	235.69 %	7.71	Excellent
TSTPS-I	-3.17 %	-0.08	Poor	-61.36 %	-4.80	Poor	96.19 %	4.77	Excellent
BARH ST-1	29.19 %	0.94	Good	37.51 %	3.65	Excellent	70.14 %	7.12	Excellent
BARH ST-2	85.09 %	1.51	Excellent	76.38 %	6.43	Excellent	51.49 %	2.57	Excellent
BRBCL	126.33 %	3.57	Excellent	152.92 %	10.53	Excellent	424.98 %	21.36	Excellent
DARLIPALLI	24.71 %	0.67	Below Average	36.03 %	2.73	Excellent	99.54 %	4.34	Excellent
NORTH KARANPURA	31.24 %	0.70	Below Average	38.21 %	2.58	Excellent	351.06 %	20.06	Excellent
NPGC	156.77 %	2.27	Excellent	5.10 %	0.29	Poor	386.70 %	18.79	Excellent
GMR	17.53 %	0.46	Poor	66.44 %	4.99	Excellent	5.60 %	0.39	Poor
MPL	79.40 %	1.60	Excellent	74.30 %	6.23	Excellent	61.44 %	3.14	Excellent
Adhunik	94.12 %	2.22	Excellent	15.23 %	1.28	Excellent	178.09 %	8.62	Excellent
JITPL	1.47 %	0.03	Poor	9.64 %	0.73	Below Average	83.64 %	5.59	Excellent
INDBHARAT	44.35 %	0.82	Average	55.54 %	2.95	Excellent	-149.52 %	-7.37	Poor
DIKCHU	141.25 %	3.94	Excellent	387.96 %	14.49	Excellent	34.78 %	2.93	Excellent
PVUNL	28.39 %	0.62	Below Average	24.55 %	2.06	Excellent	152.63 %	8.20	Excellent

Name of State	Event 1		Event 2		Event 3	
	FRP	Grading	FRP	Grading	FRP	Grading
Bihar	1.46	Excellent	0.31	Poor	3.43	Excellent
Jharkhand	0.43	Poor	0.09	Poor	-5.94	Poor
DVC	1.87	Excellent	11.16	Excellent	3.15	Excellent
OPTCL	-5.65	Poor	-1.72	Poor	1.93	Excellent
WB	1.11	Excellent	2.81	Excellent	-0.37	Poor

All concerned generating stations and entities are requested to review their frequency response performance and take necessary corrective measures to enhance frequency response performance during future frequency events.

Deliberations in the meeting

*ERLDC presented the performance of the generators and states during the May 2026 frequency events (**Annex B.2.9**) in the forum and updated on the meeting held on 10.06.2026 with ER States and generating stations to discuss the PFR performance during the May 2026 events. During the meeting, issues related to primary frequency response and secondary response was discussed, and an action plan for improvement of frequency response performance was deliberated.*

OCC Decision

- *The forum reviewed the frequency events of May 2026, including major generation loss incidents in the Northern and Western Regions, and noted the performance of Eastern Region generating stations in providing primary frequency response (PFR).*
- *OCC appreciated the generating stations which have performed well during the under-frequency event of 13.05.2026. OCC also expressed concern for the generating stations who have not adequately responded to the event.*
- *Forum advised to review the PFR performance and corrective measures to be taken for improvement of PFR and all generating stations were advised to keep their systems in FGMO mode.*

2.10 ADMS Operation in Eastern Regional States: ERLDC

- ☐ As mandated by IEGC (IEGC-2023 Regulation 36.2), Implementation of Automatic Demand Management Scheme (ADMS) is one of the defence mechanisms intended for ensuring Grid security especially during low frequency and minimizing deviations by automatically shedding load under predefined system conditions.
- ☐ For the last one month (1st May 2026 to 3rd June 2026), the summary of events to trigger ADMS operation in individual states are as follows:

SI No	State/Utility	Logic for ADMS operation	Total Implemented Quantum	Events (Nos.)	Duration (Minutes)
1	Jharkhand	System Frequency < 49.9 Hz AND deviation > 12 % or 25/50/75 MW. Block I, II & III feeders will be selected for load shedding depending on the O/D.	150 MW	198	1738
2	DVC	F < 49.9 Hz AND deviation > 12 % or 150 MW	280 MW	124	1051

3	Odisha	1. System Frequency < 49.9 Hz 2. Odisha over-drawl > 150 MW, 3. Discom OD >40 MW	400 MW	36	253
4	West Bengal	F <49.7 AND deviation > 12 % or 150 MW Block I, II, III&IV feeders will be tripped sequentially.	225 MW	6	90
5	Bihar	F <49.7 AND deviation > 12 % or 150 MW	400 MW	4	28

Although, the criteria were satisfied, the actual ADMS operation did not occur or not reported in ER. The very purpose of ADMS implementation is failing. Hence, it is requested to review and submit a report to ERLDC and ERPC after due verification. States shall review the following:

1. Whether all the load feeders included in the ADMS have healthy communication signals till end load point.
2. Exact quantum (in MW) of the total feeders which are actually reporting to the frequency and deviation signals as indicated in the logic above may be informed to ERLDC.
3. DISCOM OD related logics may be removed from the ADMS logic since the idea for ADMS is to minimize the states deviation which is not serving the purpose.
4. Healthiness of scheme and its operation to be ensured with proper testing.
5. Uniformity of logic for ADMS operation across all states in ER

The scenarios where ADMS should have been triggered / actually operated in last 1 month for each state is shown in **Annex B.2.10**. Only the days where for maximum duration the criteria had satisfied has been shown for better visibility purpose.

Deliberations in the meeting

ERLDC apprised:

- ✓ *During the recent sustained low-frequency event on 03.06.2026, the desired response from ADMS was not achieved*
- ✓ *The communication system between the SLDC ADMS system and feeder endpoints is suspected to be unhealthy for several load feeders. Consequently, despite issuance of commands from the SCADA/ADMS system, the signals did not reach the 33 kV load points, resulting in non-operation of the scheme.*
- ✓ *In Odisha, an additional input of DISCOM overdrawal, along with State overdrawal and frequency criteria, has been incorporated. As a result, ADMS did not operate even though the State overdrawal and frequency conditions were satisfied.*
- ✓ *Different states have adopted different frequency settings and intentional time delays for ADMS operation. Therefore, there is a need to harmonize the settings across States to improve operational reliability and facilitate effective monitoring of the ADMS scheme.*

OCC Decision

- *It was decided that ERLDC to formulate a common ADMS logic for all States.*
- *All concerned utilities shall ensure the healthiness of communication to feeders at 33 kV level, with the objective of enhancing the effectiveness and reliability of ADMS response especially during critical low-frequency conditions.*
- *A separate meeting will be convened in this regard by ERPC secretariat.*

2.11 High MVAR and Limiting Generating Capability in both units of MTPS: NTPC

- Recently (from mid-May 2026 onwards), it has been observed that during evening peak hours (2000 Hrs to 0100 Hrs), both units of MTPS Kanti have been operating with high reactive power output in the range of 90–102 MVAR. This is adversely impacting the active power generation capability of the units. Simultaneously, the 220 kV grid voltage has been observed to drop to around 216 kV during evening peak hours as against approximately 230 kV during daytime operation.
- Real-time communications highlighting such instances are regularly being sent by MTPS-II to ERLDC and SLDC, Patna. Copies of the communications are enclosed at **Annex B.2.11**.
- The trends of 220 kV grid voltage, reactive power generation and unit load for 01.06.2026, 02.06.2026 and 03.06.2026 are enclosed at **Annex B.2.11** for reference.
- It may be noted that on 01.06.2026, during peak hours, the reactive power output of Unit #3 reached 105 MVAR, resulting in generator operation beyond its capability curve. Consequently, active power generation had to be reduced to control the reactive power output. During the same period, 220 kV Motipur lines were opened by SLDC for reactive power management.
- Further, GT tap position has already been raised from Tap Position 3 to 4 during previous overhaul schedules. Any further increase in GT tap position is not considered advisable, as the 220 kV grid voltage remains in the higher range (~230 kV) for most of the operating period and may lead to under excitation during off-peak hours.
- Similar issue of High MVAR was also reported at NTPC Barauni since last 02 weeks.
- In view of the above, the matter may be deliberated in the OCC, and ERLDC/SLDC, Patna may be requested to take suitable measures for maintaining the 220 kV grid voltage within the prescribed limits, particularly during evening peak hours, so as to facilitate optimum active power generation from MTPS-II and ensure reliable power supply to the beneficiaries.

Deliberations in the meeting

NTPC Kanti informed:

- ✓ *During summer evening peak hours, particularly between 20:00 and 23:00 hrs, the 220 kV system voltage drops from around 220 kV to about 214 kV, requiring high reactive power injection from the generating units to maintain generator terminal voltage through AVR action.*
- ✓ *Under such conditions, the units are required to inject up to around 110 MVAR and operate close to the generator capability curve limit, raising concerns regarding unit safety and operational margin.*
- ✓ *Repeated requests are being made to SLDC for load reduction during these hours so as to improve the voltage profile and reduce reactive burden on the units.*

- ✓ The issue is being partly managed by opening the Motipur line, thereby shifting reactive absorption towards the Darbhanga end and reducing MVAR burden at the concerned location.
- ✓ No nearby generating source or reactor support is available in the vicinity, and therefore the matter may need to be addressed at ISTS/ERLDC level through suitable network arrangement and system study.
- ✓ Similar operational concerns were also reported in respect of NTPC Barauni, where units face high reactive loading during summer evening peak and, on at least one occasion, active generation had to be reduced as a safety measure.
- ✓ GT tap had earlier been changed and OEM advice has been received for one further tap change, which can only be implemented during an opportunity shutdown.
- ✓ The existing AVR set point is 15.75 kV, corresponding to about 1.0 p.u., and discussions are underway with BHEL regarding the possible implications of any revision in AVR setting.

OCC Decision

- OCC suggested NTPC Kanti that a slight reduction in AVR set point, such as from 1.0 p.u. to around 0.98 p.u., may help reduce excessive reactive injection, subject to technical examination and OEM concurrence.
- NTPC Kanti shall furnish complete details of existing AVR settings, excitation system parameters, implemented settings, and GT tap position to ERLDC for detailed examination. ERLDC was advised to study the submitted data and advise revised settings / operational measures for mitigation of the issue.
- Till finalisation of revised settings and other corrective measures, NTPC Kanti shall continue close monitoring of MVAR regulation during the critical summer period.

2.12 Shutdown proposal of Thermal generating units: ERPC

Shutdown request of NTPC Barh: NTPC

- NTPC Barh Stage-I comprises three units of 660 MW each. Unit-2 achieved Commercial Operation Date (COD) on 01.08.2023 and has been in continuous operation since commissioning. No annual maintenance or major overhauling outage has been undertaken in the unit since declaration of COD.
- As per the approved Load Generation Balance Report (LGBR) Programme for FY 2026-27, Unit-4 of Barh Stage-II was scheduled for a planned outage from **27.06.2026 to 31.07.2026**. However, the said outage is now being deferred.
- Considering the availability of the outage window arising from the deferment of the Unit-4, Barh Stage-II outage, it is proposed to undertake the overhauling of Unit-2, Barh Stage-I during the period from **01.07.2026 to 14.08.2026**.
- As per the approved LGBR Programme, Unit-1 of Barh Stage-I is scheduled for overhauling from **01.11.2026 to 15.12.2026**. In view of the proposed advancement of Unit-2 overhauling, the overhauling of Unit-1 shall be deferred. The outage period from **01.11.2026 to 15.12.2026** shall instead be utilized for carrying out the planned maintenance outage of Unit-4, Barh Stage-II.

Reschedule of NTPC Darlipalli Unit #1 OverHauling : NTPC

Unit-1 Annual Overhaul was approved in the 238th OCC meeting from 20.08.2026 for a duration of 45 days. It is requested to reschedule the OH from 10.08.2026 for 45 days, considering the availability of the Turbine OEM.

Reschedule of Shutdown of Nabinagar STPS: NTPC

- ☐ As per Load Generation Balance Report 2026-27, Maintenance Schedule of Nabinagar STPP (3x660 MW), Unit#2 is scheduled from **15th Oct'26 to 24th Nov'26**, for Boiler + Turbine-B + Gen.-B inspection.
- ☐ It is requested that the outage be allowed from **25th Nov'26 to 3rd Jan'27** for a period of **40 days**.
- ☐ This will ensure all festival days are covered and power supply to beneficiaries is not affected. Also uninterrupted workforce will be available at site to bring the unit back on schedule.

Shutdown request of BRBCL Unit-3: BRBCL

- ☐ In the LGBR, BRBCL Unit-3 is presently reflected under planned shutdown from 08.01.2027. However, approval for overhauling of Unit-3 has already been accorded in the 238th OCC meeting for carrying out the overhauling from 13.07.2026 for a period of 35 days.
- ☐ In view of the above, it is requested that BRBCL may kindly be permitted to undertake the scheduled overhauling of Unit-3 from 13.07.2026 to 16.08.2026. Further, permission may also be accorded for carrying out the statutory license renewal activities from 13.08.2026 onwards during the shutdown period.

Deliberations in the meeting

- *NTPC informed that as per the approved LGBR, Unit-4 of Barh Stage-II was scheduled for a planned outage from 27.06.2026 to 31.07.2026. However, the said outage is now being deferred and they requested to undertake the overhauling of Unit-2, Barh Stage-I in place of U#4 during the period from 01.07.2026 to 14.08.2026.*
- *Bihar representative informed that Bihar has a share of 40% from NTPC Barh St-I and July-September is the peak demand period for Bihar. In view of this, the proposed shutdown may be deferred.*
- *NTPC replied that U#4 will be in service and power will be available from U#4 during the proposed shutdown period of U#2 St-I.*
- *NTPC Darlipalli requested to prepone the shutdown of U#1 from 20 Aug, 2026 to 10 Aug, 2026. In response, SLDC Odisha submitted that the shutdown may be availed from 15 Aug, 2026.*
- *NTPC Kahalgaon requested to avail shutdown of U#1 in place U#3 from 15th July 2026 for 45 days as per LGBR.*

OCC Decision

OCC forum agreed with the following planned overhauling shutdowns:

1. NTPC Barh U#2 (in place of U#4) as approved in the LGBR.
2. NTPC Darlipalli U#1 from 15th August 2026 for 45 days.
3. NTPC Kahalgaon U#3 from 15th July 2026 for 45 days in place of U#1 as per LGBR.
4. Nabinagar STPS U#2 from 25th Nov'26 to 3rd Jan'27 for 40 days
5. BRBCL Unit-3 from 13.07.2026 to 16.08.2026 subject to consent from Indian Railways.
6. Shutdown of PVUNL Unit #1 is approved from Feb 2027 to March 2027 subject to consent from SLDC Jharkhand.

NTPC Farakka U#5 (from 1st July 2026 to 29th July 2026) stands postponed as requested by NTPC.

2.13 Agenda by NTPC Darlipalli

□ Reactive Charges revision for the week 30.03.2026 to 05.04.2026

Abnormal VAR reading was observed in meter ER 2054-A on 05.04.2026 resulting in Var charges payment 5.39 lacs. The same was cross-verified with the station meter and found to be normal. Main and check meter reading may be cross checked.

Deliberations in the meeting

It was apprised that on receipt of status of changed meters from ERLDC, the reactive charges account will be revised accordingly by ERPC at the earliest.

□ SG Ramp rate violations

- a. Approximately 40 incidents of ramp rate violations were observed from Apr-2026, resulting in a DSM loss of around 6 lacs.
- b. Relevant CERC Clause as " TRAS operation by NLDC should not result in below technical minimum or ramp rate violations as per " CERC Detailed Procedure for Security Constrained Unit Commitment (SCUC), Unit Shut Down (USD) and Security Constrained Economic Despatch (SCED) at Regional Level"

Deliberations in the meeting

ERLDC submitted:

- ✓ *The ramp rate violations of NTPC Darlipalli schedule were due to the dispatch of Market Based Ancillary Services (MBAS) quantum. ERLDC presented an example case of NTPC Darlipalli ramp violation due to MBAS which occurred on 03rd April 2026 (attached as **Annexure B.2.13**).*
- ✓ *As per extant regulations and detailed TRAS procedures, the Nodal Agency may dispatch any quantum within the MBAS-cleared quantum based on system requirements.*
- ✓ *All generating stations not to send post facto correction emails for MBAS despatched.*

NTPC Darlipalli submitted that they are placing bids within the 1% ramp rate limit; however, ramp rate violations are still being observed in the resultant schedules.

In response, ERLDC highlighted that while submitting MBAS bids, generating stations may need to consider the potential variation in other components of schedules.

OCC Decision

OCC advised ERLDC/NLDC to review the matter in accordance with the CERC clause and suggested that all generating stations may submit MBAS bids after duly considering the ramping capability of their plants.

2.14 Status of Periodic Testing of Generating Units under IEGC 2023: ERLDC

- ☐ Regulation 40 [40(2)(b) & 40(2)(c)] of the CERC (IEGC) Regulations, 2023 mandates utilities to undertake periodic testing of power system elements **every five years** or after major retrofitting and to submit annual testing plans to RPC.
- ☐ Regulation 40(3) specifies the tests required for different power system elements as follows:

Power System Element	Tests Required	Applicability
Synchronous Generator	1. Real and Reactive Power Capability Assessment. 2. Assessment of Reactive Power Control Capability as per CEA Technical Standards for Connectivity 3. Model Validation and Verification Test for complete Generator and Excitation System Model including PSS. 4. Model Validation and Verification of Turbine/Governor and Load Control or Active Power/Frequency Control Functions. 5. Testing of Governor Performance and Automatic Generation Control (AGC)	Individual generating units rated: • 100 MW and above (Coal/Lignite) • 50 MW and above (Gas Turbine) • 25 MW and above (Hydro)
Non-Synchronous Generators (Solar/Wind)	1. Real and Reactive Power Capability Test. 2. Power Plant Controller (PPC) Function Test. 3. Frequency Response Test 4. Active Power Set-Point Change Test. 4. Reactive Power (Voltage/Power Factor/Q) Set-Point Change Test	Applicable as per CEA Technical Standards for Connectivity
HVDC FACTS Devices	1. Reactive Power Controller (RPC) Capability Test. 2. Filter Bank Adequacy Assessment based on prevailing grid conditions in consultation with NLDC.	Applicable to all ISTS HVDC systems and intra-State

	3. Validation of response of FACTS devices as per settings	HVDC/FACTS installations, as applicable
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The matter was raised in 217th OCC meeting dated 24.07.2024, where OCC advised all applicable generators and HVDC/FACTS device owners to strictly adhere to the IEGC 2023 guidelines and submit the required testing data and plans to ERPC. The same matter was again discussed in 220th OCC to 224th ER OCC meeting, schedule for testing from all generating stations were collected and it was decided that generators will follow the schedule strictly and share the report subsequently.

The compiled details of generating units, HVDC/FACTS elements and their respective testing status are available at the link provided below:

<https://docs.google.com/spreadsheets/d/1m6KCKONdObMhre9-1me1kvHTEBYUdXUOISYdn5FR4fM/edit?gid=690780910#gid=690780910>

It has been observed that several generating units are yet to complete the periodic testing planned under Regulation 40 of IEGC 2023. In several cases, the previously proposed testing dates have already elapsed, while revised testing schedules have not been received. Additionally, some generating units have indicated tentative testing plans, which are yet to be executed. After completion of the testing, plants are requested to share the report along with relevant data to ERLDC.

After notification IEGC-2023 within 5 years each unit has to undergo periodic testing and few (as tabulated below) of the units have only done testing till now, this will pose an overburden as now all remaining Units has to be tested within these two years. This is to ensure that the testing schedule may be staggered as accommodation of too many units within short period may not be possible to maintain reliability.

	PFR	PSS	Reactive
JIPL	Done	Not Done	Done
Talcher St-1	Not Done	Not Done	Done
Darlipalli	Not Done	Not Done	Done
Barh-II	Not Done	Not Done	Done
Tashiding	Not Done	Done	Not Done
Jorethang	Not Done	Done	Not Done

All the ISGS, IPP, intra-state generators and owners of the transmission elements as mentioned in the link as per regulation 40(3) are requested to update the status of testing. If periodic testing has been completed by any stakeholder, then they may kindly update the status and share the testing reports along with relevant data with ERPC and ERLDC.

If test is yet to be conducted or the previously anticipated dates of test are surpassed, a fresh proposed date may be submitted to ERPC, and timeline is to be strictly followed.

Deliberations in the meeting

ERLDC deliberated that, as mandated by regulation 40(3) of the IEGC 2023, every generating unit and designated transmission element need to undergo periodic testing within a five-year from the date of notification. Accordingly, schedule for testing were collected in a google sheet & ratified in previous OCC meetings. However, following critical issues were encountered:

- Several generating units have failed to complete their planned testing within the previously proposed windows, and the dates have elapsed without any revised timelines being submitted.
- Only few nos. of regional units have completed their testing so far. Deferring the remaining vast majority of units to the final two years will create an overburden. Accommodating a large no of units for testing within a limited period will be difficult.

OCC Decision

- All ISGS, IPPs, intra-state generators, and transmission element owners must review the proposed date and update their current testing status in the shared drive.
- Generators whose testing windows have elapsed, or who previously provided tentative dates, are directed to coordinate internally and submit firm, revised schedules.
- Upon successful completion of any periodic test, the concerned utility must upload the comprehensive test report along with all high-resolution raw data logs to the designated shared folder.

2.15 Submission of Resource Adequacy Data: ERLDC

- As per the IEGC, 2023, each SLDC/utility is required to furnish Day-Ahead, Week-Ahead, Month-Ahead, and Year-Ahead Resource Adequacy (RA) data, including demand forecast, generation availability, reserve margins, and procurement plans, in accordance with the NLDC/RLDC Operating Procedure.
- Grid-India, in compliance with the Hon'ble CERC's Order dated 07.10.2024 in Petition No. 9/SM/2024, had submitted that all Eastern Region States are furnishing Day-Ahead Resource Adequacy information. Further, the Day-Ahead reporting format was revised in 2025 to incorporate reserve-related information (attached as **Annexure.B.2.15.**).
- Latest Forecast and Resource Adequacy Data receipt status at ERLDC is attached as **Annexure.B.2.15.**
- However, a review of the data being submitted through the Resource Adequacy Portal indicates that complete information is not being furnished by all States/utilities. In particular, reserve details and proposed procurement information are not being submitted in several cases. Further, variations observed between projected and actual demand and resource positions highlight the need for periodic review and improvement of forecasting methodologies to enhance the accuracy and reliability of Resource Adequacy assessments.
- In view of the above, all States/utilities are requested to ensure timely submission of complete, accurate, and updated Resource Adequacy data through the Resource Adequacy Portal. States/utilities may also review and strengthen their forecasting practices and furnish reserve and procurement details in the prescribed format to facilitate reliable resource planning, operational preparedness, and effective assessment of resource adequacy across all time horizons.

Deliberations in the meeting

ERLDC stated:

- ✓ All the Eastern Region States are submitting Day-Ahead Resource Adequacy (RA) data through the RA Portal.

- ✓ During the meeting held with CERC on 01.08.2025, emphasis was placed on the submission of complete RA data, including reserve and proposed procurement details.
- ✓ Reserve details and proposed procurement information are yet to be furnished by the States. It was further observed that variations between projected and actual demand are being noticed, particularly during sudden weather changes, indicating the need for continuous improvement in forecasting methodologies to enhance forecasting accuracy and resource planning.

OCC Decision

- OCC emphasized the significance of resource adequacy for demand-side portfolio management and outage planning.
- All ER states were advised to submit reserve quantum & tentative procurement quantum to ERLDC for better Operational planning.

2.16 State-wise Peak Demand during Non-Solar Hours & Solar Hours: ERPC

- ✓ In view of the discussions held during PSP meeting at the Ministry of Power (MoP), the forum may discuss the methodology and data requirements for capturing state-wise peak demand during non-solar hours from April 2026 onwards.
- ✓ For the purpose of data compilation, solar hours may be considered as **06:00 hrs to 18:00 hrs** and the remaining period as non-solar hours. During the winter season, solar hours may be considered as **07:00 hrs to 17:00 hrs**, in line with the methodology adopted in NLDC reports.
- ✓ The constituents may provide state-wise data on peak demand during solar & non-solar hours for further analysis and consolidation.

Deliberations in the meeting

OCC Decision

All SLDCs of ER are advised to submit their individual peak demands for Solar & Non-Solar Hours, to ERPC on monthly basis. In Summer season 06:00 hrs to 18:00 hrs need to be considered as solar hours while 07:00 hrs to 17:00 hrs to be treated as solar hours in winter and the remaining period shall be considered as non-solar hours in line with NLDC methodology.

2.17 Submission of Self Audit Compliance Report: ERLDC

- As per Clause 55 of IEGC 2023, the performance of all users, CTU, STUs, NLDC, RLDCs, SLDCs and RPCs, power exchanges, QCAs, SNAs with respect to compliance of these regulations shall be assessed periodically. IEGC mandates all entities to undertake self-audit and submit a report containing compliances, non-compliances (if any) along with sufficient information to understand how and why the non-compliance occurred, extent of damage caused by such non-compliance, steps and timeline planned to rectify the same, steps taken to mitigate any future recurrence etc., with regard to various regulations of IEGC to the respective nodal agency.
- Further as per clause 56 (2) of IEGC, to ensure compliance, all users, CTU, STUs, NLDC, RLDCs, RPCs and SLDCs, power exchanges, QCAs, SNAs shall conduct annual self-audits to review compliance of these regulations and submit the reports by 31st July of every year. Accordingly, all users of the Eastern Regional Load Despatch Centre (ERLDC) are required to submit their Self-Audit Report for FY 2025-26 to ERLDC by 31st July 2026.

Deliberations in the meeting

OCC Decision

OCC noted and advised all users/entities to abide by submission of Self Audit Compliance Report to CERC as mandated in IEGC 2023.

2.18 Dismantling of 132/33 KV Switchyard, Patratu GSS: JUSNL

- ☐ The 132/33 KV Patratu including PTPS, Switchyard area is to be utilized by PUVNL for construction of its 3rd unit generation setup installation. Now as on dated construction power of PVUNL has been shifted to PSS of JBVNL. In addition, PVUNL has requested for dismantling of old switchyard of PTPS. So there is no requirement of old switch yard in old PTPS.
- ☐ In addition, there is urgent requirement of 01 Nos 50 MVA, 132/33 KV Transformer to be shifted form old PTPS switchyard to replace the defective transformer of 132/33 KV GSS Chitra.
- ☐ There is proposal of diversion of following four lines and will be terminated to newly constructed 400/220/132/33 KV GSS Patratu New (Katia). The details are as follows:
 1. 132 KV Hatia-1 to Patratu 8C
 2. 132 KV Hatia-1 to Patratu 9C
 3. 132 KV Ramgarh(DVC) to Patratu 5C
 4. 132 KV Ramgarh(DVC) to Patratu 6C
- ☐ In view of above, approval is required for Switching OFF 5C and 6C interstate tie line of DVC and idle charge from DVC end till its diversion to newly constructed 400/220/132/33 KV GSS ,Katia, Patratu.

Deliberations in the meeting**OCC Decision**

OCC opined that the proposal may be put up in CEA meeting to be held on 16th June 2026. This was also referred to the upcoming CMETS-ER for deliberation.

2.19 Agenda from PVUNL

S No.	Agenda / Issue	Details / Background	Support Requested	240 th OCC meeting
1	Frequent AGC Communication Failure	AGC communication failure is occurring frequently, resulting in the inability of PVUNL to effectively participate in Ancillary Services. The recurring communication issues adversely impact grid support functions and operational efficiency. Reliable communication infrastructure is essential for seamless AGC operation.	STU and PGCIL may be requested to strengthen and ensure reliable AGC communication infrastructure to facilitate uninterrupted participation in Ancillary Services.	<i>Powergrid , SLDC Jharkhand and Patratu TPP will coordinate to strengthen AGC communication and formulate SOP for facilitating uninterrupted participation in Ancillary Services.</i> <i>Moreover, dual connectivity</i>

				<i>needs to be ensured by Patratu TPP.</i>
2	Strengthening of Evacuation System for Patratu Generating Station	Despite implementation of SPS at Patratu Station, one PVUNL unit is required to be taken off-bar under N-1 contingency conditions when either of the Patratu–Bero (New Ranchi) transmission lines is out of service. Patratu–Bero Line-1 has tripped twice in the last six months, highlighting the need for improved transmission reliability and evacuation capability.	JUSNL may be requested to strengthen the evacuation network from Patratu Station and improve the reliability of the Patratu–Bero transmission corridor to ensure evacuation of generation under N-1 contingency conditions without backing down any unit. The work on Patratu-Latehar line may be expedited.	<i>JUSNL was advised to submit a detailed action plan with timeline for ensuring N-1 reliability of Patratu–Bero (New Ranchi) transmission corridor responsible for power evacuation from Patratu units.</i>
3	Issuance of Additional GNA NOC by STU/SLDC.	Following the COD of Unit #2, additional GNA quantum is required for scheduling of power by ERLDC. As deliberated in the 239th OCC meeting, NOC for additional GNA is to be issued by STU. However, the same is still pending, affecting scheduling arrangements.	STU/SLDC may be advised to issue the pending NOC for additional GNA at the earliest to facilitate scheduling of power from both the commissioned units of PVUNL.	<i>SLDC Jharkhand updated that application has been received and NOC for additional GNA will be issued shortly.</i>
4	Planned Overhauling of Unit #1 (Feb 2027 to Mar 2027)	As per OEM recommendations and technical requirements, Unit #1 will be due for its first overhauling after one year of successful operation. Unit #1 achieved COD in November 2025, and the overhauling is proposed from February 2027 to March 2027.	ERPC, ERLDC and beneficiaries may kindly take note of the proposed overhauling schedule and extend necessary support for planning of system operations during the outage period.	<i>Planned Overhauling of Unit #1 is approved from Feb 2027 to March 2027 subject to consent from SLDC Jharkhand.</i>

2.20 Connectivity Scheme of NTPC Darlipali Stage-II: NTPC

- Darlipali STPP Stage-I (2×800 MW) has ISTS connectivity of 1600 MW capacity through the Darlipali–Sundergarh 765 kV, one D/C transmission line. Now, NTPC has awarded Darlipali Stage-II (1×800 MW) and has applied for dual connectivity, i.e., STU/OPTCL for 400 MW quantum and ISTS for 400 MW. To draw the OPTCL/STU share of power, OPTCL is constructing one 765 kV D/C line from Darlipali to Kolabira (OPTCL). The single line diagram of both Stage-I and Stage-II arrangements are enclosed.
- With reference to the discussions held on the NTPC Darlipali Stage-II connectivity application during the CMETS-ER meeting on 29.05.2026, Grid India raised concerns regarding the energy accounting and metering arrangement for Stage-II power accounting, as Stage-II has dual connectivity, i.e., STU and ISTS, with reference to the prevailing CEA Metering Regulations. Grid India advised NTPC to place the matter before the OCC/TCC/ERPC for detailed deliberation prior to finalization of the power evacuation scheme.
- In this regard, ERLDC is of the view that a bus sectionalizer should be provided in accordance with the current CEA Metering Regulations. However, NTPC has expressed

the opinion that, as CTUIL has informed that no separate ISTS lines are envisaged for Stage-II, ISTS power evacuation can be managed through the existing ISTS lines, which have sufficient margin to evacuate both Stage-I and Stage-II. Further, even if a bus sectionalizer is provided between Stage-I and Stage-II, it would normally remain in a closed condition.

- In view of the above, NTPC has requested Grid India to review the requirement of a bus sectionalizer. The energy measurement/accounting can be carried out based on meter readings at Stage-I and Stage-II lines, metering at the generator end, and auxiliary power consumption at the HT side of station transformers for accounting of Stage-I and Stage-II generation.

Deliberations in the meeting

NTPC presented the proposed connectivity arrangement for Darlipalli Unit-3 (Stage-II) (details at Annex B.2.20) and requested approval of the metering philosophy.

NTPC submitted:

- ✓ *The metering philosophy should be considered as per the philosophy being followed in many NTPC stations where new stage has been considered.*
- ✓ *Requirement of the additional bus section that was deliberated during the last CMETS-ER meeting was requested to be waived off.*
- ✓ *The auxiliary power system of Stage-I and Stage-II would be common and therefore installation of a bus sectionalizer solely for metering purposes may not be technically justified.*
- ✓ *No separate ISTS lines are envisaged for Stage-II as informed by CTUIL since ISTS power evacuation can be managed through the existing ISTS lines, having sufficient margin for evacuation from both Stage-I and Stage-II. Further, even if a bus sectionalizer is provided between Stage-I and Stage-II, it would normally remain in closed condition.*

ERLDC, however, highlighted that as per CEA metering regulation separate metering arrangement is required for different stages of the same generating station having different tariffs.

OCC Decision

OCC advised NTPC to adhere to the CEA metering regulation.

2.21 Certification of 400 kV STU Lines of OPTCL as Non-ISTS Lines Carrying ISTS Power: ERPC

- OPTCL had filed a petition before CERC for approval of transmission tariff for the period 2014–19 in respect of the following nine Non-ISTS lines owned by OPTCL, which are carrying ISTS power, for inclusion in the computation of PoC charges and losses in accordance with the CERC (Sharing of Transmission Charges and Losses) Regulations, 2010.
- The Hon'ble CERC, vide order dated 26.02.2025 in Case No. 288/TT/2023, has approved the tariff of seven lines and recommended their inclusion in the computation of PoC

charges. It may be noted that the Balimela PH-Upper Sileru line was considered under idle charges, and the Rengali-Rengali PGCILDC line was not considered, which appears to be inadvertent and will be apprised to the Hon'ble Commission in the subsequent tariff filing.

- ❑ In **239th OCC** meeting,
- ✓ OPTCL informed that Certification of these non-ISTS lines carrying ISTS power for the period 2014–2019 had already been completed while certification for the 2019–2024 period remains pending.
- ✓ During deliberation, it emerged that under revised sharing regulations, responsibility for certification changed after April 2019. For the period April 2019 to October 2020, certification responsibility remains with RPC and thereafter, the same is entrusted with CEA/CTU.
- ✓ ERLDC was advised to carry out the necessary power flow study to ascertain flow pattern of ISTS power through non-ISTS lines of OPTCL.
- ✓ OPTCL shall provide all necessary operational and power flow data to ERLDC with copy to ERPC. As certification of these non-ISTS lines has significant financial implications, OCC opined to expedite the process and ERLDC will submit study report in the next OCC.
- ❑ **Latest update:** ERLDC has carried out the necessary power flow study for the period April 2019 to October 2020 to ascertain the flow pattern and utilization of the 7 transmission lines toward meeting ISTS power outside Odisha.
- ❑ The utilization figures, as obtained from the ERLDC study, are enclosed for your kind reference (**Annexure A**) .

Deliberations in the meeting

OPTCL informed:

- ✓ *Tariff approval had already been granted by CERC for seven transmission lines carrying ISTS power.*
- ✓ *The Rengali Switchyard–Rengali PGCIL line had not been considered by CERC due to an apparent omission despite certification by ERLDC studies.*
- ✓ *Certification of all the seven lines up to October 2020 was requested for filing petition before CERC.*

OCC Decision

- *Based on the Webnet study submitted by ERLDC in **Annexure B.2.21**, OCC in principle agreed with the certification of 6 (six) non-ISTS lines of OPTCL as deemed ISTS for the period where the flow of ISTS power is more than 50%. The matter was referred to TCC forum for concurrence.*
- *OPTCL was advised to approach CERC for inclusion of Rengali–Rengali PGCIL D/C (220 kV) line in tariff block 2019-2024.*

2.22 Disruption of OT communication system of Eastern Regional power grid on 01-06-2026: ERLDC

- On 01-06-2026, the communication system of the Eastern Regional power grid which is vital for visualization, monitoring and control of Eastern Regional Grid, almost completely stopped functioning w.e.f 12:10 hours, leading to the disruption of the following services at ERLDC:
 1. Inter Control Centre Protocol (ICCP) links with all constituents.
 2. RTU/SAS communication from substations and generating stations reporting over IEC 60870-5-104 protocol.
 3. Phasor Measurement (PMU) and Wide Area Monitoring Systems(WAMS).
 4. VoIP communication with power stations and substations.
 5. Automatic Meter Reading(AMR) communication.
 6. Automatic Generation Control (AGC) communication with power plants.
- While complete SCADA data could be restored after midnight of 01.06.2026, PMU and PDC data were restored only on 02.06.2026, resulting in an outage of nearly 24 hours.
- The above incident led to a complete loss of real-time visibility of the Eastern Region power System, thus creating serious operational and monitoring challenges at ERLDC, NLDC, and the SLDCs of the Eastern Region.
- Considering the gravity of the situation ERLDC vide letter dated 05-06-2026 has already proposed constitution of a committee comprising concerned representatives from SLDCs, ERLDC, POWERGRID, ISGSS, IPPs & ERPC for analyzing root cause of the incident and recommending remedial measures for implementation in a time bound manner.

Deliberations in the meeting

ERLDC informed:

- *On 1st June 2026, ICCP communication between ERLDC and SLDCs was completely disrupted for nearly ten hours.*
- *As a result, RTU/SCADA data reporting, PMU data, ATC communication and other operational visibility functions were severely affected.*
- *Similar incidents had occurred earlier in 2020 and 2023, making this a recurring and serious concern.*

OCC Decision

- *OCC opined that a committee under the chairmanship of Member Secretary, ERPC, will be constituted to investigate the incident. The committee shall investigate the root cause of the communication failure, reconstruct the sequence of events, analyse fault propagation mechanisms, review network architecture and compliance with extant regulations of Communication system and recommend short-term as well as long-term corrective measures.*
- *Powergrid, ISGSS, all SLDCs, and concerned entities shall nominate communication/SCADA/cyber-security experts within one week for the investigation committee.*

2.23 Data Collection for monitoring Pan-India Captive Generating Capacity: RPSO (CEA)

- Present electricity generation figures reflect only the power generated by the utilities and do not capture the gross electricity generation from Captive Power Plants (CPPs). Accounting for generation from these sources is essential to arrive at a comprehensive assessment of total energy generation and actual power consumption beyond utility-based generation.
- In the meeting taken by **Secretary (Power)**, Govt of India on **17.12.2025**, it was decided that the State Chief Electrical Inspectors (CEIs) / State Load Despatch Centres (SLDCs) shall act as the nodal agencies for collection of **Captive Generation & Open Access** data for their respective States.
- At present, the **Central Electricity Authority (CEA)** collects details of **Captive Power Plants** having an **installed capacity of 0.5MW** and above in Format-21 prescribed under **Central Electricity Authority (Furnishing of Statistics, Returns and Information) Regulations, 2007**.
- In order to collect monthly data of captive generation, an online portal (<https://intranet.cea.gov.in/captiveNew/>) for data collection has been developed in CEA and the monthly data collection process needs to be streamlined through this portal with the involvement of **Regional Power Survey Offices (RPSOs)**.

Deliberations in the meeting

RPSO representative presented the status of data updation i.r.o CPPs in Eastern region as follows:

- ✓ *Status of Eastern Region:*

No. of entities registered on CEA-Captive Power Portal	Before 239th OCC	Before 240th OCC
	125	152

- ✓ *State-wise breakup of the number of known entities (replied in **Format-21** for FY 2024-25) which have **not yet registered** on CEA-Captive Power Portal:*

State	Nos.
Odisha	32
West Bengal	51
Jharkhand	8
Bihar	3
TOTAL	94

OCC Decision

- All SLDCs in Eastern region are advised to ensure on-boarding of the CPPs under their jurisdiction get registered on the said portal (<https://intranet.cea.gov.in/captiveNew/>) at the earliest. All SLDCs shall pursue the pending entities in their respective states for immediate registration. Present registration status of CPPs is attached at **Annex B.2.23**.
- Updated registration status of CPPs shall be reviewed periodically in OCC meetings.

2.24 Deputation of SLDC officials to Grid - India for Short Term Exposure Programme: ERLDC

- As part of the Ministry of Power's broader initiative to enhance system operations and capacity building at Load Despatch Centres, a short-term knowledge-sharing exchange programme (2 to 10 days) between RLDCs and SLDCs covering areas of System Operation, Market Operation, System Logistics and Cyber security function was introduced to foster cohesion and knowledge exchange. The objective of this programme is to facilitate peer-to-peer learning from each other and propagate best practices through hands on exposure of real time working of other LDCs.
- In this regard, SLDCs are requested to depute officials to ERLDC for the short-term exposure programme.

Deliberations in the meeting

OCC Decision

All SLDCs in Eastern region are advised to explore and nominate SLDC officials to Grid - India for Short Term Exposure in line with the initiative from Ministry of Power, Govt of India. This initiative is intended to facilitate short-term knowledge-sharing as well as propagation of best practices between RLDCs and SLDCs across the domains of System Operation, Market Operation, and System Logistics and Cyber security.

3. PART-C: ITEMS FOR INFORMATION

3.1. ER Grid performance during May 2026

The average and maximum consumption of Eastern Region and Max/Min Demand (MW), Energy Export for the month May-2026 were as follows:

AVERAGE CONSUMPTION (MU)	MAXIMUM CONSUMPTION(MU)/ DATE	MAXIMUM DEMAND (MW)	MINIMUM DEMAND (MW)	SCHEDULE EXPORT	ACTUAL EXPORT
		DATE / TIME	DATE / TIME	(MU)	(MU)
632 MU	750.7 MU, 22.05.2026	34875 MW, 22.05.2026 at 23:23 Hrs.	17389 MW, 05.05.2026 at 04:16 Hrs.	2617	3054

Deliberations in the meeting

The grid performance of ER for the month of May 2026 was highlighted.

3.2. Non-Submission of FRC data in stipulated time-frame: ERLDC

Adhering to IEGC clauses 30.8 and 30.10.(a) to 30.10.(q), generating stations within the Eastern region are required to submit essential data to ERLDC within two days of receiving a notification regarding a reportable frequency event. Additionally, according to clause 30.10.(n), all control areas within the eastern region must assess their frequency response characteristics and share the evaluation, along with high-resolution data, with the ERLDC. Therefore, timely submission of primary response data is crucial for compliance with the IEGC.

Hence all are again requested to follow the stipulated timeline and submit the data to ERLDC and also fill the google sheet below to include the email address where notifications of reportable events should be sent.

The latest data receipt status is given below: (as on **01.06.2026**):

STATIONS		14.01.2026 13:38 hrs	14.01.2026 14:05 hrs	14.01.2026 14:09 hrs	30.01.2026 11:08 hrs	13.02.2026 12:14 hrs	22.02.2026 22:56 hrs	22.02.2026 23:00 hrs	03.03.2026 11:42 hrs	13.04.2026 11:00 hrs	13.05.2026 14:09 hrs	13.05.2026 16:16 hrs	15.05.2026 11:06 hrs
FSTPP #STG 1 & 2	ISGS												
FSTPP # STG 3	ISGS												
KhSTPP #STG 1	ISGS												
KhSTPP #STG 2	ISGS												
TSTPP #STG 1	ISGS												
Barh stage-1	ISGS												
Barh stage-2	ISGS												
BRBCL	ISGS												
Darlipalli	ISGS												
North Karanpura	ISGS												
NPGC	ISGS												
TEESTA V	ISGS												
PVUNL	ISGS												
Dikchu	IPP												
IBEUL (JSW UTKAL)/INDBHARAT	IPP												
GMR	CPP												
MPL	CPP												
ADHUNIK	CPP												
JITPL	CPP												
TEESTA III	CPP												
Bihar	STATE												
Jharkhand	STATE												
DVC	STATE												
OPTCL	STATE												
WB	STATE												
Updated as on	01.06.2026												
	Received												
	Not Received												
	Plant Out												
	Data freeze at plant												

Hence all are again requested to follow the stipulated timeline and submit the data to ERLDC and also fill in the google sheet below to include the email address where notifications of reportable events should be sent.

https://docs.google.com/spreadsheets/d/1slvAOmQIEQVIMn0LnB78eKMa2sz2QYICZ-sPEpeV_jk/edit?usp=sharing

Deliberations in the meeting

OCC Decision

- All generators were advised to regularly share high resolution data against each reportable frequency event with ERLDC on time to facilitate accurate assessment of FRP for respective control areas.
- All generating utilities were also urged to update the google sheet (link mentioned above) with email address where notifications of reportable events will be shared.

3.3. Regarding Non-Submission of Forecasting Data from States: ERLDC

Clause 2 of Regulation 31 of IEGC 2023 has mandated all the SLDCs to timely submit the demand estimate data to the respective RLDC and RPC.

Current data submission status is given in the table below: Hence it is again requested to all the concerned for timely submission of demand estimation data to ERLDC. This collaboration is essential for effective planning and preparedness to meet the region's electricity demands efficiently and reliably.

Latest Forecast and Resource Adequacy Data receipt status at ERLDC is shown below:

4. PART-D: OPERATIONAL PLANNING

4.1. Major Thermal Generating Units/Transmission Element outages/shutdown in ER Grid (as on 04-06-2026)

SL No	STATION	STATE	AGENCY	UNIT NO	CAPACITY (MW)	REASON(S)	OUTAGE DATE
1	JIPL	ODISHA	JIPL	1	600	BOTTOM SCRAPPER PROBLEM.	04-Jun-2026
2	BARH	BIHAR	NTPC	3	660	Boiler Tube Leakage	01-Jun-2026
3	SANTALDIH TPS	WEST BENGAL	WBDCL	6	250	Loss of evacuation path	04-Jun-2026
4	SANTALDIH TPS	WEST BENGAL	WBDCL	5	250	Boiler Tube Leakage	03-Jun-2026
5	SAGARDIGHI	WEST BENGAL	WBDCL	5	660	Problem in AHP	31-May-2026
6	BARAUNI TPS	BIHAR	NTPC	9	250	Boiler tube leakage	04-May-2026
7	RTPS	DVC	DVC	2	600	High Turbine Vibration	14-Apr-2026

Major Generating stations Out on Reserve Shutdown due to low system demand:

SL No	STATION	STATE	AGENCY	UNIT NO	CAPACITY (MW)	REASON(S)	OUTAGE DATE
NA							

Hydro Unit Outage Report: -

S. NO	STATION	STATE	AGENCY	UNIT NO	CAPACITY (MW)	REASON(S)	OUTAGE DATE
1	BURLA HPS/HIRAKUD I	ODISHA	OHPC	5	43.65	Annual Maintenance	19-Jan-2026
2	BALIMELA HPS	ODISHA	OHPC	5	60	Repair and maintenance work	16-Jan-2025
3	BALIMELA HPS	ODISHA	OHPC	6	60	Initially unit was out due to Severe water leakage from turbine, later unit was taken under Repair and maintenance work from 00:00 hrs of 16.01.25	06-Jan-2025
4	TEESTA HPS	SIKKIM	NHPC	1	170	Sudden cloudburst at glacier fed LOHNAK Lake followed by huge	04-Oct-2023

5	TEESTA HPS	SIKKIM	NHPC	2	170	inrush of water in Teesta River and damage of Teesta III Dam & downstream Powerhouses	04-Oct-2023
6	TEESTA HPS	SIKKIM	NHPC	3	170		04-Oct-2023
7	TEESTA STG III Hep	SIKKIM	TUL	1	200		04-Oct-2023
8	TEESTA STG III Hep	SIKKIM	TUL	2	200		04-Oct-2023
9	TEESTA STG III Hep	SIKKIM	TUL	3	200		04-Oct-2023
10	TEESTA STG III Hep	SIKKIM	TUL	4	200		04-Oct-2023
11	TEESTA STG III Hep	SIKKIM	TUL	5	200		04-Oct-2023
12	TEESTA STG III Hep	SIKKIM	TUL	6	200		04-Oct-2023
13	U. KOLAB	ODISHA	OHPC	2	80	Heavy Leakage in guide vane	22-Jan-2026
14	BALIMELA HPS	ODISHA	OHPC	4	60	Could not be brought back into service due to profuse oil leakage in Cable Gland box bath at the inlet of RTT/DTT cable	04-Jun-2026
15	SANTALDIH TPS	WEST BENGAL	WBPDC	6	250	Loss of evacuation path	04-Jun-2026
16	SANTALDIH TPS	WEST BENGAL	WBPDC	5	250	Boiler Tube Leakage	03-Jun-2026

4.2. Long outage report of transmission Element (MORE THAN 01 WEEK) (As on 04.06.2026):

Transmission Element / ICT	Outage From	Reasons for Outage
220KV-FSTPP-LALMATIA-I	21-04-2021	Two nos. of tower collapsed on 29.05.2024 near to Lalmatia GSS in the Loc. No. 246 & 247. 220KV S/C Farakka-Lalmatia Transmission Line is in anti-theft charging condition from Loc no 248 (Lalmatia end) to Loc no 33. Foundation, erection, and stringing progress from loc 1 to 32
132KV-BARHI-RAJGIR-1	25-03-2023	Dismantling of tower no. 227, 228, and 229 crossing the premises of Mahabodhi Cultural centre along with Destraining of conductor of both circuits and Earth wire between tension tower no. 218-237 in same line. The lines from Barhi (DVC) will be terminated at Barachatti (BH) and new line

		to be constructed from Barachatti to Rajgir (BH)
132KV-NALANDA-BARHI(DVC)-1	25-03-2023	Dismantling of tower no. 227, 228, and 229 crossing the premises of Mahabodhi Cultural centre along with Destraining of conductor of both circuits and Earth wire between tension tower no. 218-237 in same line. The lines from Barhi (DVC) will be terminated at Barachatti (BH) and new line to be constructed from Barachatti to Nalanda (BH)
400KV-RANGPO-TEESTA-V-1	04-10-2023	Tower near gantry of Teesta V HEP collapsed during GLOF event in Oct 2023 also leading to damage in powerhouse. Tower subsequently erected on 15.06.2024. Teesta V HEP GIS damaged due to hill sinking on 20.08.2024. Presently, GIS under restoration and generation expected by July 2026.
400KV-RANGPO-TEESTA-V-2	04-10-2023	Tower near gantry of Teesta V HEP collapsed during GLOF event in Oct 2023 also leading to damage in powerhouse. Tower subsequently erected on 15.06.2024. Teesta V HEP GIS damaged due to hill sinking on 20.08.2024. Presently, GIS under restoration and generation expected by July 2026.
400KV/220KV 315 MVA ICT 1 AT NORTH KARANPURA	12-09-2024	Tripped on Differential protection
400KV/220KV 315 MVA ICT 1 AT TSTPP	01-11-2024	Tripped on PRD protection. The failed transformer has reached the vendor, the repair scope has been finalized, with the ICT likely to be available by 30.06.2026.
400KV/220KV 315 MVA ICT 2 AT MEJIA-B	20-01-2025	315 MVA ICT-2 at MTPS-B got damaged while charging from 220kV GIS bay. New procurement of ICT has been taken up & installation of the same may complete by end of Mar'28.
400KV-DIKCHU-RANGPO-2	05-08-2025	Shutdown taken for damaged insulator replacement work. While charging the line, bus bar protection operated at Dikchu. Issue in GIS chamber of Y ph Isolator between line CB and bus 2. Powder formation inside isolator chamber has occurred. Restoration work delayed.
400KV MAIN BUS - 2 AT DIKCHU	05-08-2025	Bus bar protection operated while charging 400kV Dikchu-Rangpo 2. Issue in GIS chamber of Y ph. Isolator between Rangpo ckt 2-line CB and bus 2. Powder formation inside isolator chamber has occurred. Restoration work delayed.
220KV-PATNA-KHAGAUL-1	24-09-2025	LBB relay operated during rectification of DC grounding defect by M/S KRR at GSS Khagaul. Earlier w.e.f 02-08-2025 12:06 Hrs, Tower No. 63 has bent significantly on one side.
220KV-BIDHANNAGAR-WARIA-1	29-10-2025	To control loading of 220 kV Waria-Mejia D/C (Anti-theft charged from Waria end.)

220KV-BIDHANNAGAR-WARIA-2	29-10-2025	Initially line was opened to control line loading. In between B-phase CT Blast at Bidhannagar end. Now Line is charged as anti-theft from Waria end to control loading of 220 kV Waria-Mejia D/C.
220KV-BALIMELA-UPPER SILERU-1	21-11-2025	Idle charged from U. Sileru end. Power drawl by Odisha halted due to non-concurrence by Andhra Pradesh.
400KV/220KV 315 MVA ICT 1 AT KODERMA	10-02-2026	ICT out due to rising acetylene trend.
132KV-RANGIT-SAGBARI-1	28-02-2026	Necessary jumpering modification at Sagbari station (EPDS).
400KV-TEESTA-III-RANGPO-1	15-03-2026	Rangpo: B_N, DEF, 0.68 kA
400KV/220KV 315 MVA ICT 2 AT LAPANGA	14-04-2026	Differential Protection Operated causing tripping of ICT. As reported multiple bushings damaged. Expected restoration timeline yet to be intimated by OPTCL.
220KV-RAJARHAT-NEW TOWN(AA-II)-1	18-04-2026	Shutdown to attend smoke detected from underground cable
132KV-ARRAH (PG)-JAGDISHPUR-1	23-04-2026	Site initiated Hand Trip
220KV-BUDHIPADAR-KORBA-2	24-04-2026	For modification/diversion work between loc. No. 77 (A+3m) – 80 (A+0m) (RL – 1.032 KM). Erection of 04 No. BN60 type towers and Dismantling 02 No. existing towers- 78 & 79 on request of M/s IRCON Ltd Bilaspur due to construction of new BG Uрга- Dharamjaigarh Rail Corridor.
400KV/220KV 315 MVA ICT 2 AT DSTPS(ANDAL)	03-05-2026	To be dismantled and shifted to KTPS
400KV-PATNA-NAUBATPUR(BH)-2	09-05-2026	For Destrining of Quad from AP-01 TO Ext. MC-02 OF 400kV Naubatpur -Patna (PG) ckt-02, residual quad jumpering work from AP-01 TO AP-78, Twin stringing from AP -78A to Ext MC-02 .
400KV-BUXAR-NAUBATPUR-2	12-05-2026	For stringing of twin moose conductor in 400KV Naubatpur - Buxar ckt-2 from AP- 78A/0 to M/C - 2 AP-78/0
400KV-JEYPORE-GAZUWAKA-1	15-05-2026	NH-130CD NHAI Diversion work at location 131,133, 151-153 and 179-180
400KV-JEYPORE-GAZUWAKA-2	15-05-2026	NH-130CD NHAI Diversion work at location 131,133, 151-153 and 179-180
765KV-DHARAMJAIGARH-JHARSUGUDA-2	21-05-2026	For diversion of existing 765kV D/C Jharsuguda – Dharamjaygarh Ckt 1&2, due to upcoming railway line of IRCON.

765KV-DHARAMJAIGARH-JHARSUGUDA-1	21-05-2026	For diversion of existing 765kV D/C Jharsuguda – Dharamjaygarh Ckt 1&2, due to upcoming railway line of IRCON.
220KV-KHAGARIA-NEW PURNEA-2	25-05-2026	Khagaria: RYB Fault , Ir-2.821kA ,Iy-3.130kA ,Ib-2.992kA : New Purnea :R-Y-B FAULT, Z1, FD-84.2KM, FC: IR- 2.99KA, IY- 3.07KA, IB- 3.04KA; 03 Nos of towers at Loc Nos- 320(DA+0 type), 321(DA+0 type) and 322(DB+0 type) has collapsed and 01 Nos of tower at Loc No- 323 (DB+6 type) has bent from first Cross arm during heavy cyclonic storm on 25-05-2026
220KV-KHAGARIA-NEW PURNEA-1	25-05-2026	NPRN(SITE): R-N, FD- 86.20KM, FC- 2.26KA : Khagaria :,Y ph , Zone 1, 2.322 KA; 03 Nos of towers at Loc Nos- 320(DA+0 type), 321(DA+0 type) and 322(DB+0 type) has collapsed and 01 Nos of tower at Loc No- 323 (DB+6 type) has bent from first Cross arm during heavy cyclonic storm on 25-05-2026
400/220 KV 315 MVA ICT 4 AT ARAMBAGH	29-05-2026	Tripped
400KV-NEW PURNEA-MUZAFFARPUR-1	29-05-2026	Pile Foundation TN 348

Deliberations in the meeting

Members noted.

4.3. Commissioning of new units and transmission elements in Eastern Grid in the month of May-2026

The details of new units/transmission elements commissioned in the month of May-2026 based on the inputs received from beneficiaries:

NEW ELEMENTS COMMISSIONED DURING May, 2026							
उत्पादन इकाइयाँ / GENERATING UNITS							
क्र. सं. Sl. No.	स्थान Location / Pooling Station	मालिक/यूनिट का नाम OWNER/UNIT NAME	यूनिट संख्या/स्रोत Unit No/Source	संकलित क्षमता (मेगावाट) Capacity added (MW)	कुल/स्थापित क्षमता (मेगावाट) Total/Installed Capacity (MW)	दिनांक DATE	टिप्पणी Remarks
1	Chausa, Buxar, Bihar	SJVN Thermal Pvt. Ltd/BUXAR TPP	2/Coal	660	1320	08-05-2026	First Time sync.

आई.सी.टी./जी.टी./एस.टी / ICTs/ GTs / STs							
क्र . सं. Sl. No .	एजेंसी/ मालिक Agency/ Owner	उप-केन्द्र SUB-STATION	आईसीटी संख्या ICT NO	वोल्टेज (केवी) Voltage Level (kV)	क्षमता (एमवीए) CAPACITY (MVA)	दिनांक DATE	टिप्पणी Remarks
1	SJVN Thermal Pvt. Ltd	BUXAR TPP	GT-02	400/20 kV	825	08-05-2026	
प्रेषण लाइन / TRANSMISSION LINES							
क्र . सं. Sl. No .	एजेंसी/ मालिक Agency/ Owner	लाइन का नाम LINE NAME	लंबाई (किमी) Length (KM)	कंडक्टर प्रकार Conductor Type	दिनांक DATE	टिप्पणी Remarks	
NIL							
लिलो / प्रेषण लाइन की पुनर्व्यवस्था / LILO/RE-ARRANGEMENT OF TRANSMISSION LINES							
क्र . सं. Sl. No .	एजेंसी/ मालिक Agency/ Owner	लाइन का नाम / लिलो पर Line Name/LILO at	लंबाई (किमी) Length (KM)	कंडक्टर प्रकार Conductor Type	दिनांक DATE	टिप्पणी Remarks	
1	JUSNL	220KV-DHANBAD (NKTL)-DUMKA-1	106	Single Zebra	07-05-2026 18:18	LILO of 220 KV Dumka-Govindpur Circuit-I at 400/220 KV GSS Dhanbad (NKTL)	
2	JUSNL	220KV-DHANBAD (NKTL)-GOVINDPUR-1	36	Single Zebra	07-05-2026 17:55	LILO of 220 KV Dumka-Govindpur Circuit-I at 400/220 KV GSS Dhanbad (NKTL)	
3	UPPTCL, JUSNL	132KV-NAGARUNTARI-MYORPUR-1	68.68	ACSR Panther	27-05-2026 23:59	LILO of 132 kV Rihand-Nagaruntari line at 132 kV Myorepur S/S:	
बस/लाइन रिएक्टर / BUS/LINE REACTOR							
क्र . सं. Sl. No .	एजेंसी/ मालिक Agency/ Owner	एलेमेंट का नाम Element Name	उप-केन्द्र SUB-STATION	रेटिंग (एमवीएआर) Rating (MVAr)	दिनांक DATE	टिप्पणी Remarks	

NIL						
एच.वी.डी.सी/ए.सी फिल्टर बैंक/फैक्ट्स डिवाइस संबद्ध प्रणाली / HVDC /AC Filter bank / FACTS DEVICE associated System						
क्र. सं. Sl. No.	एजेंसी/ मालिक Agency/ Owner	एलेमेंट का नाम Element Name	उप-केन्द्र SUB-STATION	वोल्टेज (केवी) Voltage Level (kV)	दिनांक DATE	टिप्पणी Remarks
NIL						
बस - बे / BUS - BAYS						
क्र. सं. Sl. No.	एजेंसी/ मालिक Agency/ Owner	एलेमेंट का नाम Element Name	उप-केन्द्र SUB-STATION	वोल्टेज (केवी) Voltage Level (kV)	दिनांक DATE	टिप्पणी Remarks
1	BSPTCL	400KV MAIN BAY OF BUXAR TPP-1 AT NAUBATPUR(BH)	NAUBATPUR (BH)	400 KV	02-05-2026 20:49	
2	SJVN Thermal Pvt. Ltd	400KV MAIN BAY OF GT-02 AT Buxar TPP	Buxar TPP	400 KV	08-05-2026 12:30	

Deliberations in the meeting

Members noted.

4.1. UFR operation during the month of May 2026

Frequency profile for the month as follows:

MONTH	MAX	MIN	% LESS IEGC BAND	% WITHIN IEGC BAND	% MORE IEGC BAND
	(DATE/TIME)	(DATE/TIME)			
May 2026	50.42 (on 03-May-26 at 08:14 Hrs.)	49.47 (on 13-May-26 at 14:09 Hrs.)	6.5	77.3	16.2

UFR details for the month as follows:

nth as follows:

Constituent	UFR Stage	Quantum Approved in MW	Quantum Installed in MW	Quantum Operated	(Quantum Not Operated)
JUSNL	Stage 1	87	85	54	31
BSPTCL	Stage 1	315	292	93	199
ODISHA	Stage 1	306	316	205	111
West Bengal (WBSEDCL)	Stage 1	377	440	226	214
West Bengal (CESC)	Stage 1	120	120	72	48
DVC	Stage 1	172	173	150	23
Total ER		1377	1427	800	627

Hence, no report of operation of UFR has been received from any of the constituents.

Deliberations in the meeting

Members noted.

Meeting ended with a vote of thanks to the chair.

Monsoon Demand dependency on Rainfall

Annex-A

Key Indicator

Monsoon
/Summer Demand Ratio

$$= \frac{\text{Average Daily Maximum Demand (Jul-Sep)}}{\text{Average Daily Maximum Demand (Apr-Jun)}}$$

ALL-INDIA DEMAND (MET)

Year	Highest Rainfall Deficit Month	Rainfall Deficit(-) / Surplus (+) (%)	Demand in Deficit Month (GW)	Previous Year Same Month (GW)	YoY Demand Growth (%)	Annual Peak Demand (GW)	Peak Month	Monsoon/Summer Avg Ratio
2018	Aug	-15%	171	159	7.5	175	Sep	1.02
2019	Jun	-33%	176	165	6.7	183	Oct	0.96
2020	Covid Impact		175	174	0.6	177	Dec	1.13
2021	Sep	-35%	191	175	9.1	200	Jul	1.07
2022	Aug	-3%	183	175	4.6	210	Apr	0.95
2023	Aug	-36%	237	183	29.5	243	Aug/Sep	1.04
2024	Jul	9%	227	209	8.6	250	May	0.94
2025	Aug	4%	217	217	0	243	May	0.99

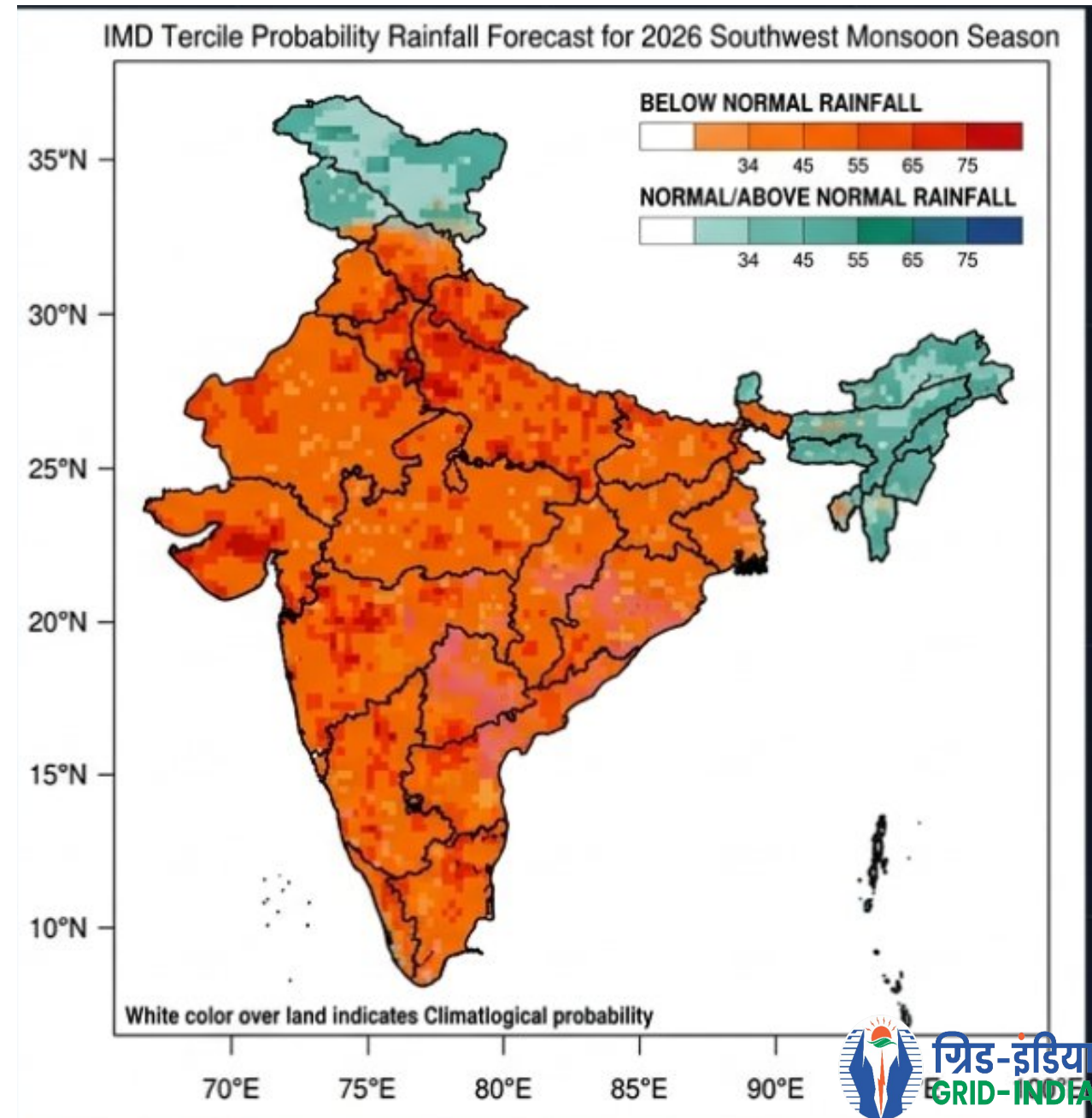
EL NINO

>1.00 → Higher demand in Monsoon

IMD Projection – Monsoon 2026

Key Forecasts

- Southwest Monsoon Rainfall (Jun–Sep 2026): **90% of** Long Period Average (LPA).
- **84% probability** of below-normal or deficient rainfall.
- **Below-normal rainfall** expected across large parts of Central, Northwest and South Peninsular India.
- **El Niño conditions** likely to develop during the monsoon season.
- Above-normal temperatures and **heatwave conditions** expected over several regions.

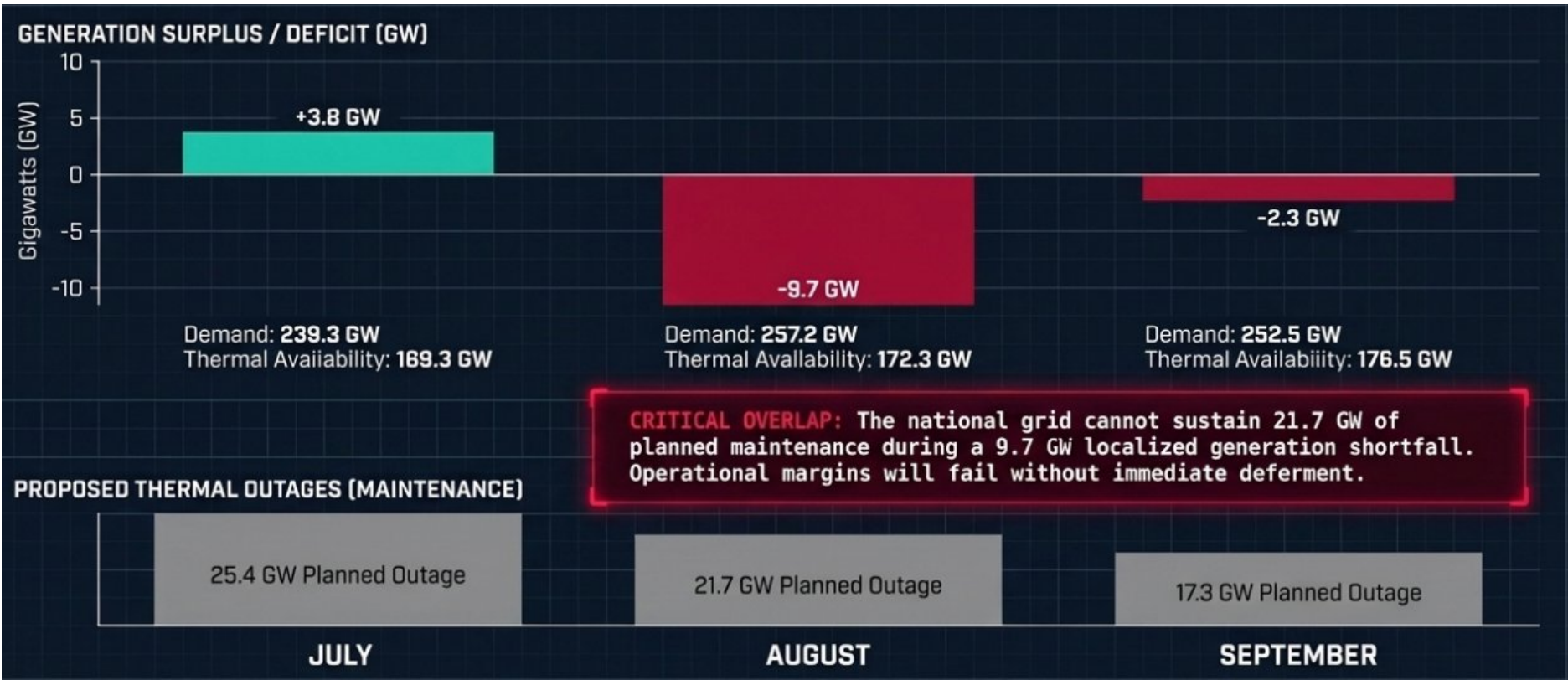


The Complication

-9.7 GW

↘ | ↗

The Crunch: This triggers skyrocketing cooling demand, projecting a **9.7 GW deficit by August**

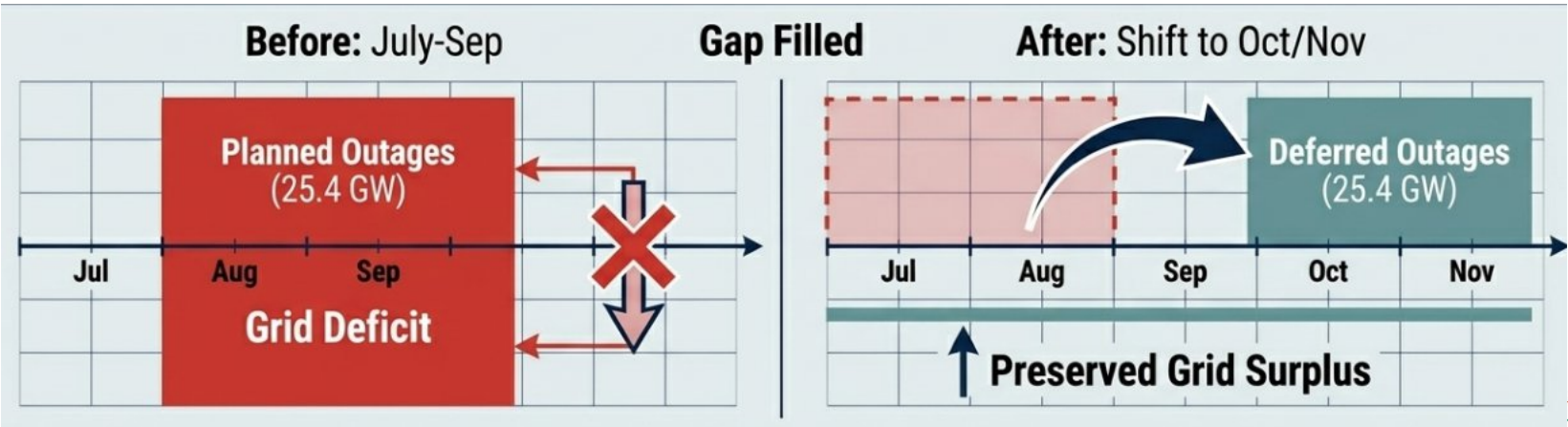


of planned thermal are currently d directly inside this ity window.

Deferred Maintenance Period (Post-Sep 2026)

DATE: 2026.05.15 | TYPE: DIRECTIVE

The Mandate: 25.4 GW of planned thermal outages are currently scheduled



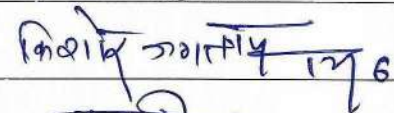
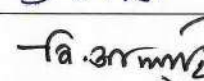
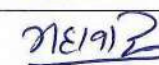
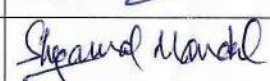
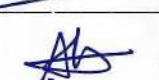
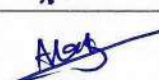
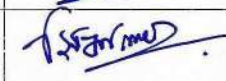
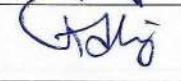
Annex-B

Participants in 240th OCC Meeting

Venue: ERPC Conference Hall, Kolkata

Time: 10:30 Hrs.

Date: 12.06.2026 (Friday)

Sl.	Name	Designation	Organisation	Contact No.	E-mail Id	Signature
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4	MANAS DAS	DGM	ERLDC GRID-INDIA	9007070925	manasdas@grid-india.in	
5	Mahavir Prasad Singh	GM (MO)	ERLDC GRID-INDIA	9729872327	mahavir@grid-india.in	
6	Prasanna Kumar Sahoo	AGM (OS)	NTPC Ltd. Bhubaneswar	9437962430	prasanna.sahoo@ntpc.co.in	
7	Adwait Kalyan	DGM (OS)	NTPC Ltd. ER-IHB Patna.	8544419320	adwaitkalyan@ntpc.co.in	
8	B N Yadav	AGM (Opn)	BRBCL/NTPC	7440226700	bnyadav 01@ntpc.co.in	
9	SHYAMAL MONDAL	AGM E	NTPC-Kolkata	5106712572	shyamal.mondal@ntpc.co.in	
10	Alok Kumar Mahato	AGM	NTPC-Kolkata	9437962430	alokmahato@ntpc.co.in	
11	Abhishek Murari	Dep. Mgr.	NTPC-Kolkata	9437962430	abhishek.murari@ntpc.co.in	
12	ALOK PRATAP SINGH	Chief Manager	Grid-India	9437962430	aloksingh@grid-india.in	
13	CHANDAN MAHAJAN	Chief Manager	Grid-India	9437962430	chandan.mahajan@grid-india.in	
14	SAJAL K BAG	ACE	SLDC, WBSETCL	7980098826	sajalkbag74@gmail.com	
15	UTSAB ADITYA	SE	WB-SLDC	9434910880	utsab.aditya@gmail.com	

Participants in 240th OCC Meeting

Venue: ERPC Conference Hall, Kolkata

Time: 10:30 Hrs.

Date: 12.06.2026 (Friday)

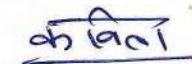
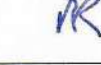
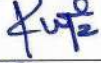
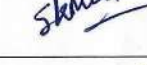
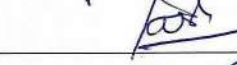
Sl.	Name	Designation	Organisation	Contact No.	E-mail Id	Signature
16	Sudip Ray Barman	AGM	NTPC NABINAGAR	9424140773	rohit SRAYBARMAN@NTPC.CO.IN	SRB
17	SOURAV MANDAL	Chief Mgr	ERLDC	9402102354	souravmandal@grid-india.in	SRM
18	MANAS MONDAL	AGM	WBPDCL	8336903808	m.mondal@wbpdcl.co.in	MM
19	RAKESH KUMAR PRADHAN	Ch. Mgr.	ERLDC	9831337570	rkpradhan@grid-india.in	RP
20	KOUSHIK BANERJEE	GM(SO)	CESC LTD.	9831003281	koushik.banerjee@rpsg.in	KB
21	DEBASMITA DE	AGM (GO)	"	9163312772	debasmita.de@rpsg.in	DD
22	RAJIV KR. SINGH	Addl. Mgr (Tech)	CESC LTD.	9831869115	rajiv.singh1@rpsg.in	RS
23	SURAJIT ROY	Chief Mgr.	HEL	8335067270	surajit.roy@rpsg.in	SR
24	MANAS GUHA	DGM(E)	DPL	8509267341	m.guha@dpl.net.in	MG
25	Rohit Kausay	Manager	PVUNL	9523384177	rohitkausay@ntpc.co.in	RK
26	Rakesh Kumar	Sr. Manager	PVUNL	9425281478	rakesh.kumar07@ntpc.co.in	Rakesh
27	AJAY Bansal	PM	GEV	9958990561	ajay.bansal1@gevernora.com	AB
28	Samal Bhattacharya	DManager	GBV	8436062436	Samal.bhattacharya1@gevernora.com	SB
29	Mahendra Malik	AGM	JSW	6376355214	mahendra.malik@jsw.in	MM
30	Sunil HN	GM	JSW	9449849774	Sunil.nagarai@jsw.in	SN

Participants in 240th OCC Meeting

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Date: 12.06.2026 (Friday)

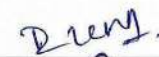
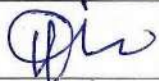
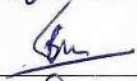
Sl.	Name	Designation	Organisation	Contact No.	E-mail Id	Signature
31	Biswanath Patel.	AGM	ISW	70166 770 30	biswanath.patel@isw.in	
32	KAVITA PARIHAR	Panel Manager	ERLDC Grid - India	9958114989	kavita@grid-india.in	
33	Rajendra Prasad	EE	TVNL	90310 49936	rajendran.prasad@tvnl.in	
34	TUNESHWAR KUMAR	SR. MANAGER	TUSNL	9431190306	tuneshwarakumar@gmail.com	
35	Raju Kachhap	Sr. Manager	SLDC Ranchi	7783087568	rajmaifone82@gmail.com	
36	P. V. RAUT.	AGM (OPN)	MPL	9223501513	rautpv@tatapower.com	
37	Rupa Kumari	SH. TD, 42B	JUSNL	7873044612	reetdheeb@gmail.com	
38	Manmath Beura	Manager (ELE.)	GRIDCO Ltd.	9337271600	ele.mbeura@gridco.co.in	
39	Sanjoya Ku. Mishra	Sr. GM (PP)	GRIDCO Ltd.	9438907414	cgm.pp@gridco.co.in	
40	Manas Ranjan Samaranaray	Sr. G. Y (G.D)	SLDC, Odisha	9438907413	samaranaray9890@gmail.com	
41	Subhas Ch Dash	CGM	SLDC, Odisha	9438907966	ele.scdash@optd.co.in	
42	S. K. KUNDU	DGM	SLDC, DVC	6202091671	santosh.kundu@dvc.gov.in	
43	D. K. Khuntia	ADD	ERPC, Kolkata	7683889161	dillip.k.khunia@erpc.gov.in	
44	K. Swati	AE	"	7903336191	Swati.cea@gov.in	
45	P. K. DE	SE	ERPC	9831620142	secomml.erpc@gov.in	

Participants in 240th OCC Meeting

Venue: ERPC Conference Hall, Kolkata

Time: 10:30 Hrs.

Date: 12.06.2026 (Friday)

Sl.	Name	Designation	Organisation	Contact No.	E-mail Id	Signature
46	P. K. Meena	SE.	ERPC	8800764820	pkmeena@nic.in	
47	A. Das	DD	"	9681214774	ampda@nic.in	
48	P. P. Jena	EE	ERPC	9776198991	ppjena.erpcc@gov.in	
49	S. K. Panda	Manager	DVC, EDC	6370134799	Santosh.Panda@dvc.gov.in	
50	P. K. MOHANTA	AGM	APWRL, JSR	8195900148	pradipk.mohanta@adhwajpower.co.in	
51	B. S. Karma	DGM	NTPC-Kanti	9431820090	BHOLEYSHANKER@NTPC.CO.IN	
52	H. K. Saw	Sr Manager	NTPC-Kanti	7008827681	hementakumarsaw@npsc.co.in	
53	S. R. TUDU	Dy. Dirct	RPSOLE)	9734907623	subhasr.tudu@gov.in	
54	B. S. Ray	DD	ERPC			
55						
56						
57						
58						
59						
60						

June 12th, 2026

OCC Meeting

Implementation of Unified Load Dispatch and Communication (ULDC) Phase-III -
SCADA/EMS upgradation Project Eastern Region SLDCs and RLDC

ERPC,Kolkata

Project Summary



Overview

Employer:

- POWERGRID for all SLDCs
- GRID-INDIA for RLDC's

Region: Eastern Region

Project Description: Implementation of Unified Load Dispatch and Communication (ULDC) Phase-III (SCADA/EMS Upgradation Project – Eastern Region)

Technical scope: Supply and Installation of Main & Backup SCADA/EMS computer system hardware and software

Contract Price:

INR 368 Cr (w/o Taxes)

INR 437.20 Cr (with Taxes)

Amendment : 10.2 Crores (w/o Taxes)

Project Start Date:

SLDC (NOA Date): 02.08.2024

ERLDC (NOA Date): 29.08.2024

Payment Milestones

- I. Supply: 85% of Supply Price shall be paid as:
 - A. SW (30% after FAT & Site delivery + 55% after field testing)
 - B. HW (70% against shipment + 15% against material receipt)
- II. Installation: 85% on pro-rata basis after installation at site.
- III. Balance 15% of Supply & Installation within 30 days of OA.

Performance bonds: Yes,

- PBG @10% of total contract price till 03rd May 2027 (for Supply and Service, AMC)
- AMC PBG @2% of contract price within 28 days after DLP valid till 03rd May 2033

Project Timelines

Project duration: 18 months(Implementation),
01 Year (DLP)
06 Years (M&S)

ER ULDC Project: Highlights

DVC system Inaugurated & SAT in advanced stage

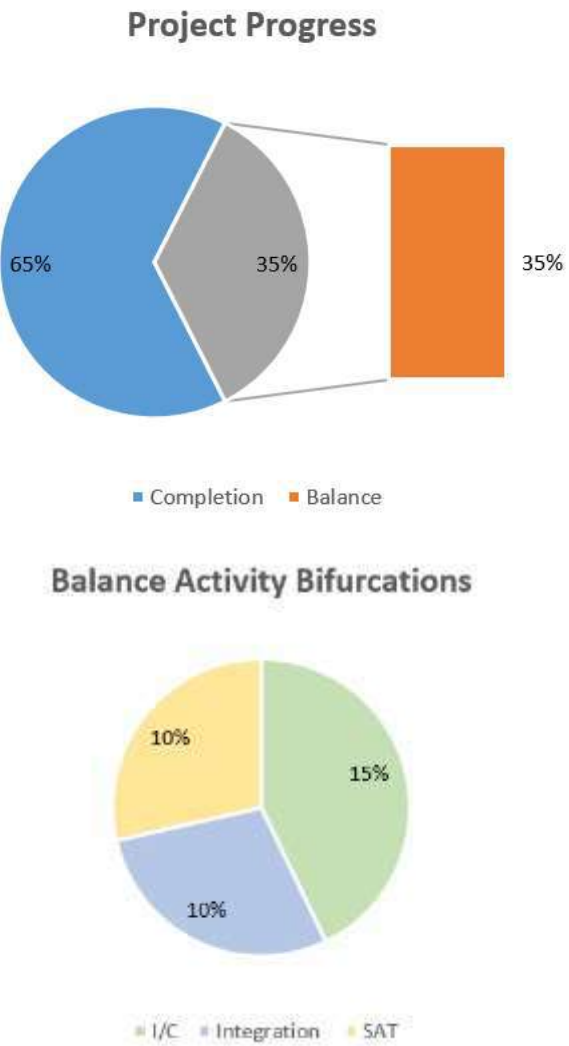
Integrated FAT completed, WB MCC Dispatched

ERLDC and WBSETCL Performance FAT Starting From 12th June

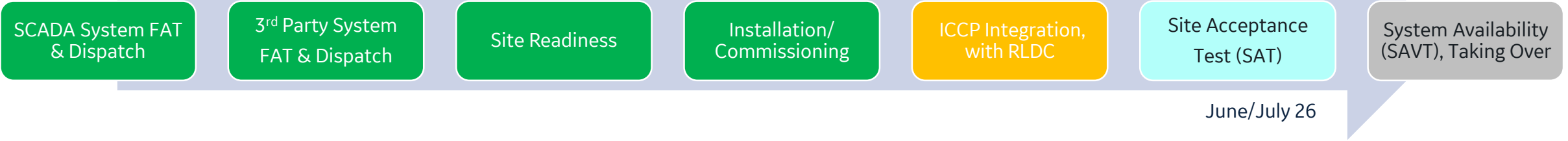
OPTCL-B, BSPTCL -B, JUSNL-M; commissioning in good progress

Overall, I&C in good swing with deputed teams at all available sites

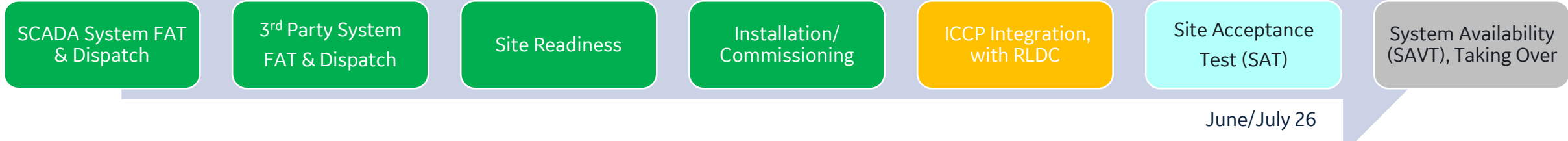
Sikkim –M: Installation activities (Furniture/VPS/VCS) getting commenced
: Civil Work is in Progress



MCC



BCC



Key Updates

- 1. SAT in advanced stage
- 2. Hardware, SCADA, DTS completed
- 3. 3rd Party SAT offered
- 4. EMS SAT offered
- 5. DG Foundation completed

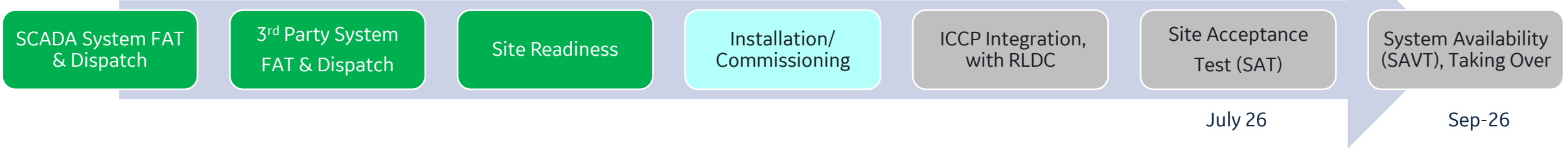
Challenges

- 1. **ICCP Integration - HOLD due to ERLDC readiness**
- 2. Customer support for the SAT closure

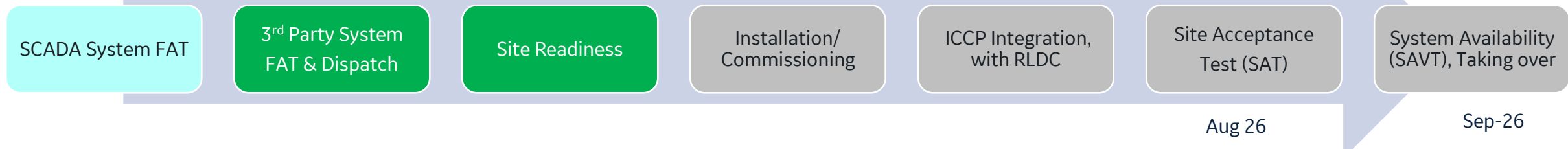
Next Steps

- 1. SAT closure
- 2. System Availability

MCC



BCC



Key Updates

- 1. MCC Dispatched and I&C is in Progress
- 2. BCC will be dispatched post completion of Performance Test
- 3. Database validated

Challenges

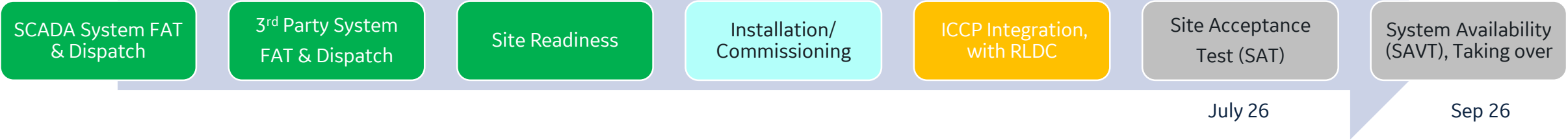
- 1. **Air conditioning in MCC** – Server powered off due to non-availability of proper air-conditioning

Next Steps

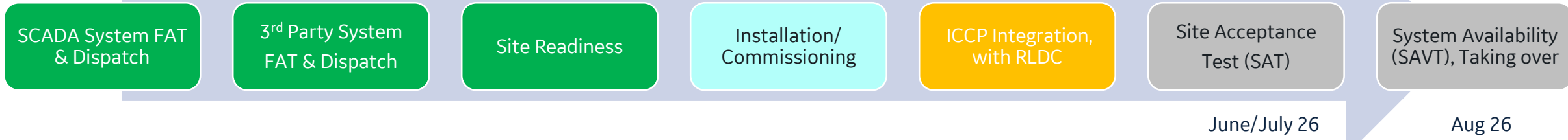
- 1. MCC: I/C activities in progress
- 2. RTU On-lining
- 3. Data end to end validation
- 4. System commissioning



MCC



BCC



Key Updates

- 1. BCC commissioning in advanced stage
- 2. MCC: Installation completed
- 3. Database validated: 286/286
- 4. RTU Reporting: 207 / 286
- 5. End to End DB Validation: 55 / 286
- 6. Customized Display completed

Challenges

- 1. **ICCP Integration - HOLD due to ERLDC readiness**
- 2. MCC: Delay in readiness of air-conditioning

Next Steps

- 1. SAT at BCC
- 2. I&C at MCC

Completed

Dependency on ERLDC

In Progress

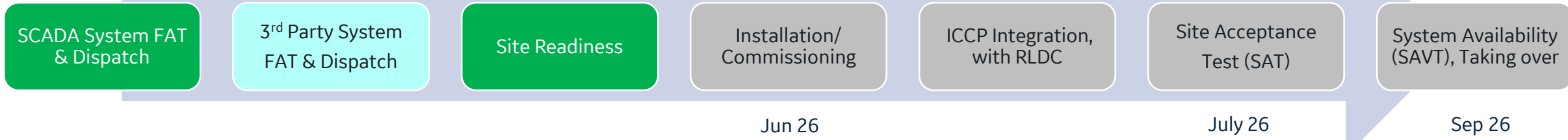
Scheduled later

Blocked

MCC



BCC



Key Updates

- 1. MCC and BCC CC HW Dispatched
- 2. 3rd Party System dispatched
- 3. MCC Furniture I&C in progress
- 4. BCC I&C will be started from 16th June
- 5. Database validated

Challenges

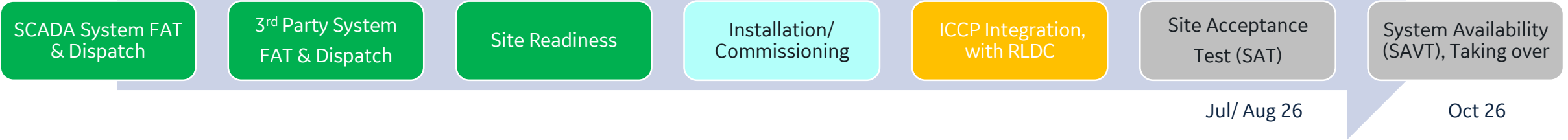
- 1. Civil work going on at MCC in Server room

Next Steps

- 1. I/C for BCC & commissioning
- 2. I/C for MCC & Commissioning
- 3. 3rd party Installations



MCC



BCC



Key Updates

- 1. MCC: commissioning in advanced stage
- 2. 3rd Party SAT scheduled (VPS/ Furniture/ APS)
- 3. Database validated
- 4. RTUs reporting
- 5. Pre-SAT SCADA completed

Challenges

- 1. BCC: Input power fluctuations/unreliable
- 2. Commencement of SAT for Hardware and SCADA.

Next Steps

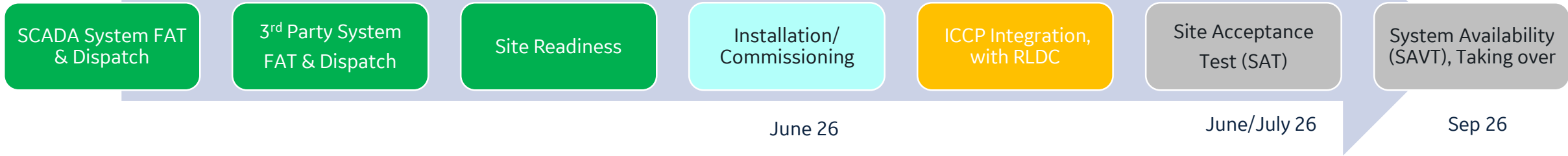
- 1. Completion of SAT on MCC
- 2. Completion of I/C activities on BCC



MCC



BCC



Key Updates

- 1. BCC: SAT started
- 2. 3rd Party SAT scheduled (APS/ Furniture)
- 3. MCC: Installation in progress
- 4. Database validated: 222/ 222
- 5. RTU Reporting in GEV System: 123 / 217
- 6. RTU Reporting in OSI System: 132 / 217
- 7. RTUTOP validated 41 / 123

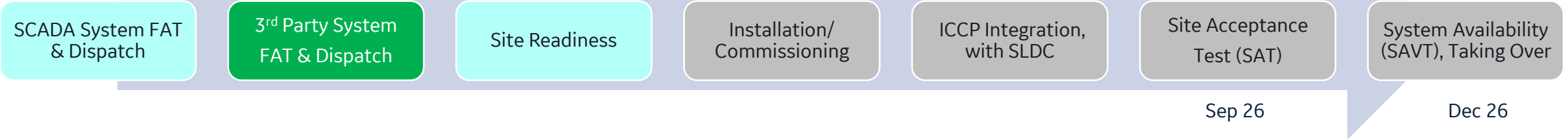
Challenges

- 1. ICCP Integration- ERLDC readiness
- 2. Customer support required to expedite the data validation & inputs on different DB
- 3. MCC: Completion of Site Civil works and Dust free environment
- 4. MCC: Material safety/ security
- 5. BCC: Input power voltage stability

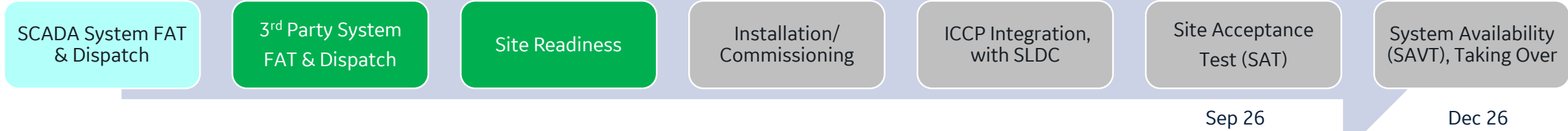
Next Steps

- 1. 3rd Party SAT to be completed for all items
- 2. Commissioning of both MCC & BCC by Jun 26

MCC



BCC



Key Updates

- 1. FAT in advanced stage (closure: 17June)
- 2. Resources deployments: 4 + 1
- 3. Database status: 77/77

Challenges

- 1. **Site readiness by ERLDC:** Dust-free environment with proper air-conditioning in Control room & Server room to be ensured by ERLDC.

Next Steps

- 1. FAT closure & Dispatch
- 2. ICCP connectivity with States
- 3. System Installation & Commissioning

Steps Ahead for Sites Available



I/C Activities completion: Jun/ Jul 2026 for available sites

ERLDC/ WB FAT in final stages; ICCP Integration by Jul 2026

- 1.Joint efforts required for DB validations
- 2.(Customer to share dedicated resource plan for Data validations)

1.SAT & system integration for each state: Jul/ Aug 2026

1.SAVT & Taking over: Oct/ Nov 2026

1.Non-ready Sites to further consider timelines of 3-4 months; post Site readiness and GEV resource alignments

Annex B.2.1.g

FEEDER LIST , STAGE 1					
GSS Name	Feeders name	Peak Load (in MW)	Integrati on status	Reporting Status	Not reporting from
Triveniganj 132/33kV	33kV JADIVA	4	Yes	Not	04.05.2026
Triveniganj 132/33kV	33kV CHHATAPUR	4	Yes	Not	04.05.2026
Purnia (Sumard) 132/33kV	33kV JALESHWAR	6	Yes	Not	26.04.2026
SKMCH 132/33kV	33kV CHANDWARA	20	Yes	Not	16.12.2022
SKMCH 132/33kV	33kV SIKENDARPUR	10	No		
Raxaul 132/33kV	33kV BIRGANJ	12	Yes	Not	24.04.2026
Raghopur 132/33kV	33kV CHAMPANAGAR	3.5	Yes	Yes	
Raghopur 132/33kV	33kV PIPRA	12.5	Yes	Yes	
Raghopur 132/33kV	33kV RAGHOPUR	11	Yes	Yes	
Raghopur 132/33kV	33kV LALGANJ	8	Yes	Yes	
Nirmali 132/33kV	33kV Marauia	5	Yes	Yes	
Muzaffarpur 132/33kV	33kV MARIPUR	7	Yes	Yes	
Muzaffarpur 132/33kV	33kV NAYATOLA	15	Yes	Yes	
Muzaffarpur 132/33kV	33kV BHIKHANPURA	16	Yes	Yes	
Mushahani 220/132/33kV	33kV CHANDWARA	20	Yes	Not	21.05.2026
Laukhi 220/132/33kV	33kV DAGMARA	1.5	No		
Laukhi 220/132/33kV	33kV MAJHARI	1.5	No		
Katlaya 132/33kV	33kV Birpur	4	Yes	Yes	
Barh 132/33kV	33kV BERHINA	7	Yes	Not	22.05.2026
Sabour 132/33kV	33kV SABOUR	12	Yes	Not	12.05.2026
Kahalgachn 132/33kV	33kV PRKPANTI	20	Yes	Not	22.05.2026
Kahalgachn 132/33kV	33kV KAHALGAON	16	Yes	Not	22.05.2026
Bodhgaya 220/132/33kV	33kV DOBHI	12	Yes	Not	16.01.2025
Bodhgaya 220/132/33kV	33kV NEW SHOBIH	24	Yes	Not	24.09.2021
Bodhgaya 220/132/33kV	33kV NEW MANPUR	15	Yes	Not	24.09.2021
Bodhgaya 220/132/33kV	33kV KHATKA CHAK	5	No		
Bodhgaya 220/132/33kV	33kV BHUSANDA.9	5	No		
Sasaram 132/33kV	33kV TAKIYA	7	No		
Sasaram 132/33kV	33kV CHEHARI	6	Yes	Not	12.05.2026
Sasaram 132/33kV	33kV SHIVSAGAR	10	Yes	Not	12.05.2026
Nawada 132/33kV	33kV WARSALIGANJ	15	Yes	Yes	
Frequency (in Hz) : 49.4 Hz	TOTAL : 315 MW	315			

Not integrated 6(30 MW)
Integrated 25 (285 MW)
Integrated and reporting 10 (97 MW)
Integrated but not reporting 15 (188 MW)

FEEDER LIST , STAGE 2					
GSS Name	Feeders name	Peak Load (in MW)	Integration status	Reporting Status	Not reporting from
Katlaya 132/33kV	33kV Raghopur	11	Yes	Yes	
Katlaya 132/33kV	33kV Jagdishpur	3	Yes	Yes	
Katlaya 132/33kV	33kV RAJBIRAJ NEPAL	12	Yes	Yes	
Hajipur 132/33kV	33kV SONEPUR	16	No		
Barsol 132/33kV	33kV SONALI	15	Yes	Not	22.05.2026
Barsol 132/33kV	33kV BARSOI	15	Yes	Not	22.05.2026
Darbhanga 132/33kV	33kV PANDASARAI	10	Yes	Not	21.05.2026
Bettiah 132/33kV	33kV BETTIAH	22	Yes	Yes	
Gangwara 132/33kV	ICV, PASHUPALAN	1	No		
Purnea 132/33kV	33kV CIRCUIT HOUSE	10	No		
Purnea 132/33kV	33kV MADHUBANI	6	Yes	Not	24.06.2022
Begusarai 220/132/33kV	33kV BEGUSARAI. I	10	Yes	Not	NA
Purnea 132/33kV	33kV LINE BAZAR	3	Yes	Not	NA
Hajipur 132/33kV	33kV KONHARA GHAT	13	No		
Katihar 132/33kV	33kV BINODPUR	7	Yes	Not	01.05.2026
Begusarai 220/132/33kV	33kV DEONA	5	No		
Balia 132/33kV	33kV BALIA	12	Yes	Yes	
Gangwara 132/33kV	33kV OLD DMCH	5	Yes	Not	07.04.2024
Muzaffarpur 132/33kV	33kV KHABRA	15	Yes		
Nawada 132/33kV	33kV HISUA	16	Yes	Not	21.05.2026
Buxar 132/33kV	33kV CHOUSA	9.3	Yes	Not	26.01.2025
Buxar 132/33kV	33kV ITARHI	9	Yes	Not	26.01.2025
Buxar 132/33kV	33kV PANDEYPATTI	5.6	Yes	Not	26.01.2025
Buxar 132/33kV	33kV PURSHOTTAMP UR	12.4	No		
Jamui 132/33kV	33kV BINDIWARA	7.68	Yes	Yes	
Sultanganj 132/33kV	33kV 235 Sultanganj	13	No		
Sultanganj 132/33kV	33kV 232 Sahkund	6	Yes	Not	06.05.2026
Sultanganj 132/33kV	33kV 230 Asharganj	5	Yes	Not	06.05.2026
Sultanganj 132/33kV	33kV 233 Shambhuganj	7	Yes	Not	06.05.2026
Jamui Old 132/33kV	33kV JAMUI-II	12	No		
Jamui Old 132/33kV	33kV Jhalha	4	Yes	Not	14.05.2026
Mohania 132/33kV	33kV BHABHUA	14	Yes	Not	01.05.2026
Bhabua (Mundeshwari) 132/33kV	33kV BHAGWANPUR MAR	15	No		
Bhabua (Mundeshwari) 132/33kV	33kV BHAGWANPUR	4	Yes	Not	26.04.2026
Banka Old 132/33kV	33kV NEHRU COLONY	8	No		
Banka Old 132/33kV	33kV BANKA	5	Yes	Not	22.05.2026
Banka Old 132/33kV	33kV GOKALA	2	No		
Banka New 132/33kV	33kV WARNE SOUTH	2	Yes	Yes	
Banka New 132/33kV	33kV KATORIYA	11	Yes	Yes	
BGCL Motama	Motama-4	11	No		
BGCL Khajarsarai	Khajarsarai	6	NA		
BGCL Khajarsarai	Nauagarh	4.5	NA		
Frequency (in Hz) : 49.2 Hz	TOTAL : 380.48 MW	380.48			

Not integrated 12 (118 MW)
Integrated 27 (236 MW)
Integrated and reporting 8 (80 MW)
Integrated but not reporting 19 (156 MW)

FEEDER LIST , STAGE 3					
GSS Name	Feeders name	Peak Load (in MW)	Integration status	Reporting Status	Not reporting from
Manihari 132/33kV	33kV AMDABAD	1.56	Yes	Yes	
Hajipur 132/33kV	33kV PASWAN CHOWK	15	Yes	Not	10.06.2025
Forbiganj 132/33kV	33kV FORBESGANJ NE	20	Yes	Yes	
Kishanganj 132/33kV	33kV POTHIA	4	Yes	Bad Quality	
Purnea 132/33kV	33kV ZERO MILE	10	Yes	Not	
Forbiganj 132/33kV	33kV FORBESGANJ OLD	20	Yes	Yes	
Chhargana 132/33kV	33kV TELPA	25	No		
SAharsha 132/33kV	33kV SAHARSHA NEW	15	No		
Katihar 132/33kV	33kV KORHA_OLD1	8	No		
Sitamarhi 132/33kV	33kV SITAMARHI	23	Yes	Not	22.05.2026
Darbhanga 132/33kV	33kV DONAR	9	Yes	Bad Quality	
Gopalganj 220/132/33kV	33kV GOPALGANJ	12	Yes	Not	08.03.2025
SAharsha 132/33kV	33kV SAHARSHA OLD	15	No		
Raxaul 132/33kV	33kV RAXAUL	20	Yes	Not	24.04.2026
Manjhaul 132/33kV	33kV CIRCUIT HOUSE	14	No		
Madhubani 132/33kV	33kV MADHUBANI	3	Yes	Not	21.05.2026
Dumraon 132/33kV	33kV DUMRAON	16	Yes	Not	21.05.2026
Ekdangarsara 132/33kV	33kV HILSA	15.1	Yes	Not	08.05.2026
Karmnasha 132/33kV	33kV DURGAWATI	5	No		
Karmnasha 132/33kV	33kV CHAND	6	Yes	Not	13.05.2026
Karmnasha 132/33kV	33kV KARAINNASA	4.5	no		
Karmnasha 132/33kV	33kV RAMGARH	4	Yes	Not	13.05.2026
Ramgarh 132/33kV	33kV RAMGARH	8	No		
Sherghati 132/33kV	33kV SHERGHATI	19	Yes	Not	06.05.2026
Tarapur 132/33kV	33kV SANGRAIMPUR	5.9	Yes	Yes	
Tarapur 132/33kV	33kV TARAPUR	7.8	Yes	Yes	
Tarapur 132/33kV	33kV BELHAR	7.7	Yes	Yes	
Sonenagar 220/132/33kV	33kV NAVINAGAR	13	No		
Sonenagar 132/33kV	33kV DEHARA	10	No		
Masaurhi 132/33kV	33kV MASAUHRI_OLD	10	Yes	Not	29.04.2026
Sabour 132/33kV	33kV Barari	4	Yes	Not	12.05.2026
Sabour 132/33kV	33kV GERADIIH	5	Yes	Not	12.05.2026
Kudra 132/33kV	33kV SHIVSAGAR	10	No		
Naugachia 132/33kV	33kV Makanpur	9	Yes	Yes	
Naugachia 132/33kV	33kV MAKANDPUR	9	No		
Piro 132/33kV	33kV PIRO	4	Yes	Yes	
Kerpa 132/33kV	33kV KERPA	8	No		
Wazirganj 132/33kV	33kV WAZIRGANJ	13.5	Yes	Not	04.05.2026
Wazirganj 132/33kV	33kV PAMAR	13	No		
Wazirganj 132/33kV	33kV FATEHPUR	14	Yes	Not	04.05.2026
Warisaliganj 132/33kV	33kV WARISALIGANJ	8.4	No		
Frequency (in Hz) : 49.0 Hz	TOTAL : 444.46 MW	444.46			

Not integrated 15 (166 MW)
Integrated 26 (276 MW)
Integrated and reporting 8 (76 MW)
Integrated but not reporting 18 (203 MW)

FEEDER LIST , STAGE 4		Peak Load (MW)	Integration status	Reporting Status	Not reporting from
GSS Name	Feeders name				
132/33 KV Gaighat	Saidpur	12	Yes	Bad Quality	
220/132/33 KV Fatuha	Katra (Dahiyaawa)	12	Yes	Not	27.02.2024
	Meena Bazar	17	Yes	Not	27.02.2024
	Fatuha	28	Yes	Not	27.02.2024
	Dina Iron	10	Yes	Not	27.02.2024
132/33kV Barh	Barh Town	20.5	Yes	Not	25.02.2026
	Pandarak	6	Yes	Not	25.02.2026
132/33 KV Bari Pahani	Bari Pahani I	12	Yes	Not	23.04.2026
	Bari Pahani II	6	Yes	Not	23.04.2026
	Schsarai	22	Yes	Not	23.04.2026
	Ramchandrapur	22	Yes	Not	23.04.2026
	Noorsarai	22	Yes	Not	23.04.2026
	Asthama (Asthawa available)	14	Yes	Not	23.04.2026
132/33 KV Harnaut	Line II Feeder (Churm) (Churm available)	6.32	Yes	Not	27.04.2026
	Kalyanbigha	3.67	Yes	Not	24.09.2023
	Harnaut	5	Yes	Not	24.09.2023
132/33 KV Ekangarsarai	Ekangarsarai	12	Yes	Not	08.05.2026
	Hijla	19.1	Yes	Not	08.05.2026
132/33 KV Katra	Pahari	24	Yes	Bad Quality	
	Sabalpur	31	Yes	Bad Quality	
	Karmalchak	20	Yes	Bad Quality	
	Ashoknagar (Technotuff)	25	Yes	Bad Quality	
	Kankarbag (Malsalami)	19	Yes	Bad Quality	
220/132/33 KV Sampatchhak	Bahadurpur (Beldarichak)	5	Yes	Bad Quality	
	Sampatchhak	9	Yes	Bad Quality	
	KudaNawada	8	Yes	Bad Quality	
132/33 KV Purnea	Maranga	11	Yes	NA	
	Madhubani	6	Yes	NA	
132/33 KV Nalanda	Nalanda	16	Yes	Yes	
132/33 KV Rajgir	Silao	12	Yes	Bad Quality	
Frequency (in Hz) : 48.8 Hz	TOTAL : 435.59 MW	435.59			

Not integrated 0 (0 MW)
Integrated 30 (435 MW)
Integrated and repting 17 (237 MW)
Integrated but not reporting 11 (181 MW)

Annex B.2.1.g

Quantum of load shedding in different stages of AUFLS FY 2026-27										
AUFLS Set Points and Percentage Quantum of Relief				BIHAR	JHARKHAND	DVC	ODISHA	WEST BENGAL	SIKKIM	TOTAL
1	Stage 1	49.4 Hz	5.00%	345	103	159	304	523	5	1439
2	Stage 2	49.2 Hz	6.00%	399	130	208	372	613	5	1727
3	Stage 3	49.0 Hz	7.00%	466	151	243	434	715	5	2015
4	Stage 4	48.8 Hz	7.00%	466	151	243	434	715	5	2015
TOTAL				1676	535	854	1544	2567	20	7195

STATE	Peak Demand Met in 2025-26 (in MW)	Demand Met Ratio	Annual Consumption in MU for 2025-26	Consumption Ratio	Constituent Contribution	Stage 1	Stage 2	Stage 3	Stage 4	TOTAL
BIHAR	8752	0.248	47435	0.231	0.240	345	399	466	466	1676
JHARKHAND	2406	0.068	15391	0.075	0.072	103	130	151	151	535
DVC	3544	0.101	24753	0.121	0.111	159	208	243	243	854
ODISHA	7302	0.207	44194	0.215	0.211	304	372	434	434	1544
WEST BENGAL	13108	0.372	72849	0.355	0.363	523	613	715	715	2567
SIKKIM	136	0.004	543	0.003	0.003	5	5	5	5	20
	35248		205165			1439	1727	2015	2015	7195

Annex B.2.2.a

Agenda of PMJTL :- Insulator Failure

Date: 11-06-2026

Details of failed/damaged(physically found)/Weakened as per TS/Replaced CLR insulator as on 11.06.2026

Name of T/L	Qty of Puncture/Flashover/Broken CLR	Identified Qty based on Physical check	Identified weakened Qty based on Thermo-scanning	Total Defective CLR Qty	Total Replaced Qty	Balance Qty yet to be replaced as on 12/05/2026	Insulator Being used for Replacement
765kV D/C Ranchi-Medinipur T/L	206	676	451	1333	1150	183 (70 Tower)	1. Make-Deccan
							Mfg Yr. 2023 & 2024
							2. Make-GKXiagh(92)
							Mfg Yr. 2021
765kV D/C Medinipur-New Jeerat T/L	201	169	559	929	716	309 (95 Tower)	1. Make-Deccan
							Mfg Yr. 2023 & 2024
							2. Make-GKXiagh(90)
							Mfg Yr. 2021
400kV D/C New Jeerat-Subhasgram T/L	98	3343	953	4394	4377	17 (5 Tower)	1. Make-Deccan
							Mfg Yr. 2023 & 2024
							2. Make OGL & BHEL (1500)
							Mfg Yr. 2022
400kV D/C New Jeerat-Jeerat T/L	20	269	204	493	480	13 (3 Tower)	1. Make-Deccan
							Mfg Yr. 2023 & 2024
							2. Make OGL (200)
							Mfg Yr. 2022
Total	525	4457	2167	7149	6367	426	

Long term plan for replacement of defective CLR insulator

- Procurement of CLR:-

CLR Rating	Qty under Procurement	Tentative date of start of progressive supply	Tentative date of end of progressive supply
400kV, 120kN	1682	Jun-26 (Supply started, 700 Nos. already supplied)	Sep-26
400kV, 160kN	8591		
400kV Total	10273		
765kV, 160kN	3357	Jun-26 (Supply started, 850 Nos. already supplied)	Oct-26
765kV, 210kN	14428		
765kV Total	17785		

- Replacement of defective CLR:-

SL No	Name of Line	Tentative date of start of Replacement work	Tentative date of end of Replacement work
1	400kV D/C New Jeerat-Subhasgram	July-26	Sep-26
2	400kV D/C New Jeerat-Jeerat	Jul-26	Sep-26
3	400kV D/C Medinipur-Chanditala	Aug-26	Oct-26
4	400kV D/C Medinipur-Kharagpur	Aug-26	Oct-26
5	765kV D/C Ranchi-Medinipur	Jun-26	Dec-26
6	765kV D/C Medinipur-New Jeerat	Jun-26	Dec-26

UFR Operation Details

Constituent	UFR Stage	Quantum Approved in MW	Quantum Installed in MW	Quantum Operated	(Quantum Not Operated)
JUSNL	Stage 1	87	85	54	31
BSPTCL	Stage 1	315	292	93	199
ODISHA	Stage 1	306	316	205	111
West Bengal (WBSEDCL)	Stage 1	377	440	226	214
West Bengal (CESC)	Stage 1	120	120	72	48
DVC	Stage 1	172	173	150	23
Total ER		1377	1427	800	627

- *This Quantum approved and installed is based on Peak demand met + Energy Consumption =30 GW.*
- *At the instant ER demand was 26.5 GW , so in proportion to that Quantum of relief around 1170MW.*
- **Net Quantum not operated 1240-800 =440 MW**
- *No ADMS operation as all stater were in Under drawl Mode.*

SCADA data availability (Feeder number wise) for UFR dated 21-05-2026

	Total No of Feeders	No of Feeders, which are YET TO BE INTEGRATED in ICCP	No of feeders, for which SCADA data is NOT UPDATING	No of feeders, for which SCADA data is UPDATING	Percentage Availability
BSPTCL	137	106	26	5	4%
DVC	62	0	3	59	95%
JUSNL	33	15	6	12	36%
OPTCL	129	21	12	96	74%
WB - WBSEDCL	198	5	15	178	90%
WB - CESC	33	0	0	33	100%
Total ER	592	147	62	383	65%

Demand Reduction Details (UFR + Demand Response)

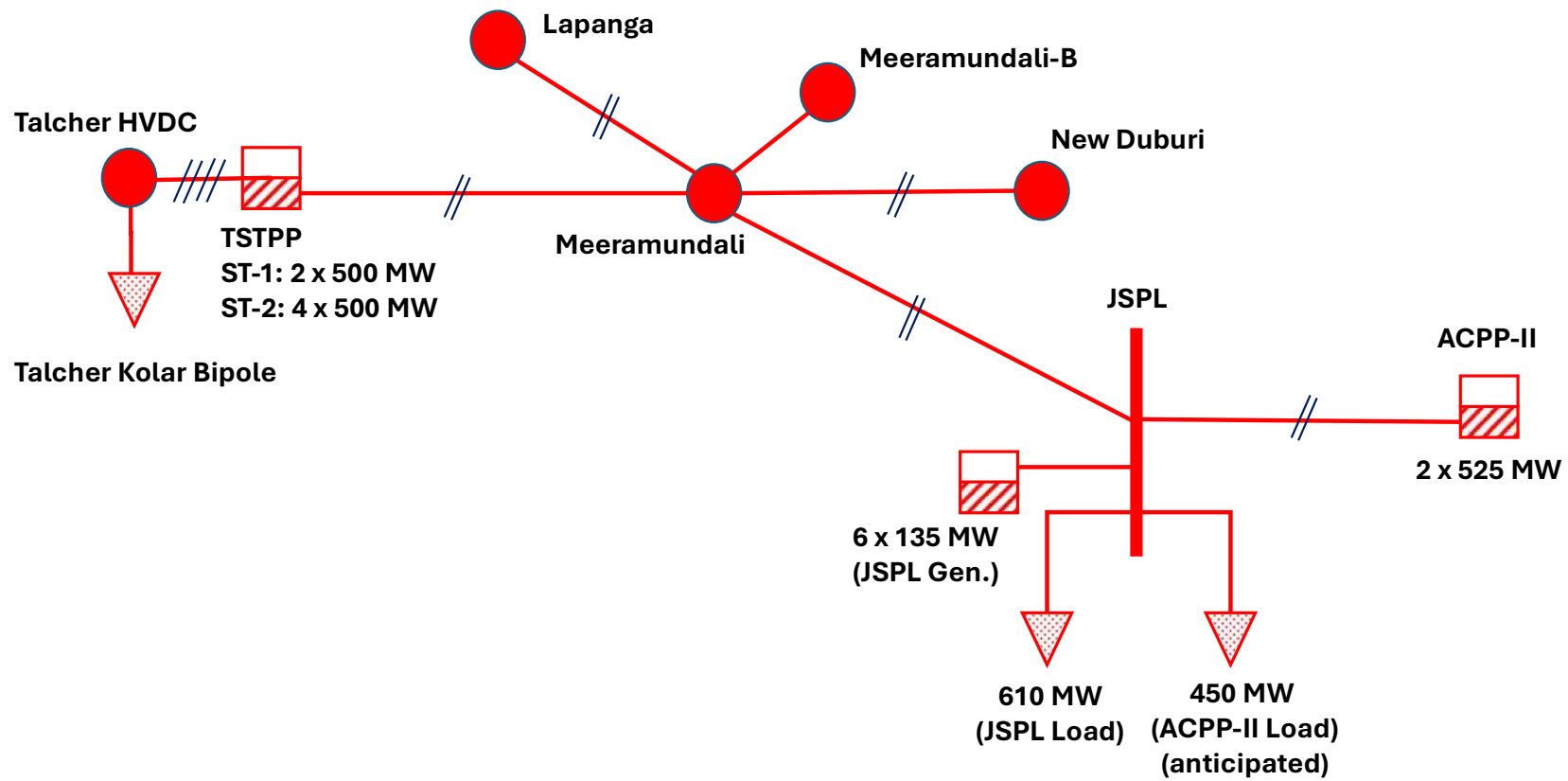
Constituent	Pre-event Demand	Post event demand	Net Change (UFR + Demand response)	UFR Operation	Demand Response
JUSNL	1710	1656	54	53	1
BSPTCL	5934	5716	218	94	124
ODISHA	6114	5803	311	205	106
West Bengal (WBSEDCL)	10361	10041	320	298	22
DVC	2610	2383	227	150	77
Total	26729	25599	1130	800	330

- UFR is based on data received from States .
- PFR is based on actual generation data.
- Net Change is based on Demand Data, *Post Event data after 1 Minute is considered, Where Demand Response very low, Scada demand data may not be updated accurately.*

Annex B.2.3

**Shutdown of
HVDC Talcher - Kolar Bipole**

**20.06.2026 to 05.07.2026
for
NHAI Diversion Work**



Seasonal Thermal Limit of 400kV Talcher -Meeramundali

Months	SPS setting (MVA)	Loading limit (MW) (Power Factor 0.95)	D/C Limit (MW)
Nov-feb	989	939.55	1108
Mar-Jun	714	678.3	800
Jul-Oct	874	830.3	980

Sensitivity of Talcher –Kolar Pole on

400kV Talcher -Meeramundali -1 : 22%
400kV Talcher -Meeramundali -2 : 13.28%

After Shutdown of Bipole:

N Condition : 400kV Talcher –Meeramundali D/C in service.
N-1 Condition: Tripping of 400kV Talcher –Meeramundali-2.

Sensitivity of 400kV Talcher -Meeramundali -2 on

400kV Talcher -Meeramundali -1 : 60%

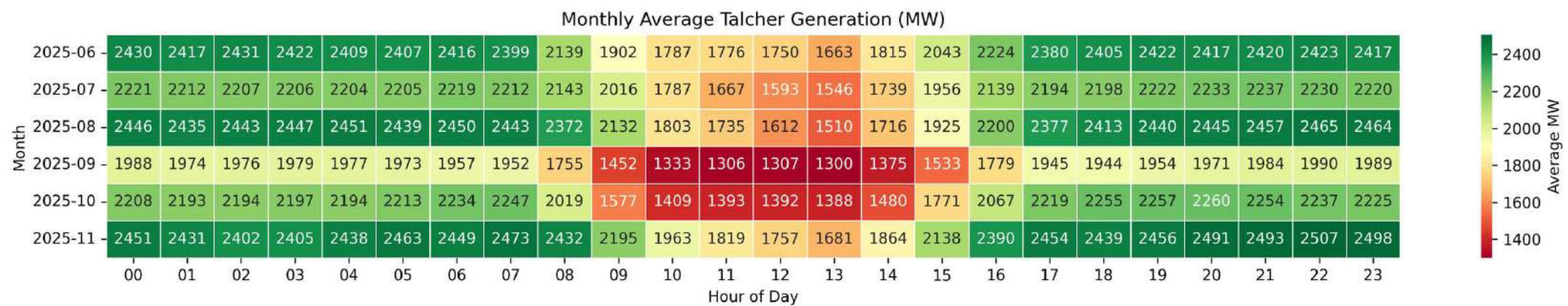
Average Odisha demand growth Y.O.Y for month of May : 8%

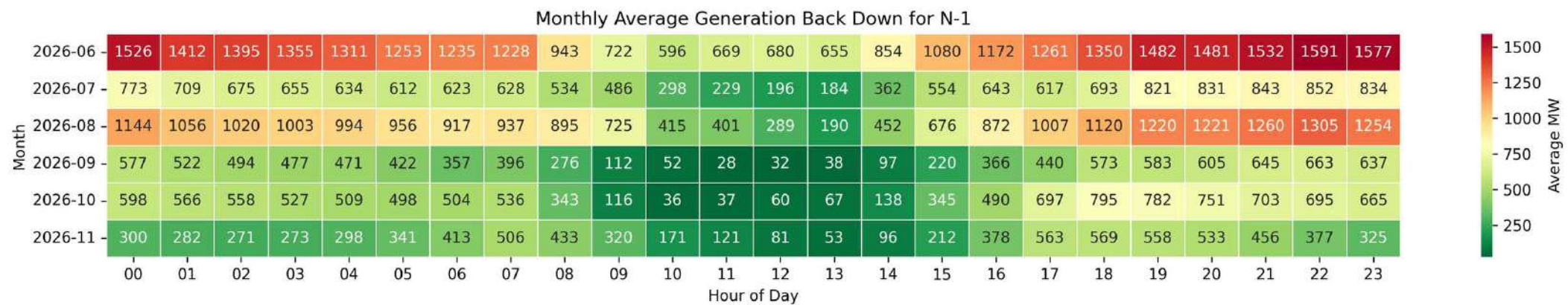
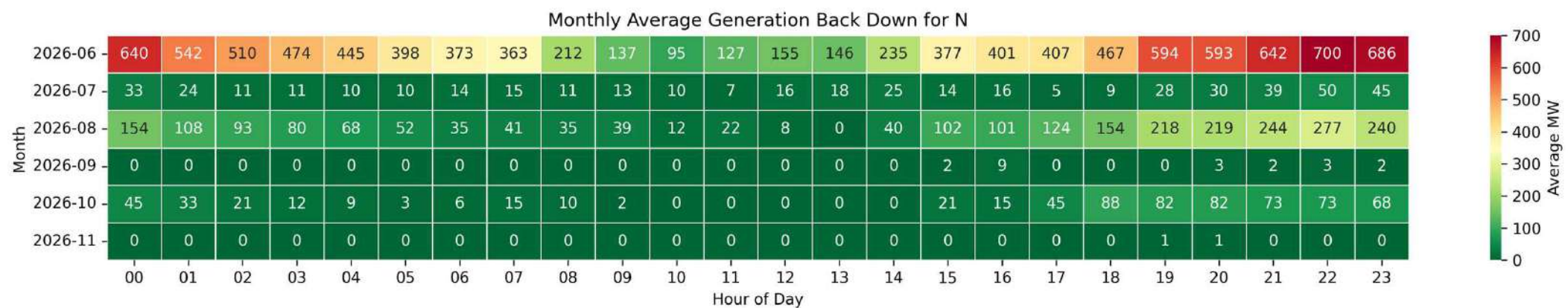
Sensitivity of Talcher Generation (When ckt-1 is out) on

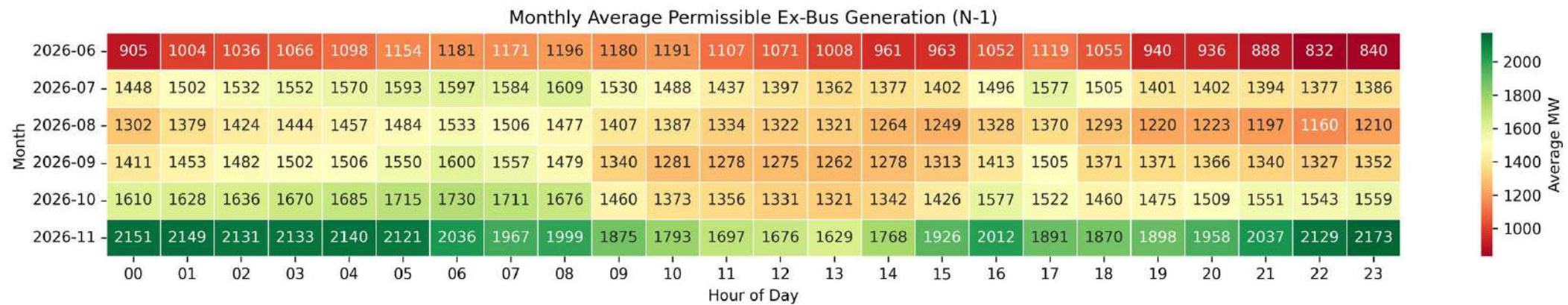
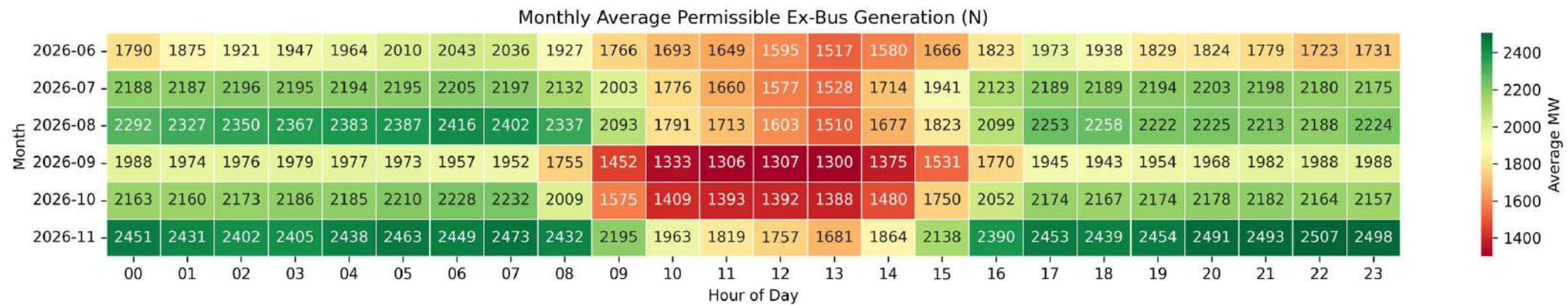
400kV Talcher -Meeramundali -1 : 29%

Sensitivity of Odisha demand on

400kV Talcher -Meeramundali -1 : 8%







Annex B.2.8

AMR Security Enhancements

Proposed additional requirements: Security and Redundancy enhancements of the AMR

Item	Current Version	New Requirement	New Version	Benefits to be achieved
AMR Firewall	CISCO- FPR 1140	Procurement of 01 additional Firewall	CISCO- FPR 3105	• High Availability Pair. • Host Standby. • Additional Malware protection and clustering. • Eliminating risk of single point failure.
AMR Router	CISCO 8200L	Procurement of 09 additional Routers.	CISCO 8200L	• 06 new routers to be installed at 6 utilities as redundant backup and controlling real time data flow of SEMs. • 03 Backup routers for PG, Generator & ERLDC master spares.
AMR Firewall – Threat protection licenses upgradation	FTD 7.0.1(build 84) VDB:338	Procurement of new licenses	FTD 7.4+ or 7.6	• This is kind of patch upgrade. • The same is suggested in the recent VAPT assessment. • More stringency in the Network policies.
AMR Switch OS upgrade	HP Aruba 6200F	Procurement of new licenses	Aruba OS-CX 10.13.0001	• This is a kind of patch upgrade. • The same is suggested in the recent VAPT assessment.

CERT-in guidelines

Guidelines on Information Security Practices for Government Entities



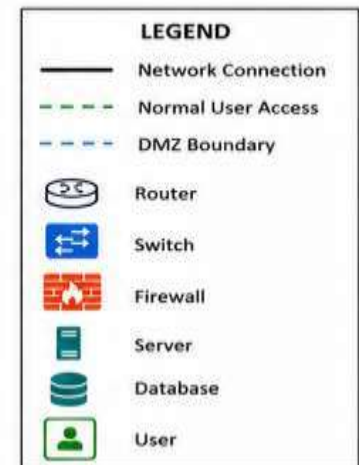
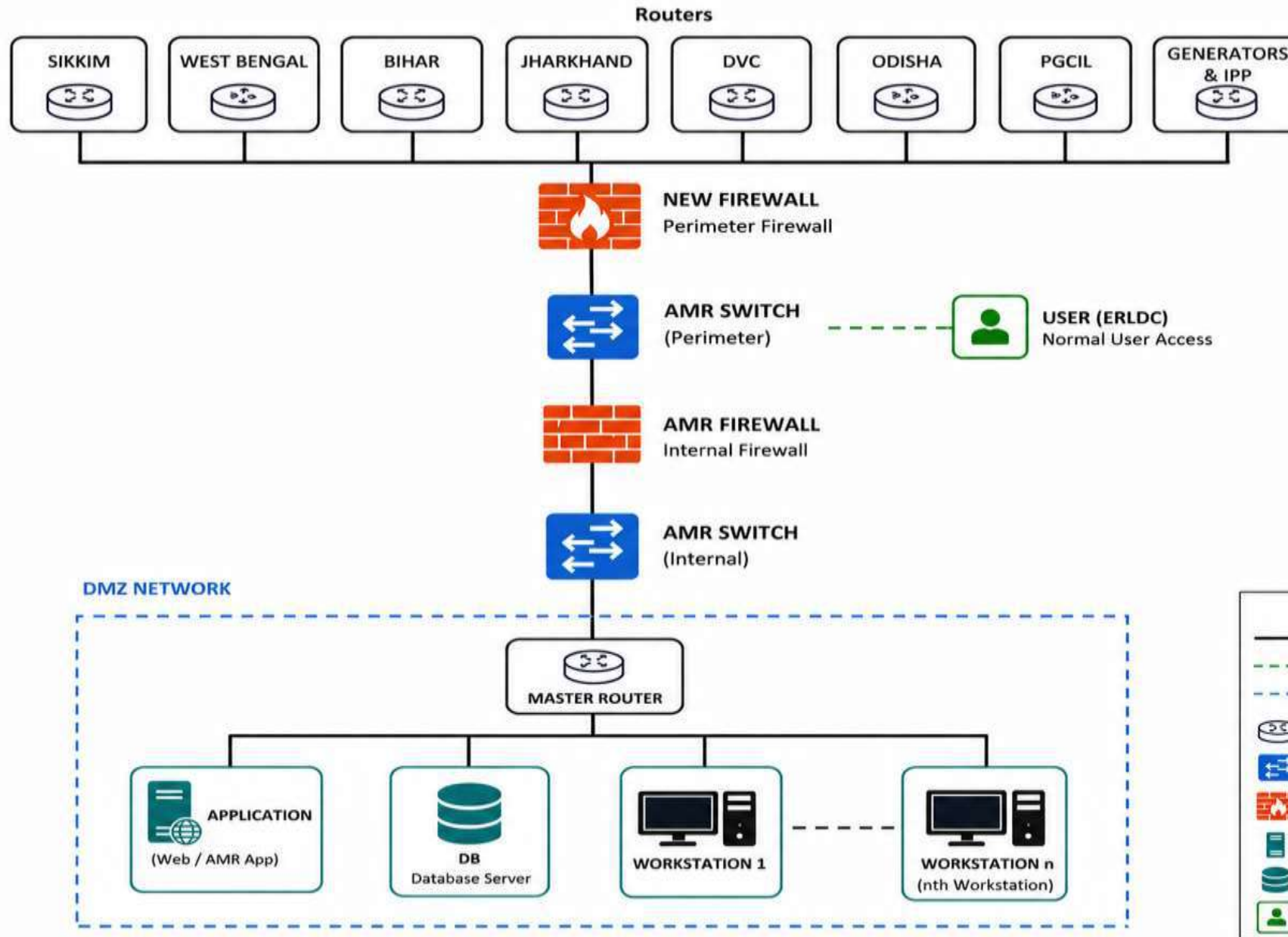
Guidelines on Information Security Practices for Government Entities

- 4.5.2. Deploy NextGen firewall (2 units) in High availability (HA) mode as a perimeter security device (external firewall). Another set of firewall from different OEM shall be used as internal firewall (in HA mode) for internal segmentation of network.

Existing Firewall & Proposed firewall

Parameter	CISCO FPR 1140 (existing)	CISCO FPR 3105 (newly proposed)
Firewall Throughput	~3.3 Gbps	10 Gbps
IPS/NGFW Throughput	~3.3–3.5 Gbps	10 Gbps
Architecture	Basic CPU	✓ Modern multi-core + hardware acceleration
TLS Inspection	~1.2 Gbps	~3.2 Gbps
Deep Inspection Load	Limited	High-scale inspection
High Availability	Basic	✓ Advanced HA (Active/Active & Active/Standby)
Clustering	✗ Not available	✓ Supported

Proposed AMR architecture



Current AMR Router setup & the additional Requirements

Utility	Layer3 Router location	Router Count
PGCIL	ERLDC, Kolkata	1
WEST BENGAL	SLDC, Howrah	1
ODISHA	SLDC, Bhubaneswar	1
JHARKHAND	Hatia-SS	1
BIHAR	SLDC, Patna	1
DVC	Maithon-SS	1
SIKKIM	ERLDC, Kolkata	1
GENERATORS & IPP	Subhasgram-PG	1
MASTER ROUTER	ERLDC	1

- We propose here to procure additional 09 routers as backup.
- Routers will be installed at SLDC/DVC/ERLDC/Subhasgram-PG location and will be configured for and improvement of network level security.
- This is required to smoothly & securely operate the 01 min real time Meter data flow for all SLDCs.

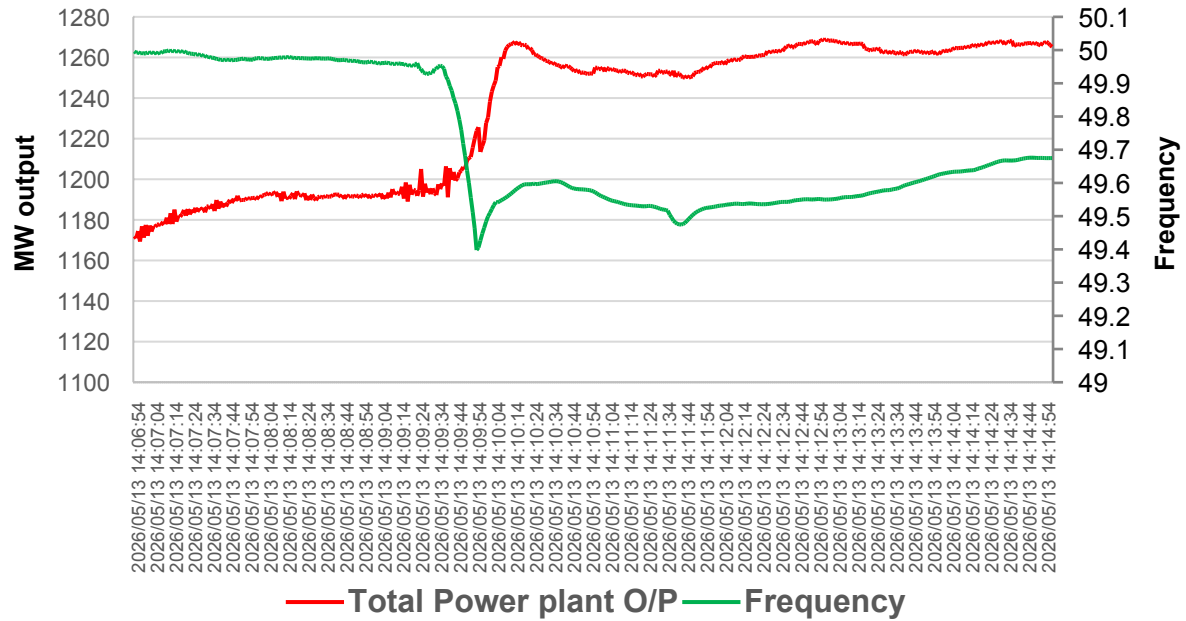
PFR Response

Constituent	Pre Event Generation	Post event Generation	Net Change (PFR Operation)	Ideal PFR Response (At least 05% of Pre event Gen)	Ideal-Actual	% Achieved w.r.t. Ideal
JUSNL	196	197	1	10	-9	10
BSPTCL	766	775	9	38	-29	23
ODISHA	3320	3330	10	166	-156	6
West Bengal	5325	5384	59	266	-207	22
DVC	3256	3378	122	163	-41	75
Total ER State	12863	13064	201	643	-442	31
Total ER ISGS	11542	11886	344	577	233	60

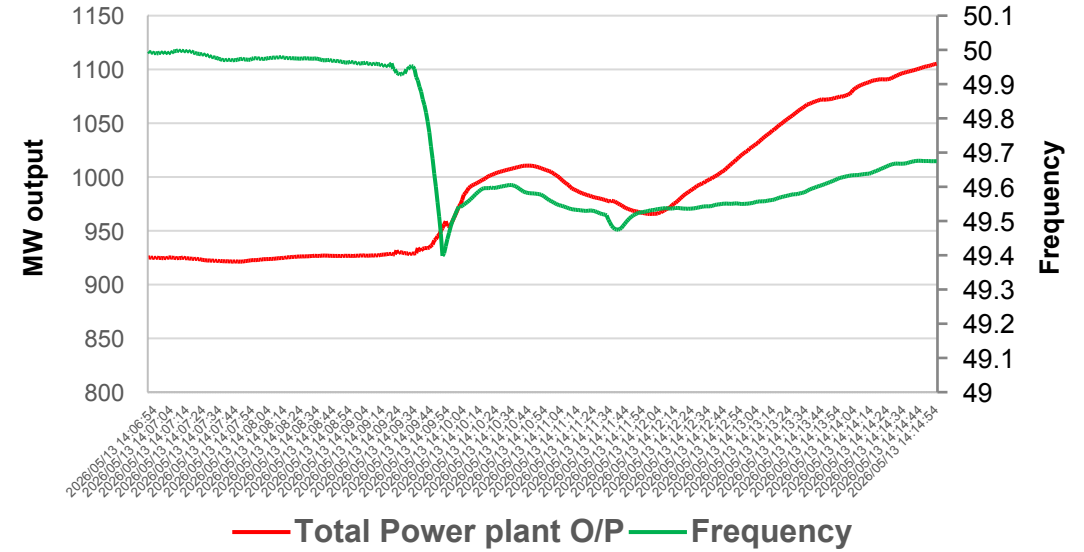
Ideal PFR response will be limited to 5% of its MCR but for simplification at least 05% of Pre event generation considered.

Sl.no	Name of Generator(ISGS+IPP)	Response	Net response	5% of present gen	% response w.r.to current gen
1	FSTPP I&II	Excellent	-37.11	-39.22	94.61%
2	FSTPP III		-15.65	-15.03	104.14%
3	KhSTPP II		-66.32	-59.75	110.98%
4	BARH ST-2		-31.78	-37.35	85.09%
5	BRBCL		-55.91	-44.26	126.33%
6	MPL		-27.27	-34.35	79.40%
7	Adhunik		-19.74	-20.98	94.12%
8	DIKCHU		-6.99	-2.471	282.49%
9	NPGC		-72.82	-46.45	156.77%
10	BARH ST-1	Poor	-20	-68	29%
11	INDBHARAT		-7.63	-17.20	44.35%
12	PVUNL		-8.47	-29.85	28.39%
13	DARLIPALLI		-19.62	-79.42	24.71%
14	NORTH KARANPURA		-19.19	-61.43	31.24%
15	JITPL		-0.8685	-59.100	1.47%
16	KhSTPP I		3.491269	-24.63	-14.17%
17	GMR		-6.1135	-34.88	17.53%
18	TSTPS-I		1.5	-47.37	-3.17%

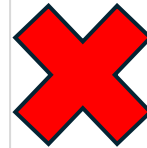
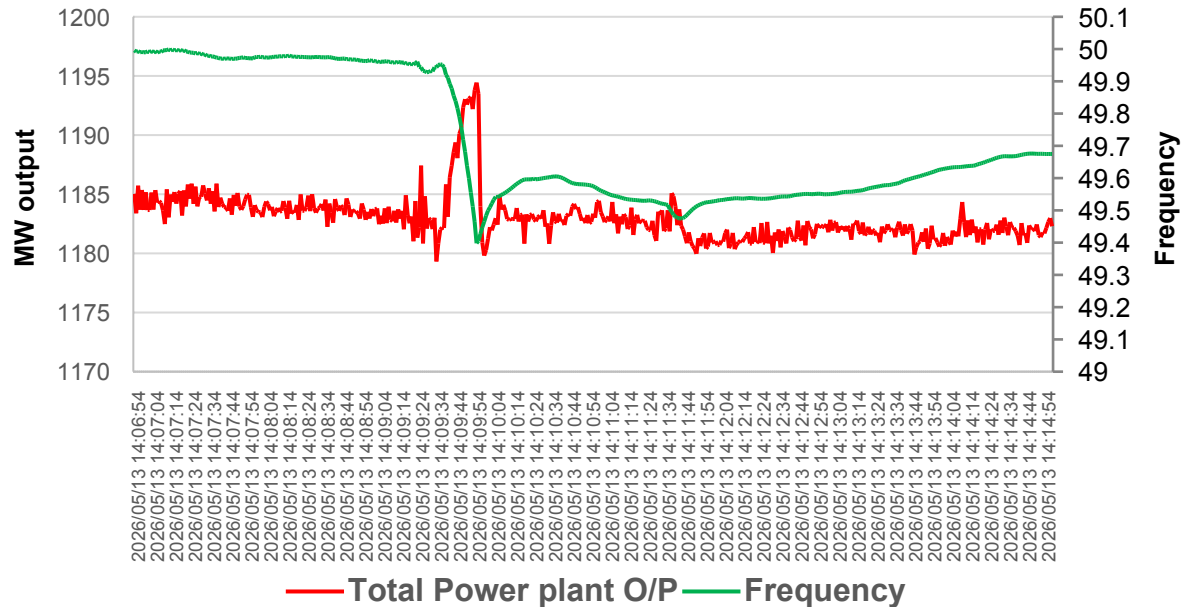
Response of KhSTPP #STG 2



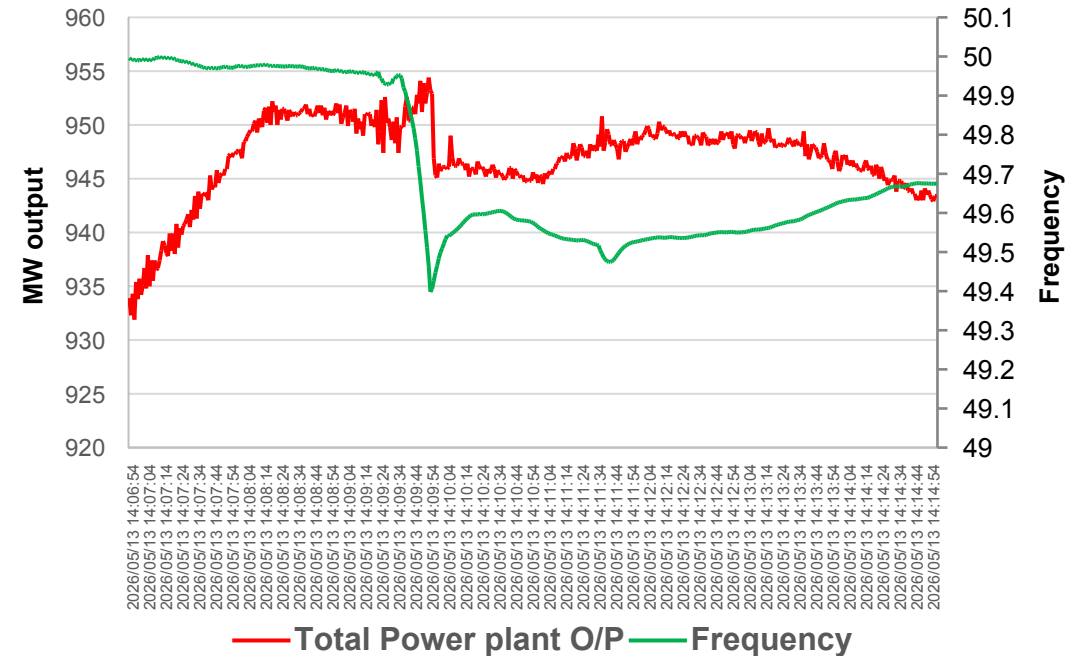
Response of NPGC



Response of JIPL



Response of TSTPP #STG 1

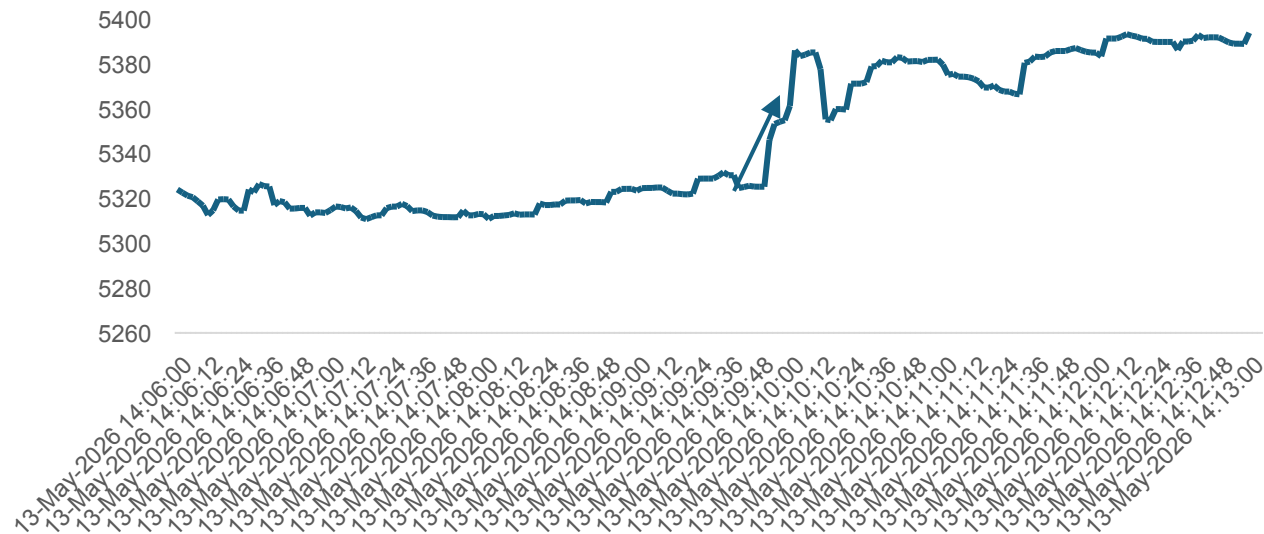
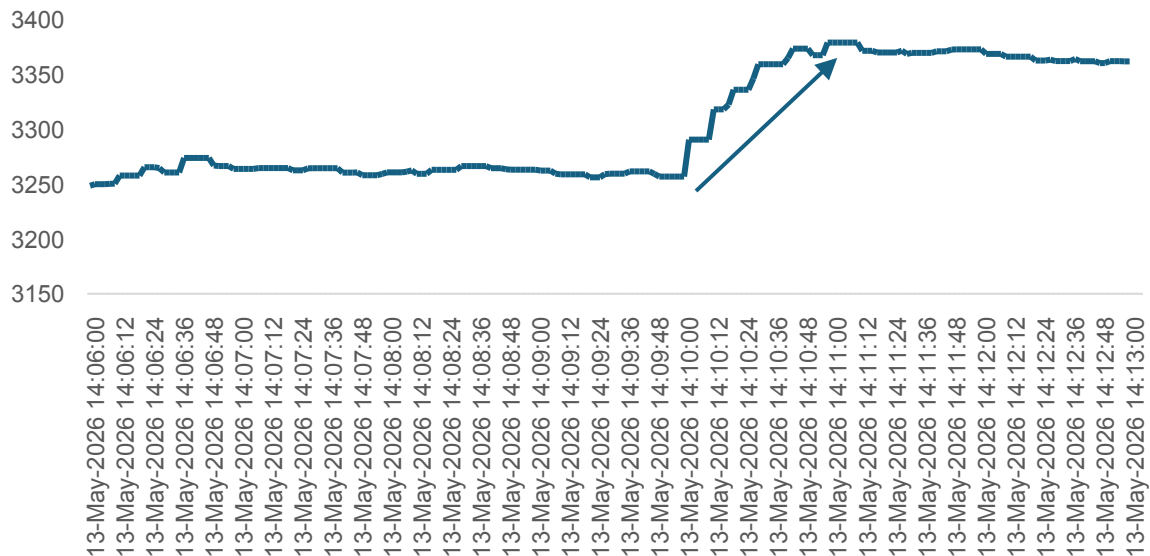


Plant Name	State/ISGS/IPP	Response (with respect to ideal droop based characteristics)
Bandel Thermal Power Station (BTPS)	West Bengal	-4.90%
Kolaghat Thermal Power Plant (KTPP)		6.83%
Bakreshwar Thermal Power Plant (BKTPP)		77.39%
Santaldih Thermal Power Station (STPS)		-29.36%
Sagardighi Thermal Power Plant (SGTPP)		23.19%
Durgapur Project Ltd (DPL)		-44.93%
Budge Budge Generating Station (BBGS)		195.28%
Haldia Energy Ltd (HEL)		114.89%
Bokaro Thermal Power Station (BTPS)	DVC	122.81%
Durgapur Steel Thermal Power Station (DSTPS)		49.63%
Koderma Thermal Power Station (KTPS)		180.73
Raghunathpur Thermal Power Station (RTPS)		-10.76%
Chandrapura Thermal Power Station (CTPS)		135.82%
Mejia Thermal Power Station (MTPS-7&8)		0.16%
Mejia Thermal Power Station (MTPS-5&6)		62.80%
Mejia Thermal Power Station (MTPS-1-3)		
Mejia Thermal Power Station (MTPS-4)		-9.32%
Mejia Thermal Power Station (MTPS-7)		
Mejia Thermal Power Station (MTPS-5)		120.79%
Mejia Thermal Power Station (MTPS-6)		4.92%
Tenughat Thermal Power Station	Jharkhand	12.99%
PVUNL		28.39%
IB VALLEY TPS	Odisha	1.66%
OPGC_STU_GEN		-6.53%
JSPL1_GR - GT_2_P		-20.75%
BALIMELA		4.97%
UPPER KOLAB		7.68%
BARAUNI TPS	Bihar	-1.39%
BIIXAR		-4.43%

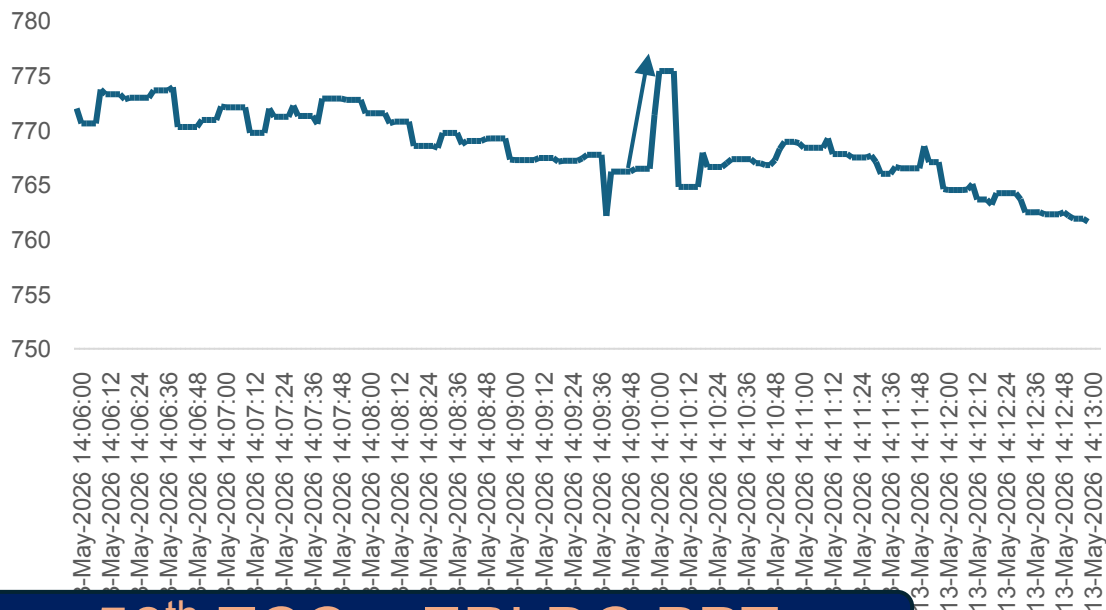
DVC

State Internal Generation

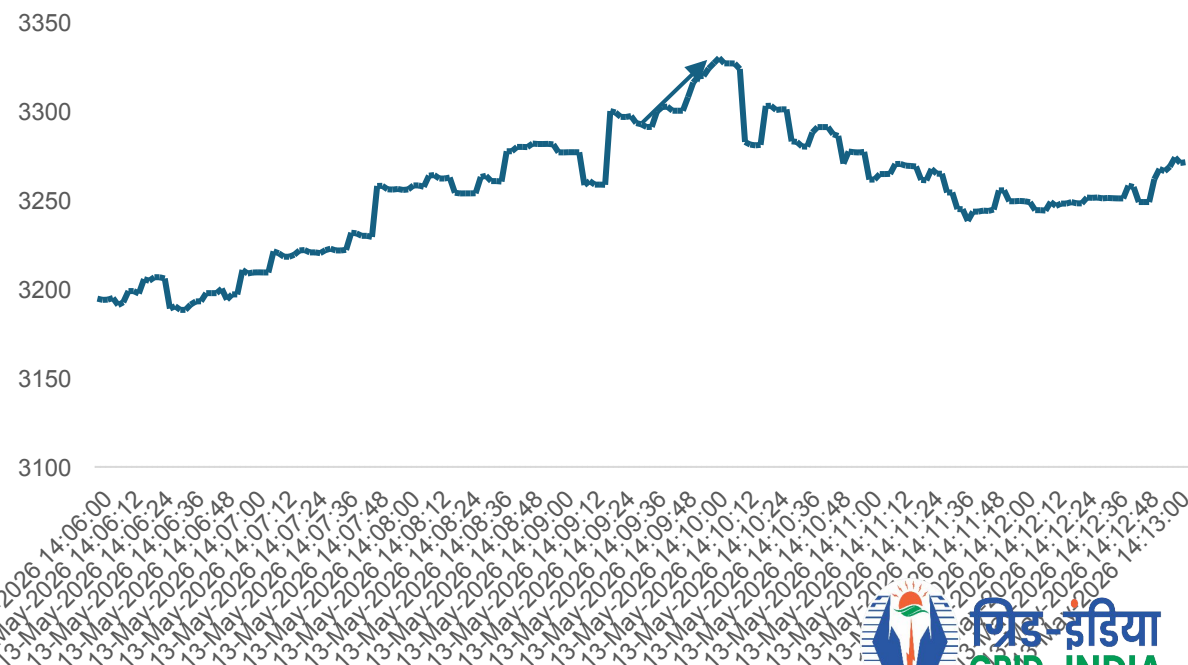
WB



BIHAR



Odisha



Annex B.2.8

Minutes of Meeting (MoM) of Review of Eastern regional entities Response during RE Generation Loss Event (8963 MW) at Khavda RE Complex (WR)

An online meeting was convened by ERLDC on 10.06.2026 at 15:00 hrs through virtual mode to review the performance of Primary Frequency Response (PFR) and Automatic Generation Control (AGC) of Inter-State and Intra-State generating stations. The meeting was attended by representatives from ER SLDCs (except Sikkim) and Generating stations and ERLDC. The meeting primarily focused on assessing the PFR and AGC performance of generating units, analysing generator response during the recent generation loss event of approximately 9 GW and discussing the issues observed along with necessary corrective actions for improving the response during similar future contingencies.

GM, SO welcomed all participants and highlighted significance of the primary frequency response and AGC in the context of high penetration of the renewables. It was also emphasized the importance of active participation by all generators in maintaining load-generation balance and supporting grid frequency control during system contingencies.

ERLDC explained the Event of 13.05.2026 briefly and then presented a summary table comprising performance analysis of generators based on: FRO (Frequency Response Obligation) and ideal response (w.r.to 5% of current generation capped to 5% of the MCR). Key observations from the response were as follows,

- Several generating stations demonstrated satisfactory to excellent response during the event.
- Cases of response withdrawal/load throw off, limited response, oscillatory response, max response received after 45 sec were observed and discussed.
- Issues with Cap Max/Min settings of AGC.
- Implementation issues of ULSP / RULSP logic.
- Some stations reported operational limitations such as:
 - ✓ Turbine vibration issues.
 - ✓ Boiler mill constraints.
 - ✓ FD fan-related limitations, Control valve (HPCV) problems
 - ✓ Units operating near technical minimum margin.
 - ✓ Absence of EHTC
 - ✓ Units operating in valve Wide open condition.
 - ✓ Generator-wise Key Observation

Sl.no	Name of Generator	Observation and Discussion
1	FARAKKA STAGE-I & II / STAGE-III:	Response received w.r.to present generation was around 95% and 104% respectively. The performance received from both the stages were satisfactory. Farakka shall share their detailed implemented FGMO logic with ERLDC within 2-3 days.

		Further frequent operation of AGC in local mode and Cap Max/Min violations were also discussed. They will review and revert to ERLDC
2	KHSTPP II	Response received w.r.to present generation was around 111%. Performance was found satisfactory.
3	BARH ST-2	Response received w.r.to present generation was around 85% but the load thrown off immediately. The representative of the plant mentioned that during part-load/technical minimum operation, Unit-5 experiences vibration issues. To mitigate these vibrations, the overload valve (OV) was previously kept open during part-load operation. In the present event, although the governor responded and the unit output increased instantaneously, the output subsequently reduced significantly due to the OV remaining open. Subsequently, the station informed that the practice of keeping the OV open during part-load operation has been discontinued and that the unit's Primary Frequency Response (PFR) performance is expected to improve during future events. At present, the plant is using the ULSP signal for AGC operation. ERLDC requested the plant to explore the possibility of implementing the RULSP signal to ensure correct AGC signal processing and response. The plant representative agreed to examine the feasibility of implementing the RULSP signal and will revert with their observations and findings. ERLDC noted the submission.
4	BRBCL	Response received w.r.to present generation was around 126%. Performance was found satisfactory.
5	MPL	Response received w.r.to present generation was around 79%. Performance was found satisfactory. At present, the plant is using the ULSP signal for AGC operation. ERLDC requested the plant to explore the possibility of implementing the RULSP signal to ensure correct AGC signal processing and response. The plant representative agreed to examine the feasibility of implementing the RULSP signal and will revert with their observations and findings
6	ADHUNIK	Response received w.r.to present generation was around 94%. But the maximum response was achieved beyond the recommended 45-second period. Adhunik was requested to review the issue.
7	DIKCHU	Performance was found satisfactory.
8	NPGC	Performance was found satisfactory.
9	BARH ST-1	Response received w.r.to present generation was around 29%. The unit's Primary Frequency Response (PFR) was also observed to be oscillatory in nature. The station attributed this behaviour to an issue in the control logic and informed that corrective actions are currently underway to rectify the logic
10	JSWEL	The primary response received during the event was only 45%. During discussion the generator informed that the unit was

		operating without maintaining any reserve margin for PFR i.en in VWO (Valve Wide Open) mode. Subsequently, the generator was advised not to operate the unit without maintaining the requisite PFR margin, as this constitutes a violation of the IEGC provisions. They were further requested to review the matter, take immediate corrective action.
11	PVUNL	Response received w.r.to present generation was around 29%. The Load thrown off immediately after the Primary response was initiated. Concerned stations agreed to review the issues and take corrective actions.
12	DARLIPALLI	Response received w.r.to present generation was around 25%. Time synchronization issue in the data submitted was found. During the discussion it was found that for U1 maximum PFR was capped at 10 MW due to some boiler temperature excursion issue. Plant representative intimated that they will be taking up the issue during their next overhauling plan in August 2026.
13	NORTH KARANPURA	Response received w.r.to present generation was around 31%. During the discussion, the station informed that a capping had been imposed on the unit's response due to turbine vibration issues. This restriction was subsequently removed during turbine overhauling in the month of May 2026. As a result, the unit's PFR performance has improved, which has also been reflected in its response during subsequent frequency events. AGC Cap Max/Min violations were also discussed. The plant representative informed that they would review the instances of these violations and revert to ERLDC with their observations and findings.
14	JITPL	Response received w.r.to present generation was around 1.5%. JIPL was requested to review, take corrective action and revert back.
15	KhSTPP I	Response received w.r.to present generation was around -14%. The station informed that it is having Mechanical governor in place and there is no EHTC (Electro Hydraulic turbine governor control) in Stage-I units, and therefore the units are unable to sustain the output following a frequency response event. The station further stated that installation of EHTC is planned during the overhaul scheduled in FY 2028–29. ERLDC advised the station to apprise the Commission of this limitation and formally intimate the Commission regarding the issue.
16	GMR	Response received w.r.to present generation was around 17.5%. U2 is having issue with control valve which they will correct in their overhauling planned in June 2026. ERLDC noted it.
17	TSTPS-I	Response received w.r.to present generation was around -3%. Sudden Load throw off was seen in the performance. The plant operators are analysing the issue and shall share any findings to ERLDC. AGC Cap Max/Min violations were also discussed. The plant representative informed that they would review the instances of

		these violations and revert to ERLDC with their observations and findings.
18	BARAUNI TPS (BIHAR):	Response received w.r.to present generation was around -1.4%. The poor PFR was attributed to the unit operating in Valve Wide Open (VWO) mode. The station informed that the unit was running in VWO mode due to an issue with the Forced Draft (FD) fan, which had gone out of service. The station further stated that the FD fan issue has since been resolved and that the unit has demonstrated satisfactory performance during other recent frequency events. ERLDC noted the submission.
19	BUXAR TPS (BIHAR):	Response received w.r.to present generation was around -4.4%. The station PFR performance was poor during the event. SLDC Bihar was requested to review the response and verify FGMO implementation.
20	WEST BENGAL INTRA-STATE GENERATORS	Response was mostly poor except for Budge-Budge, HEL and Bakreshwar. SLDC is taking up with the generators for necessary correction.
21	ODISHA	All the generators performance were poor. ERLDC requested them to take up the issue with their internal generators.
22	DVC	Most of the generators performed satisfactorily except for Mejia and Raghunathpur.
23	JHARKHAND	Response received w.r.to present generation was around 12%. Response was poor. ERLDC requested SLDC to review the issue for necessary correction.

Conclusion and Action Items:

The meeting concluded with a consensus on the need to strengthen Primary Frequency Response (PFR) and Automatic Generation Control (AGC) performance across generating stations to enhance grid resilience and ensure reliable frequency support during large-scale generation loss events.

1. ERLDC shall circulate a questionnaire to all generators seeking details of their FGMO logic, operational issues encountered and proposed corrective action plans for improving frequency response performance.
2. Generators were requested to submit Proper time-stamped data submission for PFR as well Beta calculation. Further generators were requested to intimate any implemented change in logic to ERLDC immediately.
3. SLDCs to validate generator response data, ensure compliance with FGMO implementation requirements, and collect and verify unit-wise high-resolution event data from intra-state generating stations. SLDCs were further requested to take up with intra state generators at the earliest to improve their performance.

The detailed report presented during the meeting is attached in Annexure 1

Annexure 1

PRIMARY FREQUENCY RESPONSE OF EASTERN REGION GENERATORS

Details of the event: FRC Response Analysis for Gen loss event of 8963 MW in RE Generation Complex, WR at 14:09:34 hrs. on 13.05.2026.

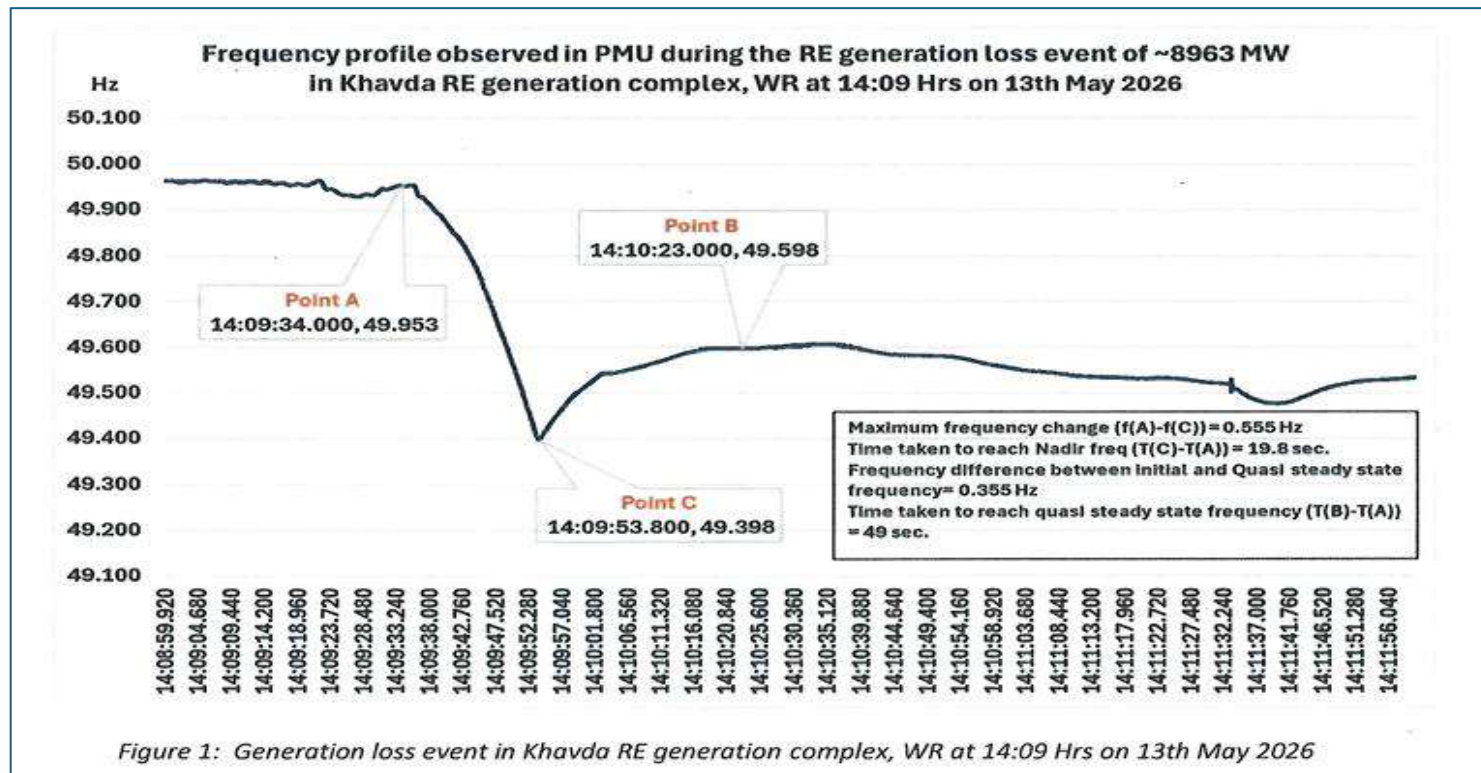
Pre-Event Time stamp (Point A): **14:09:34**

Post-Event Time stamp (Point B): **14:10:23**

Time Stamp of Nadir frequency (Point C): **14:09:53**

Total time difference for FRC calculation (A-B): **49 sec**

Total time taken to reach Nadir frequency (A-C): **19 sec**



	Name of Generator	Event: FRC Response Analysis for Gen loss event of 8693 MW in RE Generation Complex, WR at 14:09:34 hrs. on 13.05.2026							T ₄₅ =14:10:19	
Sl.no		FRP	FRP grading	Pre event gen (14:09:34)	Post event gen (14:10:23)	Net response	5% of present gen	% response w.r.to current gen	Whether Max response (5% of current gen and actual achieved max value whichever is lower) reached within 45 sec	Time taken to reach 5% of current gen and actual achieved max value whichever is lower.
1	FSTPP I&II	1.63	Excellent	-784.45	-821.56	-37.11	-39.22	94.61%	Yes 14:10:14	00:00:40
2	FSTPP III	1.83	Excellent	-300.66	-316.32	-15.65	-15.03	104.14%	Yes 14:10:17	00:00:43
3	KhSTPP II	2.55	Excellent	-1195.14	-1261.46	-66.32	-59.75	110.98%	Yes 14:10:04	00:00:30
4	BARH ST-2	1.51	Excellent	-747.03	-778.81	-31.78	-37.35	85.09%	Yes 14:09:50	00:00:16
5	BRBCL	3.57	Excellent	-885.28	-941.20	-55.91	-44.26	126.33%	Yes 14:10:11	00:00:37
6	MPL	1.6	Excellent	-687.10	-714.38	-27.27	-34.35	79.40%	No 14:10:23	00:00:49
7	Adhunik	2.22	Excellent	-419.62	-439.37	-19.74	-20.98	94.12%	No 14:10:23	00:00:49
8	DIKCHU	3.94	Excellent	-49.55	-56.55	-6.99	-2.471	282.49%	Yes 14:09:47	00:00:13
9	NPGC	2.27	Excellent	-929.11	-1001.94	-72.82	-46.45	156.77%	Yes 14:10:05	00:00:31
10	BARH ST-1	0.94	Good	-1377.40	-1397.50	-20.1	-68.87	29.19%	Yes 14:10:19	00:00:45
11	INDBHARAT	0.82	Average	-344.16	-351.79	-7.63	-17.20	44.35%	Yes 14:09:53	00:00:19
12	PVUNL	0.62	Below Average	-597.20	-605.67	-8.47	-29.85	28.39%	No 14:10:22	00:00:48
13	DARLIPALLI	0.67	Below Average	-1588.57	-1608.20	-19.62	-79.42	24.71%	No 14:10:22	00:00:48
14	NORTH KARANPURA	0.7	Below Average	-1228.68	-1247.87	-19.19	-61.43	31.24%	No 14:10:21	00:00:47
15	JITPL	0.03	Poor	-1182.01	-1182.88	-0.8685	-59.100	1.47%	Yes 14:09:53	00:00:19
16	KhSTPP I	-0.27	Poor	-492.76	-489.27	3.491269	-24.63	-14.17%	Yes 14:09:53	00:00:19
17	GMR	0.46	Poor	-697.61	-703.72	-6.1135	-34.88	17.53%	Yes 14:09:46	00:00:12

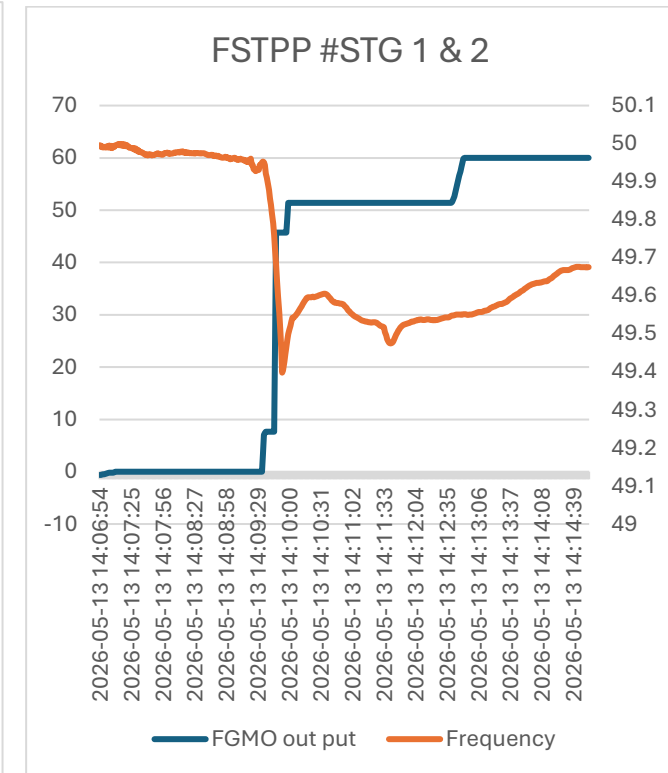
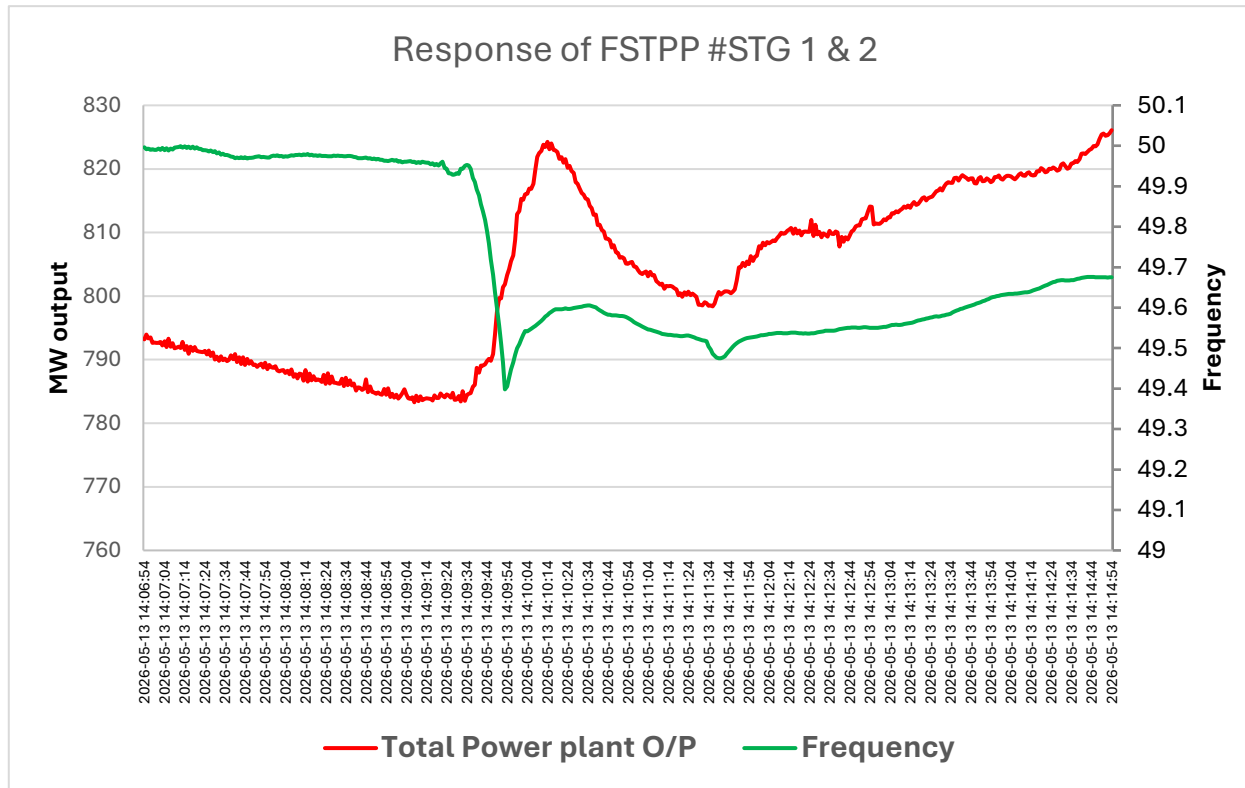
18	TSTPS-I	-0.08	Poor	-947.40	-945.90	1.5	-47.37	-3.17%	Yes 14:09:47	00:00:13
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FSTPP I&II

% response w.r.to current gen: 94.61%

Grading: Excellent

% response w.r.to FRO: 163%



Discussion:

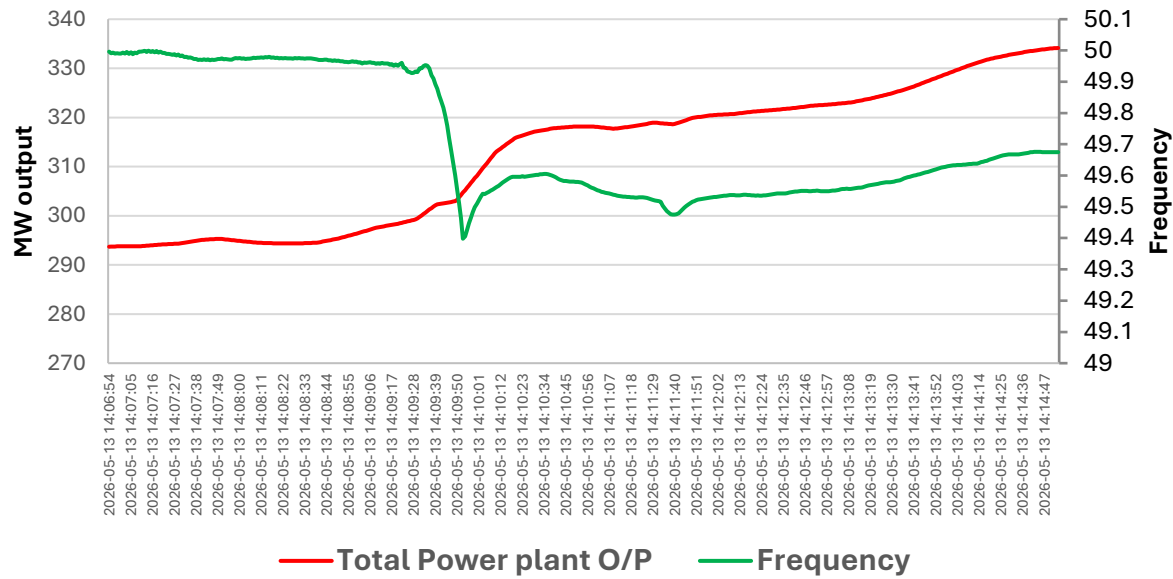
FSTPP III

% response w.r.to current gen: 104.14%

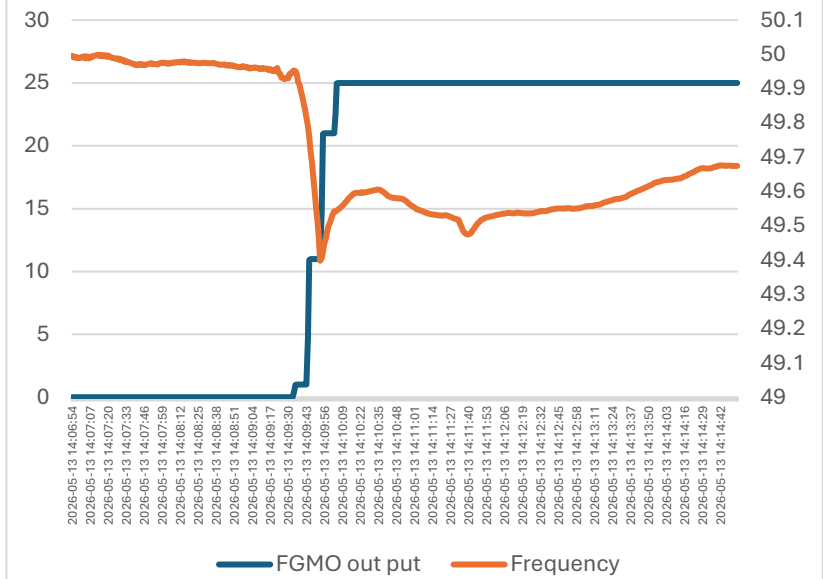
Grading: Excellent

% response w.r.to FRO: 184%

Response of FSTPP # STG 3



FSTPP #STG 3



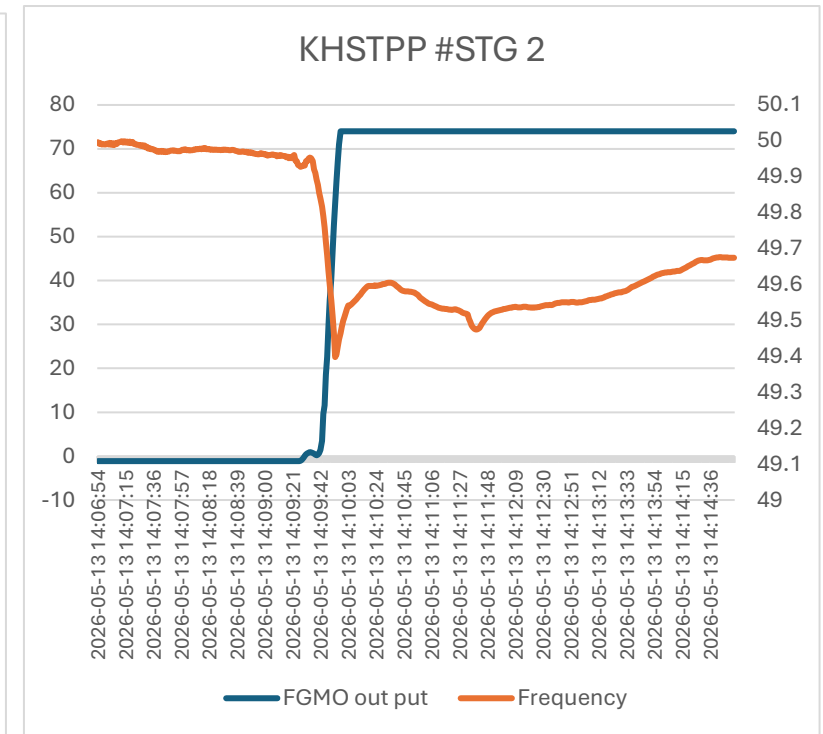
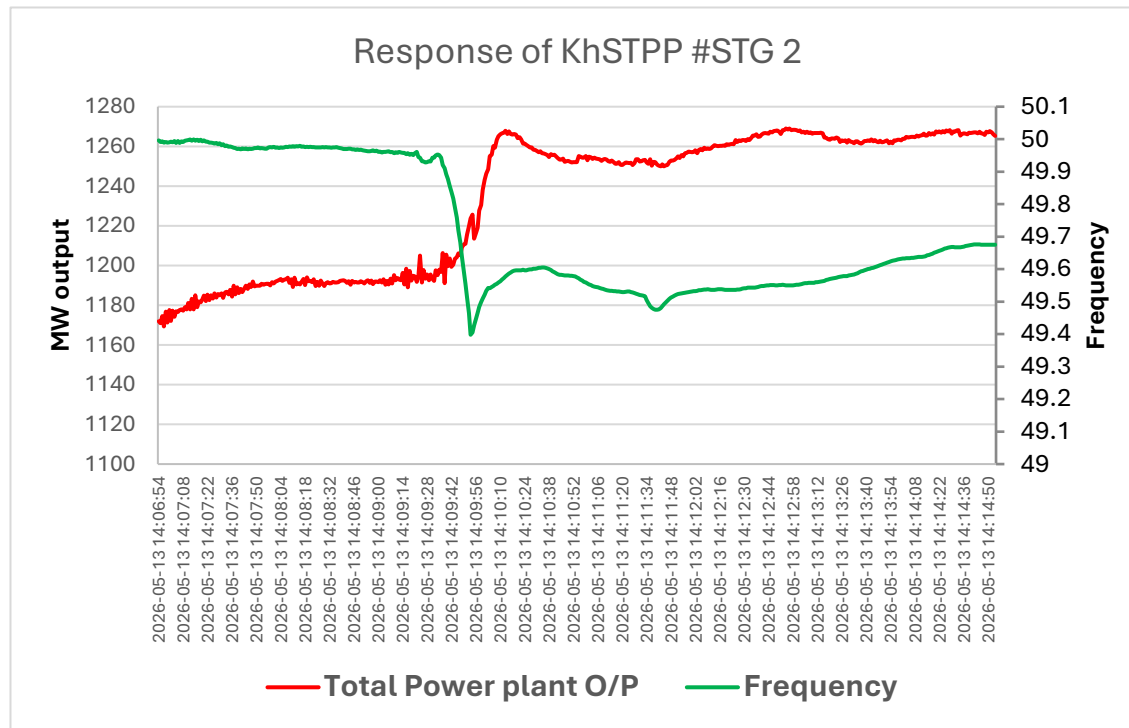
Discussion:

KhSTPP II

% response w.r.to current gen: 110.98%

Grading: Excellent

% response w.r.to FRO: 256%



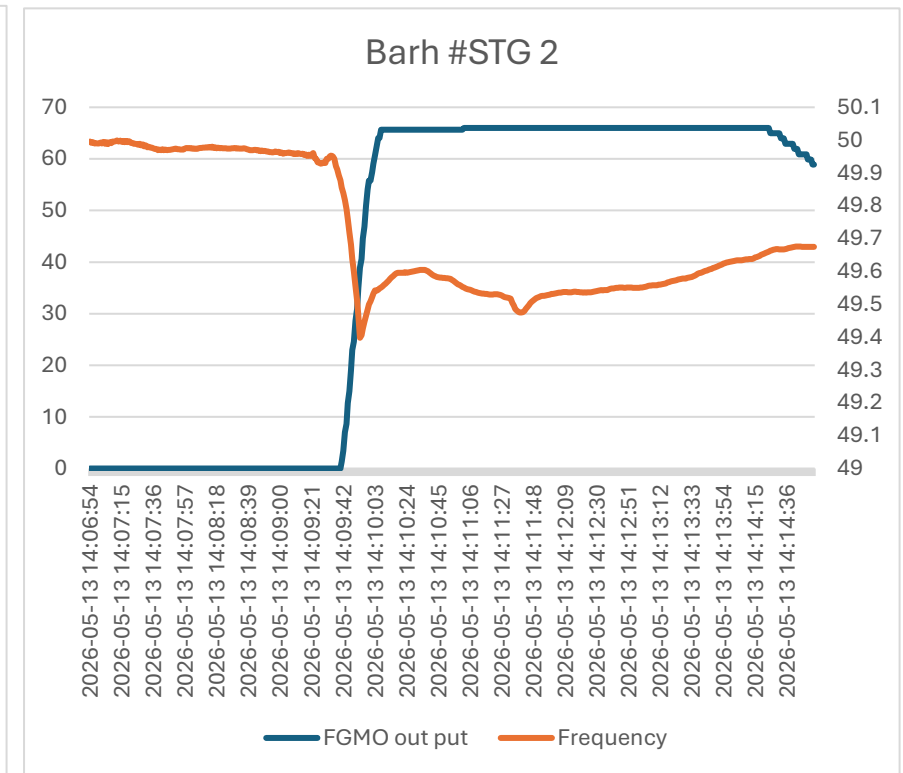
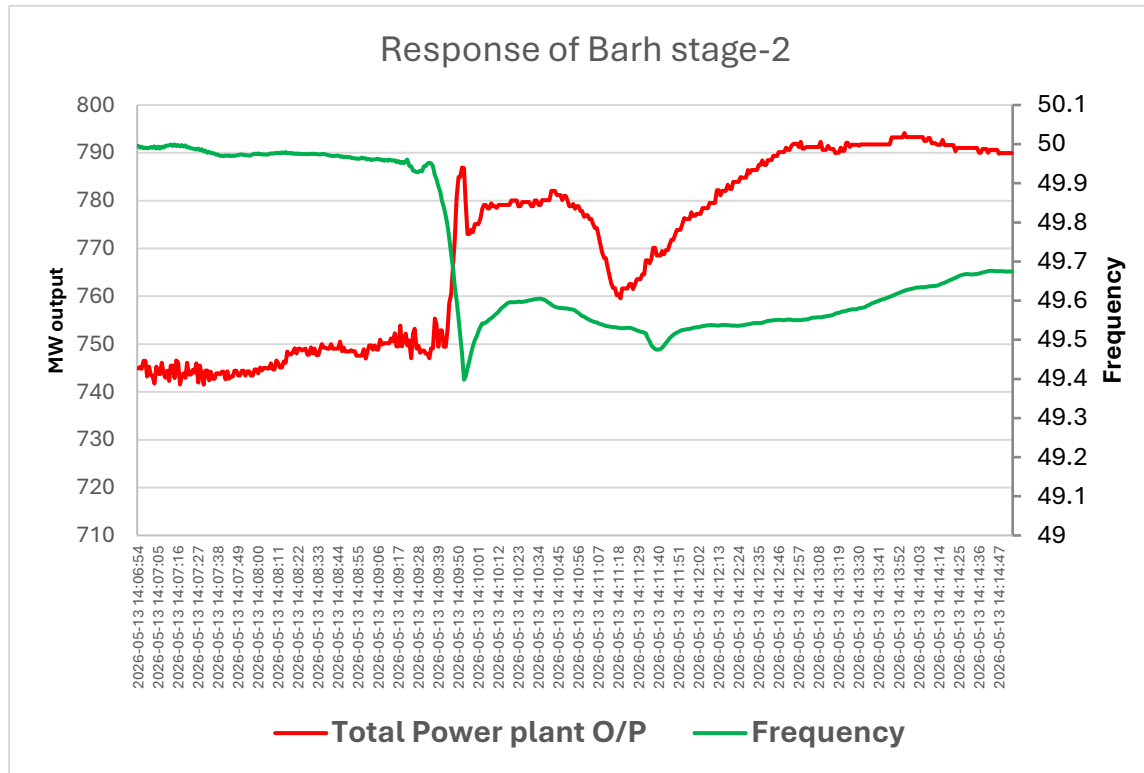
Discussion:

BARH ST-2

% response w.r.to current gen: 85.09%

Grading: Excellent

% response w.r.to FRO: 152%



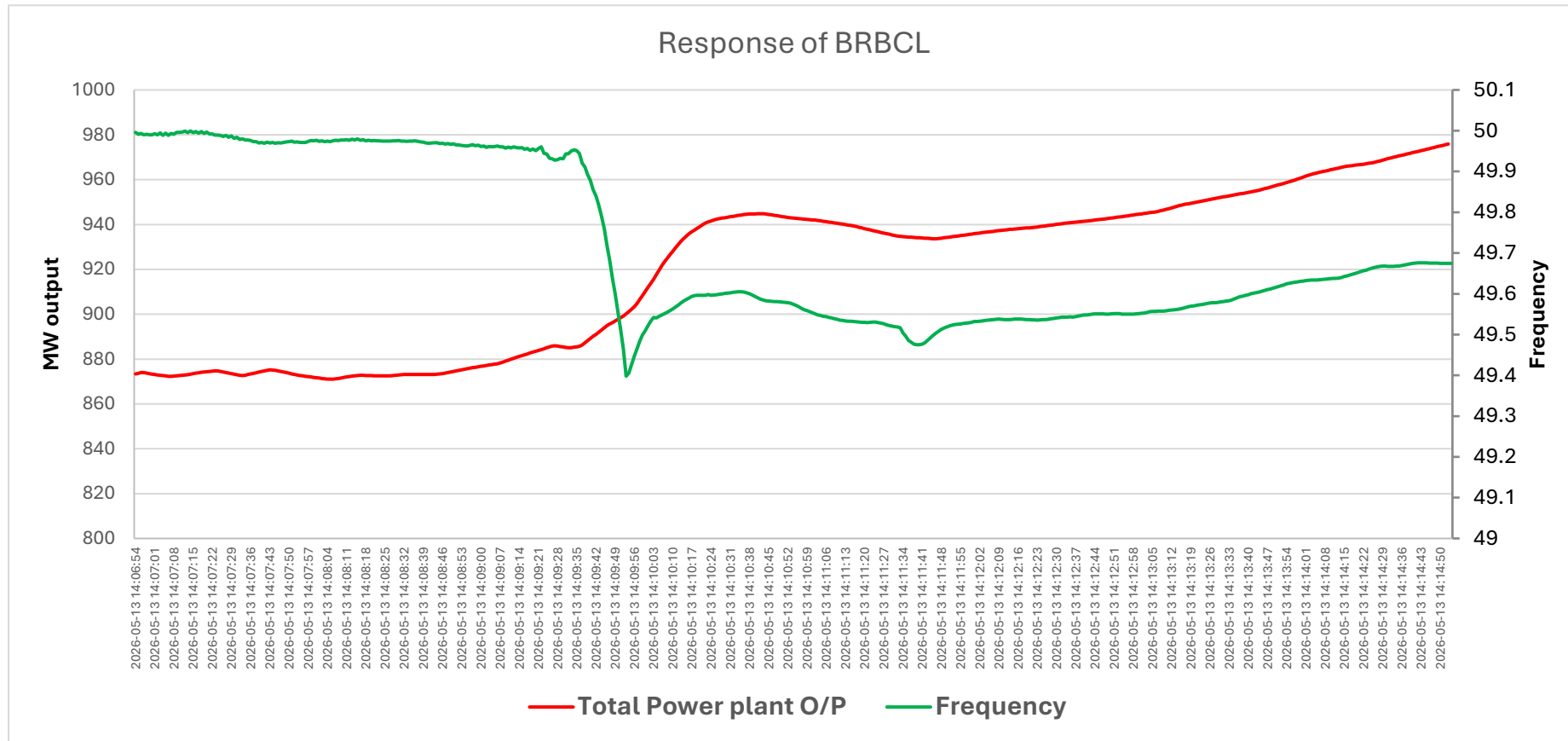
Discussion: Although the result is excellent, the response withdrawn immediately. May be clarified.

BRBCL

% response w.r.to current gen: 126.33%

Grading: Excellent

% response w.r.to FRO: 358%

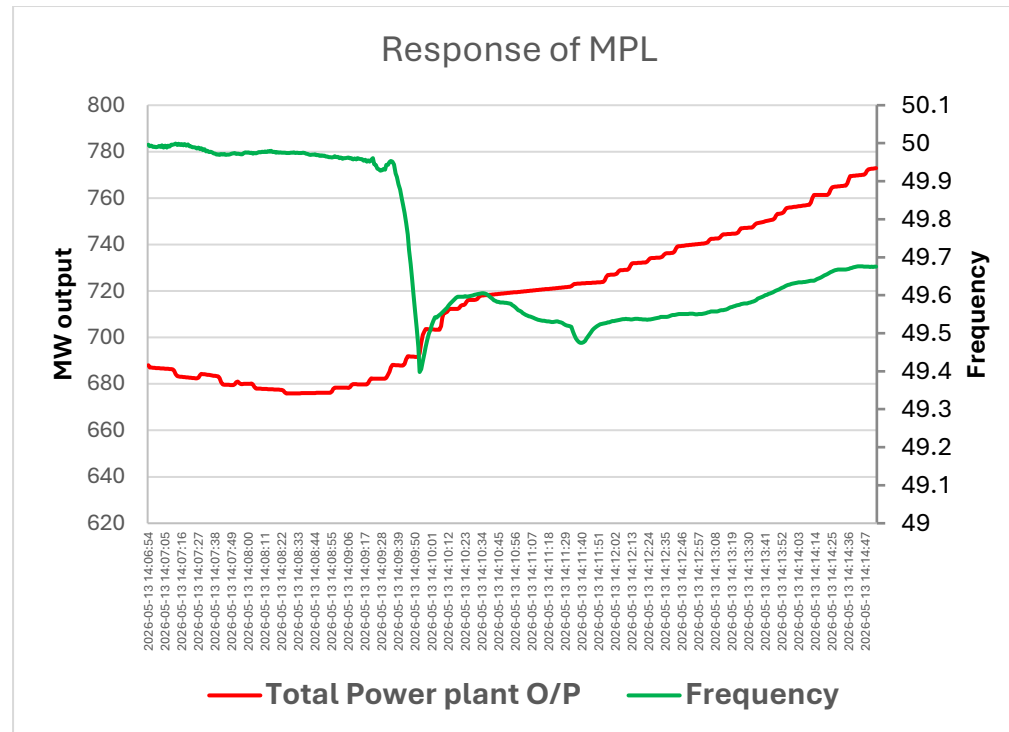


Discussion:

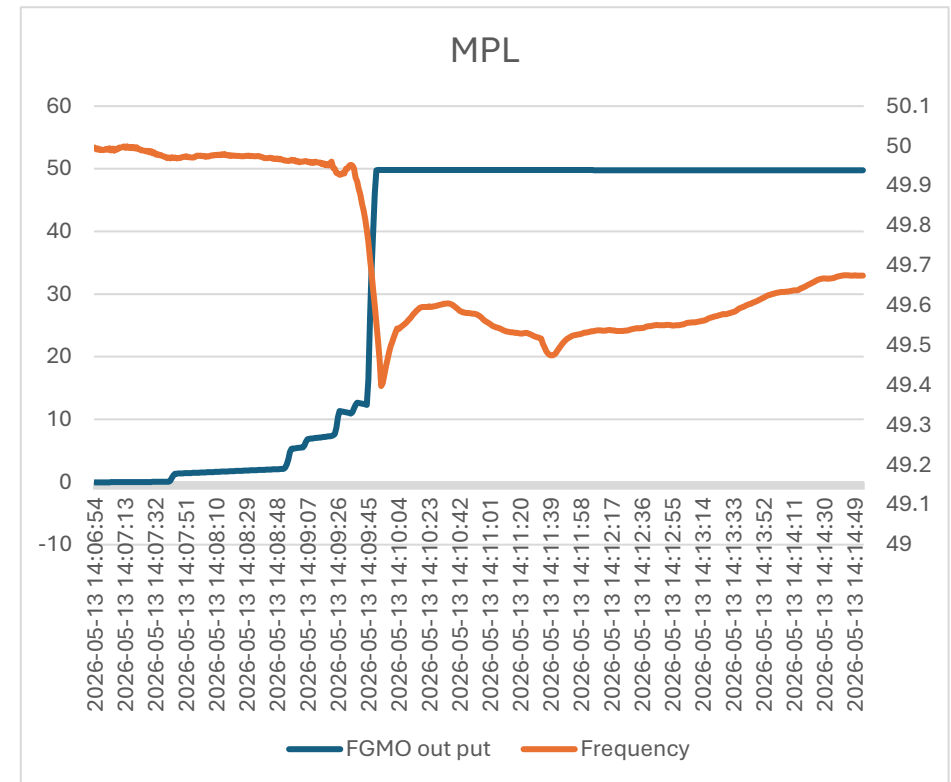
MPL

% response w.r.to current gen: 79.4%

Grading: Excellent



% response w.r.to FRO: 160%



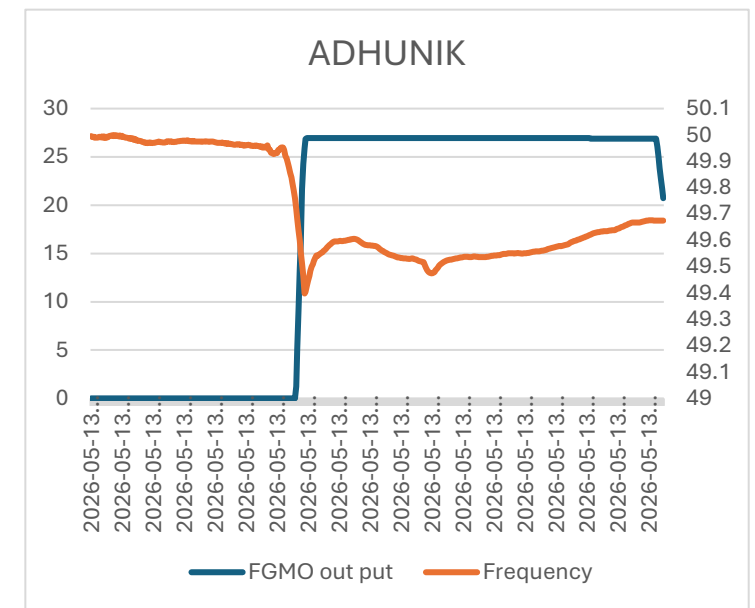
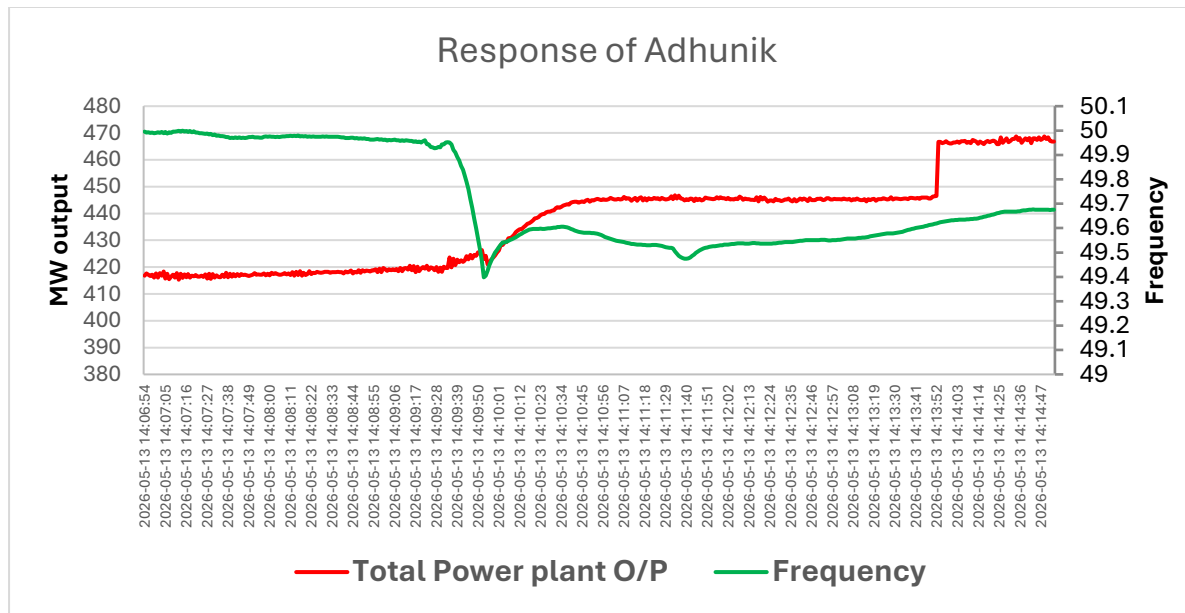
Discussion:

Adhunik

% response w.r.to current gen: 94.12%

Grading: Excellent

% response w.r.to FRO: 223%



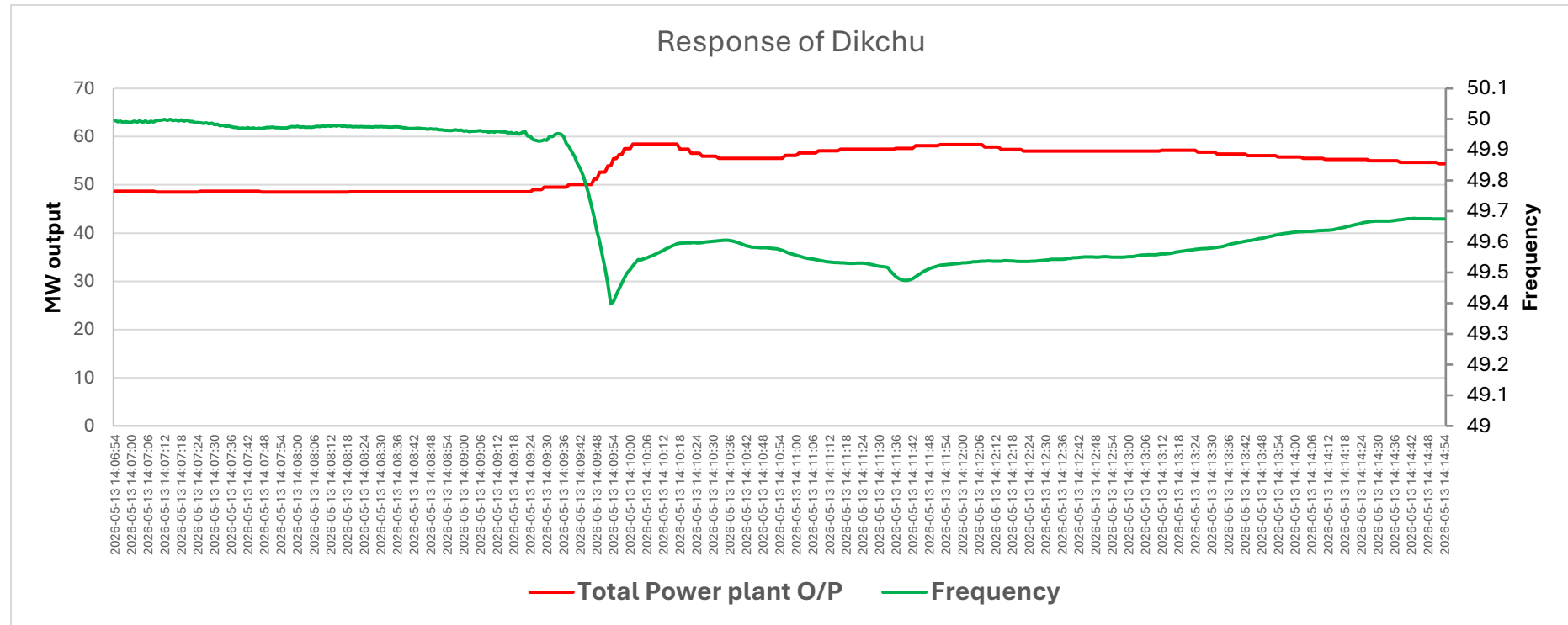
Discussion: Based on the current generation the maximum response received was 94%. But the response was received after 49 sec.

DIKCHU

% response w.r.to current gen: 282.49%

Grading: Excellent

% response w.r.to FRO: 394%



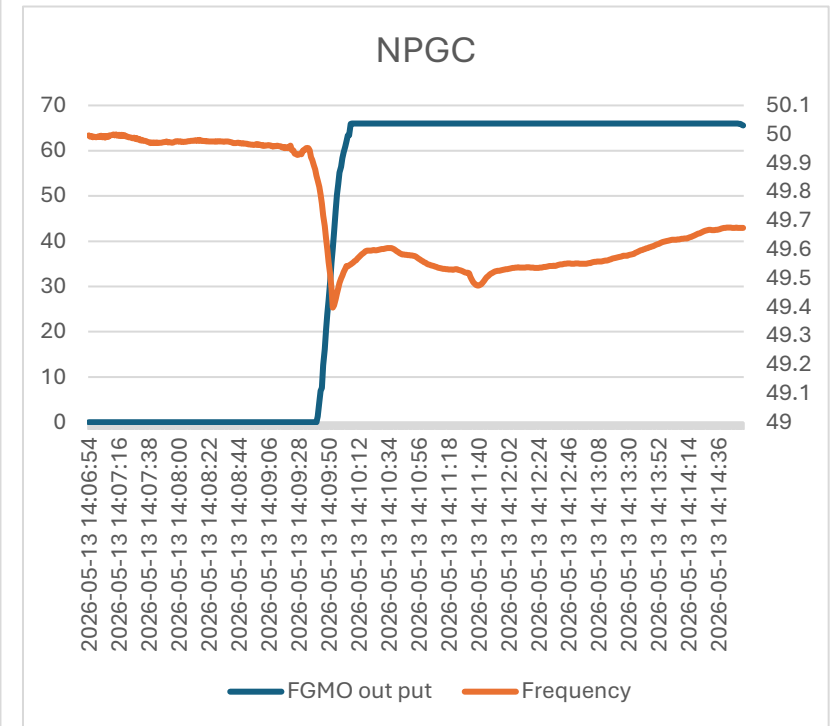
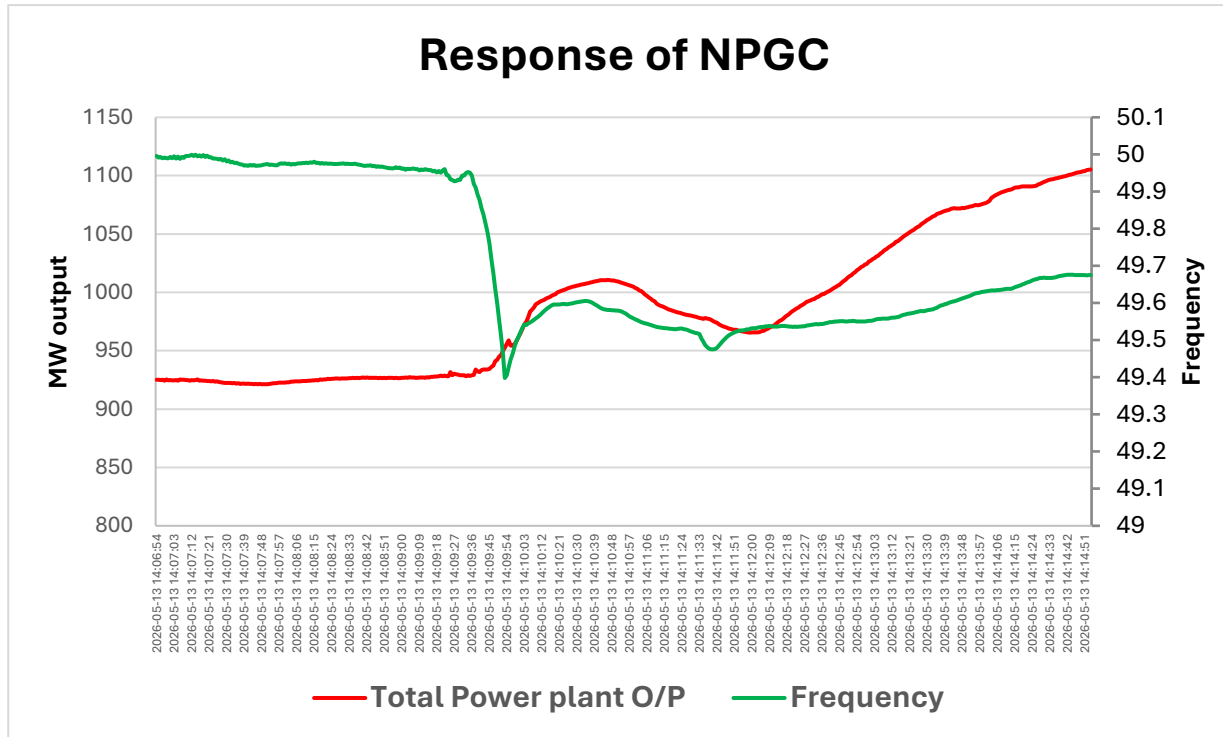
Discussion:

NPGC

% response w.r.to current gen: 156.77%

Grading: Excellent

% response w.r.to FRO: 228%



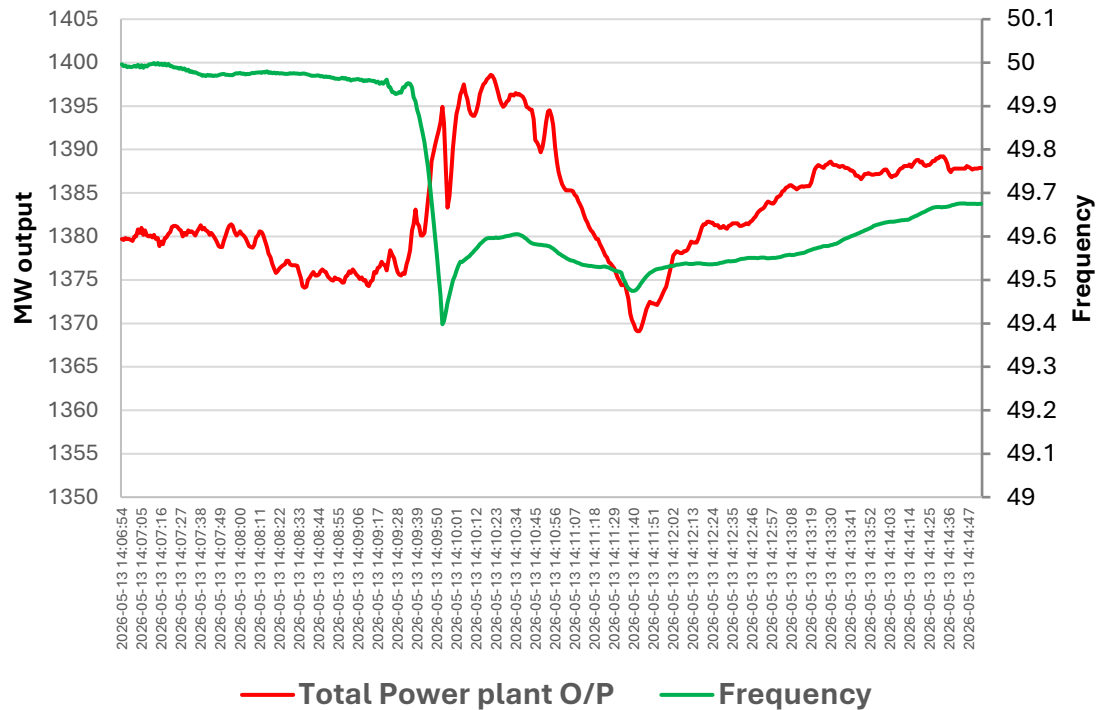
Discussion:

BARH ST-1

% response w.r.to current gen: 29.19%

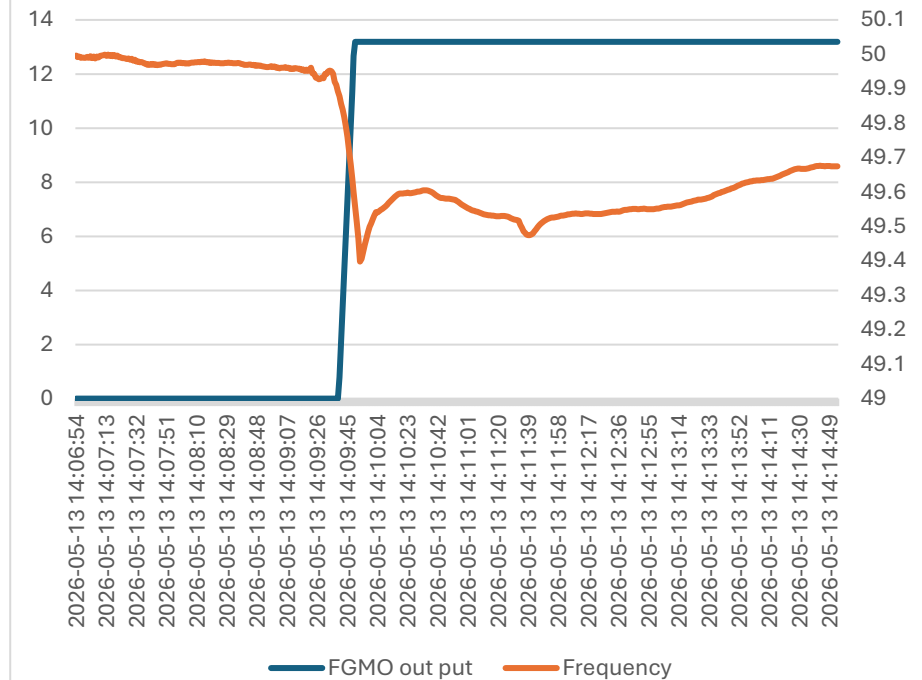
Grading: Good

Response of Barh stage-1



% response w.r.to FRO: 94%

BARH STG1



Discussion: Output is not sustaining and hence the FRP grading is Good. The % response w.r.to current gen is 29%.

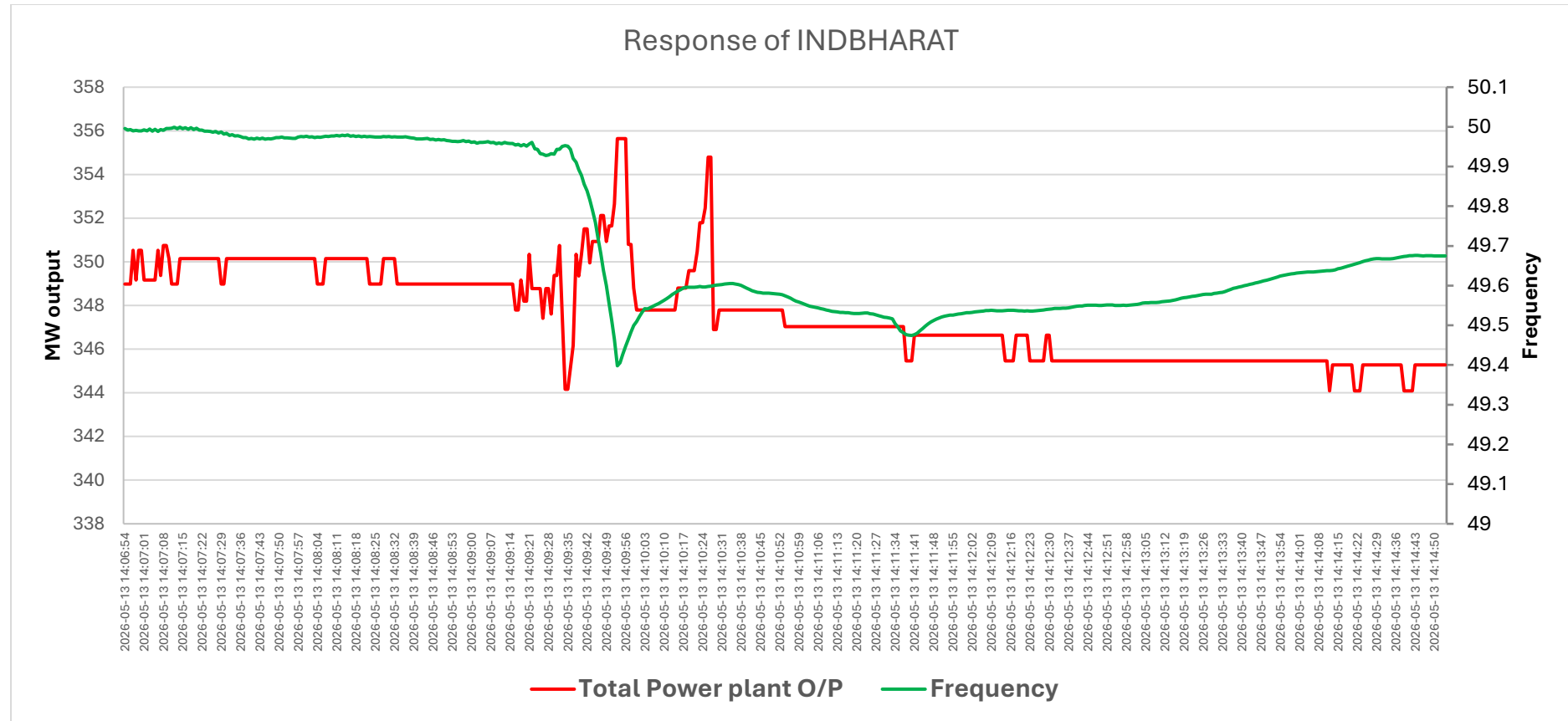
Matter also discussed in meeting dated 24.04.26. Any update on the corrective action taken and approach to CERC may be discussed.

INDBHARAT

% response w.r.to current gen: 44.3%

Grading: Average

% response w.r.to FRO: 83%



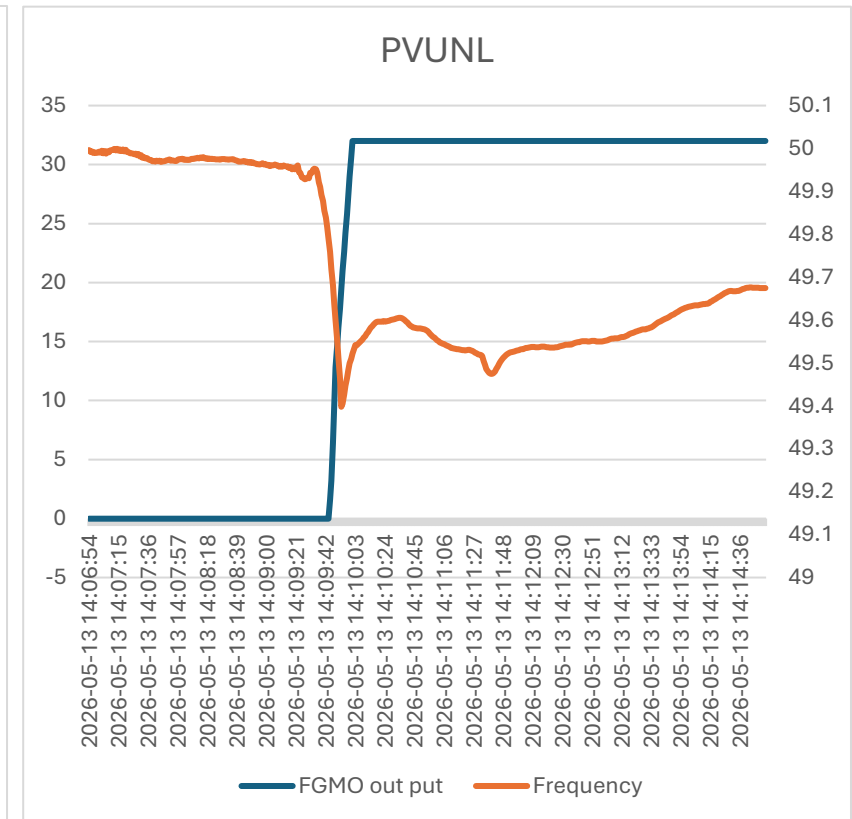
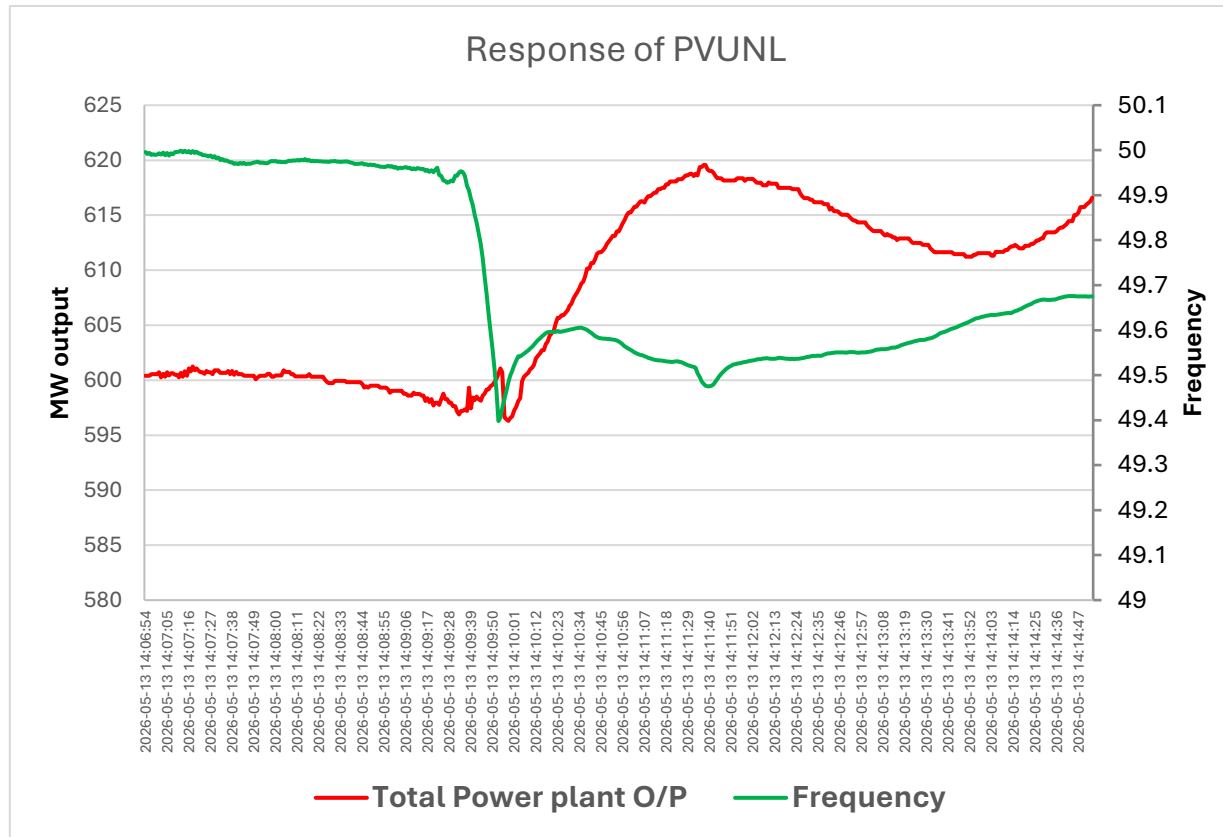
Discussion: SCADA data used as we are not receiving any Generator data. % response w.r.to current gen is 44% Response is oscillatory and not sustaining.

PVUNL

% response w.r.to current gen: 28.39%

Grading: Below Average

% response w.r.to FRO: 63%



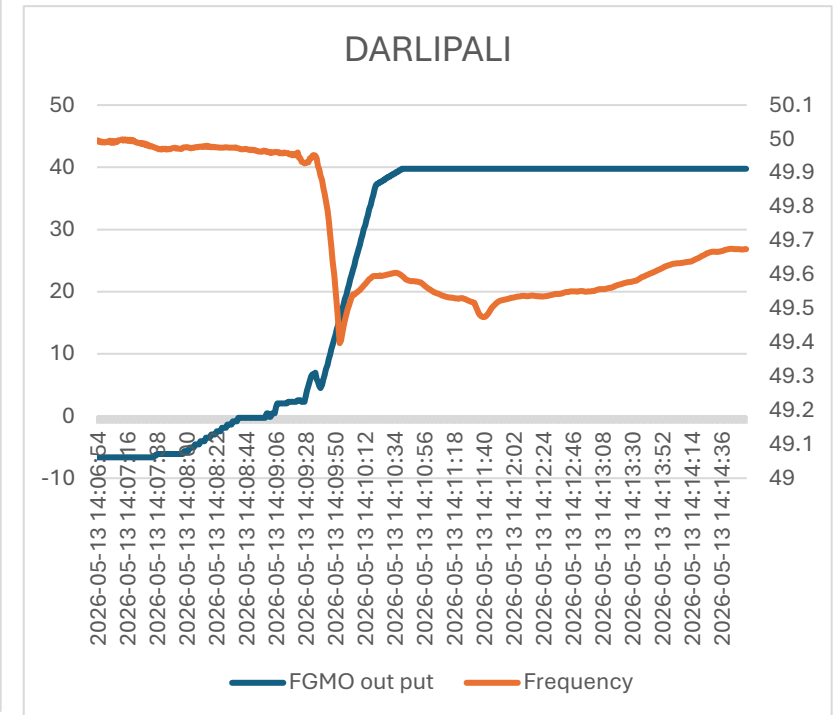
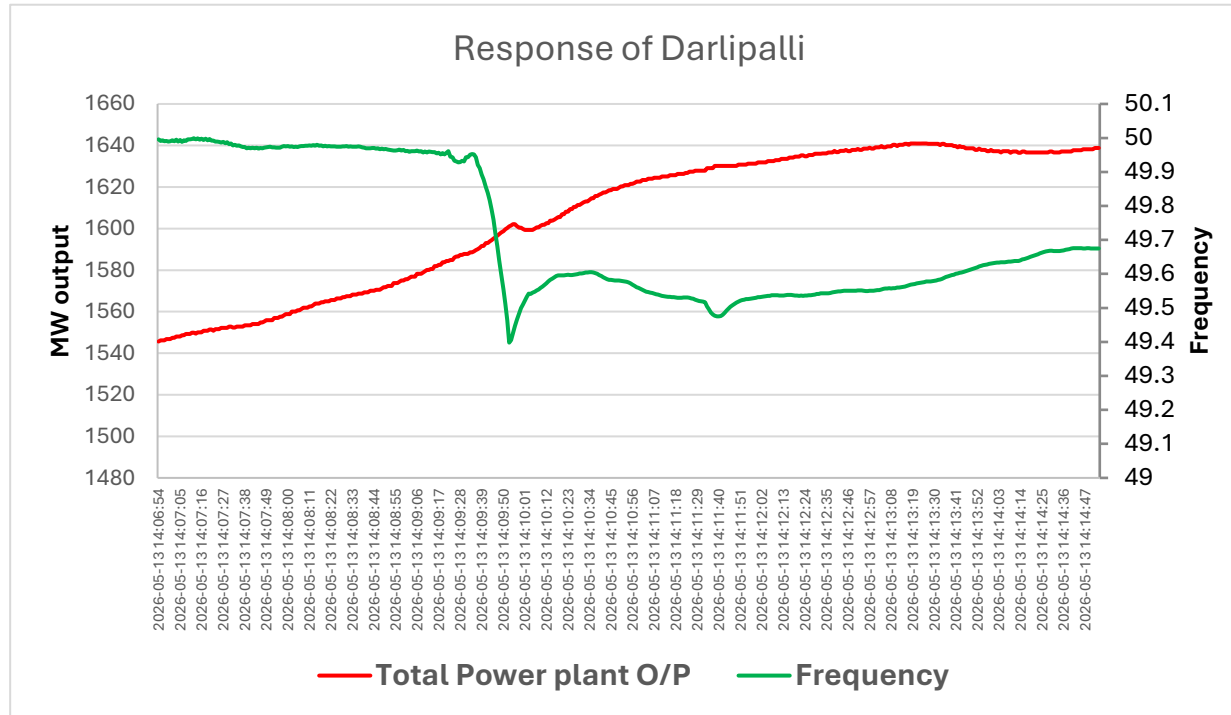
Discussion: The maximum response reached after 48 sec.% response w.r.to current gen was 28%.

DARLIPALLI

% response w.r.to current gen: 24.7%

Grading: Below Average

% response w.r.to FRO: 67%



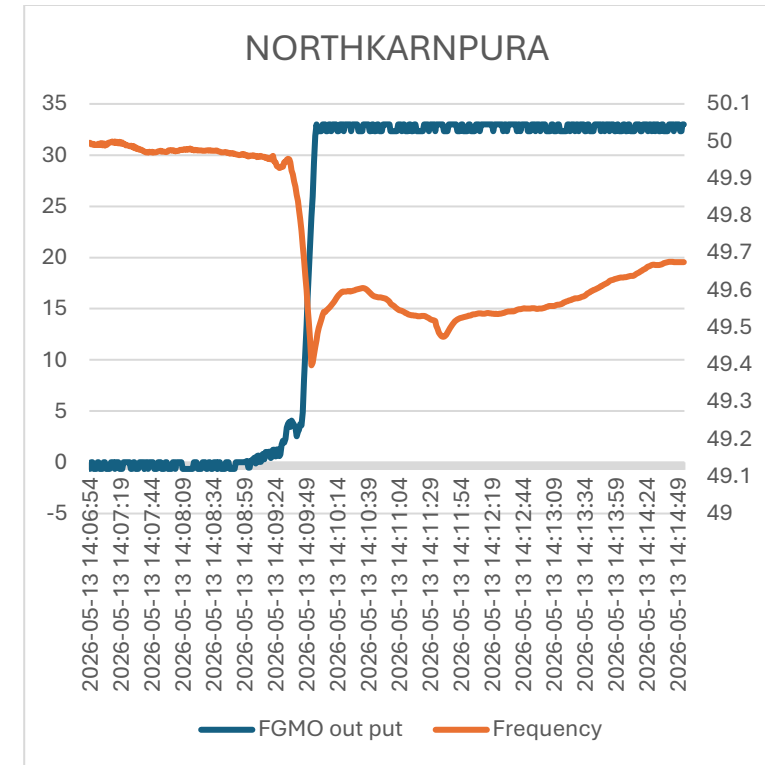
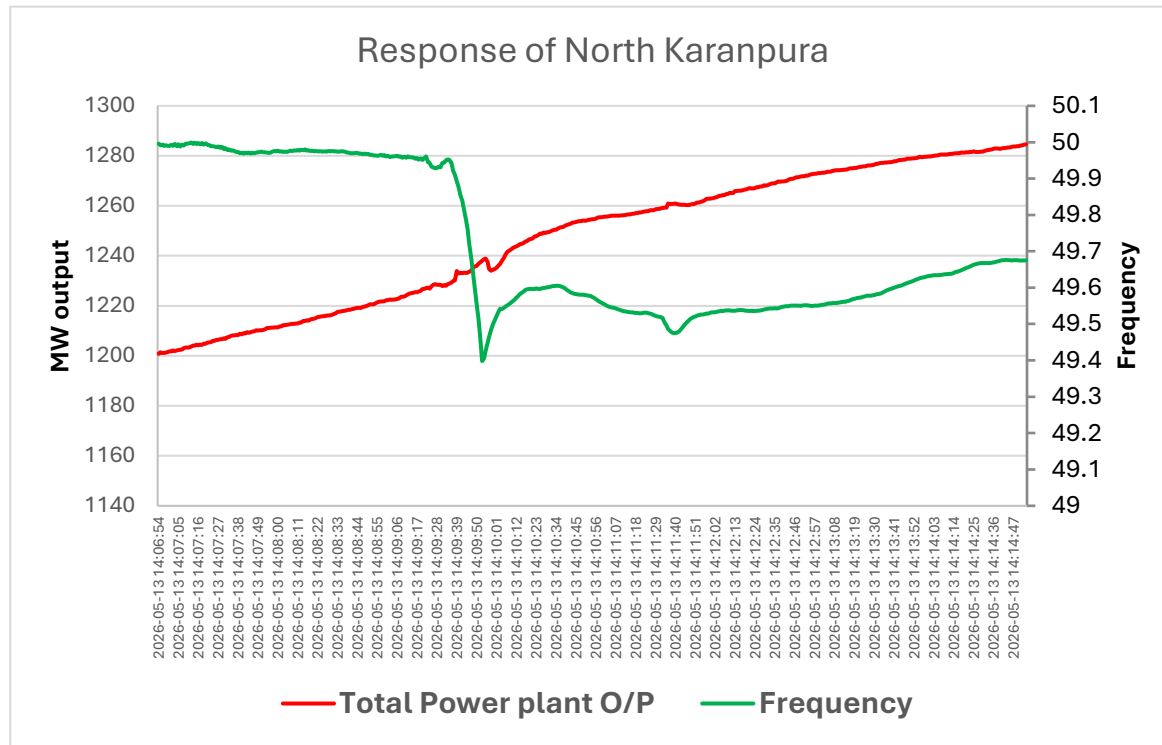
Discussion: The maximum response reached after 48 sec.% response w.r.to current gen was 24%.

NORTH KARANPURA

% response w.r.to current gen: 31.24%

Grading: Below Average

% response w.r.to FRO: 70%



Discussion: The maximum response reached after 48 sec.% response w.r.to current gen was 31%.

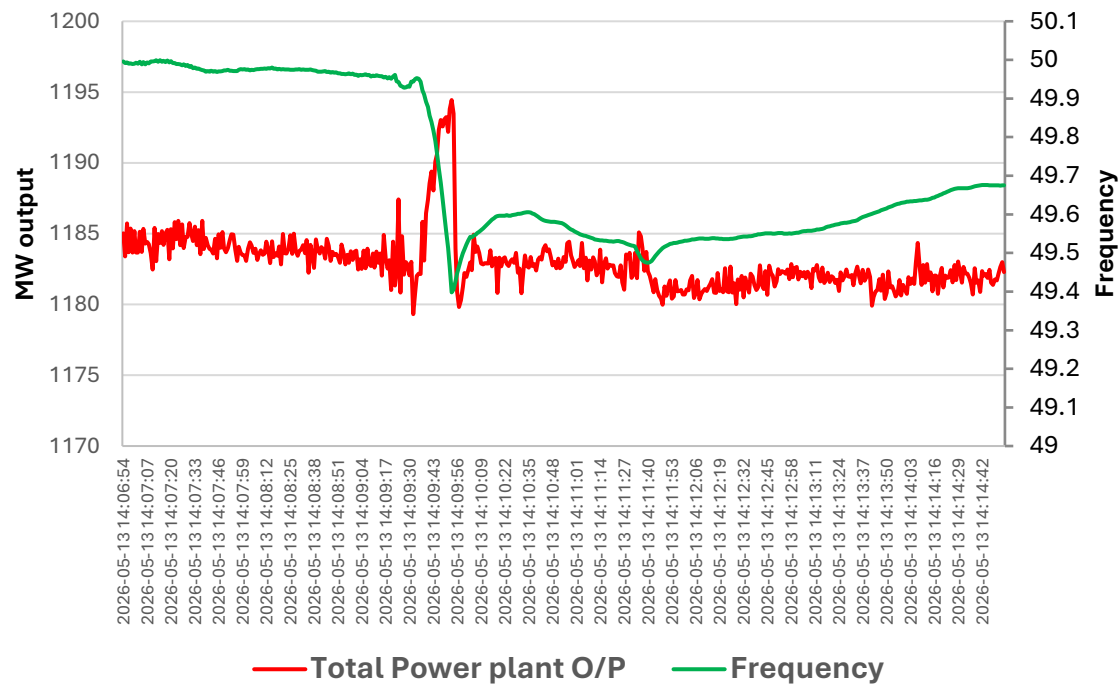
JIPL

% response w.r.to current gen: 1.47%

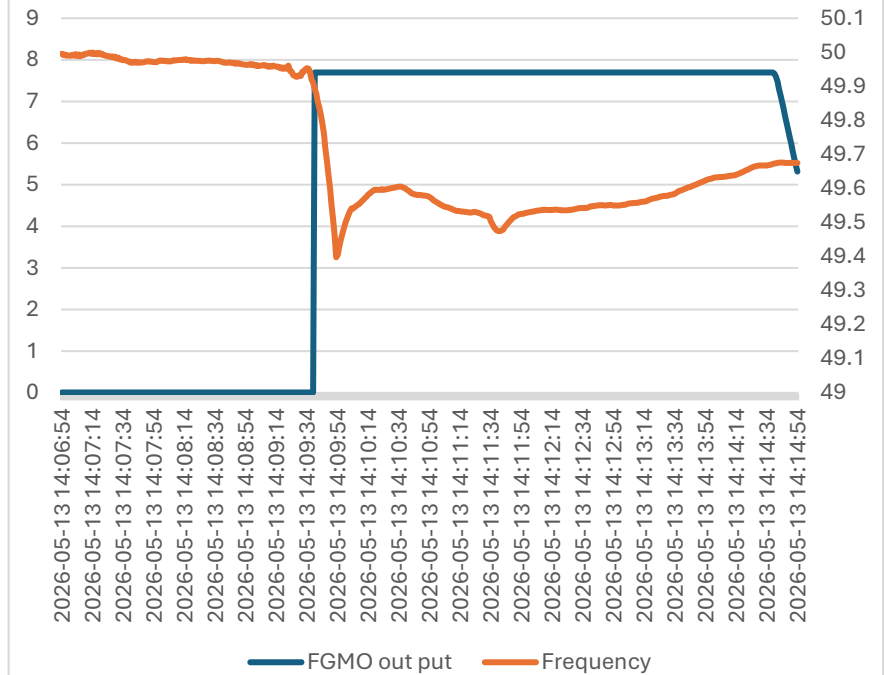
Grading: Poor

% response w.r.to FRO: 4%

Response of JIPL



JIPL



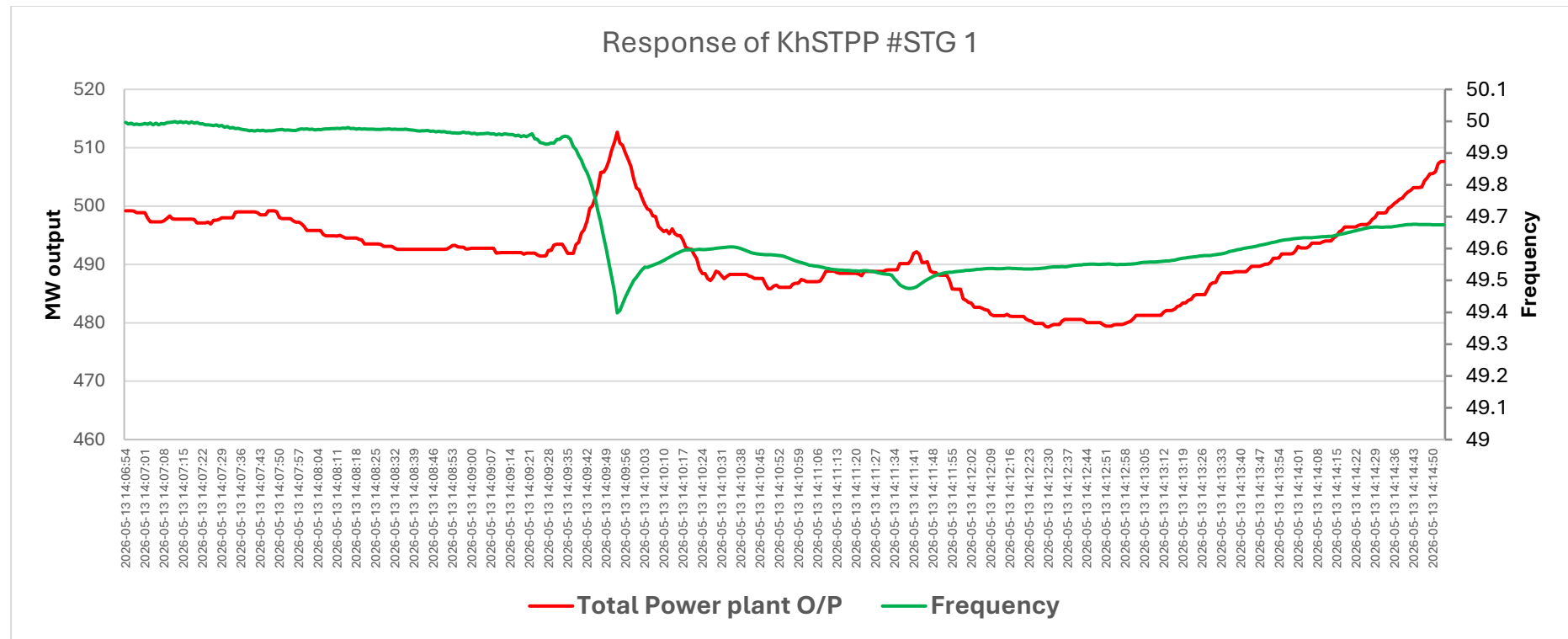
Discussion: The response achieved was poor, only 1.47 % w.r.to current gen.

KhSTPP I

% response w.r.to current gen: -14.17%

Grading: Poor

% response w.r.to FRO: -27%



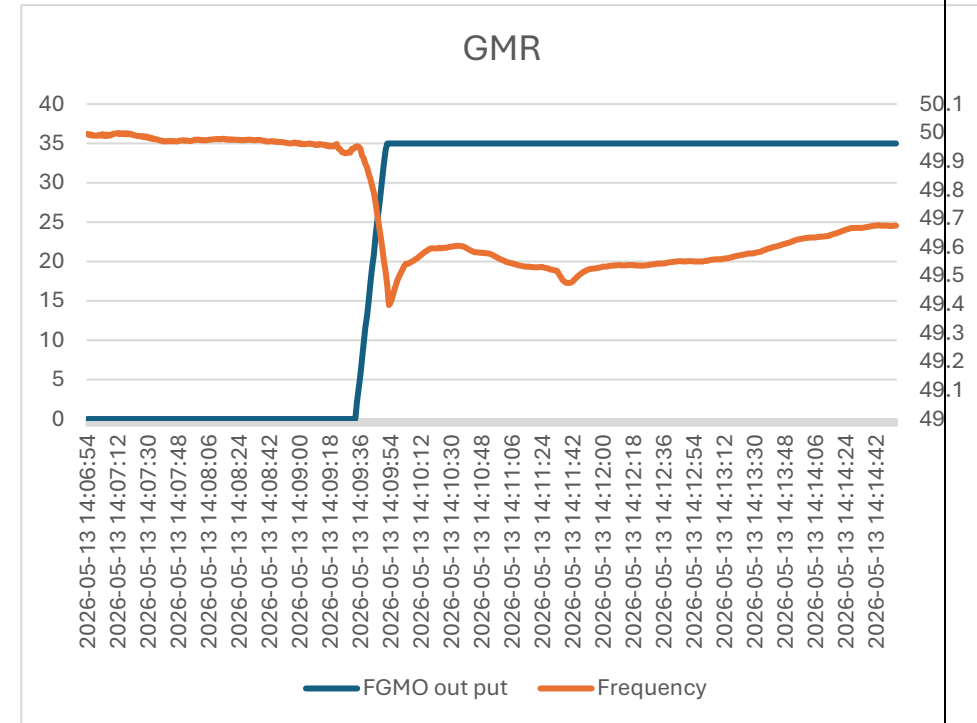
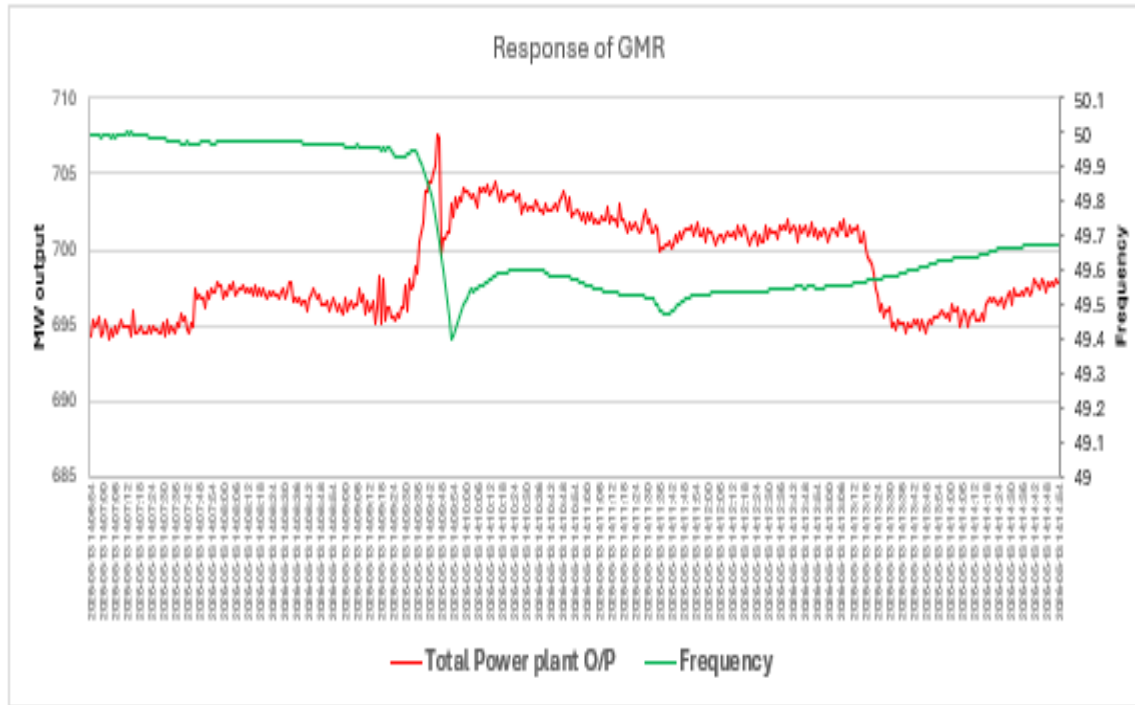
Discussion: KhSTPP I could not hold the response as per requirements and hence the FRP turned out to be poor. Matter discussed in meeting on 24.04.26. Any update on the same to be discussed regarding approach to CERC.

GMR

% response w.r.to current gen: 17.53%

Grading: Poor

% response w.r.to FRO: 47%

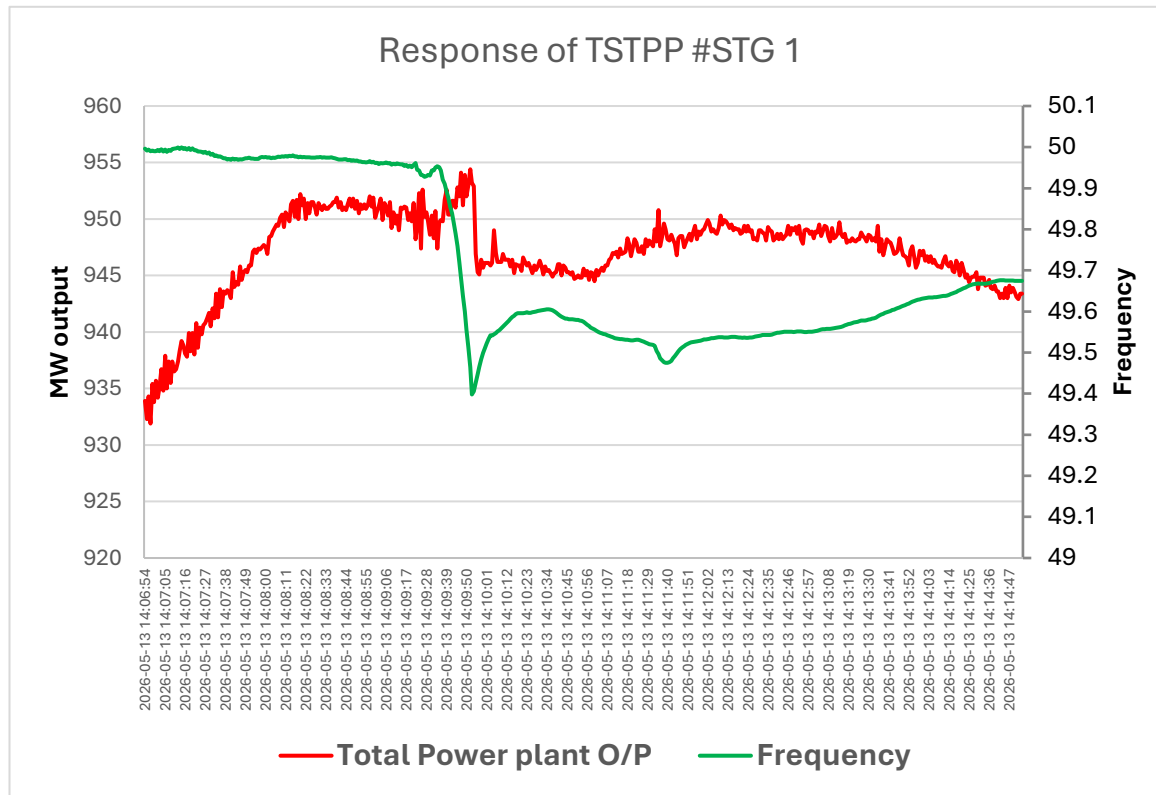


Discussion: Output oscillatory, withdrawn immediately. 17% response was achieved w.r.to current generation.

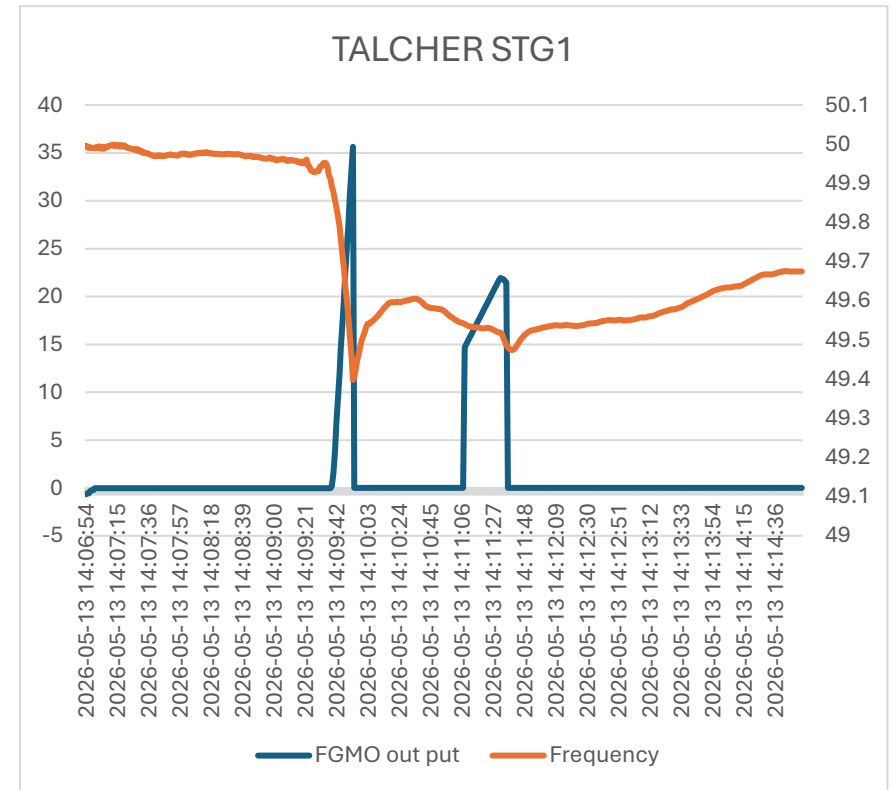
TSTPS-I

% response w.r.to current gen: -3.17%

Grading: Poor



% response w.r.to FRO: -9%

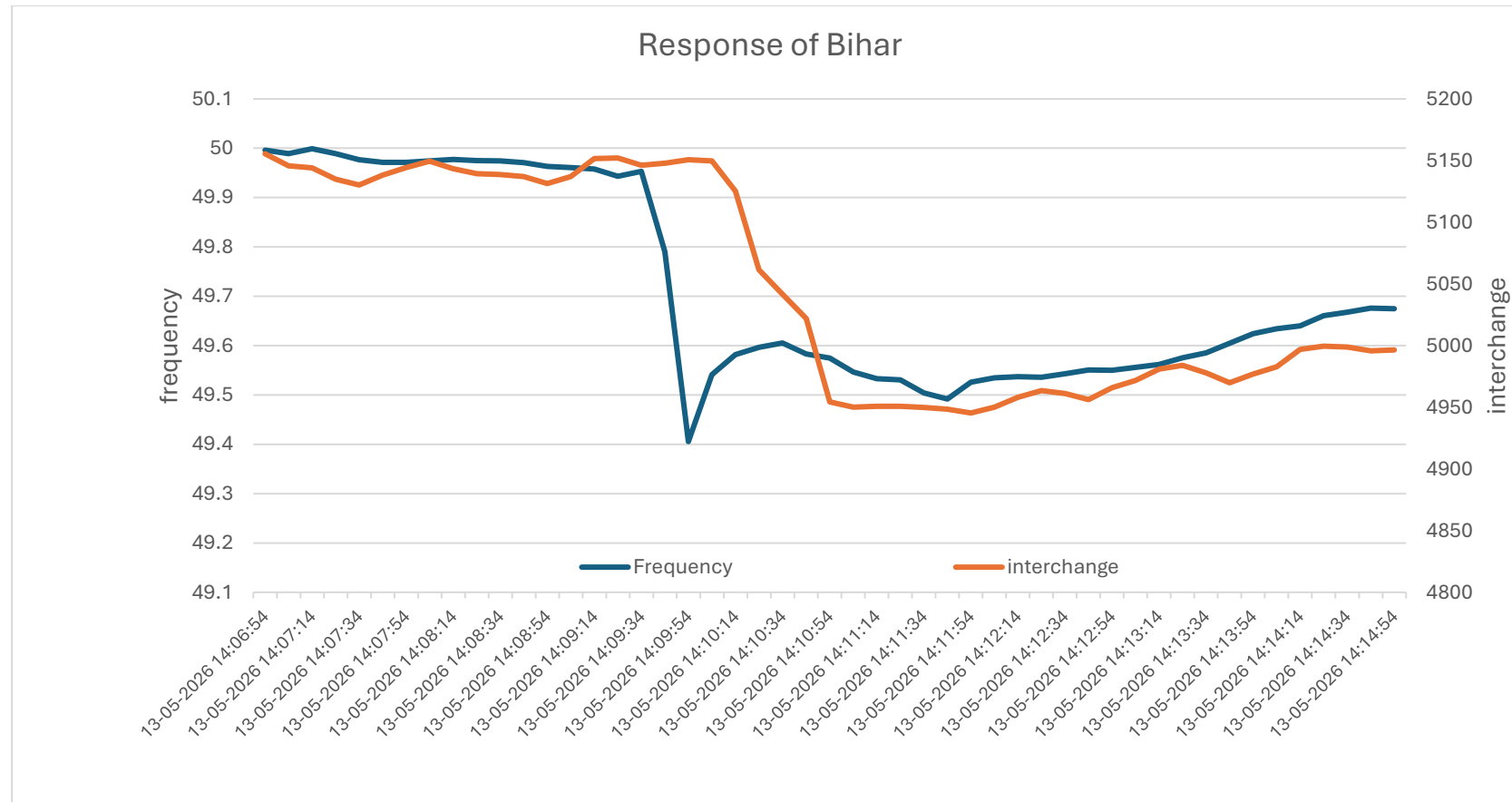


Discussion: Output was withdrawn immediately and hence grading was poor. The % response was achieved w.r.to current generation was -3%.

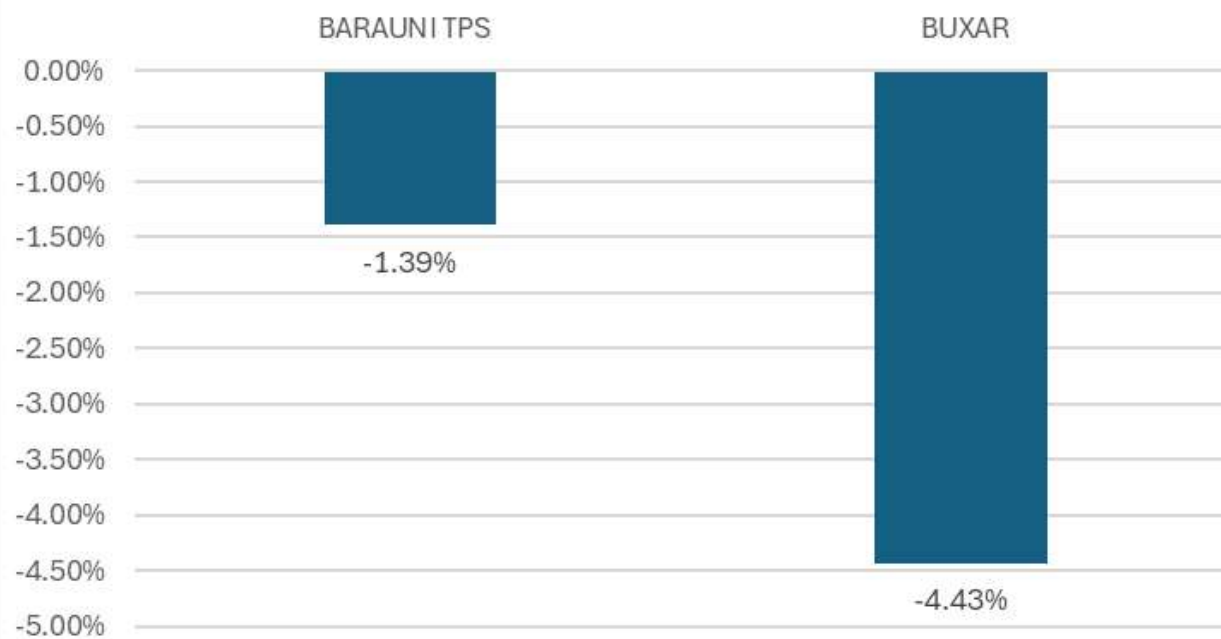
STATES

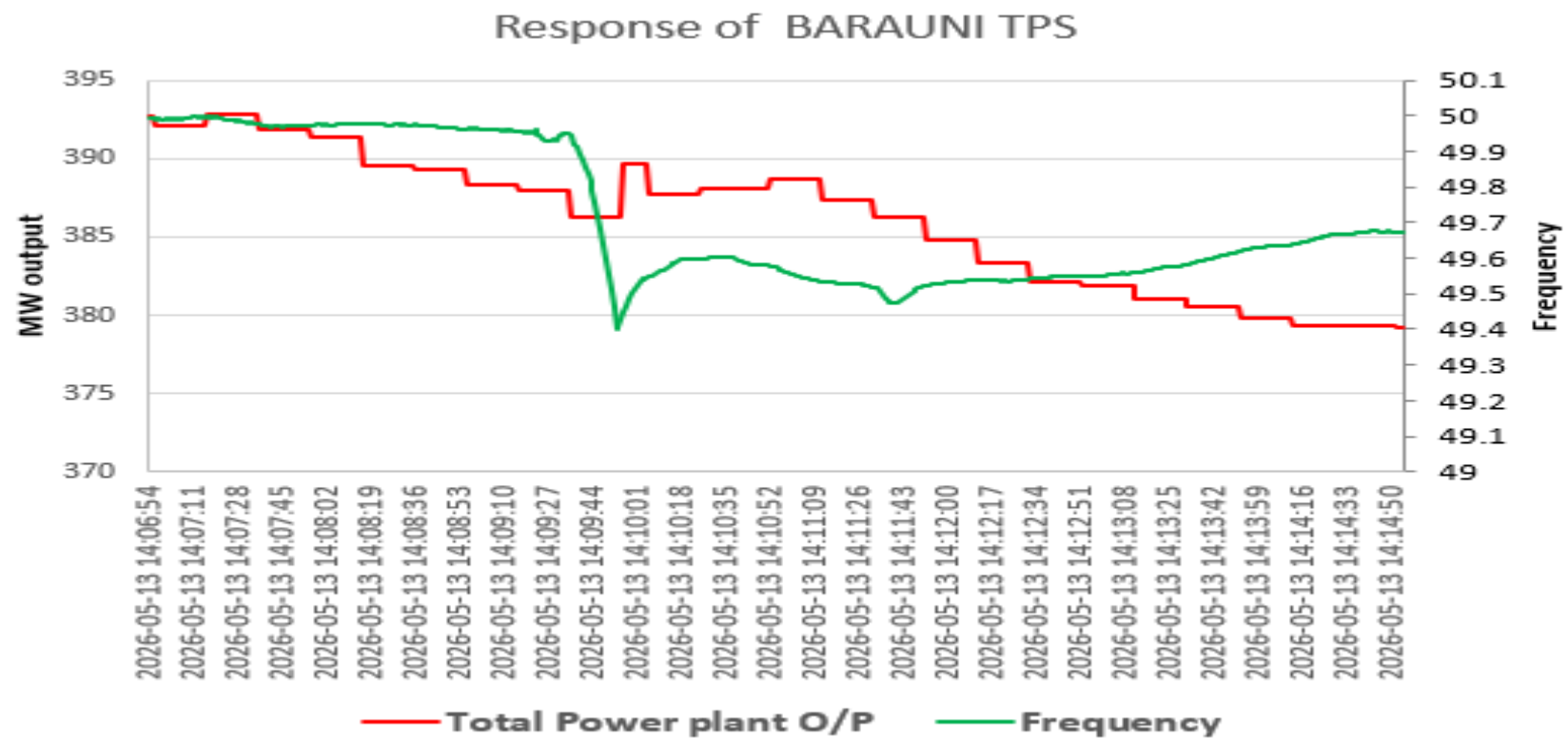
Name of State	Event 1	
	FRP	Grading
DVC	1.87	Excellent
WB	1.11	Excellent
Bihar	1.46	Excellent
Jharkhand	0.43	Poor
OPTCL	-5.65	Poor

Bihar

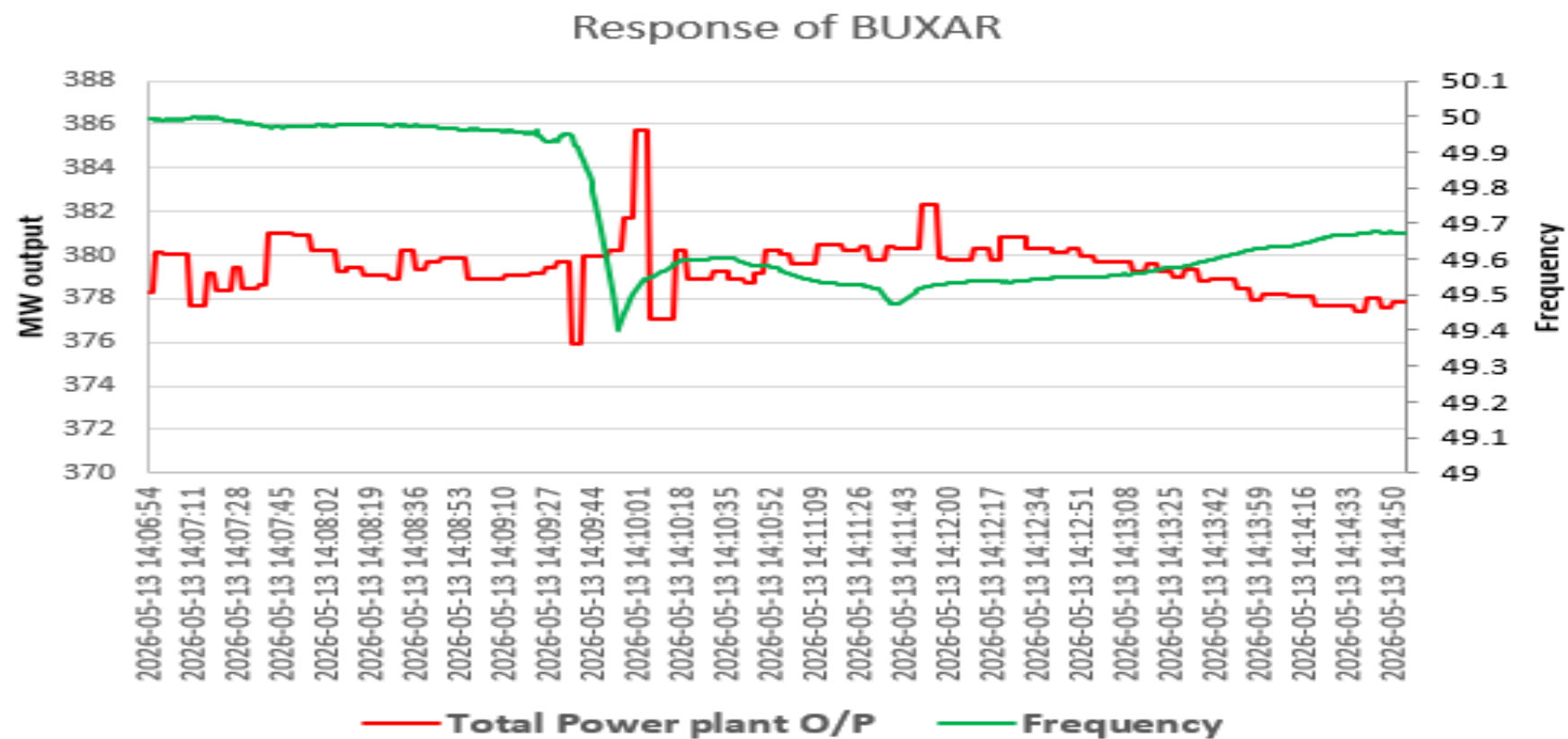


Bihar intra state generator performance



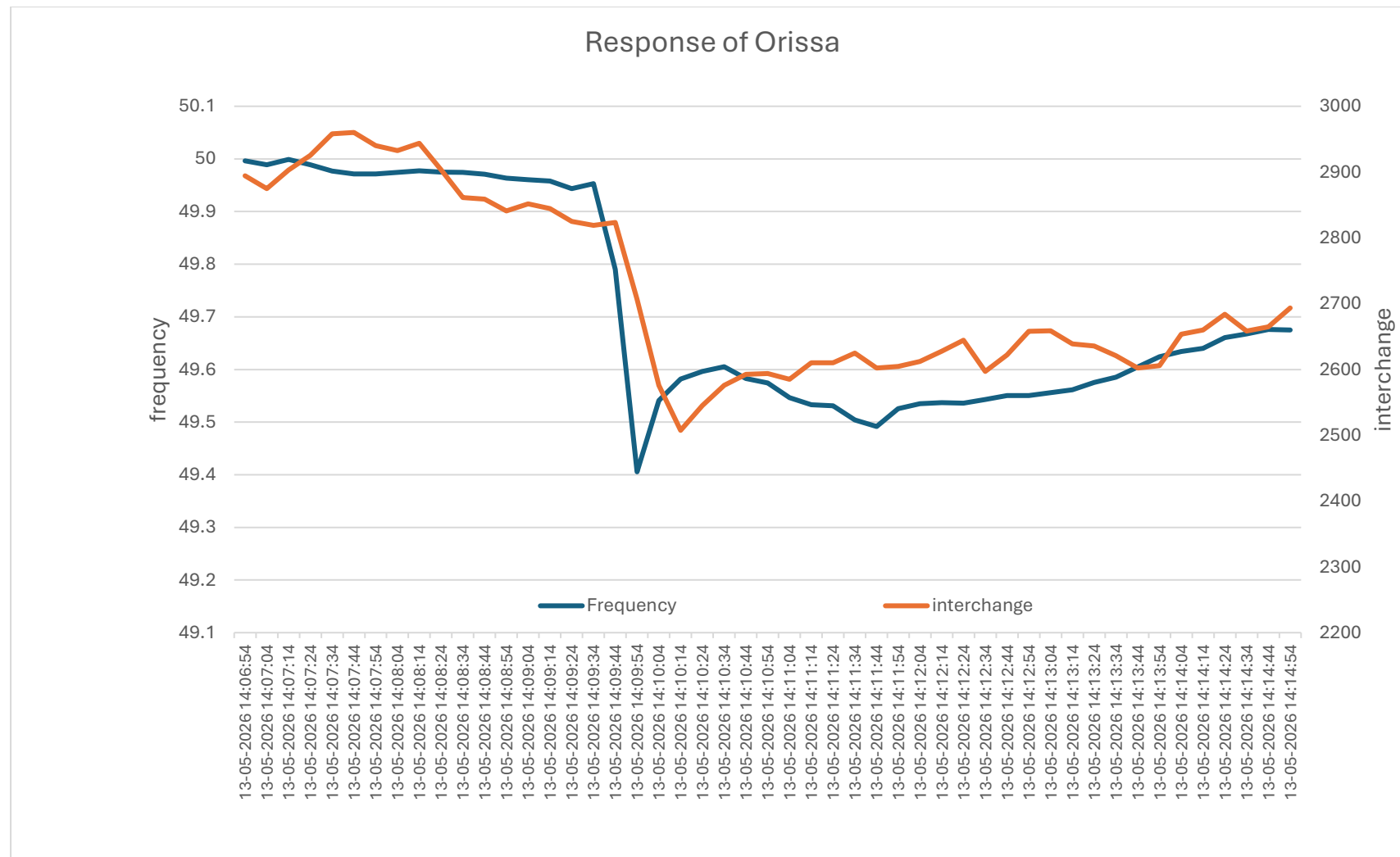


Responses received = -1.39 %



Responses received = -4.43 %

OPTCL



Odisha intra state generator performance

IB VALLEY TPS

OPGC_STU_GEN

JSPL1_GR - GT_2_P

BALIMELA

UPPER KOLAB

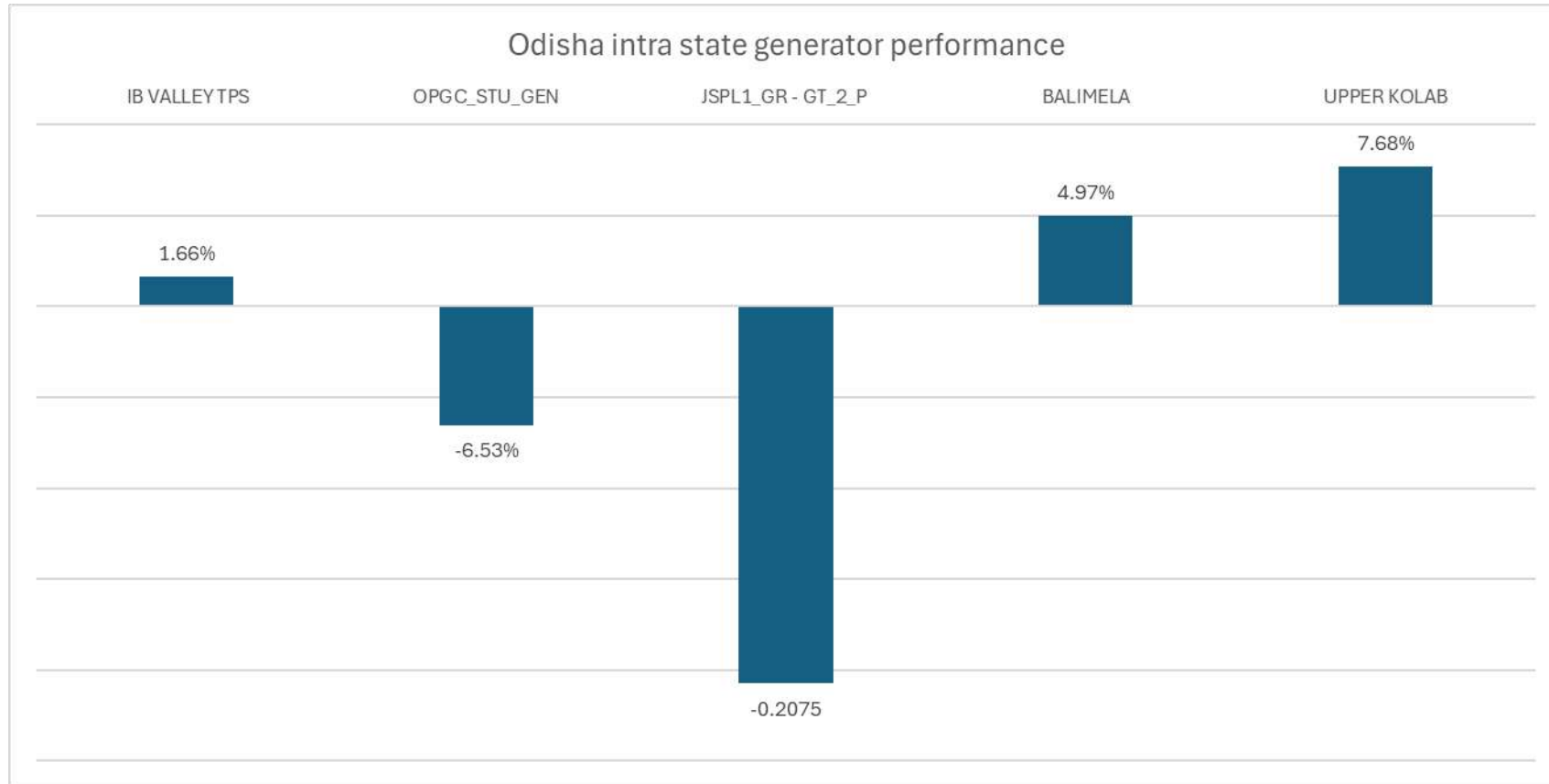
1.66%

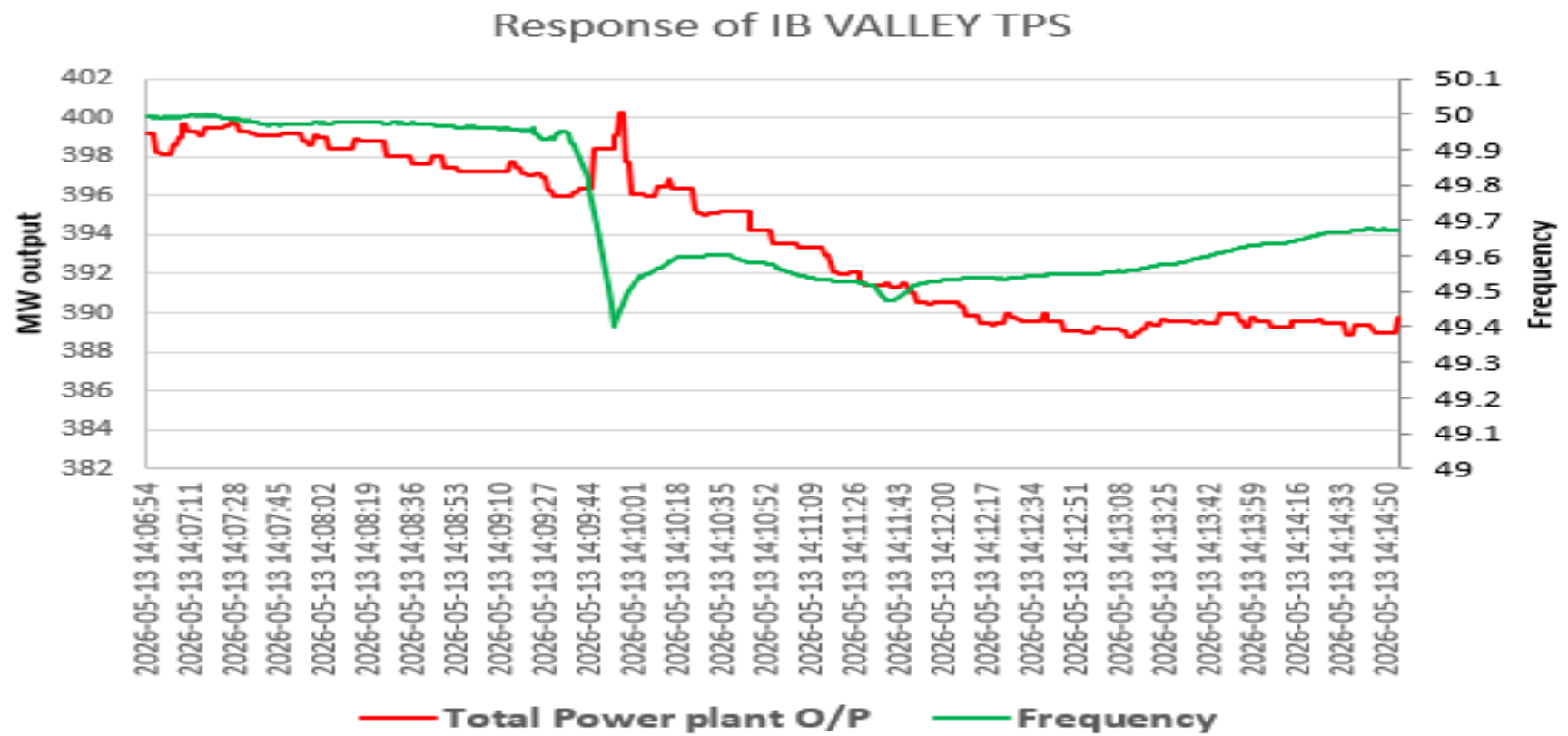
-6.53%

-0.2075

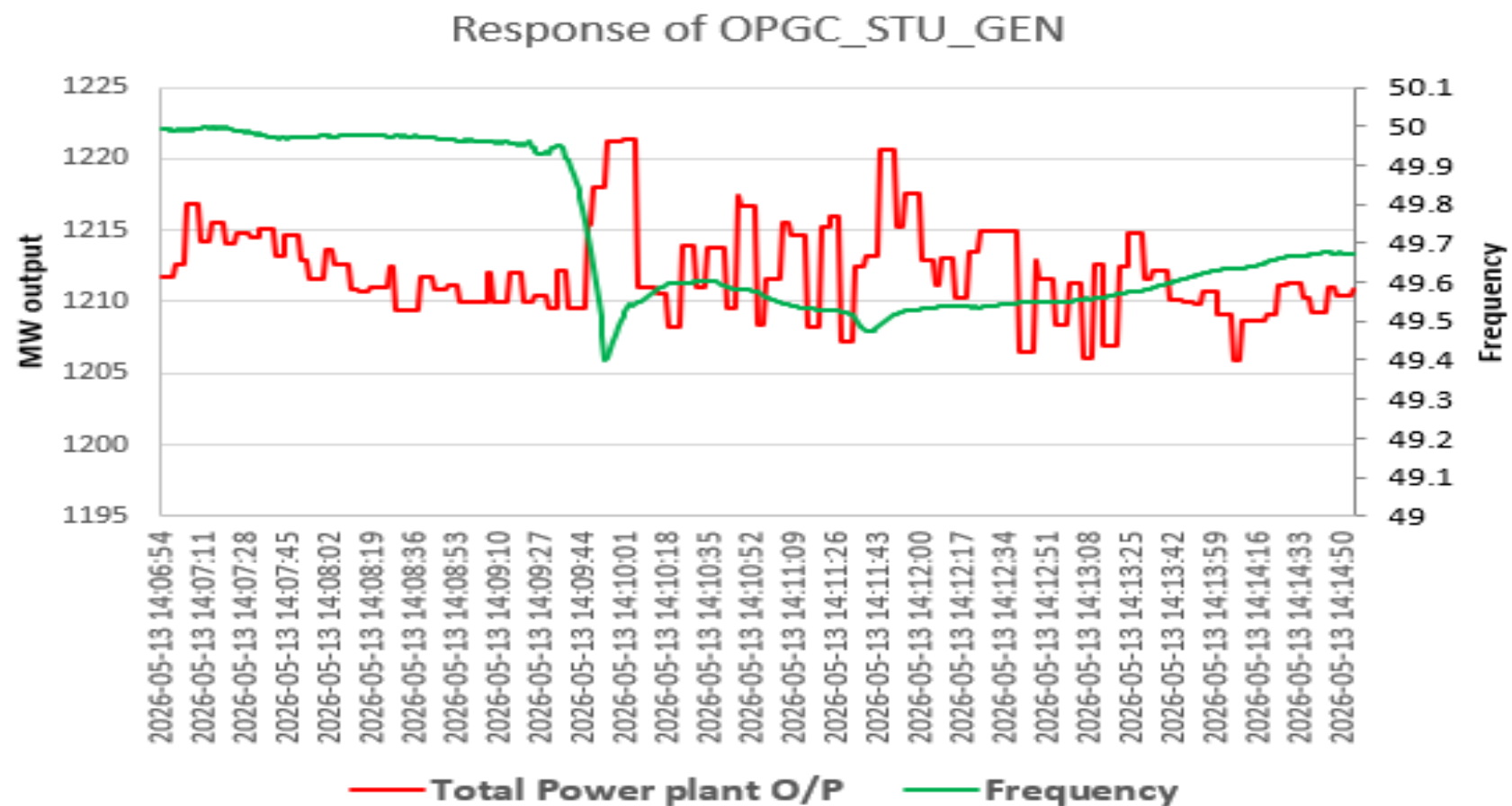
4.97%

7.68%

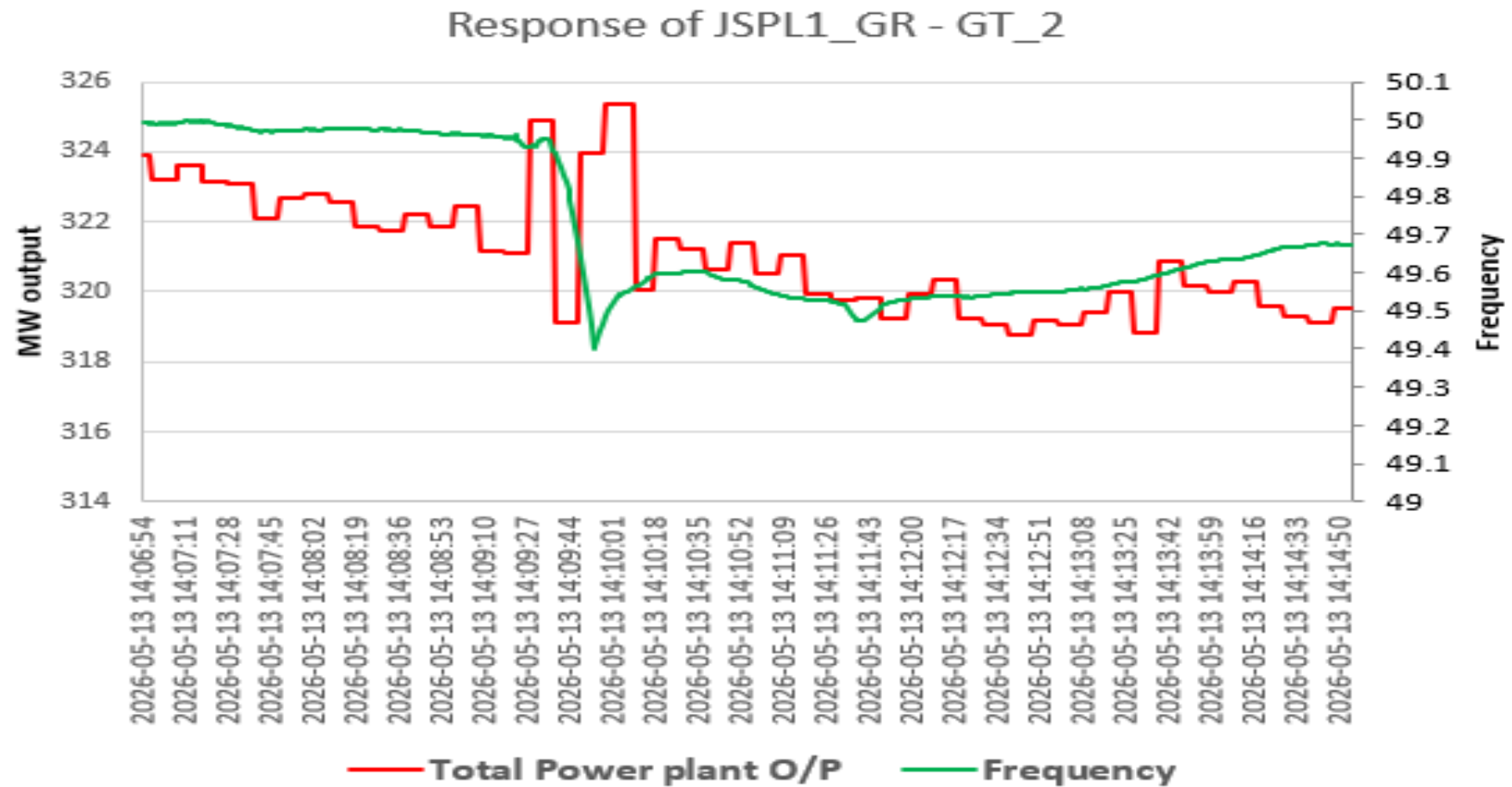




Responses received = 1.66 %

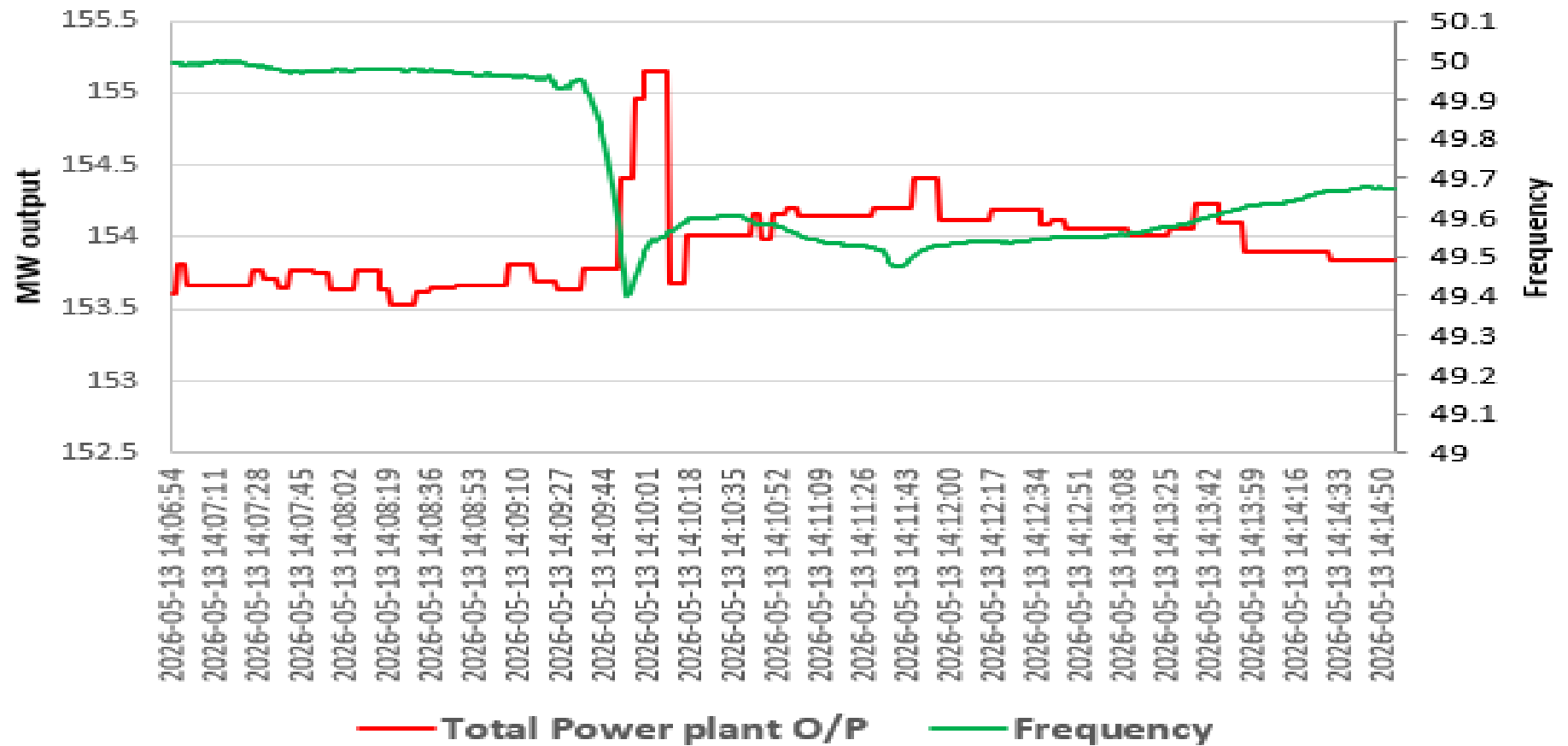


Responses received = -6.53 %

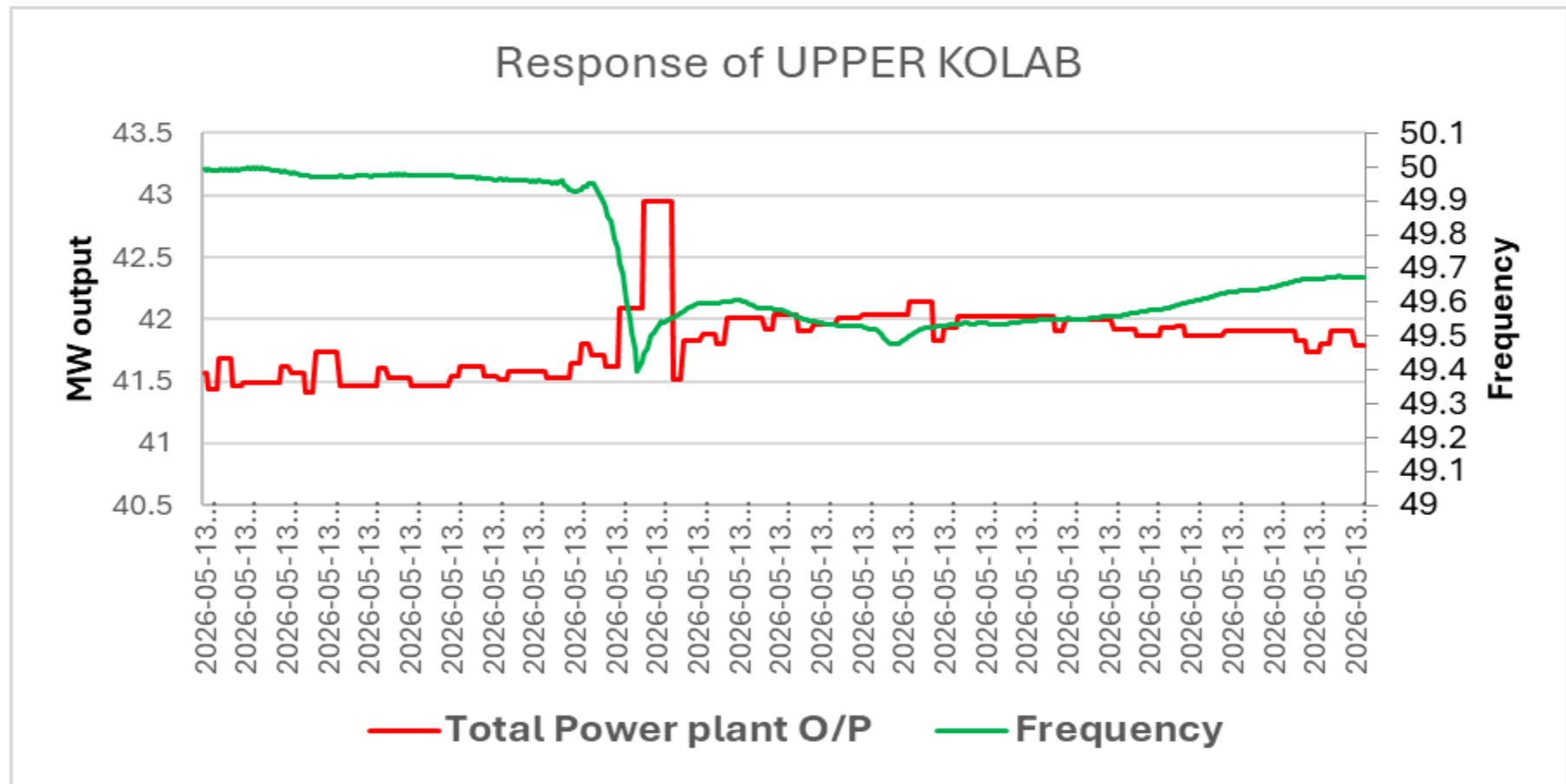


Responses received = -20.75 %

Response of BALIMELA

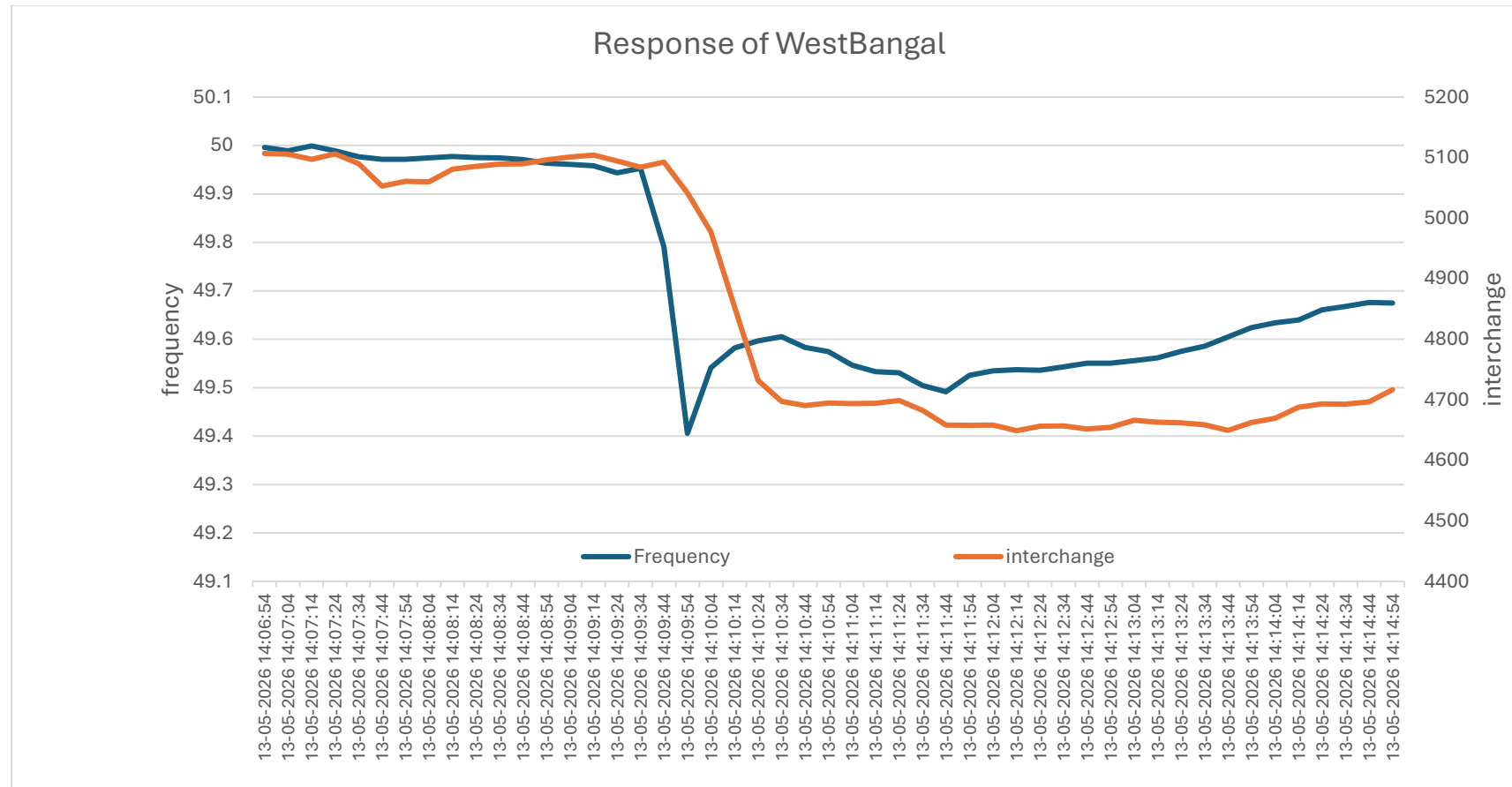


Responses received = 4.97 %

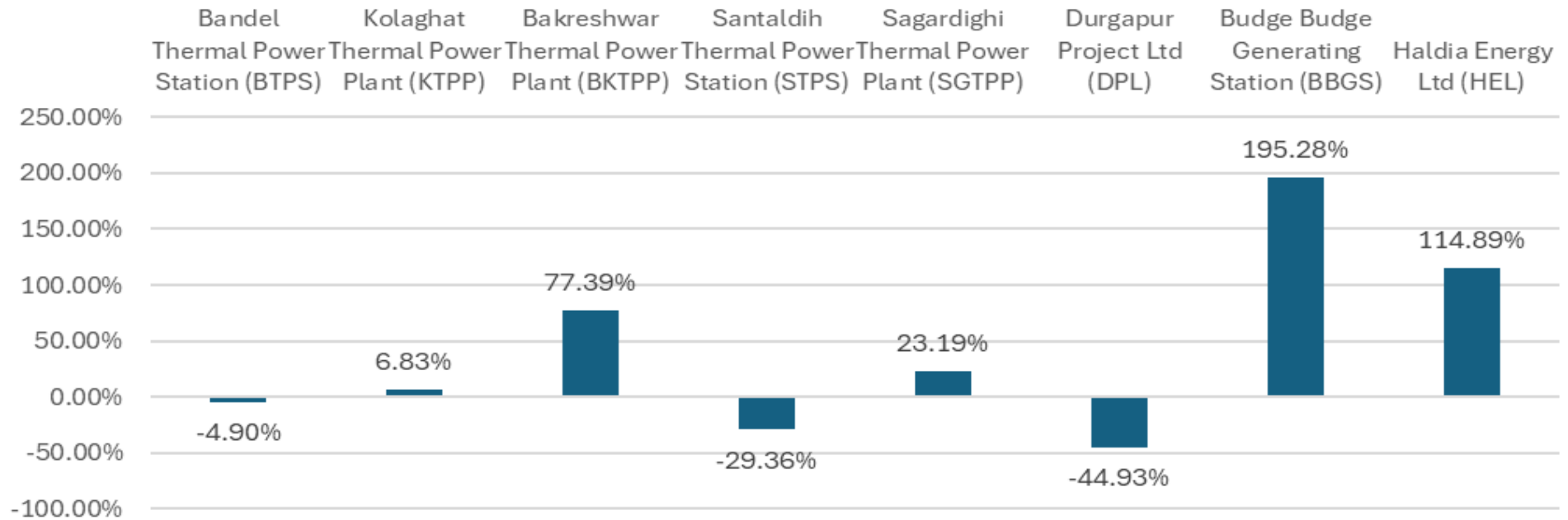


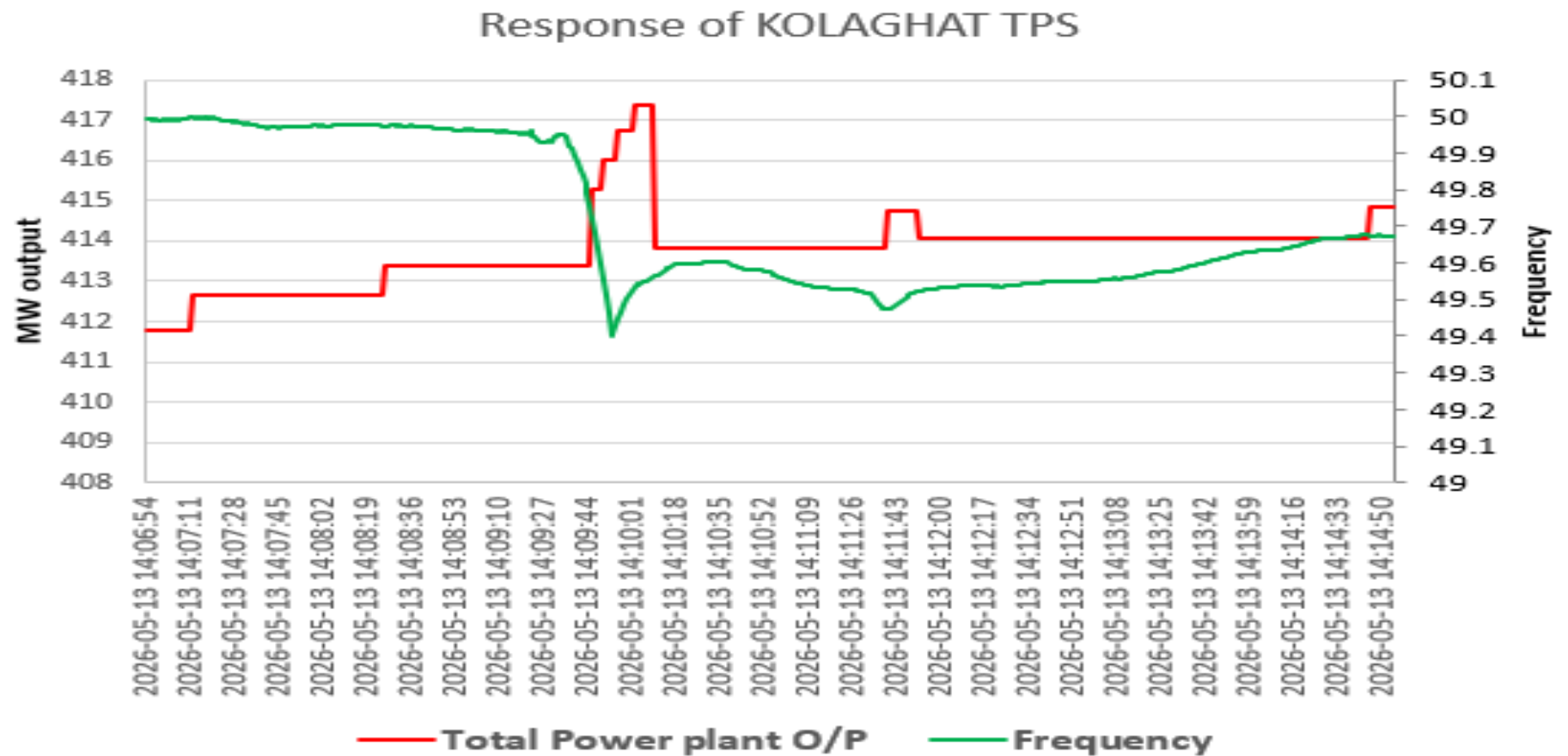
Responses received = 7.68 %

West Bengal

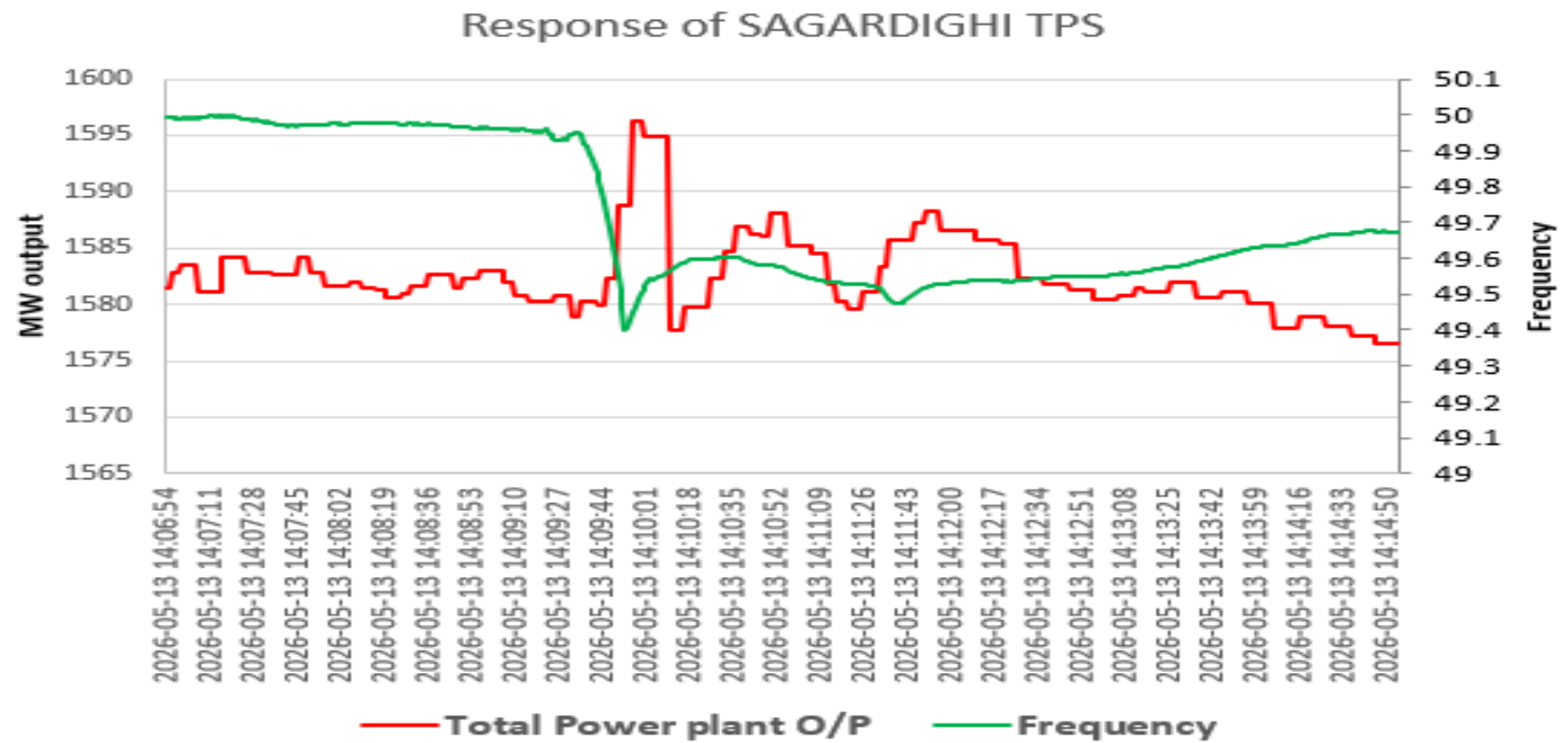


West Bengal intra state generator performance



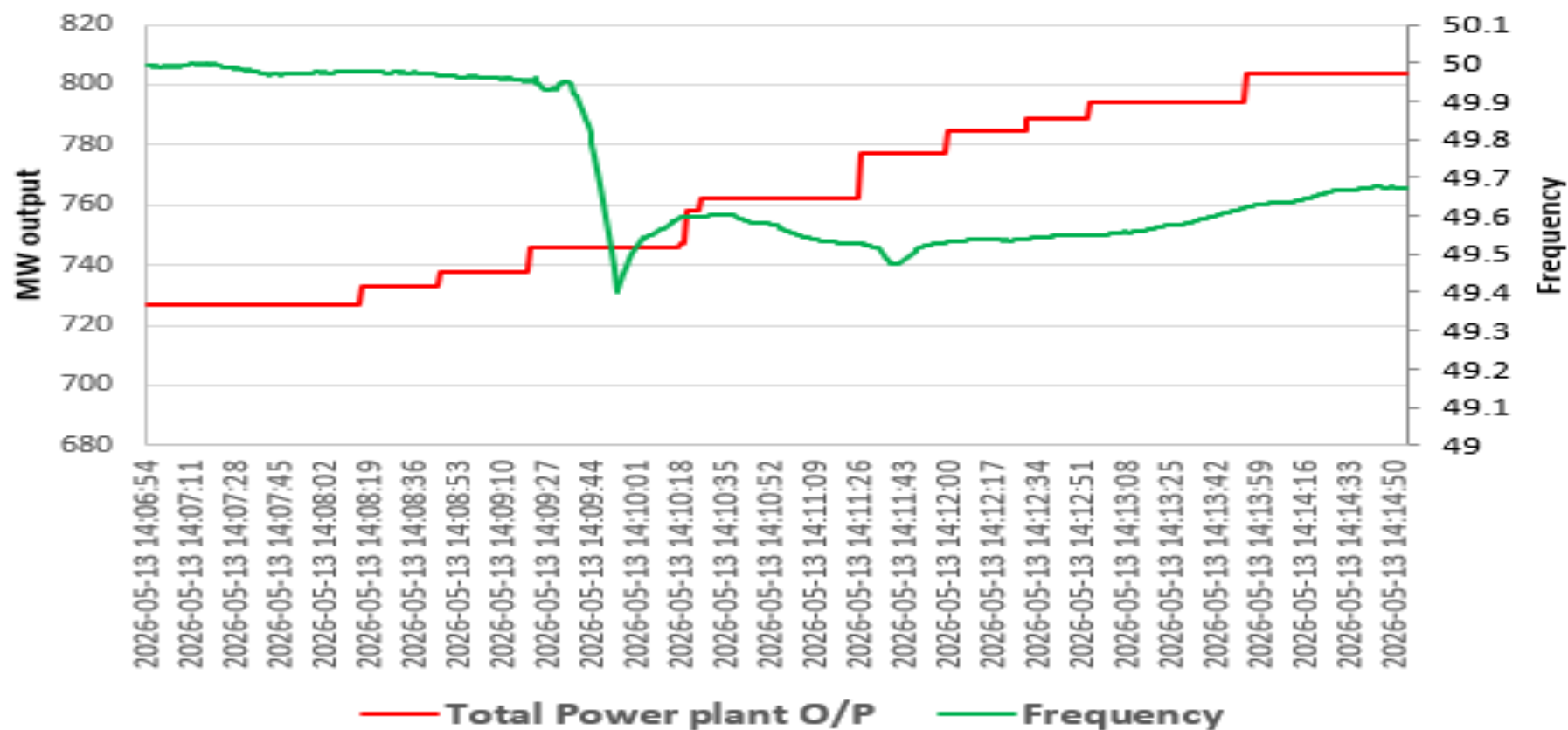


Responses received = 6.83 %

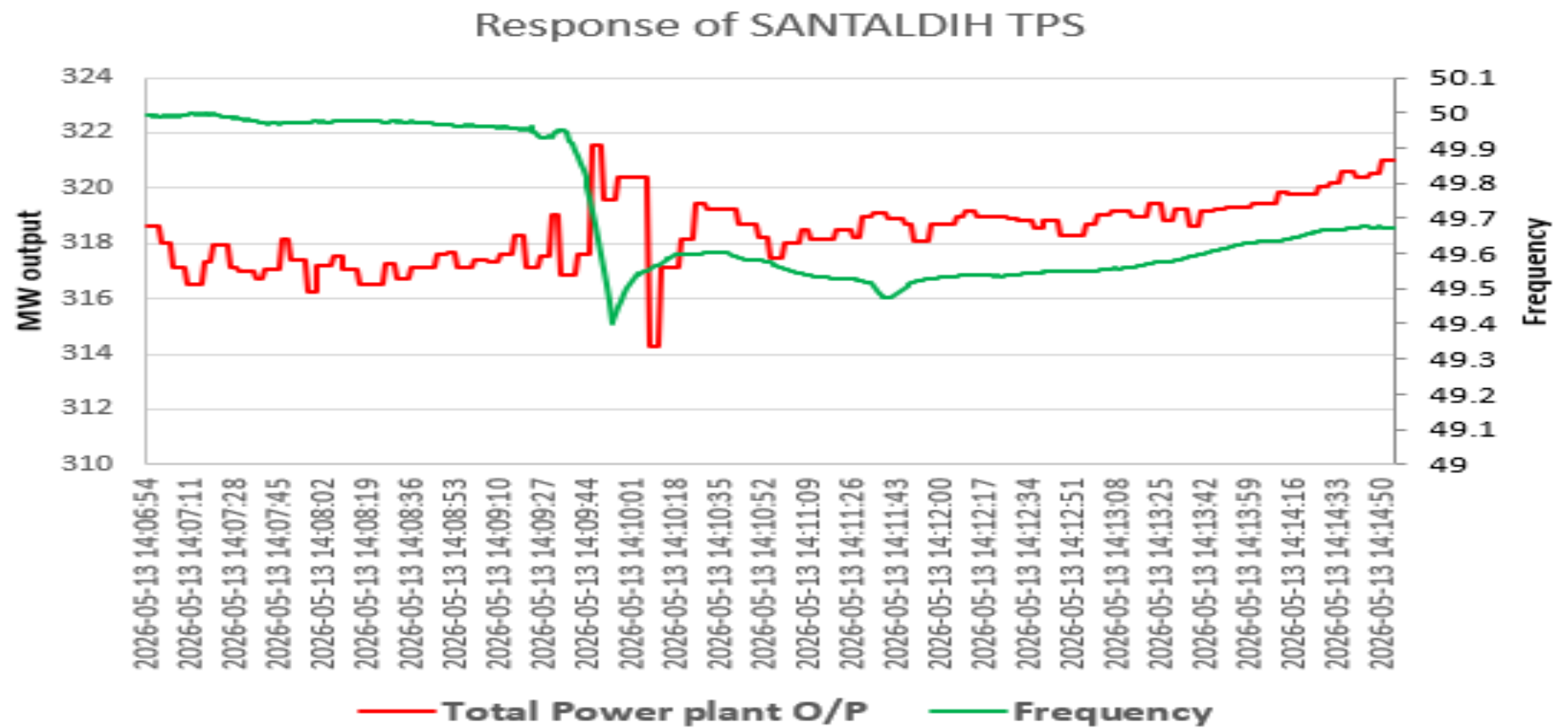


Responses received = 23.19 %

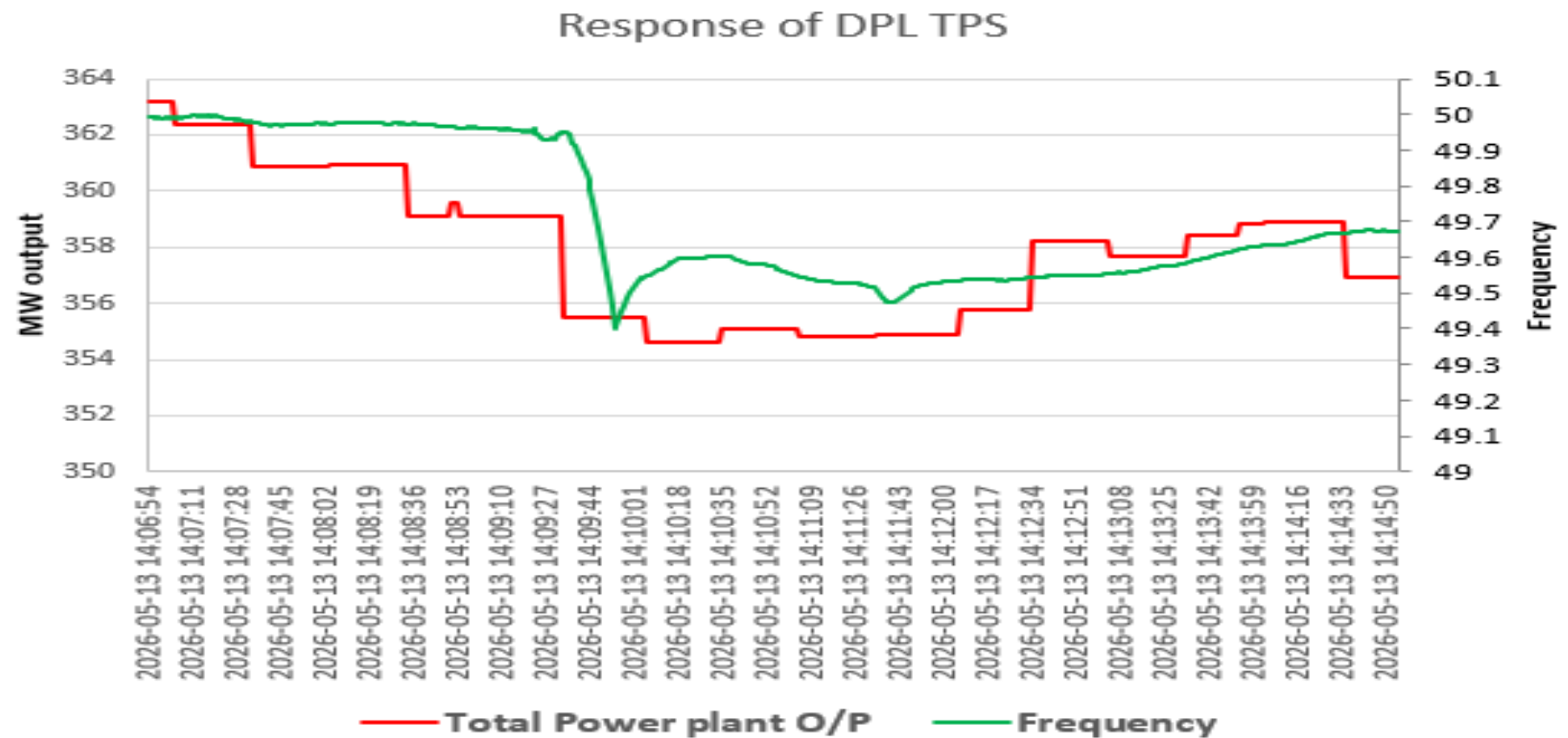
Response of BAKRESWAR TPS



Responses received = 77.39 %

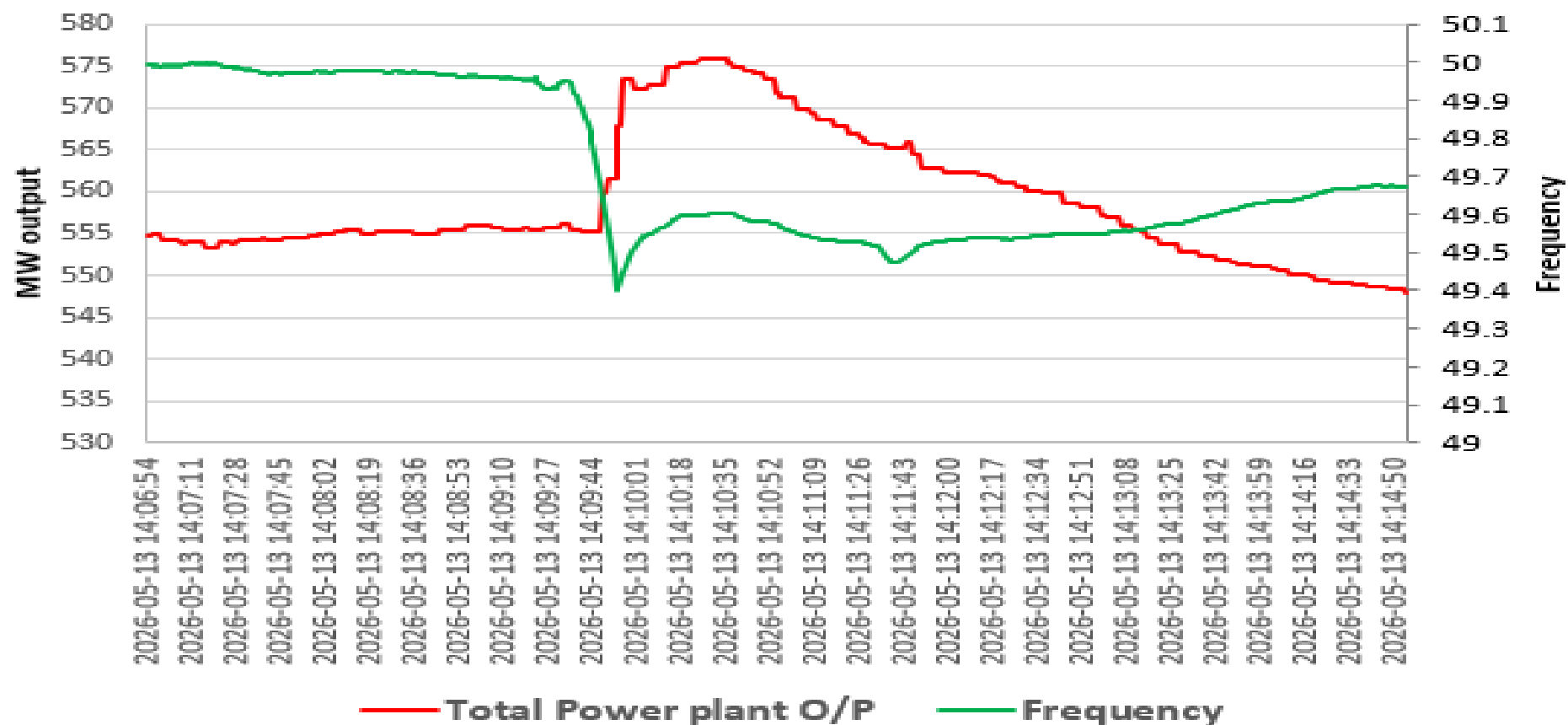


Responses received = -29.39 %

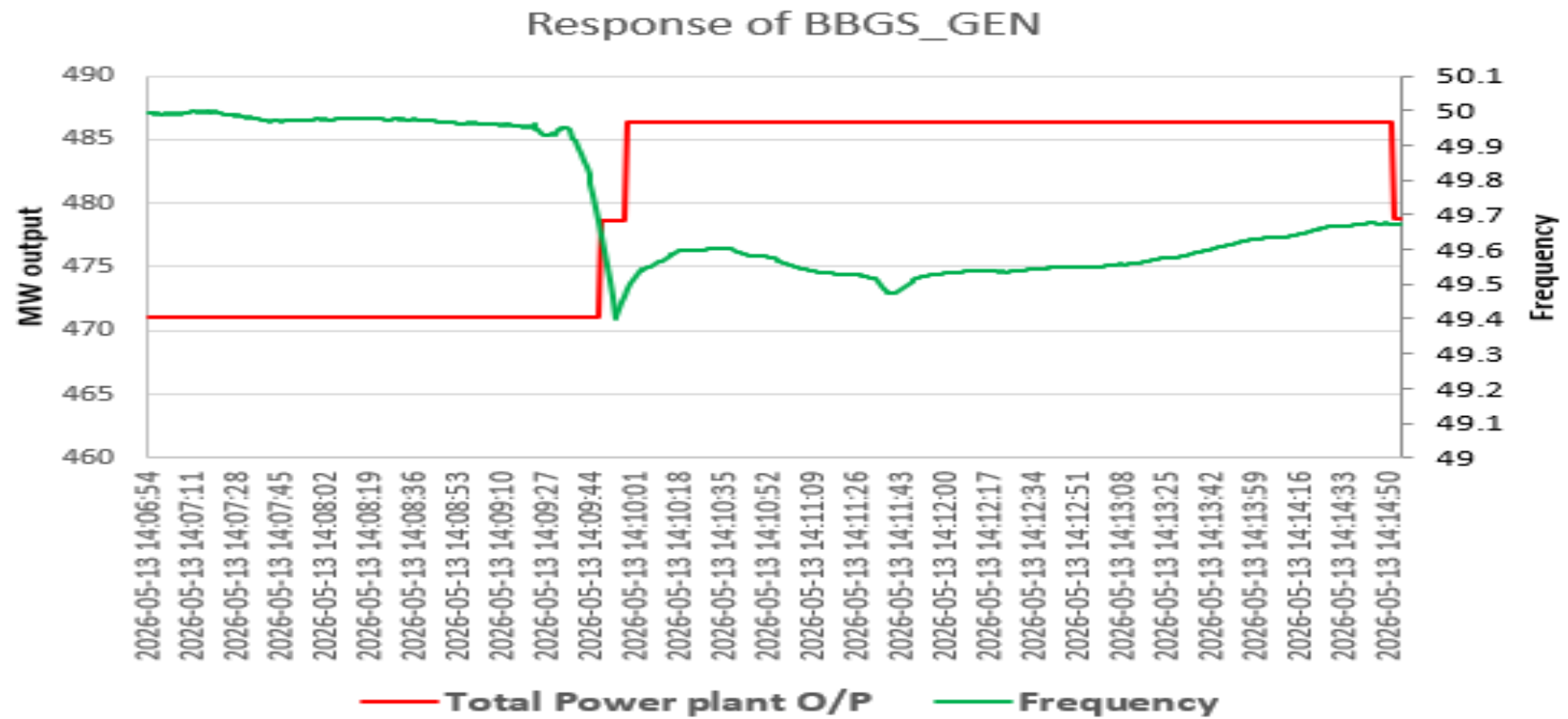


Responses received = -44.93 %

Response of HEL-GEN

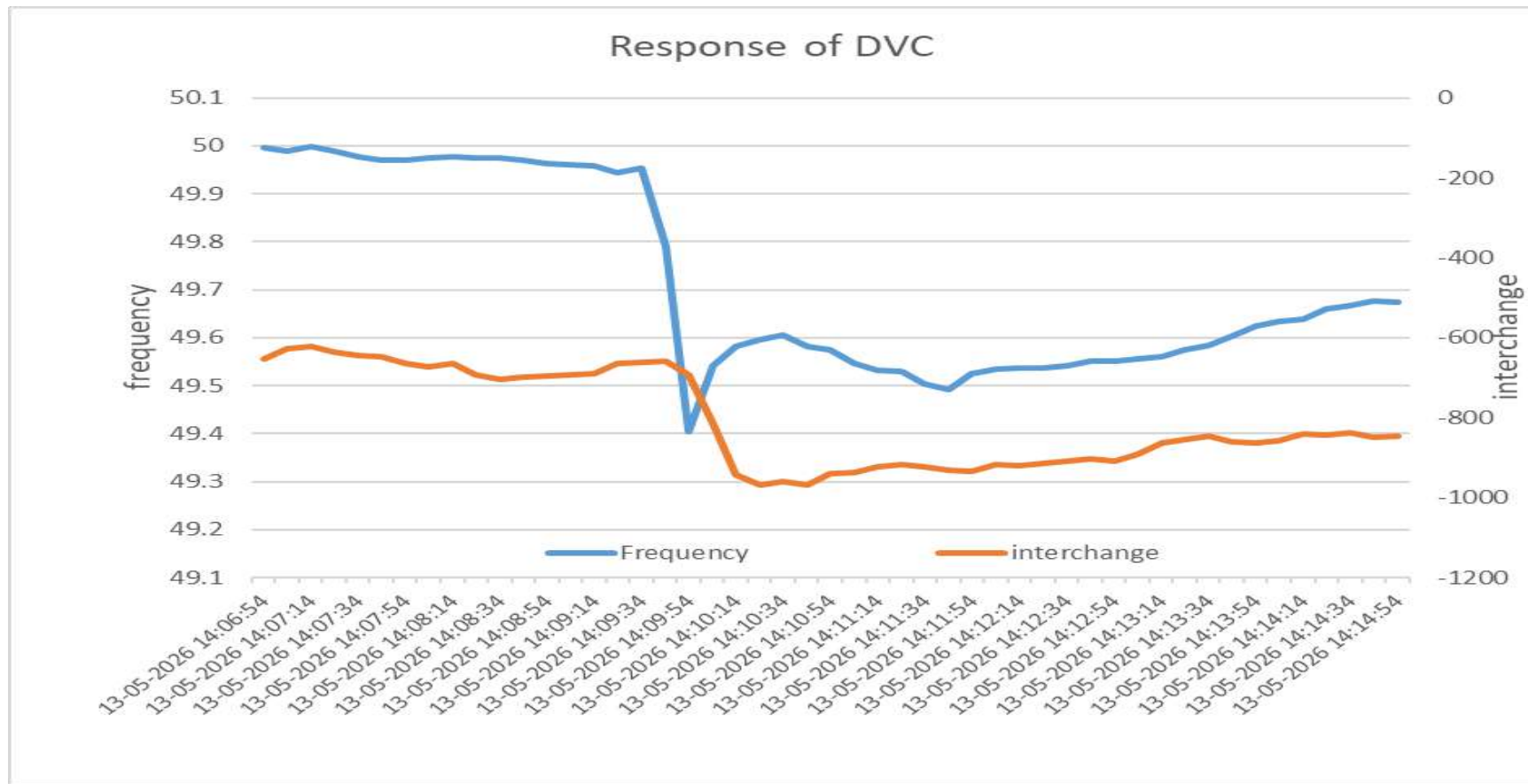


Responses received = 114.89 %

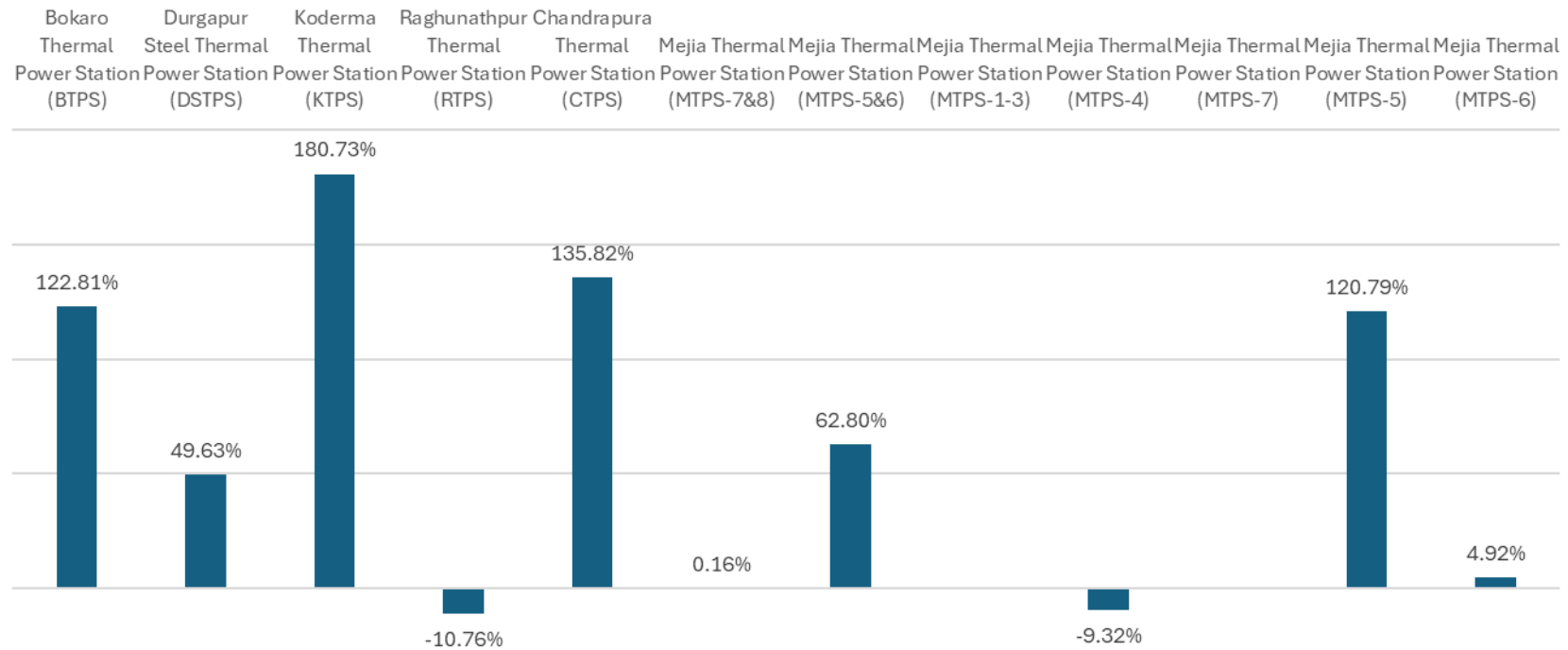


Responses received = 195.28 %

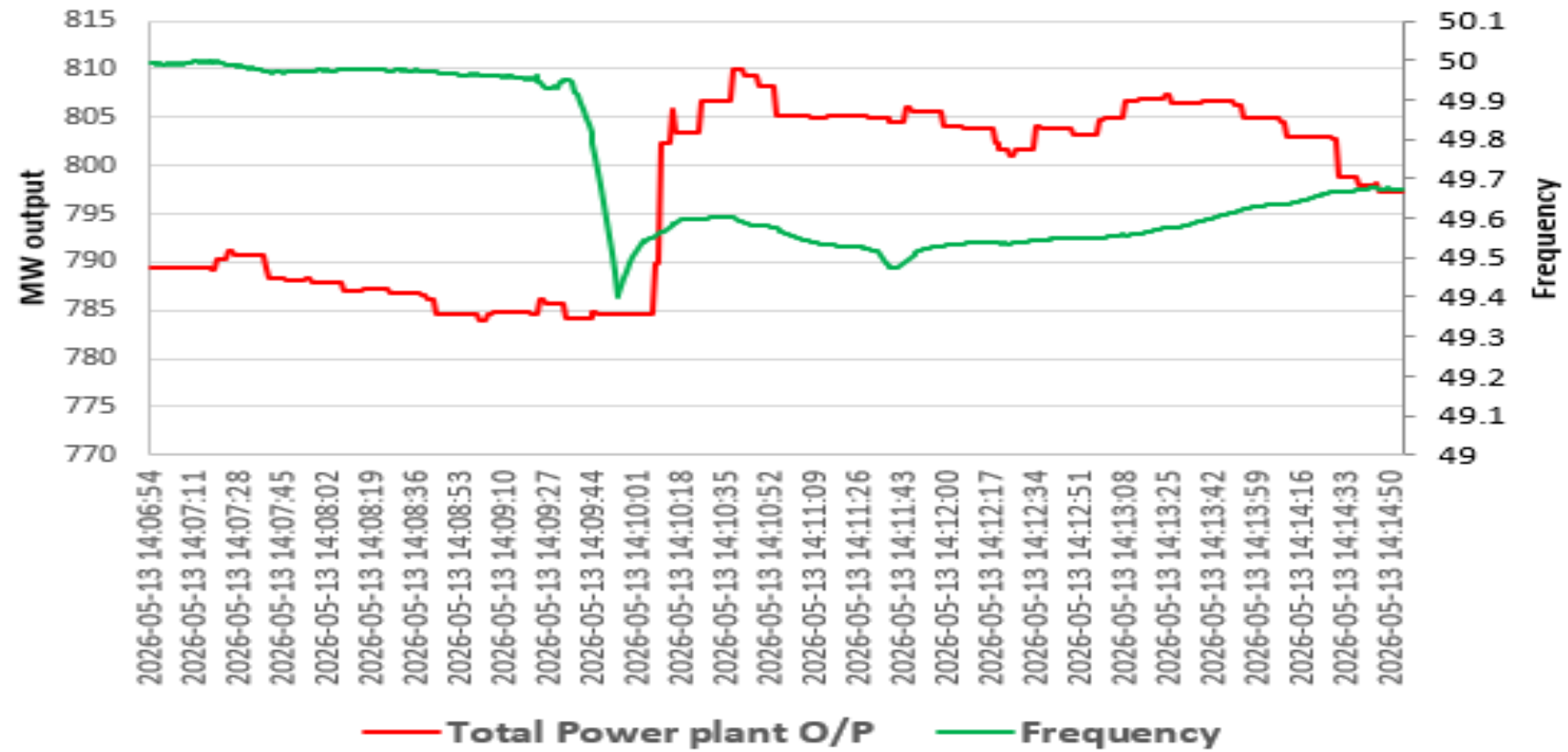
DVC



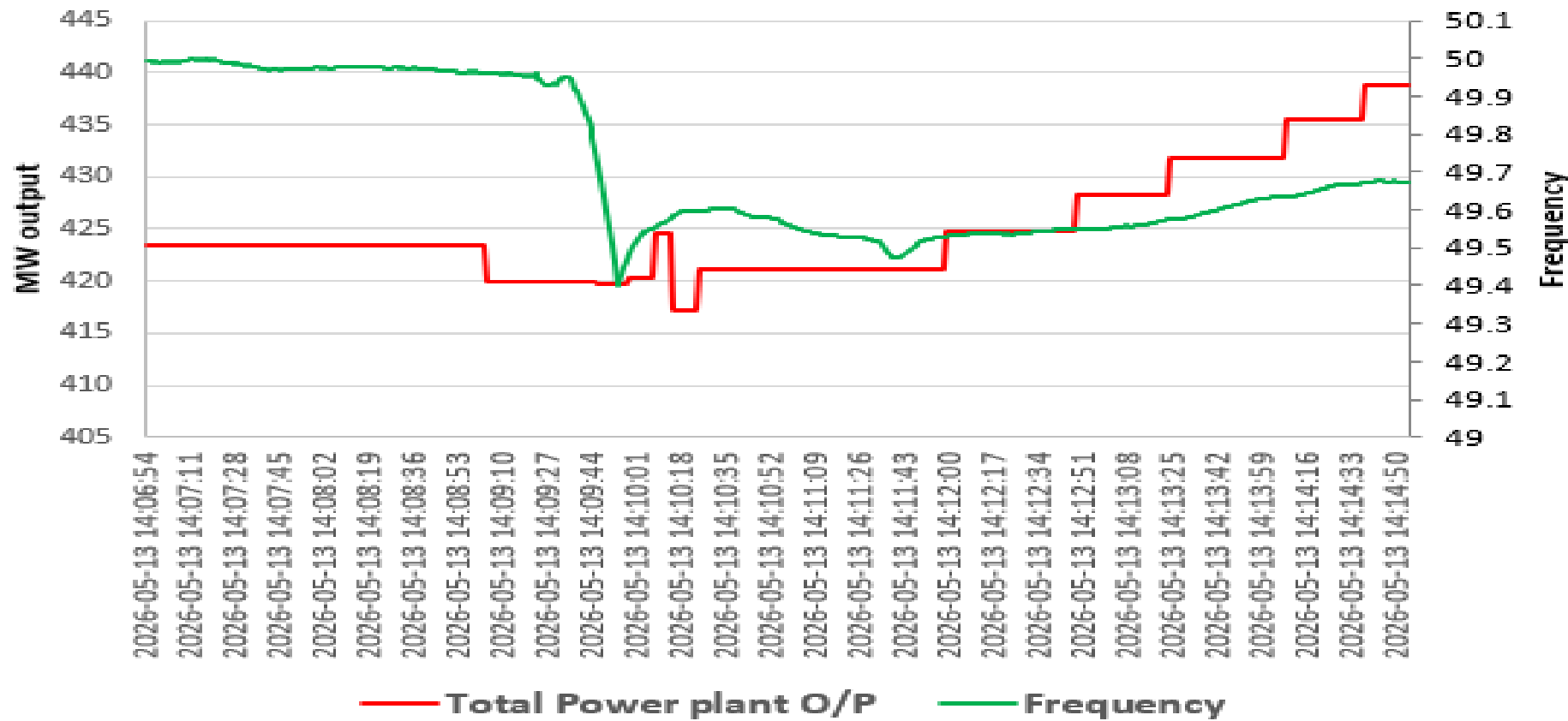
DVC intra state generator performance

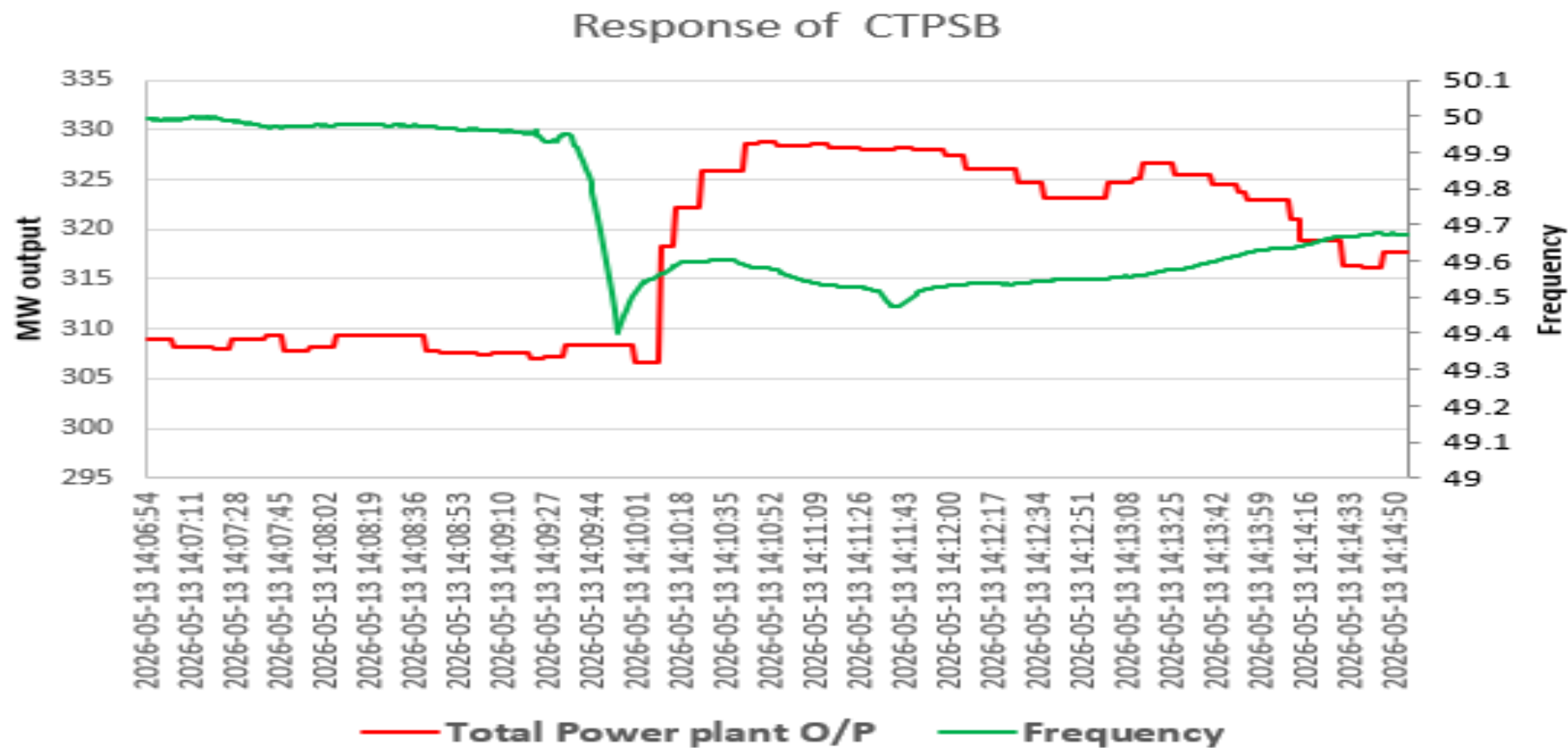


Response of MEJIA TPS

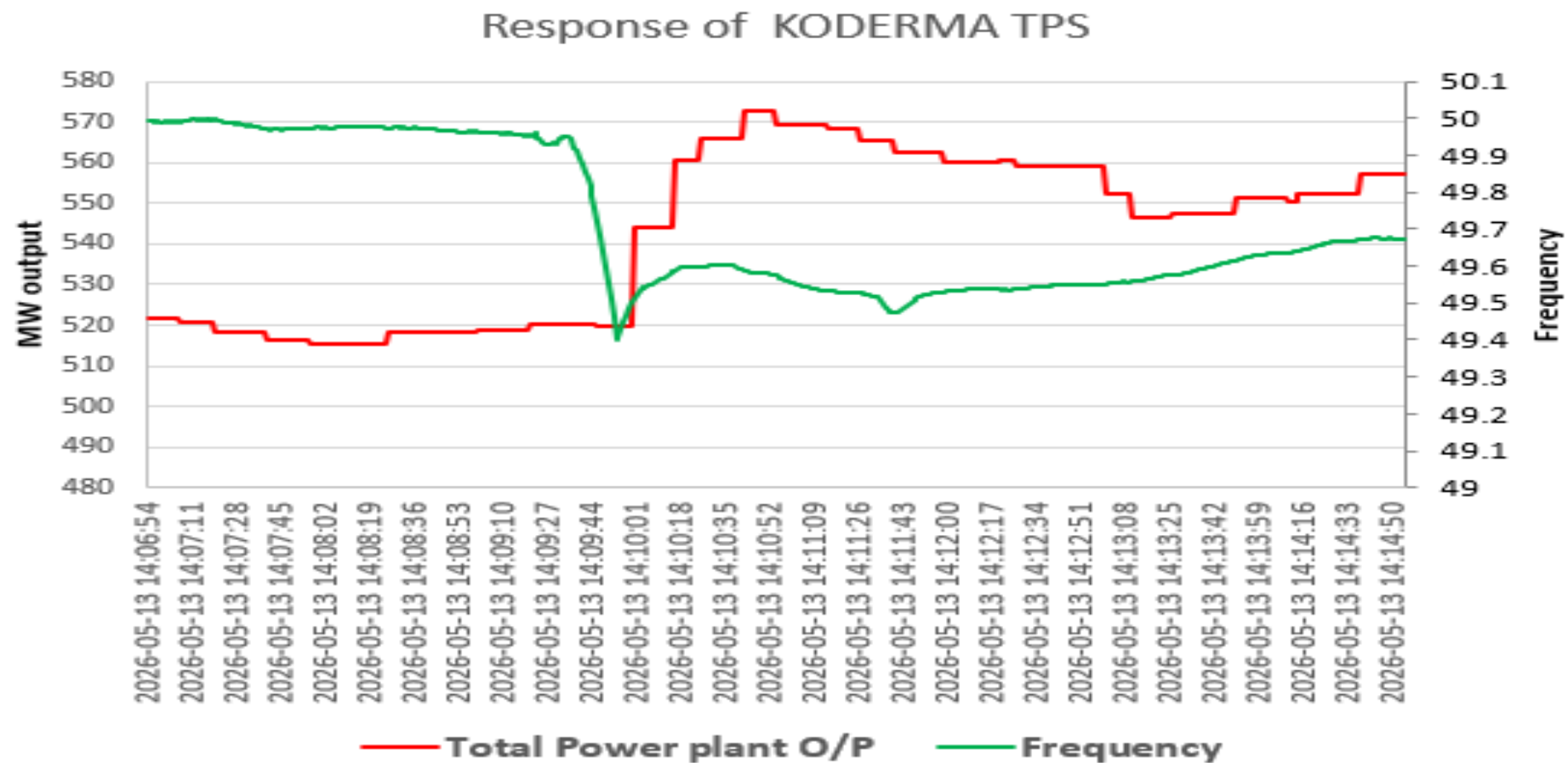


Response of MEJIAB TPS



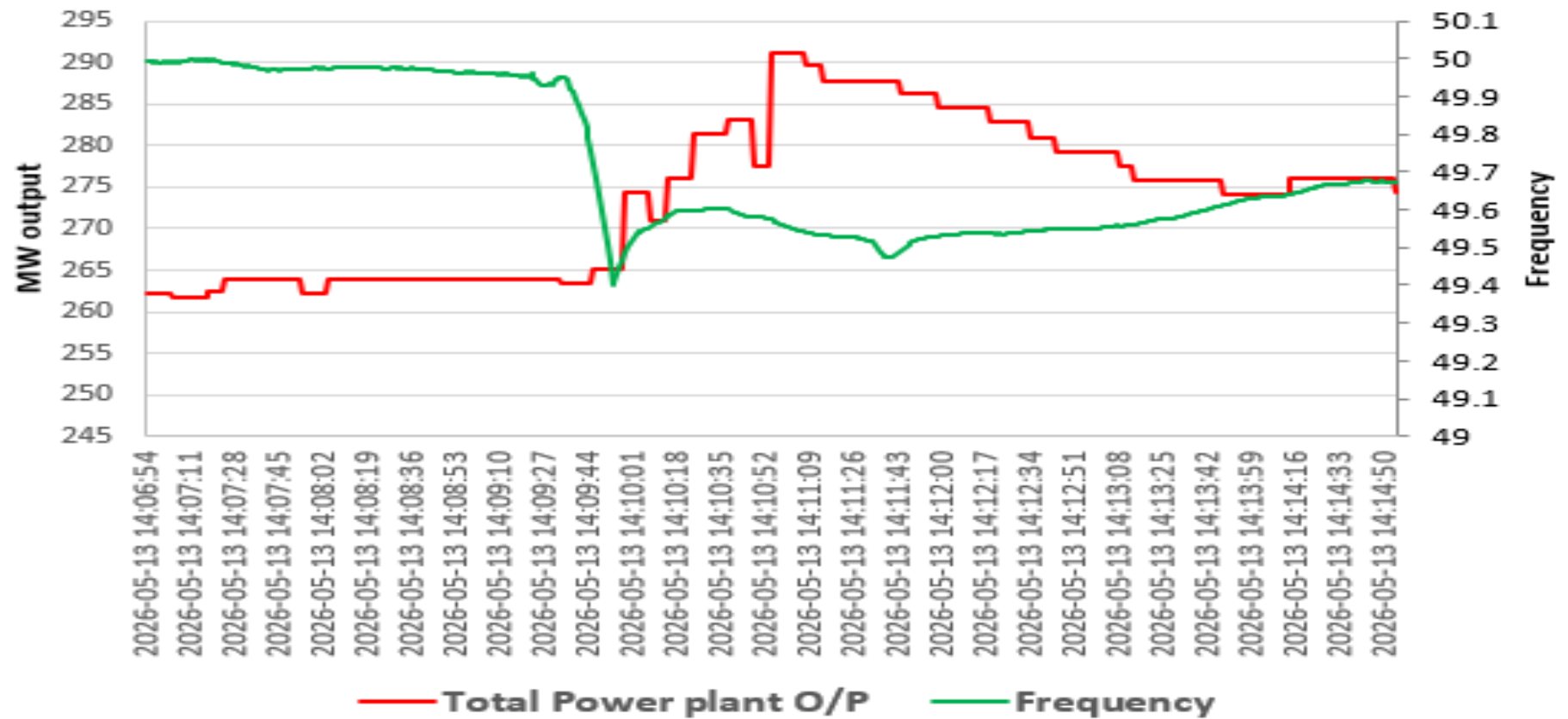


Responses received = 135.82 %



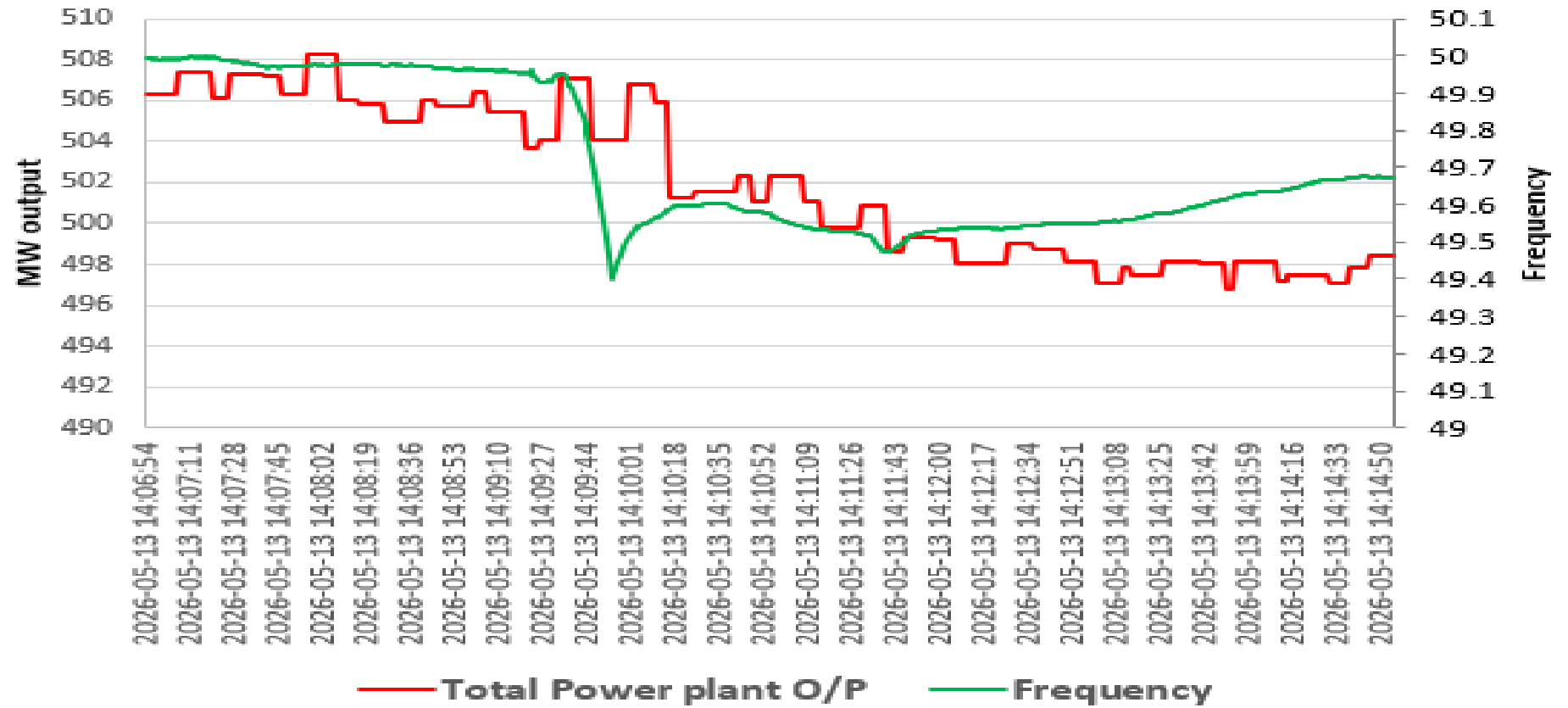
Responses received = 180.73 %

Response of BOKARO A GEN

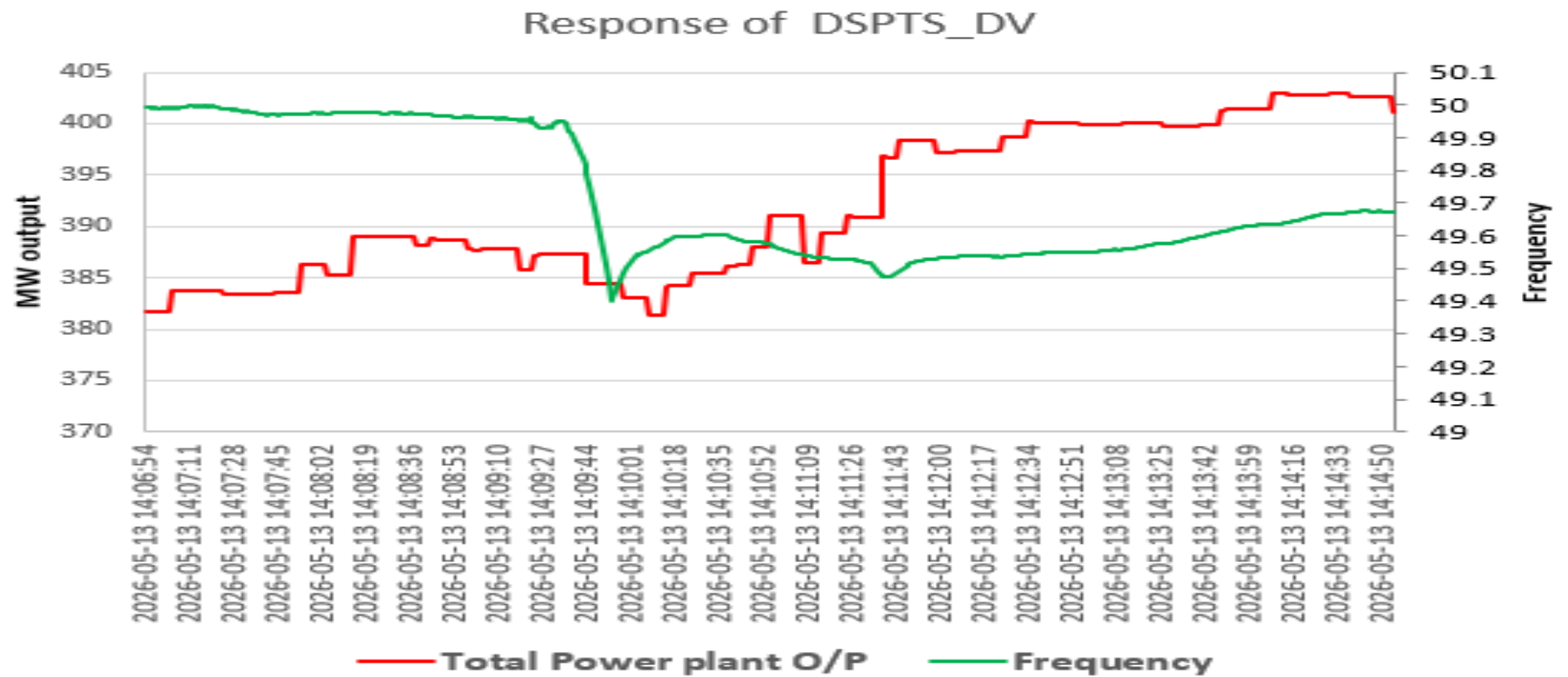


Responses received = 122.81 %

Response of RAGUNATHPUR

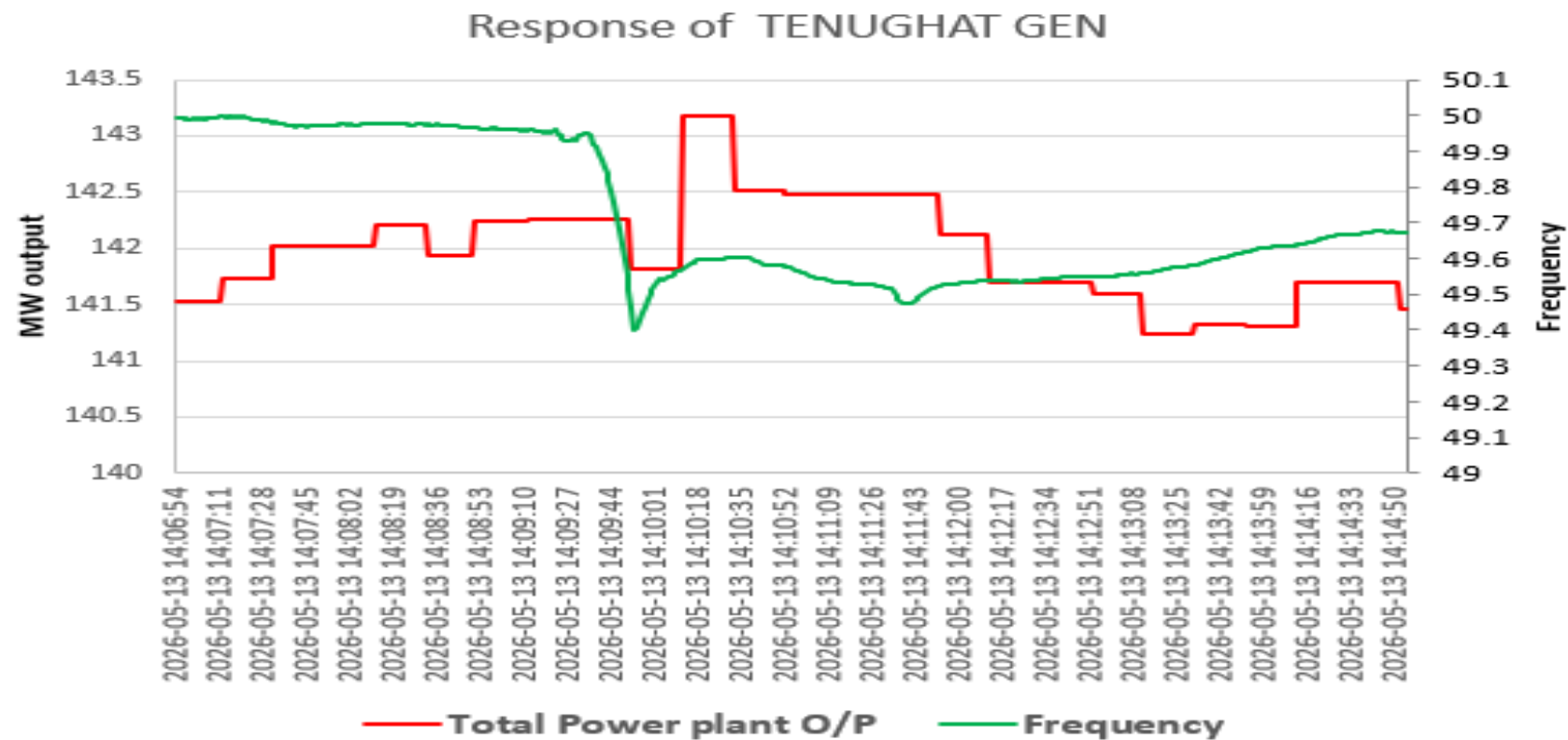


Responses received = -10.76 %



Responses received = 49.63 %

Jharkhand



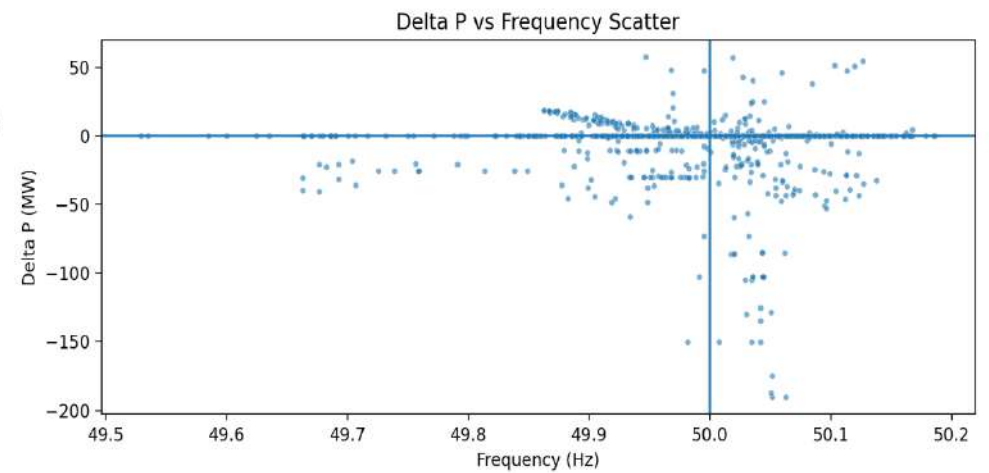
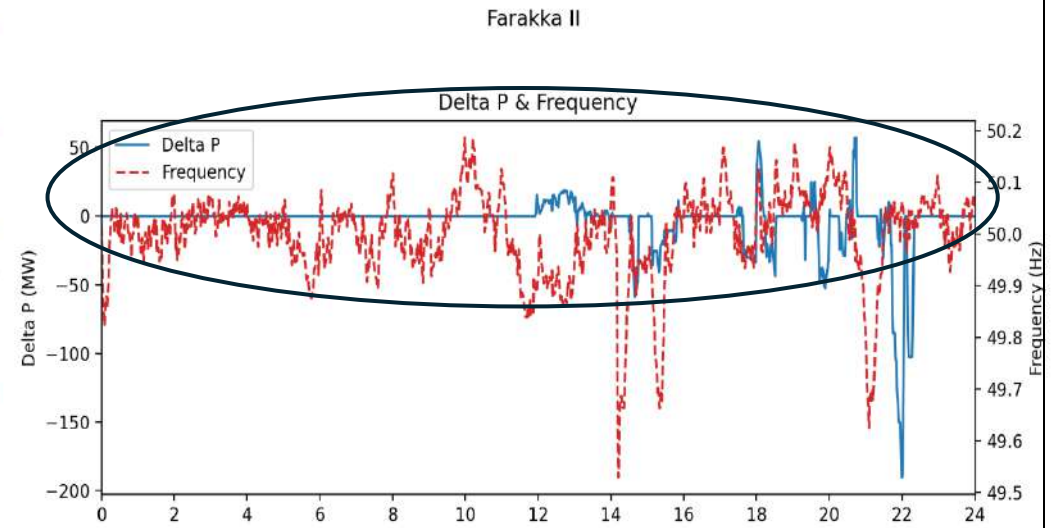
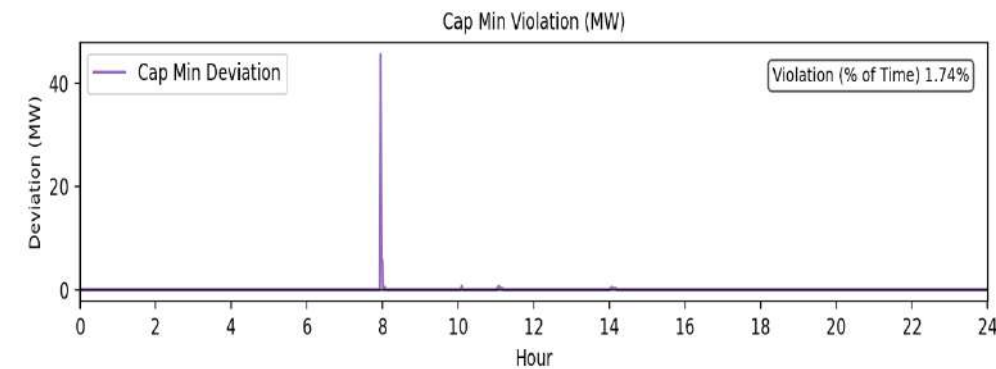
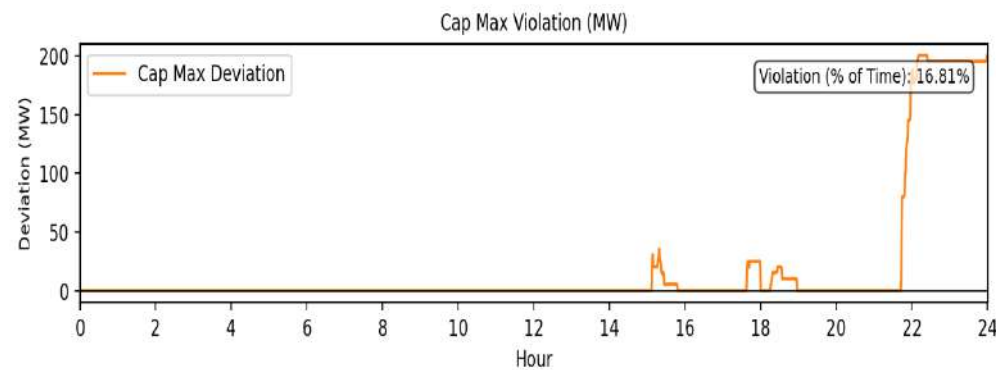
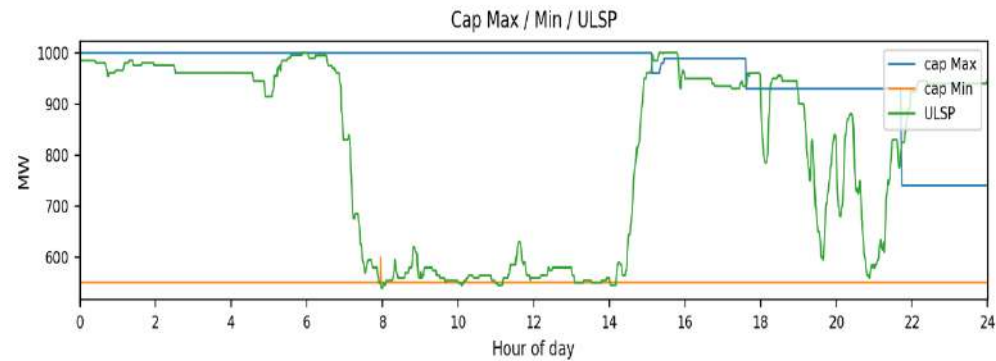
Responses received = 12.99 %

AGC PERFORMANCE OF EASTERN REGION

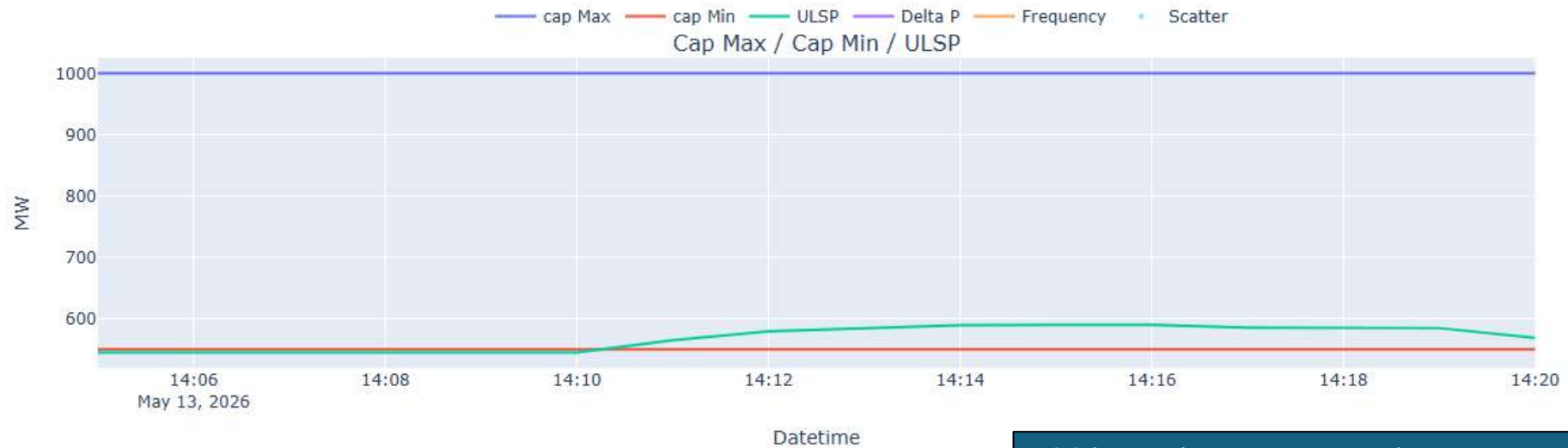
AGC in Local Mode

Farakka II

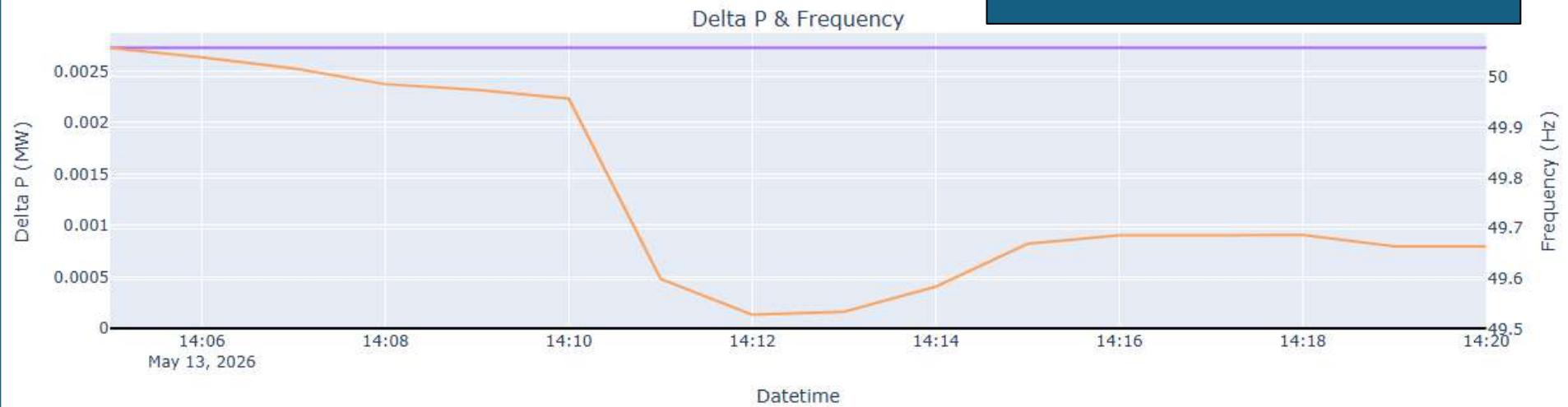
AGC is kept in local mode for most of the time



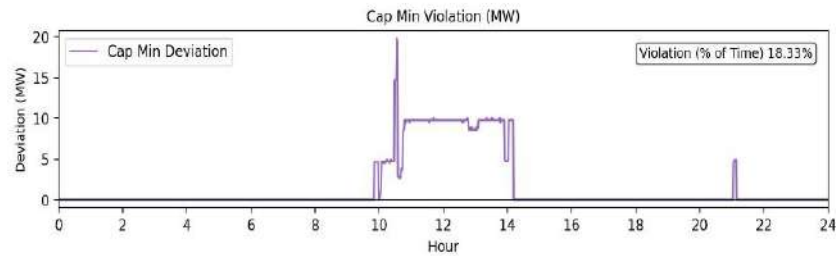
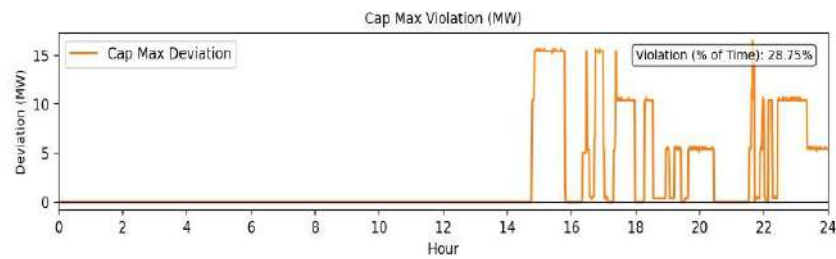
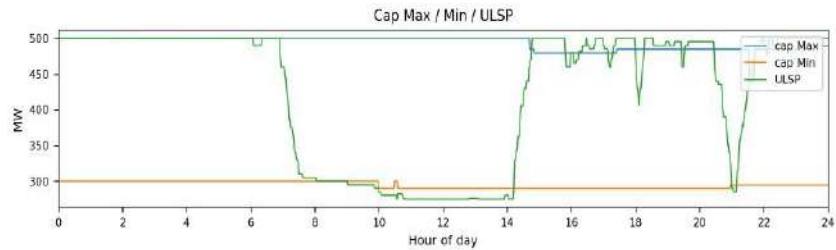
Farakka II



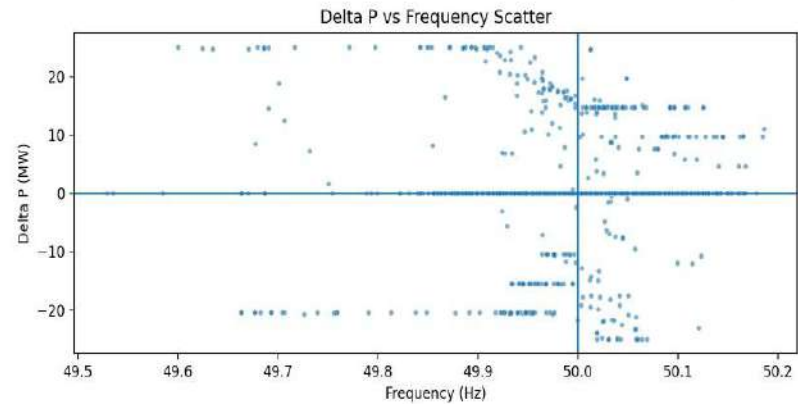
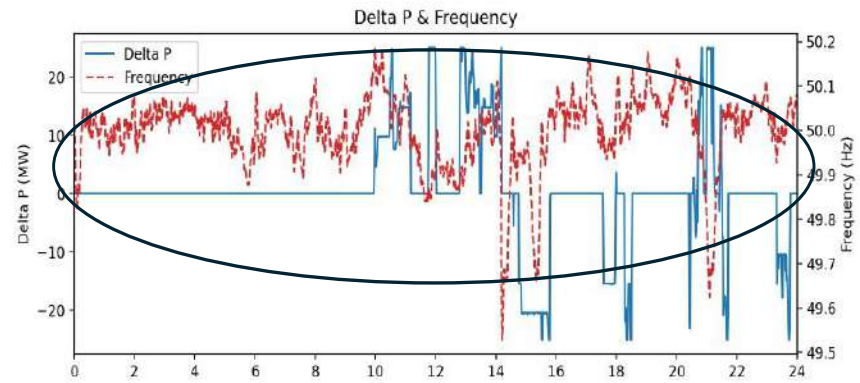
AGC is kept in local mode during the event



Farakka III



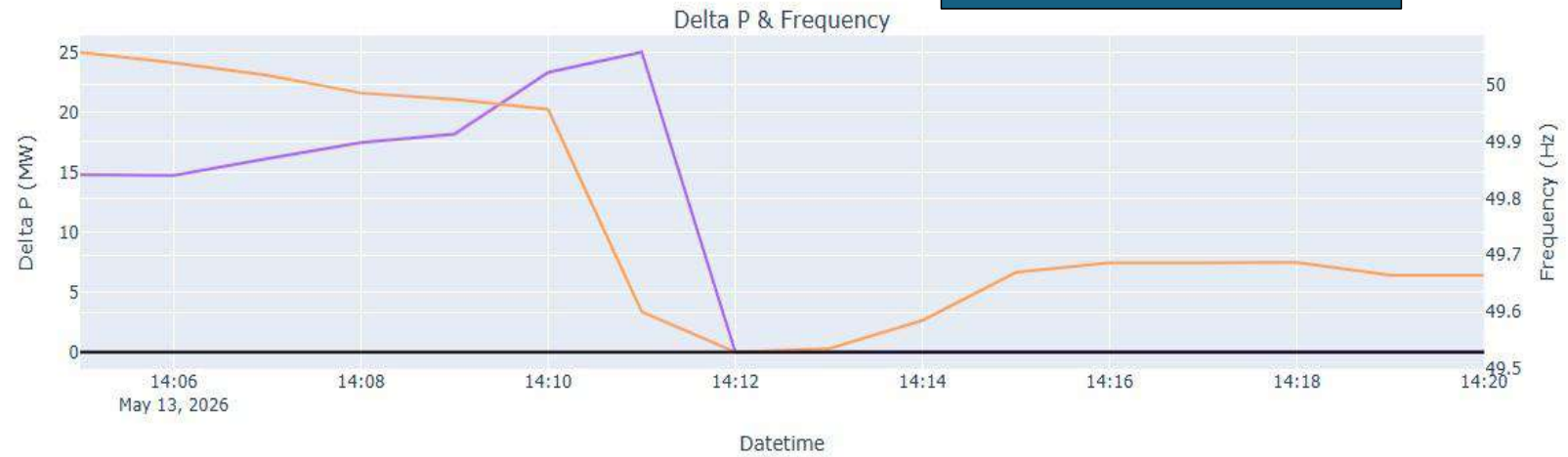
AGC is kept in local mode for most of the time



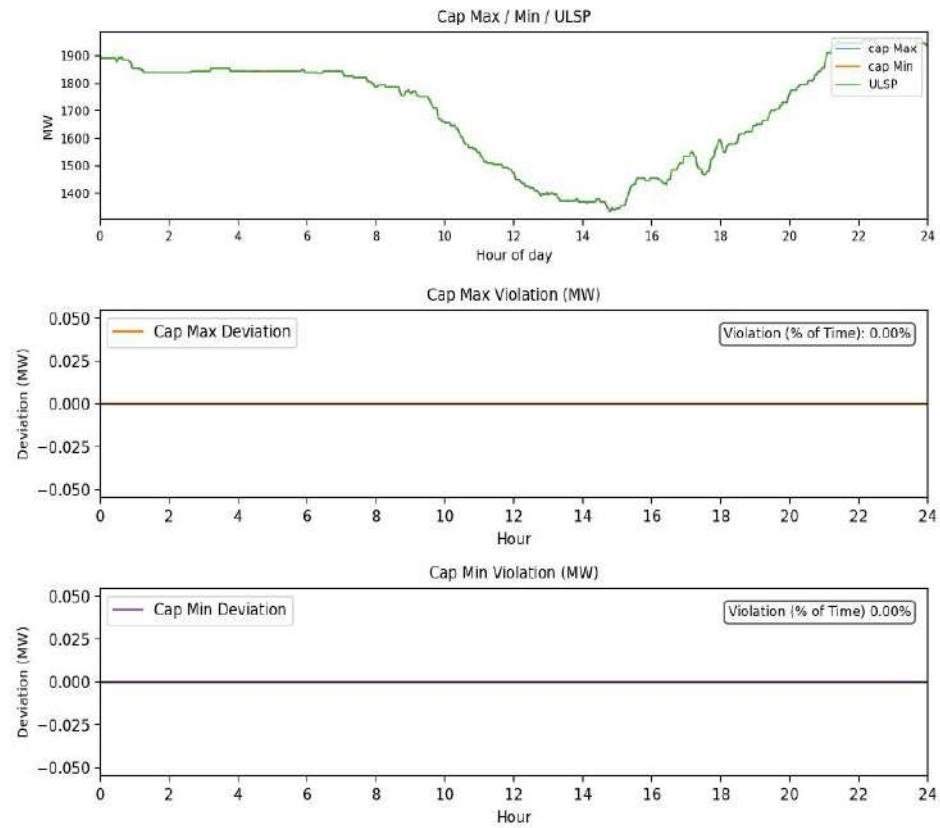
Farakka III



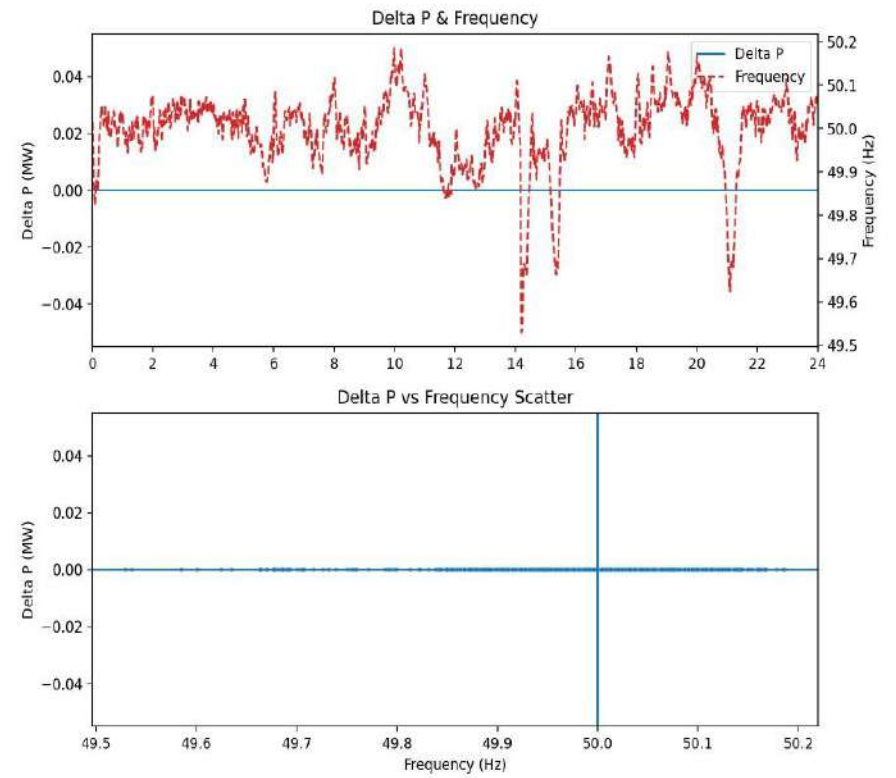
AGC is kept in local mode after the event



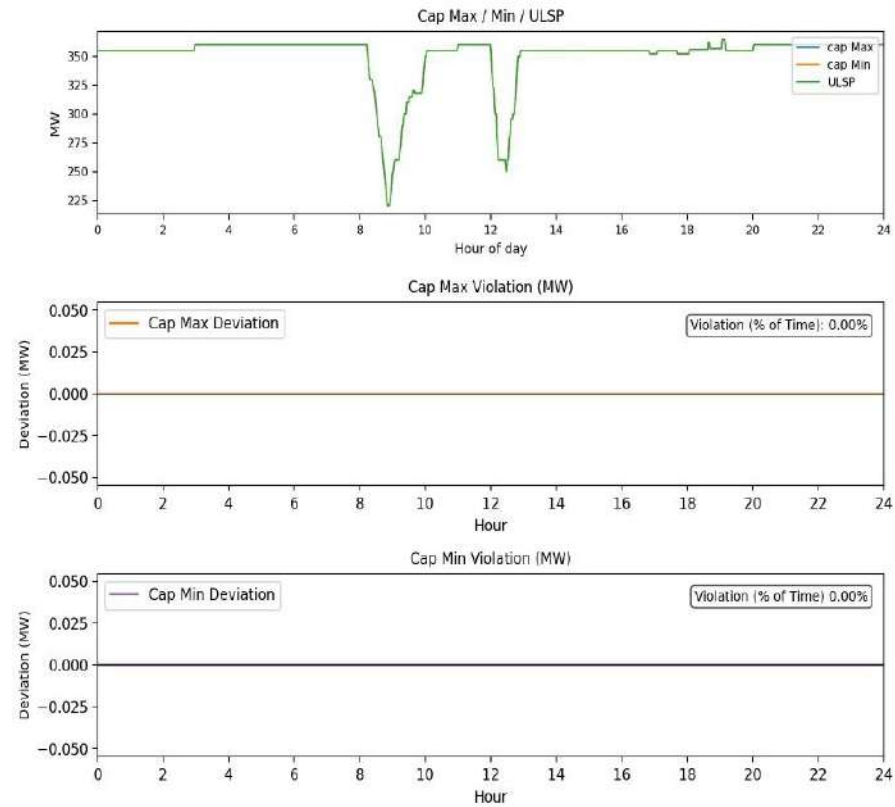
Barh I



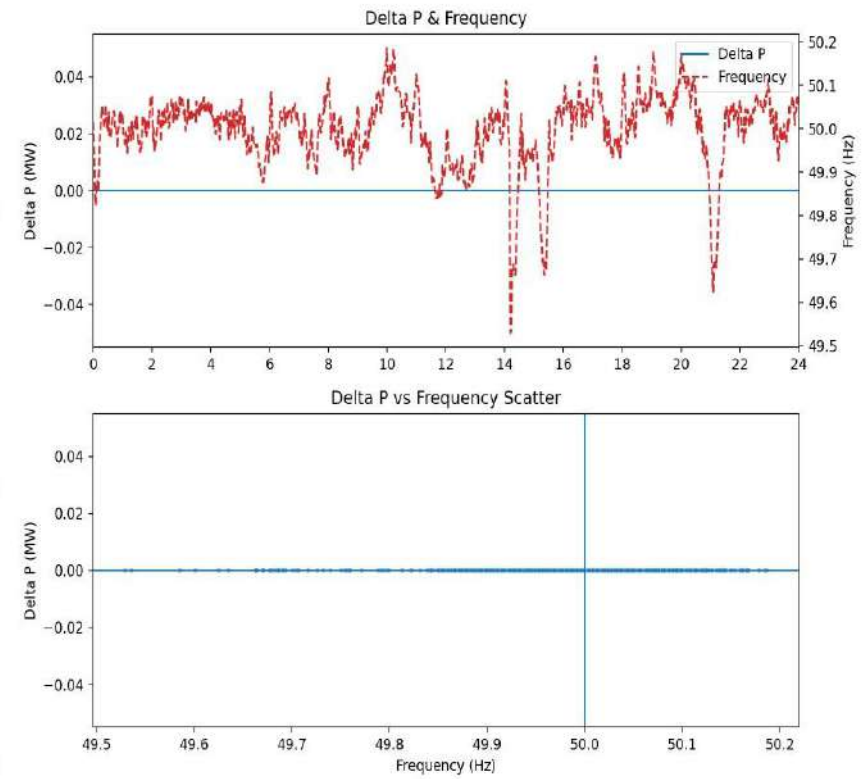
Barh I



JSWEUL



JSWEUL



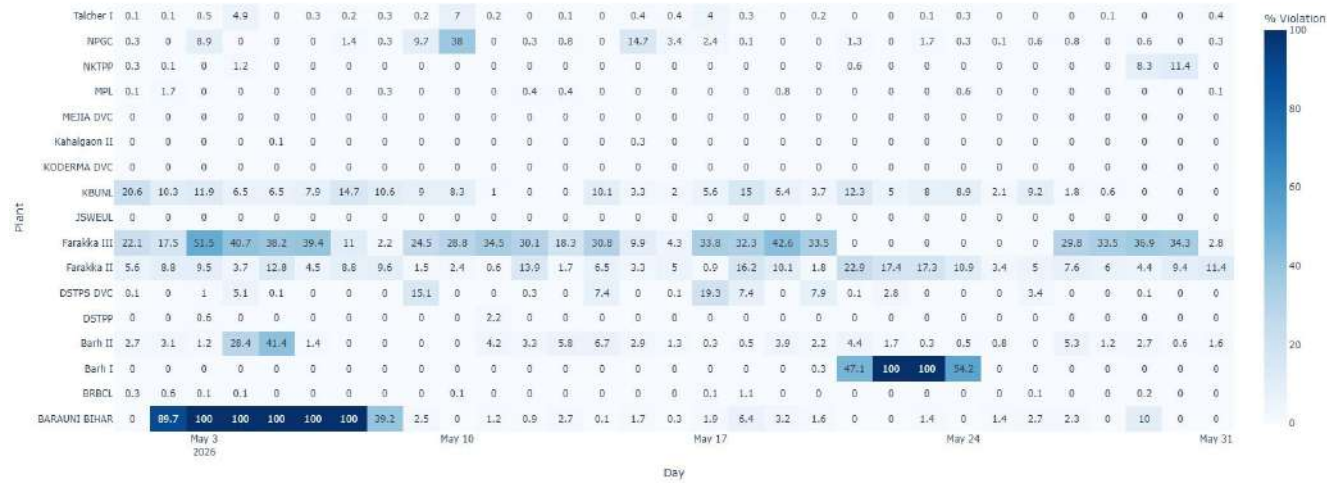
Undesired operation

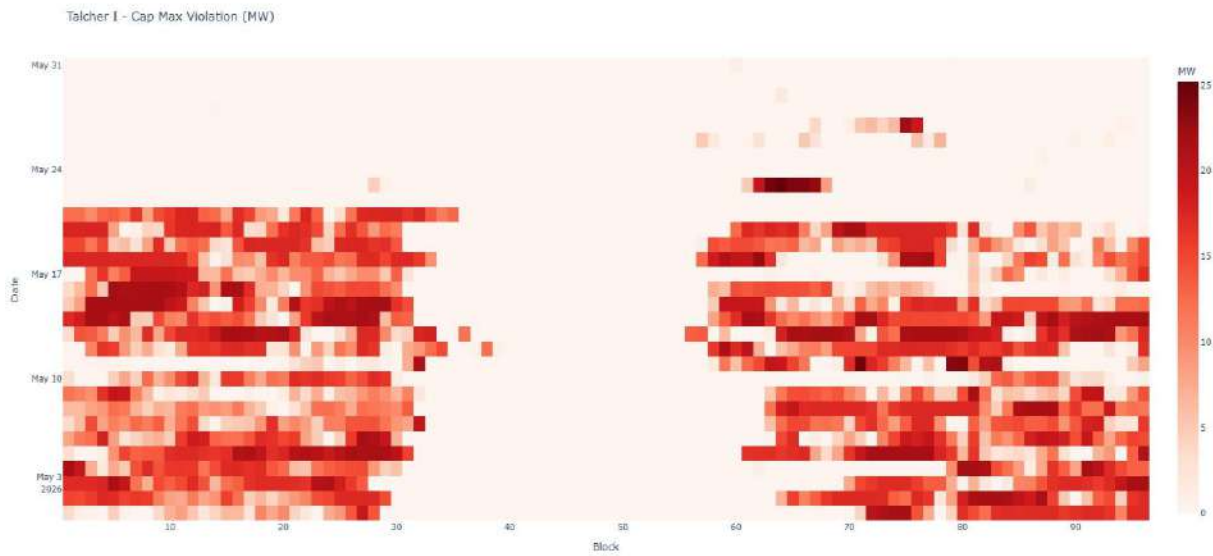




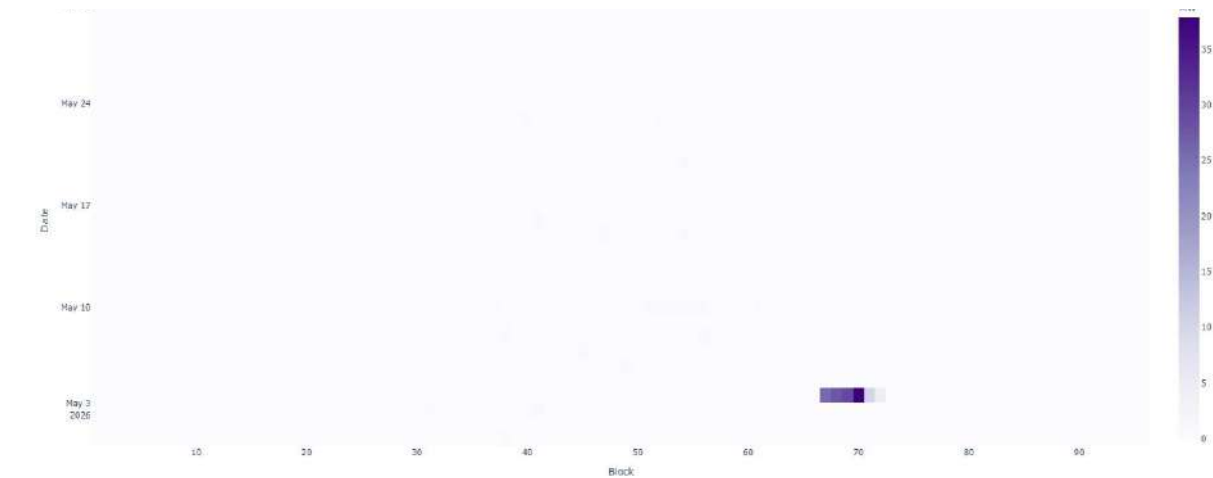
Monthly Analysis (May,2026)

Cap Max Violation (% of Time) Day-wise

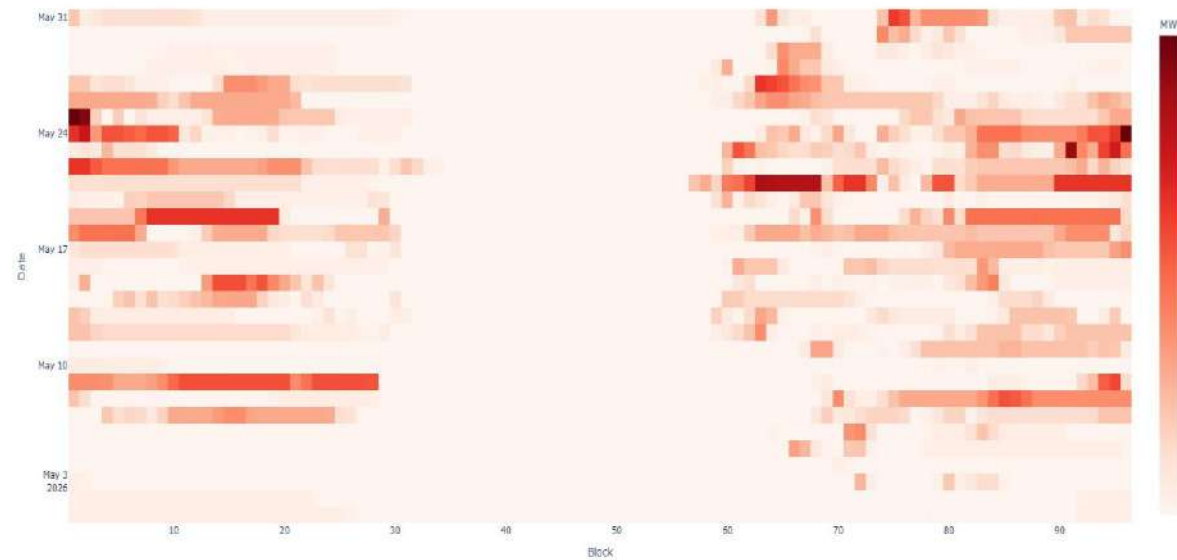




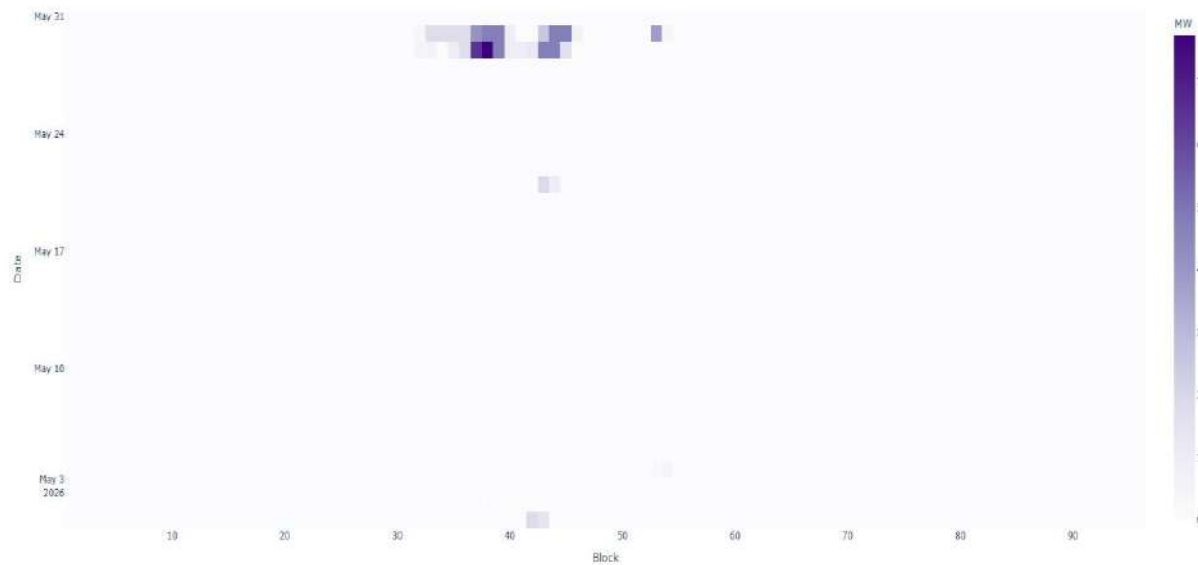
Up to 21st May Cap
Max violation up to 25
MW during Non -Solar
Hour

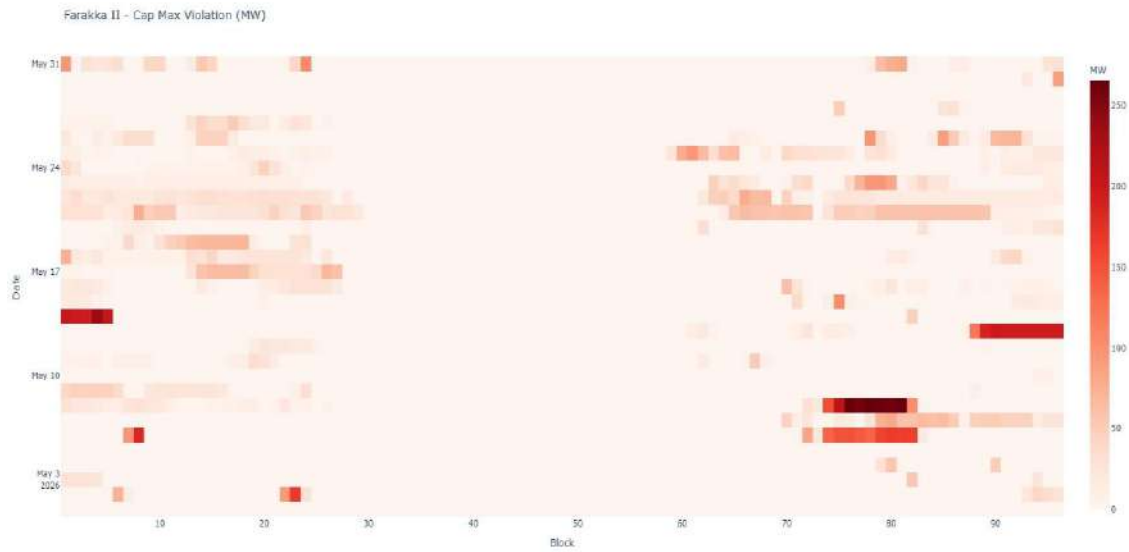


NKTPP - Cap Max Violation (MW)

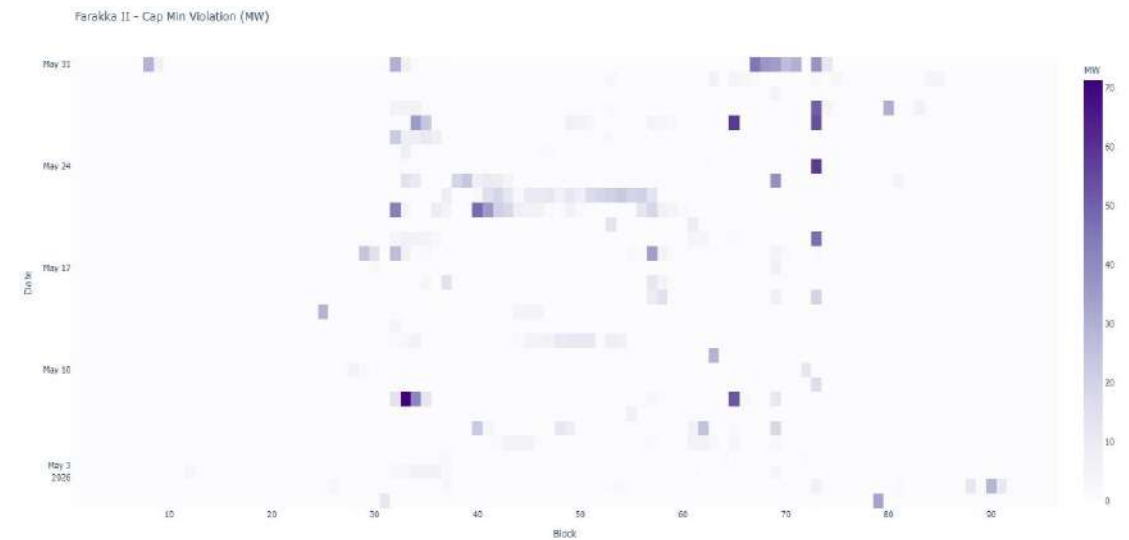


Cap Max violation up to 55 MW during Non -Solar Hour

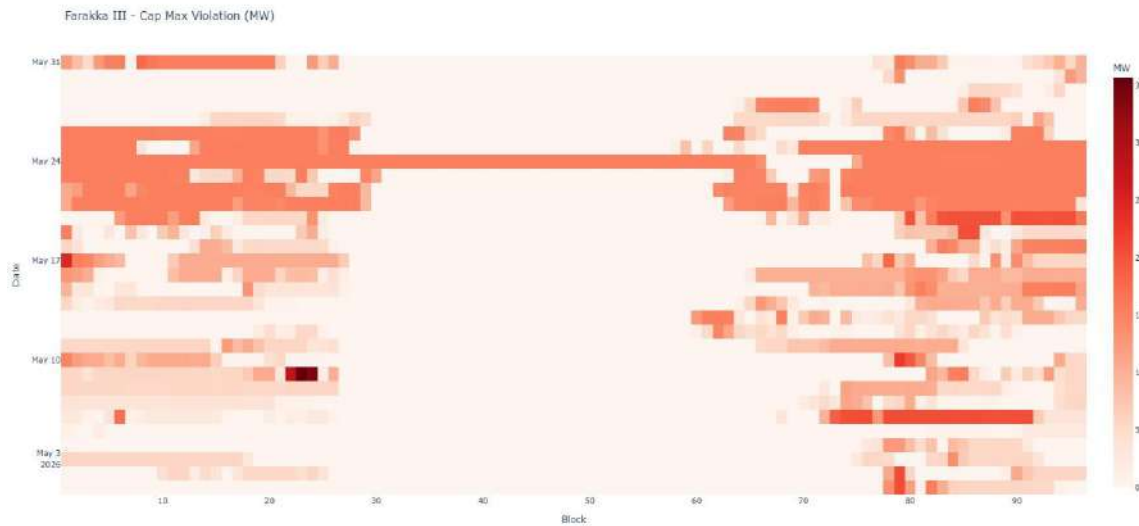




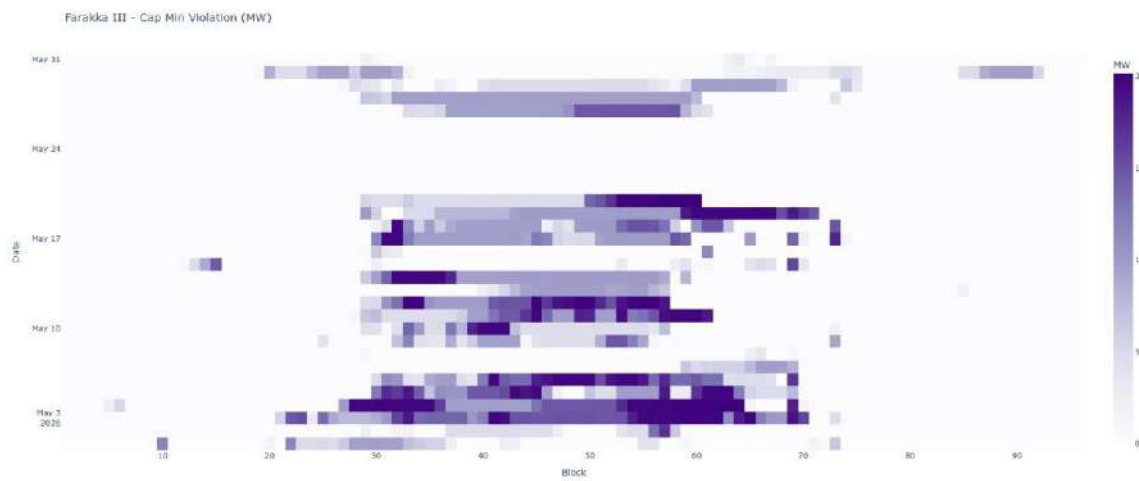
Average Cap Max violation up to 60 MW during Non-Solar Hour



Average Cap Min violation up to 40 MW during Solar Hour



Cap Max violation up to 35 MW during Non-Solar Hour

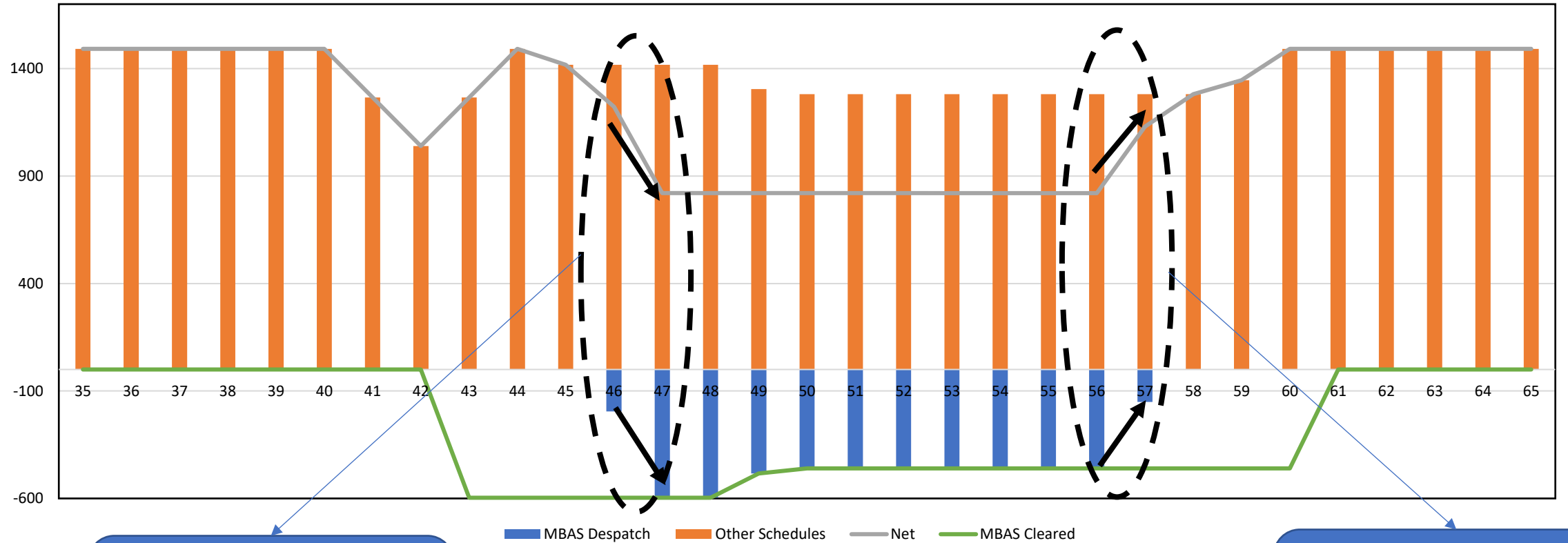


Cap Min violation up to 20 MW during Solar Hour

Thank YOU

Annexure B.2.13

Example Of Ramp Rate Violation of Darlipalli STPS on dated 03-04-2026

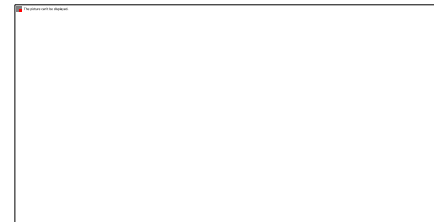
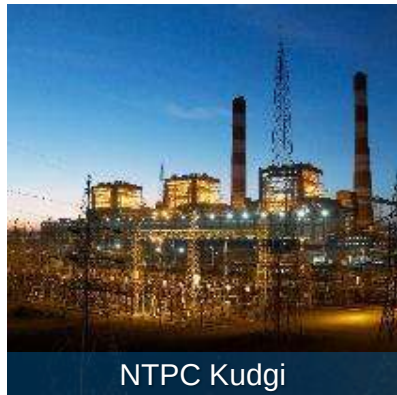
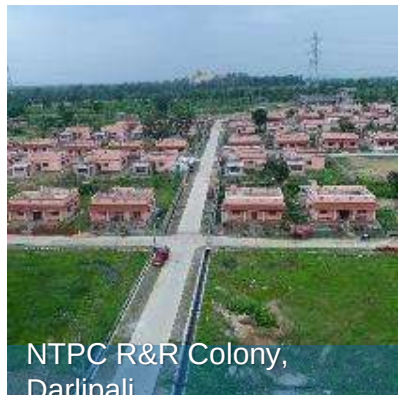


Ramp Violation: 46 TB to 47 TB
Ramp In other Schedule: 0 MW
Ramp Down given in MBAS schedule:
402MW
Declared Ramp Down: 226 MW

Ramp Violation: 56 TB to 57 TB
Ramp In other Schedule: 0 MW
Ramp Up given in MBAS schedule:
309.20MW
Declared Ramp: 226 MW

1. NLDC shall dispatch MBAS based on system requirements, subject to the quantum cleared in the market.
2. Generating Stations may place MBAS bids taking into account their ramping capabilities like energy market .

Annex B.2.20

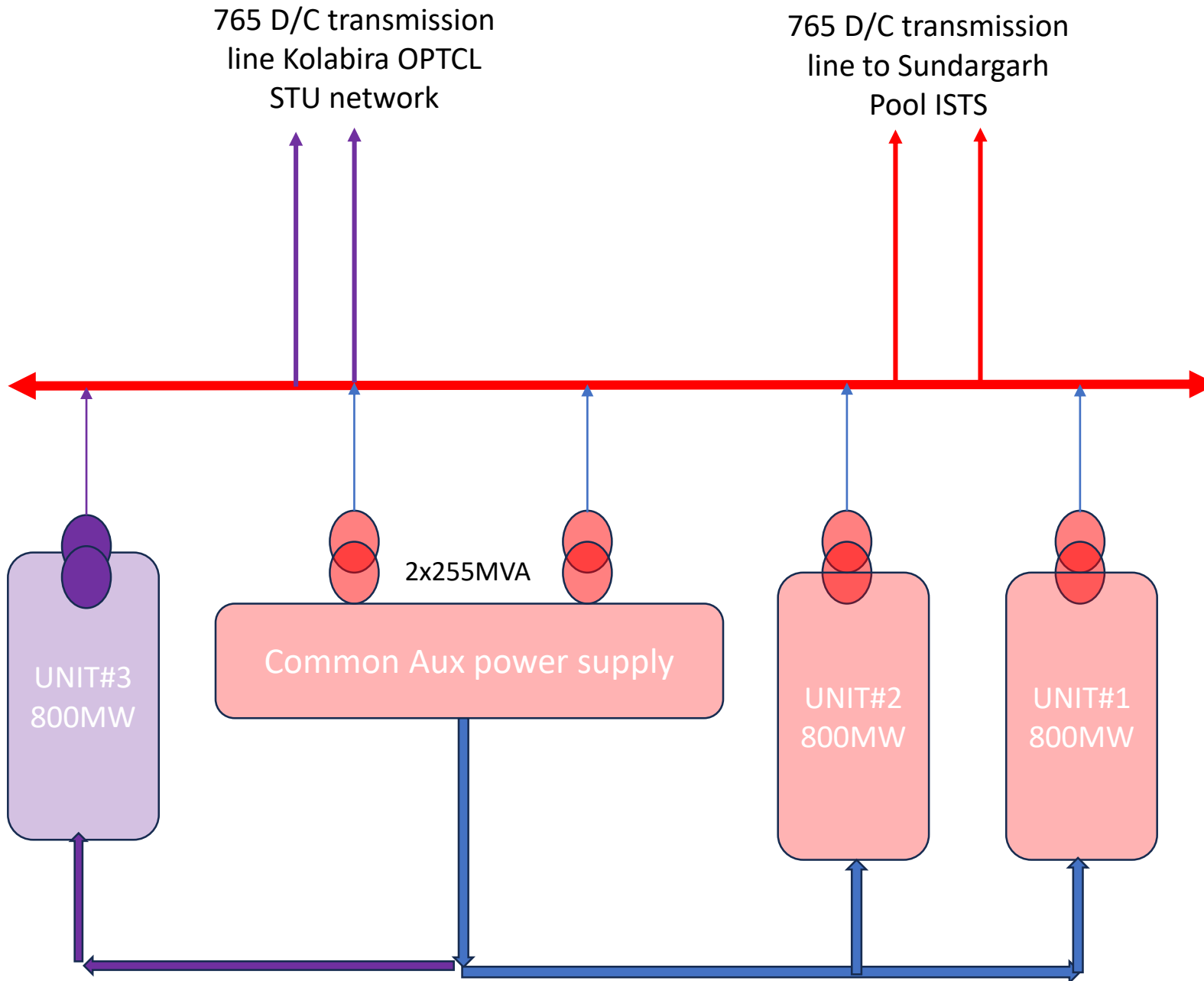


Darlipali Stage-II (1x800MW) Connectivity and Metering

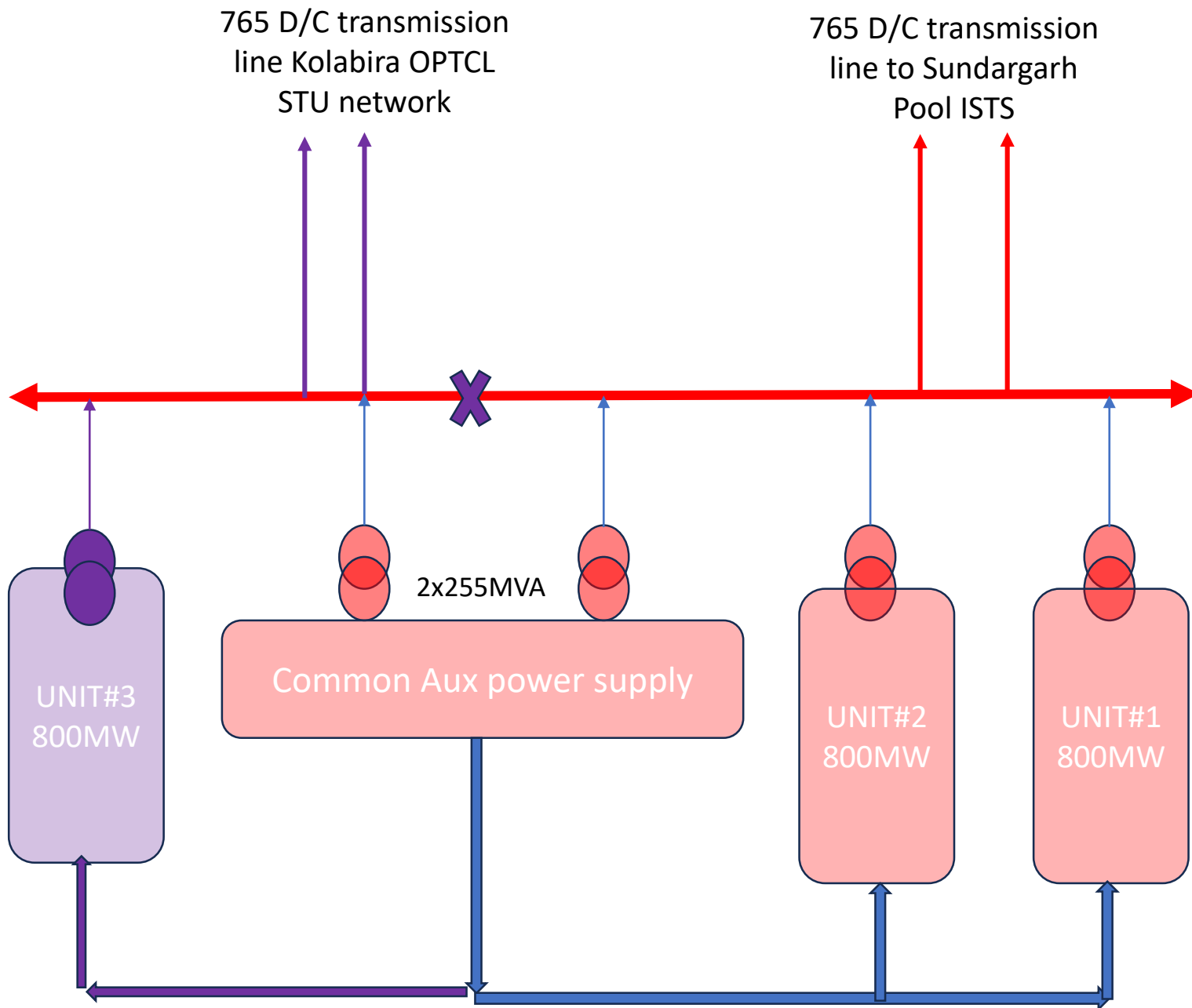


Follow us on:  /ntpc1  /ntpcltd1  /ntpclimited  /Company/ntpc  /ntpclimited

1. NTPC has undertaken Darlipali Stage-II (1X800MW) unit within the existing premises of Stage-I (2 x 800 MW units).
2. Stage-I is currently connected to the Inter-State Transmission System (ISTS) via One number 765 kV Double Circuit (D/C) line to the Sundargarh pool ISTS Substation.
3. NTPC has submitted dual connectivity for Stage-II i.e STU/OPTCL for 400MW & ISTS for 400MW to CTUIL under Regulation 4.1(a). Furthermore, NTPC requested grant of connectivity at the existing switchyard of Darlipali stage-I generating station which is already connected to the ISTS, as per the provisions of Regulation 5.6 of the CERC Grant of Connectivity and GNA Regulations.
4. CTUIL informed, for Stage-II ISTS quantum (400MW) existing ISTS Transmission line is sufficient & no further line is required even for full quantum of 800MW .
5. During CMETS meeting discussion, GRIDCO informed to build 1no 765KV, D/C line from Darlipali STPP to Kolabira OPTCL 765KV/400KV S/S to draw their share of power & further informed they will not charge transmission charges for any ISTS power.



- For stage-I also 50% of power is allocated to Odisha & balance is allocated to other ISTS beneficiaries and Odisha also drawing its share through ISTS Line.
- The plant Auxiliary supply is drawn from 2 nos. 765/132kV Tie transformers with 132KV switchyard for station TF & other plant load requirements.
- The new unit is connected directly to stage-I generation switchyard through a dedicated GT bay. However, the auxiliary supply system for stage-I and II is common and shall be drawn through the existing 132KV switchyard of stage-I. Attached SLD .



- The connectivity application for Stage-II submitted earlier in feb'24 for 800MW ISTS connectivity & power evacuation was accepted with existing ISTS Tr.lines. Grid India has not raised requirement of Bus sectionaliser b/w stage-I & II at the time .This application had cancelled due to delay in EC/FC for the project.
- Now Grid-India has suggested to provide Bus sectionaliser b/w Stage-I & II. As the existing ISTS lines serving to evacuate Stage-I as well as ISTS quantum of stage-II, the Bus sectionaliser shall be always kept in closed & it is technically redundant.
- Energy accounting can be calculated based on meter reading at outgoing lines, metering at generator end & auxiliary power consumption at HT side of TT for energy accounting of Stage-I & Stage-II generation.

NTPC humbly requests the following:

- 1) Connectivity of unit#3 may be allowed at existing bus of stage-I as per the provisions of the CERC Grant of connectivity and GNA regulations 2022.
- 2) Requirement of Additional Bus section between stage-I and II may be waived off.
- 3) Detailed Metering scheme shall be as approved in the ERPC-CCM meetings at a later date.
- 4) Metering philosophy may be considered in line with other multi stage NTPC generating stations where new units are connected to common bus with out bus sectionaliser & sharing the common Auxiliary supply systems.

THANK YOU

Additional pointers

1) In case of metering of B/s:

Stage-I injection = ISTS lines+ B/S

Stage-II injection = STU lines + B/S

Only possible B/s location is between TT 1 and 2

However, since TT 1 and 2 are together feeding Aux. loads of 3 units and B/S is always close, this philosophy does not represent the true share of loading on TT's for the 2 stages.

2) In one of NTPC project having such arrangement with bus section as metering periphery between ISTS and STU systems, the STU connection section is being construed as generation embedded in the STU network leading to regulatory ambiguity.

Annexure B.2.21

Utilization of Transmission lines

ODISHA

Name of Line	Percentage utilization to meet ISTS power (outside ODISHA control area)							Remarks
	2019-20 Quarter 1	2019-20 Quarter 2	2019-20 Quarter 3	2019-20 Quarter 4	2020-21 Quarter 1	2020-21 Quarter 2	2020-21 Quarter 3	
400kV Indravati(GR)- Indravati(PG)	56.25%	42.63%	50.86%	64.14%	59.73%	67.33%	64.14%	-
400kV Rengali(PG)-Keonjhar	86.82%	88.60%	78.67%	88.01%	84.61%	83.16%	88.01%	-
400kV Keonjhar-Baripada	86.82%	88.60%	78.67%	88.00%	84.61%	83.15%	88.00%	-
400kV Baripada-Kharagpur	99.76%	99.91%	99.43%	25.12%	99.42%	99.17%	25.12%	-
220kV Jeynagar-Jeypore each CKT	90.02%	71.15%	69.81%	84.30%	82.98%	86.95%	84.30%	-
220kV Joda-JSPL	100.00%	84.14%	83.38%	84.27%	83.80%	85.54%	84.27%	-
132kV Joda-Kendposi	-	-	-	-	-	-	-	<i>Line in open condition</i>

*The figures indicated in the table corresponds to the percentage utilization to meet ISTS power (outside ODISHA control area).

Annex B.2.23

Status of Eastern Region

	Before 239 th OCC	Before 240 th OCC
No. of entities registered on CEA-Captive Power Portal	125	152

State-wise breakup of the number of known entities (replied in Format-21 for FY2024-25) which have **not yet registered** on CEA-Captive Power Portal

State	Nos.
Odisha	32
West Bengal	51
Jharkhand	8
Bihar	3

Entities pending for registration – Odisha

Sl. No.	Name Of Entity	District	State
1	JINDAL STEEL AND POWER LIMITED	Anugul	Odisha
2	Jindal Stainless Limited	Jajapur	Odisha
3	HINDALCO INDUSTRIES LIMITED	Sambalpur	Odisha
4	SMC Power Generation Ltd. Unit-II	Jharsuguda	Odisha
5	SHREE GANESH METALIKS LIMITED	Sundergarh	Odisha
6	Shri Jagannath Steel and Power Limited	Koenjhar	Odisha
7	BR Steel & Power Pvt Ltd	Sambalpur	Odisha
8	SREE METALIKS LIMITED	Kendujhar	Odisha
9	TATA Steel Sponge Iron Joda	Koenjhar	Odisha
10	JINDAL STEEL AND POWER LIMITED	Koenjhar	Odisha
11	Hindustan Aeronautics Limited	Koraput	Odisha
12	Ardent Steel Private Limited	Kendujhar	Odisha
13	Indian Metals and Ferro Alloys Ltd	Rayagada	Odisha
14	Mangalam Timber Products Ltd	Nabarangapur	Odisha
15	Essel Mining and Industries Limited	Kendujhar	Odisha
16	sudhakar pvc products pvt ltd	Ganjam	Odisha
17	BOUDH DISTILLERY PVT.LTD.	Boudh	Odisha
18	LARSEN AND TOUBRO LIMITED	Sundergarh	Odisha
19	TATA STEEL FAP BAMNIPAL	Kendujhar	Odisha
20	MSP SPONGE IRON LTD	Keonjhar	Odisha
21	BARSUA IRON MINES - SAIL	Sundargarh	Odisha
22	M/s RSB Transmissions (I) Ltd - Forging Unit	Cuttack	Odisha
23	Apollo hospitals enterprise limited	Khordha	Odisha
24	THE RAMCO CEMENTS LIMITED	Jajapur	Odisha
25	VIKRAM PVT LTD	Sundergarh	Odisha
26	IDL EXOLOSIVE LTD	Sundargarh	Odisha
27	Sagar Cements Ltd.	Jajpur	Odisha
28	DLF Cybercity	Khordha	Odisha
29	MANGANESE GROUP OF MINES	Kendujhar	Odisha
30	Arya Iron & Steel Company Pvt Ltd	Koenjhar	Odisha
31	TRB Iron Ore Mines Jindal Steel and Power	Sundargarh	Odisha
32	Ultra Tech Cement Limited	Jharsuguda	Odisha

Entities pending for registration – West Bengal

Sl. No.	Name Of Entity	District	State
1	Shyam Sel & Power Ltd	Barddhaman	West Bengal
2	PMD SAIL DSP	Barddhaman	West Bengal
3	Matix Fertilisers and Chemicals	Purba Bardhaman	West Bengal
4	HIMADRI SPECIALITY CHEMICAL LTD	Hooghly	West Bengal
5	SHYAM STEEL INDUSTRIES LTD	Paschim Bardhaman	West Bengal
6	KESORAM RAYON (unit of Cygnet Industries Ltd)	Hoogly	West Bengal
7	Gloster Ltd	Howrah	WEST BENGAL
8	GILLANDERS ARBUTHNOT AND CO LTD UNIT GIS COTTON MILL	Hugli	WEST BENGAL
9	Sukhjit Starch industries	Malda	WEST BENGAL
10	TATA Metaliks Limited	West Midnapur	WEST BENGAL
11	BENGAL BEVERAGES PVT LTD - (UNIT-II)	Hugli	WEST BENGAL
12	Epic Agro Products Pvt. Ltd.	Howrah	WEST BENGAL
13	BENGAL BEVERAGES PVT LTD	Hugli	WEST BENGAL
14	HUL Haldia	Purba Medinipur	WEST BENGAL
15	NATIONAL INSTITUTE OF TECHNOLOGY DURGAPUR	Barddhaman	WEST BENGAL
16	Airport Authority Of India	North Twenty Four Parganas	WEST BENGAL
17	SHREE RENUKA SUGARS LTD	PURBA MEDINIPUR	WEST BENGAL
18	Mother Dairy Calcutta	Hoogly	WEST BENGAL
19	KHARAGPUR WORKSHOP	Paschim Medinipur	WEST BENGAL
20	Danguajhar Tea Garden	Jalpaiguri	WEST BENGAL
21	Himalayan Builders	Darjeeling	WEST BENGAL
22	Tega Industries Ltd	South 24 Prgs.	WEST BENGAL
23	Geo Nutri Foods Pvt. Ltd.	Howrah	WEST BENGAL
24	Garden Reach Shipbuilders and Engineers Ltd(Main Unit)	Kolkata	WEST BENGAL
25	M/s Weighmatic (I) Pvt. Ltd	Kolkata	WEST BENGAL
26	Websol Energy System Limited	South 24 Prgs.	WEST BENGAL
27	ABP PVT LTD	Bankura	WEST BENGAL
28	EXIDE INDUSTRIES LTD.- Shyamnagar unit	North Twenty Four Parganas	WEST BENGAL

29	NILKAMAL LIMITED	Bankura	WEST BENGAL
30	DAMDIM TEA ESTATE A UNIT OF - M/S. AMALGAMATED PLANTATIONS PVT. LTD.	Jalpaiguri	WEST BENGAL
31	UltraTech Cement Limited (Dankuni Cement Works)	Hoogly	West Bengal
32	BRITANNIA INDUSTRIES LTD	South Twenty Four Parganas	WEST BENGAL
33	Maithan steel and power Ltd	Bardhaman	WEST BENGAL
34	SHARP FERRO ALLOYS LTD	Bardhaman	WEST BENGAL
35	TATA STEEL LTD , BEARINGS DIVISION	West Midnapur	WEST BENGAL
36	Durgapur Cement Works (Unit of Birla Corporation Ltd.)	Paschim Bardhaman	WEST BENGAL
37	Khadim India Ltd.	North Twenty Four Parganas	WEST BENGAL
38	Shyam Steel Manufacturing Ltd.	Bankura	WEST BENGAL
39	Ajantashoes India Pvt Ltd	Kolkata	West Bengal
40	IISER KOLKATA	Nadia	WEST BENGAL
41	Ajanta Shoes Pvt. Ltd.	Howrah	WEST BENGAL
42	MARCUS OILS AND CHEMICALS PVT. LTD.	Purba Medinipur	WEST BENGAL
43	Berger Paints India Ltd	Hugli	WEST BENGAL
44	Hindustan Unilever Limited	Kolkata	WEST BENGAL
45	BIRLA CORPORATION LIMITED	South Twenty Four Parganas	WEST BENGAL
46	Nezone Strips Limited	Hoogly	WEST BENGAL
47	RDB RASAYANS LIMITED	PURBA MEDINIPUR	WEST BENGAL
48	Adani Wilmar Limited	Purba Burdwan	WEST BENGAL
49	The Jay Shree Chemicals and Fertilisers	North Twenty Four Parganas	WEST BENGAL
50	Ajanta shoes (I) pvt.ltd	South 24 Prgs.	WEST BENGAL
51	Eveready Industries India Ltd.	Kolkata	WEST BENGAL

Entities pending for registration – Jharkhand

Sl. No.	Name Of Entity	District	State
1	SAIL , BOKARO STEEL PLANT	Bokaro	Jharkhand
2	Rungta Mines Limited Chaliyama Steel Plant	Saraikela-Kharsawan	Jharkhand
3	Hindustan Copper Limited	East Singhbhum	Jharkhand
4	UCIL Jadugoda mines	East Singhbhum	Jharkhand
5	Timken India Limited, Bara	East Singhbhum District	Jharkhand
6	Uranium Corporation of India Limited, Bagjata Mines	Purbi Singhbhum	Jharkhand
7	Carbon Resources Private Limited	Giridih	Jharkhand
8	Indo Danish Tool Room	Seraikela kharsawan	Jharkhand

Entities pending for registration – Bihar

Sl. No.	Name Of Entity	District	State
1	New Bihar Cement Plant	Aurangabad	Bihar
2	MAGADH SUGAR AND ENERGY LTD UNIT NSSM	Pashchim Champaran	Bihar
3	Majhauria Sugar Industries	West Champaran	Bihar